

General Relativity

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1 Introduction

General Relativity represents the pinnacle of classical physics and, as such, it is a rather complicated task writing lecture notes. On the one hand, the literature is full of very complete textbooks, see *e.g.* [Car19, HaEl73, Wal84] and, to my utmost surprise it is one of the few topics of human knowledge without even the shortest note written by Valter Moretti. Hence I had to start really from scratch and I soon realized that I had to give two courses merged in a single one. On the other hand, this is one of the few modern theories where one can avoid sacrificing mathematical rigor or physical clarity. Here you can merge the best of both worlds, but the risk that one of the two overtakes the other is huge. This means that I cannot take any detour: First of all I shall introduce you to the realm of differential geometry and then I shall use its full power to discuss Einstein's theory, from its original motivations to its main applications: gravitational waves, cosmology and black holes. I think I will not say more because a lengthy, may be even a little bit historical motivation leading to the formulation of general relativity is boring and many of you will probably skip it to get straight to the core of these notes (rather what I ask during the exam). After all, if the physical and mathematical motivations leading to general relativity will not be clear from my lecture notes, why I am giving them in the first place? So short introduction it is and let us get started.

2 Mathematical Background

Differential geometry is one of the main branches of modern mathematics. Its origin can be traced to Riemann, although influential works due to Beltrami, Gauss, Lobachevsky have paved the way to its formulations. For many decades it has been seen more as a mere curiosity since applications requiring a generalization of the ordinary Euclidean $(\mathbb{R}^d, \delta)$ or Minkowskian space $(\mathbb{R}^d, \eta)$, $d \geq 2$, were scarce. The coming or rather the advent (today it is the 08/12/2024, hence it makes sense) of General Relativity changed completely this attitude since it represents a physical theory whose natural language is that of differential geometry. Yet the goal of these notes is not to breed new differential geometers. This topic will be considered as a tool and, as such, it will be introduced and discussed only to the extent where it allows for a better understanding of general relativity.

Yes, I know that, among you, some if not many wish to know more about this topic and I can picture clearly in my mind your smirking face. In this case I can recommend delving in [Lee13] and in [BEE96, ONe83]. I am recommending books on semi-Riemannian/Lorentzian geometry since there are so many books on Riemannian geometry that any you will chose is certainly fine.

That said, let us get down to business, shall we? Differential geometry revolves around the concept of a differentiable manifold which is the mathematical transliteration of the necessity of describing generic geometric objects whose global structure can be wide but, locally, they must resemble in a suitable sense to $\mathbb{R}^d$, $d \geq 1$. The reasons for these requirements are several and, to a certain extent, they are imbued by pragmatism and physical sense. On the one hand we wish to translate the idea that each observer perceives the underlying space of configurations as being indistinguishable from $\mathbb{R}^d$ at least locally. This will see a precise formulation in the equivalence principles which will be discussed at a later stage. On the other hand, $\mathbb{R}^d$ is the natural arena of all analytic tools that we have learned in calculus. These are the backbone of all physical models and it would be unrealistic getting rid of them completely. We must be able at least to a certain extent to recover them and to make use of them.

In the following, we shall follow this guideline introducing the notion of manifold. We shall proceed in steps starting from a basic concept, which can be heuristically rephrased as *the building blocks of manifolds must be open sets*.

Definition 1. A **topological space** is a pair (M, τ) where M is a set and τ a collection of subsets of M such that

1. $\emptyset$ and $M \in \tau$,
2. $\forall \tau_1, \tau_2 \in \tau$, then $\tau_1 \cap \tau_2 \in \tau$,
3. if I is an arbitrary, not necessarily finite collection of indices, then, if $\tau_s \in \tau$ for all $s \in I$, it holds that $\bigcup_{s \in I} \tau_s \in \tau$.

If one of our beloved readers knows a different definition of topological space, which is based on the notion of closed sets, we highlight that the de Morgan's laws guarantee the equivalence with Definition 1. Let us now look at a pair of examples:

Example 2. Definition 1 is satisfied by $\mathbb{R}^n$, $n \in \mathbb{N}$, considering the collection of all open balls $B_{\delta, x_0} \doteq \{x \in \mathbb{R}^n \mid d(x - x_0) < \delta\}$, where $x_0 \in \mathbb{R}^n$ and where $\delta \in \mathbb{R}$. Here $d(x - x_0)$ is the Euclidean distance. The *standard topology* of $\mathbb{R}^n$ is constructed as the collection of all possible, countable unions of the sets B_{δ, x_0} varying among all $x_0 \in \mathbb{R}^n$ and among all values of δ .

Observe that topologies are huge and there is a good reason for that, namely we want to be able to account for all possible notions of open subset of a set M . One might think that, given M , the associated topology is unique, but this is not the case. As an example consider $\mathbb{R}$ and

- $\tau_1 = \{\emptyset, M\}$,
- $\tau_2 = \{\emptyset, M, (0, \infty), (-\infty, 0]\}$.

Both τ_1 and τ_2 abide by Definition 1, but they contain way less elements than the standard one in Example 2. The reason is that these are coarser topologies and this entails drastic differences when studying analytic objects defined on top of these spaces as any veteran of functional analysis should have by now ingrained in his soul. Following the example of finite dimensional vector

spaces where the linear structure guarantees that all elements can be reconstructed starting only from a finite number of vectors by taking linear combinations, we introduce a useful notion.

Definition 3. *Given a topological space (M, τ) as per Definition 1, we call **basis for (M, τ)** a collection of open subsets $\mathcal{B} \subset \tau$ such that, $\forall x \in M$ and $\forall U \in \tau$ such that $x \in U$, there exists $V \in \mathcal{B}$ such that $x \in V$ and $V \subseteq U$.*

Definition 3 entails that

1. for all $U \in \tau$, U can be realized as the union of elements in $\mathcal{B}$. One can show that this condition is actually equivalent to Definition 3 itself.
2. $\forall x \in M$, $\exists U \in \mathcal{B}$ such that $x \in U$ and, if $x \in U_1 \cap U_2$, then $\exists U_3 \in \mathcal{B}$ such that $U_3 \subset U_1 \cap U_2$ and $x \in U_3$.

Remark 1. *In view of Definition 3 we can infer that, considering $\mathbb{R}$, all open subsets of the form (a, b) form a basis of the standard topology as per Definition 1. The reader is invited to work out the generalization on $\mathbb{R}^n$. In addition, observe that, if we consider the topologies τ_1, τ_2 introduced above, a basis of the topology is in the first case τ_1 itself while, in the second one, $\mathcal{B} = \{(0, \infty), (-\infty, 0]\}$.*

The following proposition, whose proof is left to the reader, entails that a topology for a set M can be actually built only by assigning its base.

Proposition 4. *Let M be a set and let a collection $\mathcal{B}$ of subsets of M be assigned as being a basis for a topology τ of M . Then τ is constructed as the collection of all possible unions of elements lying in $\mathcal{B}$.*

The generality of Definition 1 is also its greatest weakness, since it considers at the same level different topologies, some of which have properties which are by far not desirable. The following example clarifies why we shall specialize this notion in our path to defining a manifold first and a spacetime later.

Example 5. Consider $\mathbb{R}$ and let X be the set constructed considering two copies of $\mathbb{R}$ and identifying a point x in the first copy of $\mathbb{R}$ with the same point x in the second one, provided that $x \neq 0$. The ensuing object is known as *the line with two origins* as the zeros of the two copies of $\mathbb{R}$ are still separate entities. We can build a topology for X using Proposition 4 and considering once all intervals of the form $(a, b) \subset \mathbb{R}$ provided that $0 \notin (a, b)$, while, if 0 lies therein, we consider two copies, one containing the origin of the first copy of $\mathbb{R}$, the other that of the second one.

The main deficiency of this example lies in the fact that the topology defined does not allow to separate points, namely we cannot find two sets in τ such that the origin of the first copy of $\mathbb{R}$ lies therein, but not that of the second one. Heuristically, translating this feature in a physical language, two different observers sitting at the separate origins cannot be realized as such by our notion of open subsets. Since topologies are the building blocks to define the notion of continuity

of functions and since these are used in physical models as the building block for constructing observables, the lack of separability between points is an unwanted feature. Different physical phenomena occurring at these points would not be distinguishable.

Definition 6. Let (M, τ) be a topological space. We say that it is **Hausdorff** (or T_2 – Trennung 2) if, for every $p, q \in M$ such that $p \neq q$, $\exists U, V \in \tau$ such that $p \in U$, $q \in V$ and $U \cap V = \emptyset$

Henceforth we shall be only considering topologies which are Hausdorff and a closer look to the example outlined above unveils that the standard topology of $\mathbb{R}^n$, $n \geq 1$, see Example 2, lies in this class. On the contrary the topologies on $\mathbb{R}$, denoted by τ_1 and τ_2 , do not separate points and, for this reason, we discard them. Definition 6 helps in restricting the class of topological spaces of our interest but it does not suffice. As next step, we codify the heuristic statement that we look for spaces such that every observer/point therein can find a set which is indistinguishable from an open subset of $\mathbb{R}^n$, $n \geq 1$.

Definition 7. An n -dimensional topological variety $M \equiv (M, \tau)$, $n \geq 1$, is a topological, Hausdorff space as per Definition 1 and 6, such that $\forall p \in M$, $\exists U \in \tau$, homeomorphic to an open subset of $\mathbb{R}^n$.

Remark 2. We recall that, given two topological spaces (M_1, τ) and (M_2, τ') , an homeomorphism is a continuous map $f : M_1 \rightarrow M_2$ such that $f^{-1} \in C^0(M_2; M_1)$. Continuity is equivalent to stating that the pre-image of every open set is an open set. In the case of Definition 7, the rôle of (M_1, τ_1) is played by (U, τ_U) where the elements of τ_U are the intersection between U and all elements lying in τ , the topology of M .

Topological varieties are a rather interesting topic and we refer to [Lee11] for additional information. While the Hausdorff property is easy to justify from a pragmatic viewpoint, less transparent is the following condition, which is nonetheless fundamental at a technical level as we shall clarify in the following sections.

Definition 8. Let (M, τ) be a topological space as per Definition 1. We say that it is **second countable** if there exists a countable collection $\mathcal{U} \equiv \{U_i\}_{i \in \mathbb{N}}$ of elements of τ , such that every $U \in \tau$ can be written as the union of elements lying in $\mathcal{U}$.

We define instead the main mathematical structure used in general relativity.

Definition 9. A topological variety $M \equiv (M, \tau)$ as per Definition 7 is called a **(smooth) n -dimensional manifold**, $n \geq 1$, if it is second countable as per Definition 8 and if it is endowed with a differentiable structure, namely

1. there exists a collection of indices I and of open subsets $U_\alpha \in \tau$, $\alpha \in I$, as well as of homeomorphisms $\varphi_\alpha : U_\alpha \rightarrow O_\alpha \subset \mathbb{R}^n$ such that $\bigcup_{\alpha \in I} U_\alpha = M$. The collection of open subsets $\{U_\alpha\}_{\alpha \in I} = M$ is an **atlas** of M .
2. $\forall \alpha, \beta \in I$, the maps $\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \mathbb{R}^n$ are smooth. These are called **transition maps**,

3. The collection $\mathcal{F} \doteq \{U_\alpha, \varphi_\alpha\}_{\alpha \in I}$ is **maximal**, namely, if there exists (U, φ) with $U \in \tau$, $\varphi : U \rightarrow O_\varphi \subset \mathbb{R}^n$ and $\varphi \circ \varphi_\alpha^{-1} \in C^\infty(O_\alpha \cap O_\varphi; O_\alpha \cap O_\varphi)$, $\varphi_\alpha \circ \varphi^{-1} \in C^\infty(O_\alpha \cap O_\varphi; O_\alpha \cap O_\varphi)$, then $(U, \varphi) \in \mathcal{F}$.

Remark 3. Observe that an alternative, yet equivalent definition of maximal atlas $\mathcal{F}$ is that there does not exist any other atlas of M containing it properly.

It is worth mentioning that, in Definition 9, smoothness of all maps is a rather strong condition from a mere mathematical viewpoint, which is justified on physical grounds. Strictly speaking, one could replace it with a milder counterpart, namely being C^k , $k \geq 0$. In this case we say that M is *k-differentiable manifold*. In addition, we emphasize the following nomenclature, subordinated to Definition 9:

- Each pair $(U_\alpha, \varphi_\alpha) \in \mathcal{F}$ is called a **local chart** of M , not to be misunderstood with local coordinates!
- Given a local chart $(U_\alpha, \varphi_\alpha)$ of M and a collection of n -linearly independent maps $r_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, n$, we call $x_i : r_i \circ \varphi_\alpha : U_\alpha \rightarrow \mathbb{R}$, $i = 1, \dots, n$, a **coordinate system** on U_α .

Remark 4. A reader might be surprised by the definition of coordinate system since, in many books, local charts and coordinate systems are thought as synonyms. Yet the author prefers to avoid this habit since it hides a freedom which plays an important rôle in many models. On top of that, if one thinks of $\mathbb{R}^n$ as a manifold of its own, one can choose $U = \mathbb{R}^n$ and $\varphi = id$. Clearly, in this instance, we would not call $(\mathbb{R}^n, id)$ a coordinate system.

The effectiveness of a definition lies first and foremost in the set of examples abiding by it that one can construct. Yet, applying Definition 9 might be quite tricky since one has to prove that a given atlas is maximal. The following propositions helps us avoiding this quandary. The reader should bear in mind that our goal is to study general relativity and not differential geometry. Hence we shall omit the proofs.

Proposition 10. [Lee13, Prop 1.17] Let M be a topological variety as per Definition 7. Then

1. Every smooth atlas $\mathcal{A}$ for M is contained in a unique, maximal smooth atlas $\mathcal{F}$, which is known as the smooth structure determined by $\mathcal{A}$,
2. Two smooth atlases for M determine the same smooth structure if their union is a smooth atlas for M .

Even more effective is the *smooth manifold chart lemma*:

Lemma 11. [Lee13]–[Lem. 1.35] Let M be a set and let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in I}$, I being a countable index set, be a collection of subsets of M such that $\bigcup_{\alpha \in I} U_\alpha = M$ and

1. $\forall \alpha \in I$, the map $\varphi_\alpha : U_\alpha \rightarrow \varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$ is a bijection between open subsets,

2. $\forall \alpha, \beta \in I$, the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open subsets of $\mathbb{R}^n$,
3. whenever $U_\alpha \cap U_\beta \neq \emptyset$, $\varphi_\alpha \circ \varphi_\beta^{-1} \in C^\infty(\varphi_\beta(U_\alpha \cap U_\beta); \varphi_\alpha(U_\alpha \cap U_\beta))$,
4. $\forall p, q \in M$ with $p \neq q$, either there exists U_α containing both p and q or there exist U_α and U_β such that $p \in U_\alpha$, $q \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$.

Then M admits a unique smooth manifold structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

It is worth observing that Proposition 10 identifies a smooth manifold via a two-step procedure. In the first one, a set M is endowed with the structure of a topological variety and, subsequently, a smooth structure is identified. In Lemma 11 these two operations are combined in a single body. Observe that, in this statement, the Hausdorff property is a hidden consequence of condition 4., while M is second countable on account of condition 3.. We are now ready to give a few examples of manifolds. It is worth mentioning that one can construct several exotic examples of manifolds, but their applicability to general relativity is rather limited. One should not discard them a priori, nor the author thinks or claims that they are useless. Simply they will not play a rôle in these notes (all right, all right... they are useless if I do not ask them at the exam...).

Example 12. Consider $\mathbb{R}^n$, $n \geq 1$. These are manifolds endowing them with the standard topology τ as per Example 2. This is a topological Hausdorff space considering for every $U \in \tau$ the homeomorphism $\text{id} : U \rightarrow U$, where id is the identity map. Per construction this is also second countable. We can apply Proposition 10 to conclude that $\mathbb{R}^n$ is a manifold, as it should be!!! The smooth structure identified is called *standard*.

Observe that a similar conclusion can be drawn for $\mathbb{C}^n$, $n \geq 1$, provided one identifies it with $\mathbb{R}^{2n}$. There exists a notion of *complex manifold* which will not be discussed in these notes, but it is a natural extension of Definition 9. In addition we emphasize that we have never stated a uniqueness result for smooth structures and, even on $\mathbb{R}$, this is not the case. The following exercise helps clarifying this point:

Exercise 13. Let $\mathbb{R}$ be seen as a topological variety with respect to the standard topology as per Example 2. Show that, if one replaces in Example 12 the identity map with $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}$ such that $x \mapsto \varphi(x) = x^3$, it identifies a smooth structure different from the standard one.

In order to discuss additional examples, a rather convenient trick, given a set, is to endow it with a topology which is induced by the existence of a map from such a set into a known topological space.

Definition 14. Let X be a set and let (Y, τ_Y) be a topological space as per Definition 1. Given $f : X \rightarrow Y$, we call **inverse image topology** on X the collection τ_X of subsets such that $U \in \tau_X$ if $\exists V \in \tau_Y$ with $U = f^{-1}(V)$. If X is a subset of Y and f is the injection map, then τ_X is called **induced topology**.

Definition 15. Let (X, τ_X) be a topological space as per Definition 1 and let Y be a set. Given $f : X \rightarrow Y$, we call **direct image topology** on Y the collection τ_Y of subsets such that $V \in \tau_Y$ if $\exists U \in \tau_X$ with $U = f^{-1}(V)$. If the map f is surjective, τ_Y is called **quotient topology**.

The net advantage of these two definitions is that the new topologies individuated inherit the structural properties of the first one, in particular that of being Hausdorff and second countable.

Example 16. The (unit) 2-sphere $\mathbb{S}^2$ is a smooth manifold. To show it, we realize it as

$$\mathbb{S}^2 \doteq \{(x^1, x^2, x^3) \in \mathbb{R}^3 \mid (x^1)^2 + (x^2)^2 + (x^3)^2 = 1\},$$

namely as a subset of $\mathbb{R}^3$. In other words there exists an injective map $\iota : \mathbb{S}^2 \rightarrow \mathbb{R}^3$ which entails that, applying Definition 14, $\mathbb{S}^2$ is a topological variety endowed with the induced topology from the standard one on $\mathbb{R}^3$. In order to apply Proposition 10, we need to identify an atlas. We exploit the Mercatore projection. Let us consider two antipodal points $P = (0, 0, 1)$ and $Q = (0, 0, -1)$. An atlas $\mathcal{A}$ can be constructed from the open subsets (U_1, U_2) , where $U_1 \doteq \mathbb{S}^2 \setminus \{P\}$, while $U_2 \doteq \mathbb{S}^2 \setminus \{Q\}$. We define the maps $\varphi_i : U_i \rightarrow O_i \subset \mathbb{R}^2$, $i = 1, 2$,

$$\begin{aligned}\varphi_1(x^1, x^2, x^3) &= (y^1, y^2) = \left(\frac{2x^1}{1-x^3}, \frac{2x^2}{1-x^3} \right), \\ \varphi_2(x^1, x^2, x^3) &= (z^1, z^2) = \left(\frac{2x^1}{1+x^3}, \frac{2x^2}{1+x^3} \right).\end{aligned}$$

These are two homeomorphisms and we need to check that the transition maps $\varphi_2 \circ \varphi_1^{-1} : O_1 \cap O_2 \sim \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2$ and $\varphi_1 \circ \varphi_2^{-1} : O_1 \cap O_2 \sim \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2$ are smooth. A direct computation shows that $\varphi_2(\varphi_1^{-1}(y^1, y^2)) = (z^1, z^2)$ where

$$z^1 = \frac{4y^1}{(y^1)^2 + (y^2)^2}, \quad z^2 = \frac{4y^2}{(y^1)^2 + (y^2)^2}.$$

A similar result holds true mutatis mutandis for $\varphi_1 \circ \varphi_2^{-1}$. We have therefore identified an atlas for $\mathbb{S}^2$, the sought conclusion.

Exercise 17. Prove that all spheres $\mathbb{S}^n$, $n \geq 1$ are smooth manifolds.

Example 18. All real projectives planes $\mathbb{RP}^n$, $n \geq 1$ are smooth manifolds. These are defined as the equivalence classes

$$\mathbb{RP}^n = \frac{\mathbb{R}^{n+1} \setminus \{0\}}{\sim} \doteq \{[x] \mid x = (x^1, \dots, x^{n+1}) \in \mathbb{R}^{n+1} \setminus \{0\} \text{ and } x \sim x' \text{ if } \exists \lambda \neq 0 \mid x' = \lambda x\}.$$

Observe that $\mathbb{R}^{n+1} \setminus \{0\}$ is a topological variety if endowed with the induced topology from $\mathbb{R}^{n+1}$ via the canonical embedding map $\iota : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{R}^{n+1}$. We can now consider the surjective projection map $\pi : \mathbb{R}^{n+1} \setminus 0 \rightarrow \mathbb{RP}^n$, such that $\pi(x) = [x]$ for all $x \in \mathbb{R}^{n+1} \setminus \{0\}$. Applying Definition 15 we endow $\mathbb{RP}^n$ with the quotient topology $\tau_{\mathbb{RP}^n}$ induced from that of $\mathbb{R}^{n+1} \setminus \{0\}$.

In order to apply Proposition 10, consider the collection of open subsets lying in $\tau_{\mathbb{RP}^n}$ such that

$$U_i \doteq \{[x] \in \mathbb{RP}^n \mid x = (x^1, \dots, x^{n+1}) \text{ and } x^i \neq 0\},$$

for all $i = 1, \dots, n$. Observe that $\{U_i\}_{i=1}^n$ covers M and define the maps $\varphi_i : U_i \rightarrow O_i \subseteq \mathbb{R}^n$ such that, for all $i = 1, \dots, n$

$$\varphi_i([x]) = \left(\frac{x^1}{x^i}, \dots, \frac{x^{n+1}}{x^i} \right),$$

where the i -th entry is omitted. Consider two overlapping open subsets U_i and U_j and the transition functions $\varphi_i \circ \varphi_j^{-1} : U_i \cap U_j \rightarrow \mathbb{R}^n$. Using the notation ${}_i\zeta_h \doteq \frac{x^h}{x^i}$ with $i \neq h$, we obtain

$$\begin{cases} {}_j\zeta_h = \frac{{}_i\zeta_h}{{}_i\zeta_j} & h \neq i, j \\ {}_j\zeta_i = \frac{1}{{}_i\zeta_j} \end{cases},$$

which are smooth functions. We can therefore draw the sought conclusion.

Exercise 19. Prove that the following groups are smooth real manifolds:

- $GL(n, \mathbb{R}) = \{A \in \mathcal{M}(n, \mathbb{R}) \mid \det(A) \neq 0\}$ for all $n \geq 1$,
- $O(n) \equiv O(n; \mathbb{R}) = \{U \in \mathcal{M}(n, \mathbb{R}) \mid U^T U = U U^T = \mathbb{I}_{n \times n}\}$, $n \geq 1$, where $\mathbb{I}_{n \times n}$ is the identity matrix.
- $U(n) = \{U \in \mathcal{M}(n, \mathbb{C}) \mid U^* U = U U^* = \mathbb{I}_{n \times n}\}$, $n \geq 1$, where $\mathbb{I}_{n \times n}$ is the identity matrix.

2.1 Tangent Space

Having established in the previous section the notion of a smooth manifold, see Definition 9, we observe that, in general, the linear structure of $\mathbb{R}^n$, $n \geq 1$, is lost, since a manifold need not to be a vector space. This is a byproduct of the existence only of local homeomorphisms between a manifold M and $\mathbb{R}^n$. While recovering this peculiar global structure of $\mathbb{R}^n$ is impossible, we can replace it with a local counterpart.

To this end, we need to give an alternative representation and interpretation to the linear structure of $\mathbb{R}^n$. The following discussion is very analytic in spirit and, in the literature, there are several alternative approaches which we will not discuss. We assure our beloved readers that they are all equivalent and, should you be asked during the exam, you are completely free to use your preferred one.

We start by considering $\mathbb{R}^n$, $n \geq 1$, endowed with the standard Euclidean coordinates $x_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, n$ as well as with standard basis e_i , $i = 1, \dots, n$. Given a vector $v \in \mathbb{R}^n$, we can decompose it as $v = v^i e_i$, where we are employing Einstein's summation convention. Associated to this vector, we can define a directional derivative

$$D_v : C^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n), \quad f \mapsto D_v(f) \doteq \sum_{i=1}^n v^i \frac{\partial f}{\partial x^i}.$$

In other words we have established a 1:1 correspondence between vectors in $\mathbb{R}^n$ and directional derivatives, that is

$$v \equiv D_v.$$

This suggests, on the one hand, that we can employ the symbol v both to denote an element of $\mathbb{R}^n$ and a directional derivative. Going one step further, we are entitle to actually define a vector on $\mathbb{R}^n$ as a linear map $v : C^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n)$ abiding by the Leibnitz rule:

$$\begin{cases} v(af + bh) = av(f) + bv(h), & \forall f, h \in C^\infty(\mathbb{R}^n), \quad \forall a, b \in \mathbb{R} \\ v(fh) = v(f)h + fv(h), & \forall f, h \in C^\infty(\mathbb{R}^n) \end{cases}.$$

This correspondence has the merit of being extensible to smooth manifolds exploiting the fact that the notion of derivative is intrinsically local. Yet, to make this heuristic comment precise, we need first of all to define what is a k -times differentiable (smooth) function, $k \geq 1$.

Definition 20. *Let M, N be two smooth manifolds as per Definition 9. Given a map $f : M \rightarrow N$, we say that it is **k-times differentiable** (resp. **smooth**), $k \geq 1$, if, for every $(U, \psi) \in \tau_M$ and for every $(V, \varphi) \in \tau_N$, $\varphi \circ f \circ \psi^{-1} \in C^k(\psi(U); \varphi(V))$, $k \geq 0$ (resp C^∞).*

We have all the ingredients to generalize to a smooth manifold M the notion of a vector seen as a derivative of a function. First and foremost we need to observe that the corresponding notion need to be local since differentiability of a function is tied to local charts as highlighted by Definition 20. As starting point, we observe that, when considering the derivative of a function at a given point of an underlying manifold, we are only interested in its behaviour close to such point. This can be encoded in the notion of **set of germs** at $p \in M$

$$\mathcal{F}_p \doteq \left\{ [f] \mid f \in C^\infty(M) \text{ e } f \sim f', \text{ se } \exists U_p \subset M \mid (f - f')|_{U_p} = 0 \right\}, \quad (1)$$

where U_p is an open subset of M containing p . We should bear in mind that, to each germ $[f] \in \mathcal{F}_p$, we can associate a distinguished real number $[f] \mapsto f(p)$.

Proposition 21. *The collection of germs $\mathcal{F}_p$ at a point $p \in M$ as per Equation (1) is an algebra over the real numbers. In other words, it is a real vector space endowed with a distributive, bilinear map*

$$\cdot : \mathcal{F}_p \times \mathcal{F}_p \rightarrow \mathcal{F}_p, \quad [f] \cdot [f'] \doteq [ff'].$$

Proof. First of all, we need to verify that $\mathcal{F}_p$ is a real vector space. This is a direct consequence of the existence of the well-defined maps $[f] + [f'] \doteq [f + f']$ and $\lambda[f] = [\lambda f]$ for all $[f], [f'] \in \mathcal{F}_p$ and for all $\lambda \in \mathbb{R}$. The map $+$ is commutative and it admits the neutral element, $[0]$. In addition, for all $[f] \in \mathcal{F}_p$ the inverse is $[-f]$. Also the property of $\cdot$ being distributive is a consequence of its definition, hence we omit proving it explicitly, leaving it as an exercise  □

Given a generic algebra $\mathcal{A}$, especially noteworthy are the following subsets:

Definition 22. *Let $(A, \cdot)$ be an algebra over the real numbers. We call **subalgebra** a linear subspace $B \subset A$ closed under the action of $\cdot$. A subalgebra is called a **left ideal** if*

1. $a + b \in B$ and $\lambda a \in B$ for all $a, b \in B$ and for all $\lambda \in \mathbb{R}$,

2. $c \cdot a \in B$ for all $a \in B$ and for $c \in A$.

Similarly we can define a **right ideal** though the last condition is replaced by the requirement $a \cdot c \in B$. An ideal which is both left and right is called *bilateral*.

Henceforth we shall write for simplicity *ideal* in place of *bilateral ideal*

Lemma 23. *The set $\tilde{\mathcal{F}}_p \doteq \{[f] \in \mathcal{F}_p \mid f(p) = 0\}$ is an ideal of $\mathcal{F}_p$.*

Proof. One can verify by direct inspection that $\tilde{\mathcal{F}}_p$ is a linear subspace of $\mathcal{F}_p$ closed under the action of $\cdot$. In order to establish that $\tilde{\mathcal{F}}_p$ is an ideal, observe that for all $[f] \in \mathcal{F}_p$ and for all $[f'] \in \tilde{\mathcal{F}}_p$, it holds that $[f] \cdot [f'] = [ff']$ and $[f'] \cdot [f] = [f'f]$. At $p \in M$ they both take the value $f(p)f'(p) = 0 = f'(p)f(p)$ and hence they lie in $\tilde{\mathcal{F}}_p$. $\blacksquare$

A notable consequence of introducing $\tilde{\mathcal{F}}_p$ as per Lemma 23 is that it induces a filtration in terms of the sets $\tilde{\mathcal{F}}_p^k$ ($k \in \mathbb{N}$):

$$\tilde{\mathcal{F}}_p^k \doteq \text{span}_{\mathbb{R}} \left\{ [h] \in \tilde{\mathcal{F}}_p \mid \exists [f_i] \in \tilde{\mathcal{F}}_p, i = 1, \dots, k \text{ such that } [h] = [f_1] \cdot \dots \cdot [f_k] \right\}.$$

These are all real, linear and finite combinations of k -products of elements lying in $\tilde{\mathcal{F}}_p$. It descends

$$\dots \subset \tilde{\mathcal{F}}_p^k \subset \dots \subset \tilde{\mathcal{F}}_p^2 \subset \tilde{\mathcal{F}}_p \subset \mathcal{F}_p.$$

We can use these sets to establish a notion of vector at a generic point of a smooth manifold.

Definition 24. *Given a smooth manifold M as per Definition 9, a **tangent vector** v at $p \in M$ is a derivative operator on $\mathcal{F}_p$ as per Equation (1), namely a map $v : \mathcal{F}_p \rightarrow \mathbb{R}$ such that*

1. $v([f] + \lambda[f']) = v([f]) + \lambda v([f']), \quad \forall [f], [f'] \in \mathcal{F}_p \text{ e } \forall \lambda \in \mathbb{R}$
2. $v([f] \cdot [f']) = v([f]) f'(p) + f(p) v([f']). \quad \forall [f], [f'] \in \mathcal{F}_p$

The collection of all tangent vectors is denoted by $\mathbf{T}_p M$, the **tangent space** of M at p .

Remark 5. *Given a smooth manifold M and $p \in M$, if we denote by $[k] \in \mathcal{F}_p(M)$ the germ generated by the constant function, then, for every $v \in T_p M$, it holds that $v([k]) = v(k[1]) = kv([1])$. Yet*

$$v([1]) = v([1] \cdot [1]) = [1](p)v([1]) + v([1])[1](p) = 2v([1]) \implies v([1]) = 0.$$

This suggests that, in characterizing the collection of all tangent vectors at a point of a manifold, we can work with $\tilde{\mathcal{F}}_p$ rather than with $\mathcal{F}_p$. As a matter of fact, given any $f \in C^\infty(M)$, we can consider the equivalence class $[f - f(p)] \in \tilde{\mathcal{F}}_p$ since, for all $v \in T_p M$, $v([f]) = v([f - f(p)])$.

We can establish the main structural result concerning the tangent space at point p of a smooth manifold M .

Theorem 25. *Let M be a smooth manifold as per Definition 9. The following two statements are equivalent:*

1. *there exists an isomorphism between $T_p M$ and the space dual to $\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}$,*

$$2. \dim(M) = \dim\left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right) = \dim\left(\left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right)^*\right) = \dim(T_p(M)).$$

Proof. 1) Let $v \in T_p M$. Per restriction, this is a real valued linear function on $\tilde{\mathcal{F}}_p$. Furthermore, if $[f] \in \tilde{\mathcal{F}}_p^2$, there must exist $[f'], [f''] \in \tilde{\mathcal{F}}_p$ such that $[f] = [f'] \cdot [f'']$. This entails that $v([f]) = v([f'] \cdot [f'']) = v([f'])f''(p) + f'(p)v([f'']) = 0$. In other words we have identified via v a real valued linear map on $\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}$, that is $v \in \left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right)^*$.

In order to prove the converse statement, let us consider $h \in \left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right)^*$. We define $v_h : \mathcal{F}_p \rightarrow \mathbb{R}$ as $v_h([f]) \doteq h([f] - [f(p)])$. This is a slight abuse of notation since we should be considering $\{[f] - [f(p)]\}$, that is the class lying $\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}$, whose representative is $[f] - [f(p)]$. Since $\tilde{\mathcal{F}}_p^2$ is an ideal of $\tilde{\mathcal{F}}_p$ we can work at a level of representative. While v_h is linear per construction, the Leibnitz rule needs to be verified. Taking $[f], [f'] \in \mathcal{F}_p$ it holds that

$$\begin{aligned} v_h([f] \cdot [f']) &= h([ff'] - f(p)f'(p)) = h([ff'] - f(p)f'(p) \pm f(p)([f'] - f'(p)) \pm [f]f'(p)) = \\ &= h(([f] - f(p))([f'] - f'(p)) + f(p)([f'] - f'(p)) + ([f] - f(p))f'(p)) = \\ &= h(([f] - f(p))([f'] - f'(p))) + f(p)h([f'] - f'(p)) + h([f] - f(p))f'(p) = \\ &= f(p)v_h([f']) + v_h([f])f(p), \end{aligned}$$

where, in the last equality, we have used that $([f] - f(p))([f'] - f'(p)) \in \tilde{\mathcal{F}}_p^2$. A direct calculation shows that these maps are one the inverse of the other, which entails the sought conclusion

2) We prove that $\dim\left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right) = \dim(M)$. Let (U, φ) be a local chart such that $p \in U$ and let $(x^1, \dots, x^n)$ be n -coordinates such that $x^i = r^i \circ \varphi$ con $i = 1, \dots, n$, where $r^i : \mathbb{R}^n \rightarrow \mathbb{R}$ are the Cartesian coordinates on $\mathbb{R}^n$. Standard results in calculus entail that if U were a convex open subset of $\mathbb{R}^n$ centered at a point p , it holds that

$$\begin{aligned} F(q) &= F(p) + \sum_{i=1}^n \frac{\partial F}{\partial r^i} \Big|_p (r^i(q) - r^i(p)) + \\ &+ \sum_{i,j} (r^i(q) - r^i(p)) (r^j(q) - r^j(p)) \int_0^1 dt (1-t) \frac{\partial^2 F}{\partial r^i \partial r^j} \Big|_{p+t(q-p)}, \end{aligned}$$

where F is any function lying in $C^2(U)$. In particular, if we consider $f \in \tilde{\mathcal{F}}_p$, we can apply this formula to $F \doteq f \circ \varphi^{-1}$ obtaining that, in a neighbourhood of p

$$f(q) = (f \circ \varphi^{-1}) \circ \varphi(q) = \sum_{i=1}^n \frac{\partial f \circ \varphi^{-1}}{\partial r^i} \Big|_{\varphi(p)} (x^i(q) - x^i(p)) + h, \quad (2)$$

where $h \in \tilde{\mathcal{F}}_p^2$. From this identity it descends that $\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}$ is generated by linear combinations of the monomials $x^i - x^i(p)$ where we omit the arbitrary point q . In other words we have shown that $\dim\left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right) \leq \dim(M)$. In order to see that these monomials are a basis, suppose that $\sum_{i=1}^n a_i(x^i - x^i(p)) \in \tilde{\mathcal{F}}_p^2$ with $a_i \in \mathbb{R}$ for all $i = 1, 2, \dots, n$. This implies that

$$\sum_{i=1}^n a_i(x^i - x^i(p)) \circ \varphi^{-1} = \sum_{i=1}^n a_i(r^i - r^i(\varphi(p))) \in \tilde{\mathcal{F}}_{\varphi(p)}^2,$$

which entails, in turn, that

$$\left. \frac{\partial}{\partial r^j} \right|_{\varphi(p)} \left(\sum_{i=1}^n a_i(r^i - r^i(\varphi(p))) \right) = 0.$$

This can happen only if all coefficients a_i vanish, namely $x^i - x^i(p)$ con $i = 1, \dots, n$ is a basis of $\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}$. This entails that $\dim\left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right) = \dim(M)$. Furthermore, since $\dim(M) < \infty$, the found isomorphism entails that $\dim\left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right) = \dim\left(\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}\right)^* = \dim(T_p(M))$, where we have used what proven at point 1). $\square$

Remark 6. Given $f \in C^\infty(M)$, for every $v \in T_p M$, we shall write $v(f)$ rather than $v([f])$. This a small abuse of notation, which is adopted in the literature.

To conclude this section, we make precise the standard notation used when discussing derivatives of a function defined on a smooth manifold.

Definition 26. Let M be a smooth n -dimensional manifold as per Definition 9 and let (U, φ) be a local chart with associated coordinates $x^i = r^i \circ \varphi$, $i = 1, \dots, n$. Here $r^i : \mathbb{R}^n \rightarrow \mathbb{R}$ denote the standard Cartesian coordinates on $\mathbb{R}^n$. Given $p \in M$, for every $i = 1, \dots, n$ we set $\frac{\partial}{\partial x^i} \Big|_p \in T_p M$ as

$$\left(\frac{\partial}{\partial x^i} \Big|_p \right) ([f]) = \frac{\partial [f \circ \varphi^{-1}]}{\partial r^i} \Big|_{\varphi(p)}, \quad (3)$$

for all $[f] \in \mathcal{F}_p$.

Observe that, in the literature, one always omits the equivalence class and the definition is applied to any $f \in C^\infty(U)$. Furthermore the left hand side of Equation (3) is called the *directional derivative* of f at $p \in M$ along the i -th coordinate. This is denoted by $\frac{\partial f}{\partial x^i} \Big|_p$.

Furthermore

- $\frac{\partial f}{\partial x^i} \Big|_p$ depends only on the germ of f at p and, therefore, it identifies an element of $T_p M$.

The set of all these directional derivatives forms a basis of $T_p M$ dual to that of $\frac{\tilde{\mathcal{F}}_p}{\tilde{\mathcal{F}}_p^2}$, made

out of the equivalence classes whose representatives are the monomials $x^i - x^i(p)$. Notice that $x^i \in C^\infty(U)$ for all $i = 1, \dots, n$. Furthermore it holds per construction that

$$\frac{\partial(x^k - x^k(p))}{\partial x^i} = \delta^{ik},$$

where δ_{ik} is the Krönecker delta.

- Having identified a basis of the tangent space of M at p , we can decompose each $v \in T_p M$ as

$$v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p.$$

Observe that, usually, the point p is left understood and one can verify that, for all $j = 1, \dots, n$,

$$v^j = \sum_{i=1}^n v^i \frac{\partial(x^j - x^j(p))}{\partial x^i} \Big|_p.$$

Exercise 27. Given a smooth manifold M as per Definition 9, a local chart (U, φ) as per Definition 26 as well as another chart (V, ψ) such that $p \in V$ with coordinates $y^1, \dots, y^n$, show that

$$\frac{\partial}{\partial y^j} \Big|_p = \sum_{i=1}^n \frac{\partial x^i}{\partial y^j} \Big|_p \frac{\partial}{\partial x^i} \Big|_p.$$

Having established the notion of tangent space at a point of a manifold M as a canonical structure, it is natural to wonder whether, given two smooth manifolds and a smooth map between them, this induces also a relation between the tangent spaces.

Definition 28. Let M, N be smooth manifolds as per Definition 9 and let $\psi \in C^\infty(M; N)$ as per Definition 20, we call **differential (map) or push-forward**

$$d\psi \equiv \psi_* : T_p M \rightarrow T_{\psi(p)} N, \quad v \mapsto d\psi(v), \quad (4)$$

where $p \in M$ and where $d\psi(v)(f) = v(f \circ \psi)$ for all $f \in \mathcal{F}_{\psi(p)}(N)$.

Observe that, to be picky with the notation, we should be writing $d\psi_p$ to highlight the dependence on the application point. We will refrain from doing it since we will be extending the notion of differential in the next section and the dependence on p will disappear. The differential enjoys some notable properties:

1. Under the assumptions of Definition 28, let (U, φ) and (V, χ) be two charts of M and N respectively. In addition let $(x^1, \dots, x^n)$ and $(y^1, \dots, y^m)$ be two coordinate systems on U and V . Given a map $\psi \in C^\infty(M; N)$ such that there exists $p \in U$ with $\psi(p) \in V$, it holds that

$$d\psi \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \sum_{j=1}^m \frac{\partial y^j \circ \psi}{\partial x^i} \Big|_p \frac{\partial}{\partial y^j} \Big|_{\psi(p)},$$

where $J^{ij} \doteq \frac{\partial y^j \circ \psi}{\partial x^i}$ is called the *Jacobian (matrix)*.

2. The *chain rule* holds true, namely, given a third smooth manifold X as well as $\psi \in C^\infty(M; N)$ and $\Upsilon \in C^\infty(N; X)$, then, for all $p \in M$

$$d(\Upsilon \circ \psi)_p = d\Upsilon_{\psi(p)} \circ d\psi_p. \quad (5)$$

Having introduced the tangent space at a point and having proven in Theorem 25 that it is a finite dimensional vector space, linear algebra techniques entail that we can introduce its dual vector space $\mathbf{T}_p^*M \doteq (\mathbf{T}_pM)^*$, called the **cotangent space** at $p \in M$.

Remark 7. Having realized the cotangent space at $p \in M$ as the dual vector space to T_pM , we can exploit some standard results of linear algebra. In particular, given a local chart (U, φ) of M and a collection of coordinates $\{x^i\}_{i=1, \dots, n=\dim M}$, Definition 26 entails that $\left\{ \frac{\partial}{\partial x^i} \Big|_p \right\}_{i=1, \dots, n}$ is a basis of T_pM . Hence we can introduce the standard dual basis $\{dx^i|_p\}_{i=1, \dots, n}$ defined via the dual pairing

$$dx^j(\partial_i) = \delta_i^j, \quad (6)$$

where δ_i^j is the Kronecker delta and where, for simplicity of the notation, we omitted the subscript referring to the point p and we introduce the notation

$$\partial_i \equiv \frac{\partial}{\partial x^i}.$$

Observe that using the symbol dx^i is not in contradiction with Definition 28. As a matter of fact, recall that, a coordinate is a map $x^i : M \rightarrow \mathbb{R}$ whose differential is $dx^i : T_pM \rightarrow T_{x^i(p)}\mathbb{R}$ such that, for all $v \in T_pM$ and for all $f \in C^\infty(\mathbb{R})$

$$dx^i(v)(f) = v(f \circ x^i).$$

Yet, since one can introduce the decomposition $v = \sum_{k=1}^n v^k \partial_k$, $v^k \in \mathbb{R}$ for all $k = 1, \dots, n$, then

$$v(f \circ x^i) = \sum_{k=1}^n v^k \frac{\partial f \circ x^i}{\partial x^k}.$$

Comparison with Equation (6) requires first of all to fix an index $j \in \{1, \dots, n\}$ and to set $v^k = \delta_j^k$, so that $v = \partial_j$. Given $f \in C^\infty(\mathbb{R})$ and adapting Equation (2) to the case under scrutiny, we expand f as $f(x^i(p)) = f(0) + f'(0)(x_i - x_i(p)) + \mathcal{O}((x_i - x_i(p))^2)$. This yields that

$$dx^i(\partial_j)(f) = f'(0) \frac{\partial(x^i - x^i(p))}{\partial x^j} = f'(0) \delta_j^i.$$

Remark 8. In our presentation the cotangent space is introduced exploiting the algebraic properties of the tangent space at a point $p \in M$. Yet, it is important to highlight that T_p^*M is a geometric structure and its points are closely related to differential forms. We will not delve into these details leaving on interested reader to the chair of the course of Mathematical Methods in Theoretical Physics.

Since the cotangent space at a point of a smooth manifold is constructed as the dual of the tangent space, every linear operation defined on the latter can be extended per duality on the former. This applies in particular to the differential introduced in Definition 28.

Definition 29. Let M, N be two smooth manifolds and let $\phi \in C^\infty(M; N)$. We call **pull-back** (of ϕ) the map $\phi^* : T_{\phi(p)}^* N \rightarrow T_p^* M$ such that, given $\omega \in T_{\phi(p)}^* N$

$$\phi^* \omega(v) = \omega(\phi_*(v)), \quad \forall v \in T_p M,$$

where ϕ_* is the push-forward map as per Definition 28.

Remark 9. The structures introduced in this section are often referred to in the physics literature with different names. More precisely, given a smooth manifold M as per Definition 9, any vector lying in $T_p M$ is called contravariant while elements of $T_p^* M$, covariant vectors. The author of these notes pays is due respect to tradition, but he will certainly stick to tangent and cotangent vectors.

2.2 Vector Fields

In the previous section we have introduced the notion of tangent and cotangent space at a given point $p \in M$. Yet, it is convenient to consider the union of all these vector spaces as the point on the underlying manifold varies.

Definition 30. Let M be a smooth manifold as per Definition 9. We call

- **tangent space (or bundle)** of M $TM \doteq \bigsqcup_{p \in M} T_p M$,
- **cotangent space (or bundle)** of M , $T^*M \doteq \bigsqcup_{p \in M} T_p^* M$.

Remark 10. Observe that a point in TM is a pair (p, v) where $p \in M$, while $v \in T_p M$. Hence one can introduce a canonical projection map

$$\pi_{TM} : TM \rightarrow M, \quad (p, v) \mapsto \pi_{TM}(p, v) = p. \quad (7)$$

Similarly a point in T^*M is a pair (p, ω) where $p \in M$, while $\omega \in T_p^* M$ and we can introduce the projection map

$$\pi_{T^*M} : T^*M \rightarrow M, \quad (p, \omega) \mapsto \pi_{T^*M}(p, \omega) = p. \quad (8)$$

The following theorem is a cornerstone in differential geometry since it clarifies why the tangent and cotangent bundles are an important canonical structure associated to a manifold

Theorem 31. Let M be an n -dimensional smooth manifold as per Definition 9, $n \geq 1$. Then both TM and T^*M are smooth $2n$ -dimensional manifolds.

Proof. We shall focus on TM since the analysis of T^*M is identical mutatis mutandis. Let $\mathcal{F}$ be the maximal atlas of M and let (U, φ) be a local chart therein. Given any coordinate system on U , $x^1, \dots, x^n$, where $x^i = r^i \circ \varphi$, $i = 1, \dots, n$ we can introduce the map

$$\tilde{\varphi} : \pi_{TM}^{-1}(U) \rightarrow \mathbb{R}^{2n}, \quad (p, v) \mapsto \tilde{\varphi}(v) = (x^1(p), \dots, x^n(p), dx^1(v), \dots, dx^n(v)),$$

where π_{TM} is the projection map in Equation (7). Observe that, per construction, $\tilde{\varphi}(\pi_{TM}^{-1}(U))$ is an open subset of $\mathbb{R}^{2n}$ which is invertible on its image. More precisely, given a point $(x^1, \dots, x^n, v^1, \dots, v^n) \in \tilde{\varphi}(\pi_{TM}^{-1}(U))$ it holds that

$$\tilde{\varphi}^{-1}(x^1, \dots, x^n, v^1, \dots, v^n) = (p, v) = \left(\varphi^{-1}(x^1, \dots, x^n), \sum_{j=1}^n v^j \frac{\partial}{\partial x^j} \Big|_{\varphi^{-1}(x^1, \dots, x^n)} \right).$$

Consider in addition any $(V, \psi) \in \mathcal{F}$ such that $U \cap V \neq \emptyset$. Given a coordinate system $y^1, \dots, y^n$ on (V, ψ) where $y^j = \eta^j \circ \psi$, $j = 1, \dots, n$ and calling

$$\tilde{\psi} : \pi_{TM}^{-1}(V) \rightarrow \mathbb{R}^{2n}, \quad (p, v) \mapsto \tilde{\psi}(v) = (y^1(p), \dots, y^n(p), dy^1(v), \dots, dy^n(v)),$$

it holds that

$$\begin{aligned} \tilde{\psi} \circ \tilde{\varphi}^{-1} : \tilde{\varphi}(\pi_{TM}^{-1}(U \cap V)) &\rightarrow \tilde{\psi}(\pi_{TM}^{-1}(U \cap V)) \\ \underbrace{(x^1, \dots, x^n)}_x, \underbrace{(v^1, \dots, v^n)}_v &\mapsto (\tilde{\psi} \circ \tilde{\varphi}^{-1})(x, v) = \left(y^1(\varphi^{-1}(x)), \dots, y^n(\varphi^{-1}(x)), v^j \frac{\partial \eta^1}{\partial r^j} \Big|_x, \dots, v^j \frac{\partial \eta^n}{\partial r^j} \Big|_x \right) \end{aligned}$$

Since, in each entry this is either a linear combination or a composition of smooth maps, we can conclude that $\tilde{\psi} \circ \tilde{\varphi}^{-1} \in C^\infty(\tilde{\varphi}(\pi_{TM}^{-1}(U \cap V)); \tilde{\psi}(\pi_{TM}^{-1}(U \cap V)))$. We can now apply Lemma 11 to infer the sought conclusion. $\square$

Exercise 32. Complete and integrate the proof of Theorem 31 by showing that

1. for any local chart (U, φ) on M $\pi_{TM}^{-1}(U)$ is diffeomorphic to $U \times \mathbb{R}^n$,
2. T^*M is a smooth $2n$ -dimensional manifold.

The previous analysis has allowed us to establish the following two concepts:

1. at any point of a smooth manifold M , we can recover the notion of vector thanks to the differentiable structure,
2. the collection of all vectors at all points of a smooth manifold is in turn a smooth manifold.

Putting together these two notions allows us to codify the heuristic idea of assigning at all points of a given smooth manifold M a vector by encoding a regularity condition on the way such assignment varies with respect to the point of M .

Definition 33. Let M be a smooth manifold as per Definition 9. We call **(smooth) vector field** on M a map $X \in C^\infty(M; TM)$ such that $\pi_{TM} \circ X = id|_M$, where $id|_M$ is the identity map on M , while $\pi|_{TM}$ is the projection map as per Equation (7). The collection of all smooth vector fields is denoted by $\Gamma(TM) \equiv \Gamma^\infty(TM)$.

A similar definition can be formulated with reference to the cotangent space.

Definition 34. Let M be a smooth manifold as per Definition 9. We call **(smooth) differential form (or 1-form)** on M a map $\omega \in C^\infty(M; T^*M)$ such that $\pi_{T^*M} \circ \omega = id|_M$, where $id|_M$ is the identity map on M , while $\pi|_{T^*M}$ is the projection map as per Equation (8). The collection of all smooth differential 1-forms is denoted by $\Gamma(T^*M) \equiv \Gamma^\infty(T^*M) \equiv \Omega^1(M)$.

The following result is quite useful to characterize smooth vector fields.

Theorem 35. Given an n -dimensional smooth manifold M , $n \geq 1$, as per Definition 9 and a map $X : M \rightarrow TM$ such that $\pi_{TM} \circ X = id|_M$, where $id|_M$ is the identity map on M , the following statements are equivalent:

1. $X \in \Gamma(M; TM)$
2. given a local chart (U, φ) and a coordinate system thereon, $(x^1, \dots, x^n)$, if $a^i : U \rightarrow \mathbb{R}$, $i = 1, \dots, n$ is a collection of functions such that $X|_U = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$, then $a^i \in C^\infty(U)$,
3. whenever V is an open subset of M and $f \in C^\infty(V)$, then $X(f) \in C^\infty(V)$ where $X(f) : V \rightarrow \mathbb{R}$ is such that

$$X(f)(p) = X_p(f). \quad \forall p \in V$$

Proof. 1) $\implies$ 2): Given $X \in \Gamma(M; TM)$, for any open subset $U \subseteq M$ $X|_U \in C^\infty(U; TU)$ where $TU \doteq TM|_U$. Consequently, under the given assumptions, for any $i = 1, \dots, n$, $dx^i \circ X|_U = a^i \in C^\infty(U)$ where the last statement is a consequence that we are composing smooth functions.

2) $\implies$ 3): Following Definition 20 $X(f)|_U$ is open on every chart (U, φ) such that $U \subset V$. Yet Condition 2) entails that $X(f)|_U = \sum_{i=1}^n a^i \frac{\partial f}{\partial x^i}$ which is a linear combination of products of smooth functions and therefore it is smooth itself.

3) $\implies$ 1): ; We need to show that $X|_U \in C^\infty(U; TU)$ for every chart (U, φ) of M . In turn this is equivalent to proving that the composition between $X|_U$ and any coordinate map on $\pi^{-1}(U)$ is smooth. Yet this is automatic since we are composing smooth maps, namely, for all $i = 1, \dots, n$, $x^i \circ \pi \circ X|_U = x^i$ and $dx^i \circ X|_U = X(x^i)$ where $x^i \in C^\infty(U; \mathbb{R})$. $\square$

We conclude the section by stating a notable algebraic property of all smooth vector fields which will play a distinguished rôle in setting up the mathematical background of general relativity.

Definition 36. Let M be a smooth manifold as per Definition 9 and let $\Gamma(TM)$ be as per Definition 33. The map

$$\begin{aligned} [\cdot, \cdot] : \Gamma(TM) \times \Gamma(TM) &\rightarrow \Gamma(TM) \\ (X, X') &\mapsto [X, X'], \end{aligned}$$

where

$$[X, X']_p(f) = X_p(X'(f)) - X'_p(X(f)), \quad \forall f \in C^\infty(M)$$

is called **Lie bracket** between two vector fields.

Corollary 37. *Given the Lie brackets as per Definition 36, the following properties hold true:*

- $[\cdot, \cdot]$ is antisymmetric,
- it satisfies the Jacobi identity,
- for all $f, f' \in C^\infty(M)$ it holds that $[fX, f'X'] = ff'[X, X'] + f(X(f'))X' - f'(X'(f))X$.

The proof is omitted as it is a direct consequence of the definition itself.

Exercise 38. Discuss whether it is possible to formulate a counterpart of Theorem 35 and of Definition 36 for $\Gamma(T^*M)$.

Exercise 39. Let M be a smooth n -dimensional manifold, $n \geq 1$, as per Definition 9 and let $X, Y \in \Gamma(TM)$. Given any local chart (U, φ) of M and a coordinate system $x^1, \dots, x^n$ thereon. Prove that, given $[X, Y] = [X, Y]^\mu \partial_\mu$, it holds that

$$[X, Y]^i = X^j \partial_j Y^i - Y^j \partial_j X^i, \quad (9)$$

where $X = X^i \partial_i$ and $Y = Y^j \partial_j$.

Remark 11. *Given two smooth manifolds M, N as per Definition 9, the structure of their tangent and cotangent space entails that the content of Definitions 28 and 29 can be extended. More precisely, given $f \in C^\infty(M; N)$, we call **differential (or push-forward)** the map $df : TM \rightarrow TN$ such that the following diagram is commutative:*

$$\begin{array}{ccc} TM & \xrightarrow{df} & TN \\ \pi_{TM} \downarrow & & \downarrow \pi_{TN} \\ M & \xrightarrow{f} & N \end{array}$$

Yet, contrary to what one might expect, acting with the push-forward on a vector field $X \in \Gamma(TM)$ as per Definition 33 does not automatically identify a counterpart on TN . As a matter of fact, $Y = df(X)$ is defined only on $f[M] \subseteq N$ and, unless f is both injective and surjective, then $Y \notin \Gamma(TN)$. If f is at least injective, then $df(X) \in \Gamma(TN|_{f[M]})$. Another viable option is to consider compactly supported smooth vector fields $X \in \Gamma_0(TM)$, that is $X \in C_0^\infty(M; TM)$. In this case, if f is injective, then $df(X) \in \Gamma_0(TN|_{f[M]})$ and, considering the canonical extension as 0 outside its support, $df(X) \in \Gamma_0(TN)$.

It is convenient to classify maps between manifolds in terms of the properties of the associated differential.

Definition 40. Given two smooth manifolds M, N as per Definition 9 and a map $f \in C^\infty(M; N)$, we say that

- f is an **immersion** if the differential $df : TM \rightarrow TN$ is injective for all $p \in M$, namely $\ker(df|_{T_p M}) = \{0\}$. In this case we say that $f[M]$ is an **(immersed) submanifold** of N ,
- f is an **embedding (or imbedding)** if it is an immersion and it identifies an homeomorphism between M and $f[M]$. In this case we say that $M \equiv f[M]$ is an **(embedded or regular) submanifold** of N
- f is a **submersion** if the differential $df : TM \rightarrow TN$ is surjective at all $p \in M$.
- f is a **diffeomorphism** if $f \in C^\infty(M; N)$ and $f^{-1} \in C^\infty(N; M)$. In this case we write $M \sim N$.

If $M = N$, diffeomorphisms form a group under composition which is denoted by $\text{Diff}(M)$.

Exercise 41. Let M be a smooth manifold as per Definition 9 such that there exists $f \in \text{Diff}(M)$, see Definition 40. Show that, for all $X, Y \in \Gamma(TM)$, the push-forward is a Lie-algebra homomorphism, namely

$$f_*[X, Y] = [f_*X, f_*Y].$$

Observe that an embedding always identifies a diffeomorphism between M and its image $f[M]$.

Example 42. We give four explicit realizations of the maps defined above so to make our readers acquainted with these new concepts:

1. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be such that

$$(x, y) \mapsto f(x, y) = ((2 + \cos(2\pi x)) \cos(2\pi y), (2 + \cos(2\pi x)) \sin(2\pi y), \sin(2\pi x)).$$

To compute the differential, let us introduce the Cartesian coordinates $(u^1, u^2, u^3) \in \mathbb{R}^3$ and let us observe that a generic element of $T_q \mathbb{R}^3$, $q \in \mathbb{R}^3$, can be written as $c_1 \partial_{u^1} + c_2 \partial_{u^2} + c_3 \partial_{u^3}$.

$$df : T_{(x,y)} \mathbb{R}^2 \rightarrow T_{f(x,y)} \mathbb{R}^3, \quad v = a \partial_x + b \partial_y \mapsto df(v).$$

Recalling that, for every $h \in C^\infty(\mathbb{R}^3)$, $df(v)(h) = v(h \circ f) \equiv v(h(f(x, y)))$, we can set $u \doteq f(x, y) \in \mathbb{R}^3$ and, therefore,

$$df(v)(h) = (a \partial_x u + b \partial_y u) \nabla_u h,$$

where ∇_u denotes the gradient. Since

$$\begin{aligned} \partial_x u &= (-2\pi \sin(2\pi x) \cos(2\pi y), -2\pi \sin(2\pi x) \sin(2\pi y), 2\pi \cos(2\pi x)), \\ \partial_y u &= (-2\pi(2 + \cos(2\pi x)) \sin(2\pi y), 2\pi(2 + \cos(2\pi x)) \cos(2\pi y), 0), \end{aligned}$$

in order for $df(v)(h)$ to vanish for all choices of $h \in C^\infty(\mathbb{R}^3)$, it must hold that $(a \partial_x u + b \partial_y u) \nabla_u h = 0$. In turn, this implies $a \partial_x u + b \partial_y u = 0$ which holds true only if $a = b = 0$.

Hence the differential df is injective for all values of (x, y) and therefore f is an *immersion*. Yet this is not an embedding since $f[\mathbb{R}^2]$ is the torus obtained revolving the circle $(u^2 - 2)^2 + (u^3)^2 = 1$ along the u^1 -axis. This is a closed and bounded region in $\mathbb{R}^3$, hence, by Heine-Borel theorem, it is compact. Consequently f cannot be an homeomorphism since it would be a map preserving topological properties and $\mathbb{R}^2$ is not compact.

2. Let us consider the map $f : \mathbb{S}^2 \rightarrow \mathbb{R}^3$ such that $f(x^1, x^2) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$ where $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$. For simplicity we consider only one of the two-charts covering $\mathbb{S}^2$. Strictly speaking we are working with $\mathbb{S}^2 \setminus \{N \cup S\}$, where N, S are the north and south pole meant as two antipodal points on $\mathbb{S}^2$. To compute the differential, let us introduce the Cartesian coordinates $(x, y, z) \in \mathbb{R}^3$ and let us observe that a generic element of $T_q \mathbb{R}^3$, $q \in \mathbb{R}^3$, can be written as $c_1 \partial_x + c_2 \partial_y + c_3 \partial_z$.

$$df : T_{(\theta, \varphi)} \mathbb{S}^2 \rightarrow T_{f(x, y)} \mathbb{R}^3, \quad v = a \partial_\theta + b \partial_\varphi \mapsto df(v).$$

Recalling that, for every $h \in C^\infty(\mathbb{R}^3)$, $df(v)(h) = v(h \circ f) \equiv v(h(f(\theta, \varphi)))$, we can set $u \doteq f(\theta, \varphi) \in \mathbb{R}^3$ and, therefore,

$$df(v)(h) = (a \partial_\theta u + b \partial_\varphi u) \nabla_u h,$$

where ∇_u denotes the gradient in $\mathbb{R}^3$. A direct computation shows that

$$\begin{aligned} df(v) &= (a \partial_\theta u + b \partial_\varphi u) \nabla_u = \\ &= (a \cos \theta \cos \varphi - b \sin \theta \sin \varphi) \partial_x + (a \cos \theta \sin \varphi + b \sin \theta \cos \varphi) \partial_y - a \sin \theta \partial_z. \end{aligned}$$

Observe that $df(v) = 0$ if and only if $a = b = 0$, hence f is an immersion. Furthermore it is a smooth map whose inverse is ruled by Equation (15). This identifies a smooth map from $f[\mathbb{S}^2]$ to $\mathbb{S}^2$ and hence f is a *smooth embedding*.

3. Let $\mathbb{B}^3 = \{x \equiv (x^1, x^2, x^3) \in \mathbb{R}^3 \mid \|x\|^2 = \sum_{i=1}^3 (x^i)^2 < 1\} \subset \mathbb{R}^3$ and let $f : \mathbb{B}^3 \rightarrow \mathbb{S}^2$ be the map which, in Cartesian coordinates, reads

$$f(x, y, z) = (x, y, \sqrt{1 - x^2 - y^2}).$$

Switching to spherical coordinates (r, θ, φ) which is legitimate since the origin is not in the image of f , we have that $\mathbb{S}^2$ is the locus $r = 1$. Writing a generic $v \in T_p \mathbb{B}^3$ as $v = a \partial_x + b \partial_y + c \partial_z$, following the same strategy as in the previous points, it turns out that

$$df(v) = (a \partial_x f + b \partial_y f + c \partial_z f) \nabla_{(\theta, \varphi)},$$

namely

$$df(v) = (a \cos \theta \cos \varphi + b \cos \theta \sin \varphi - c \sin \theta) \partial_\theta + \left(b \frac{\cos \varphi}{\sin \theta} - a \frac{\sin \varphi}{\sin \theta} \right) \partial_\varphi.$$

The divergence at $\sin \theta = 0$ is a byproduct of the angular coordinates since $\theta \neq 0, \pi$. One can realize by direct inspection that $df(v)$ is surjective and therefore f is a *submersion*.

4. Let $\mathbb{B}^n = \{x \equiv (x^1, \dots, x^n) \in \mathbb{R}^n \mid \|x\|^2 = \sum_{i=1}^n (x^i)^2 < 1\}$. Consider the map

$$f : \mathbb{B}^n \rightarrow \mathbb{R}^n, \quad x \mapsto \frac{x}{\sqrt{1 - \|x\|^2}}.$$

The map is smooth since $\|x\|^2 < 1$ and it admits an inverse which reads

$$f^{-1} : \mathbb{R}^n \rightarrow \mathbb{B}^n \quad y \equiv (y^1, \dots, y^n) \mapsto F^{-1}(y) = \frac{y}{\sqrt{1 + \|y\|^2}}.$$

Also f^{-1} is a smooth map since the denominator is never vanishing and a direct calculation shows that both $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity maps on $\mathbb{R}^n$ and on $\mathbb{B}^n$ respectively. Therefore f is a *diffeomorphism* and $\mathbb{B}^n$ is *diffeomorphic* to $\mathbb{R}^n$.

We state a very important theorem whose proof can be found in [Lee13, Thm 4.14] which provides an efficient criterion to classify certain classes of maps between smooth manifolds.

Theorem 43. [Global Rank Theorem] *Let M, N be two smooth manifolds as per Definition 9 and let $f \in C^\infty(M; N)$ be of constant rank, that is the rank of df is the same at every $p \in M$. Then*

- *If f is surjective, then it is a submersion,*
- *If f is injective, then it is an immersion,*
- *If f is a bijection, then it is a diffeomorphism.*

Exercise 44. Rewrite Example (42) using the global rank theorem.

Remark 12. *Under the same assumptions as in Remark 11 we can construct a dual map which acts on the cotangent spaces. More precisely we call **pull-back** the map $f^* : T^*N \rightarrow T^*M$ such that the following diagram is commutative*

$$\begin{array}{ccc} T^*M & \xleftarrow{f^*} & T^*N \\ \pi_{T^*M} \downarrow & & \downarrow \pi_{T^*N} \\ M & \xrightarrow{f} & N \end{array}$$

As one can observe from the directions of the arrows, commutativity is not meant here in the canonical way. This peculiar feature is reflected by the fact that, contrary to vector fields, differential forms behave well under pull-back. More precisely given $\omega \in \Gamma(T^*N)$, we define $f^*\omega \in \Gamma(T^*M)$ as the 1-form such that, for all $p \in M$,

$$f^*\omega|_p(v) = \omega_{f(p)}(f_*(v)), \quad \forall v \in T_pM. \quad (10)$$

It is rather important to stress that this behaviour is not inherited by compactly supported differential forms, namely there is no guarantee that, given $\omega \in \Gamma_0(TN)$, then $f^*\omega \in \Gamma_0(TM)$. This is related to the fact that f is only assumed to be smooth, but it does not need to be a proper map, since $\text{supp}(f^*\omega) = f^{-1}[\text{supp}(\omega)]$.

Exercise 45 (Very Important). Given two smooth manifolds M, N as per Definition 9 and a map $f : C^\infty(M; N)$, prove that, given a local chart (U, φ) with coordinates $(x^1, \dots, x^n)$, $n \geq 1$, and given a local chart (V, ψ) with coordinates $(y^1, \dots, y^m)$, $m \geq 1$, under the assumption that $V \subseteq f[U]$, it holds that

$$f^*\omega = \sum_{i=1}^n v_i(x) dx^i,$$

where, writing $\omega(y) = \omega_j(y) dy^j$

$$v_i(x) = \left. \frac{\partial(y^j \circ f)}{\partial x^i} \right|_x \omega_j(f(x)).$$

Remark 13. *Embedded submanifolds enjoy a notable local, characterization property, see [Lee13, Thm. 5.8] for a proof. Let M be a regular n -dimensional submanifold of a k -dimensional manifold N , with $0 < n \leq k < \infty$. At every point $p \in M$, one can find a local chart $(U, \varphi) \subset N$ and a coordinate system thereon, $\{x^1, \dots, x^k\}$ such that $U \cap M$ is the locus $x^{n+1} = \dots = x^k = 0$. In addition $\{x^1, \dots, x^n\}$ is a coordinate system on $U \cap M$ and, for every $q \in U \cap M$, $T_q M = \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right\}$.*

2.3 Tensor Fields and the Metric Tensor

In the previous section, we have introduced the tangent space as a canonical structure associated to a smooth manifold M . In turn, this prompted the identification of the cotangent space using standard tools of linear algebra, more precisely the existence of an isomorphism between a finite-dimensional vector space V and its dual V^* . Following the same rationale we can exploit other notable operations to identify more advanced structures. As before, we shall first start working at an arbitrary fixed point of M , subsequently extending the whole construction globally. The operation that we will be exploiting is the *tensor product*. Readers who wish to revise this concept are encouraged to refer to [Lang02]. Readers who think that I am mean by pointing to this reference are absolutely right, but this is the best textbook on algebra.

Definition 46. *Let M be a smooth n -dimensional manifold, $n \geq 1$, as per Definition 9 and let $p \in M$. We say that τ is a **(k,l)-tensor** at p with $k, l \geq 0$ if*

$$\tau \in \mathcal{T}_p^{(k,l)}(M) \doteq \underbrace{T_p M \otimes \dots \otimes T_p M}_k \otimes \underbrace{T_p^* M \otimes \dots \otimes T_p^* M}_l,$$

where, if $k = l = 0$, then we set $\tau \in \mathbb{R}$.

The collection of all (k, l) -tensors is therefore a vector space of dimension n^{k+l} . In addition, if we consider a local chart (U, φ) such that $p \in U$ and a coordinate systems $x_1, \dots, x_n$ centered at p , Definition 26 and Remark 7 entail that $\left(\frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right)$ and $(dx^1|_p, \dots, dx^n|_p)$ are a basis of

$T_p M$ and $T_p^* M$ respectively. Dropping for convenience of the notation the subscript p , it turns out that

$$\frac{\partial}{\partial x^{i_1}} \otimes \cdots \otimes \frac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_l}, \quad i_1, \dots, i_k, j_1, \dots, j_l = 1, \dots, n$$

is a basis of $\mathcal{T}_p^{(k,l)}$. On top of this structural result, we introduce two maps which allow to construct tensors which are either symmetric or antisymmetric upon exchange of their arguments. As we shall see the former play a distinguished rôle, especially in the definition of the key geometric tensors in general relativity.

Definition 47. Let M be a smooth n -dimensional manifold, $n \geq 1$, as per Definition 9 and let $p \in M$. Let $\mathcal{T}_p^{(k,0)}(M)$. We call

- **symmetrization map**

$$\text{Sym} : \mathcal{T}_p^{(k,0)}(M) \rightarrow \mathcal{T}_p^{(k,0)}(M) \quad \tau \mapsto \text{Sym}(\tau) = \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} \eta_\sigma(\tau),$$

where $\mathfrak{S}_\sigma$ is the set of all permutations of k -indices. The map η_σ is such that, given any basis $\{\partial_i\}$, $i = 1, \dots, n$, of $T_p M$ and having decomposed

$$\tau = \tau^{i_1 \dots i_k} \partial_{i_1} \otimes \cdots \otimes \partial_{i_k}, \quad \tau^{i_1 \dots i_k} \in \mathbb{R},$$

then

$$\eta_\sigma(\tau) = \tau^{i_{\sigma(1)} \dots i_{\sigma(k)}} \partial_{i_1} \otimes \cdots \otimes \partial_{i_k}.$$

We say that τ is **symmetric** if $\tau = \text{Sym}(\tau)$ and we denote it by $\tau \in \mathcal{T}_p^{(k,0)s}(M) \equiv \mathcal{T}_p(M)^{\otimes_s k}$.

- **anti-symmetrization map**

$$\text{Alt} : \mathcal{T}_p^{(k,0)}(M) \rightarrow \mathcal{T}_p^{(k,0)}(M) \quad \tau \mapsto \text{Alt}(\tau) = \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} \epsilon(\sigma) \eta_\sigma(\tau),$$

where $\mathfrak{S}_\sigma$ is the set of all permutations of k -indices while $\epsilon(\sigma)$ is the sign of the permutation.

We say that τ is **anti-symmetric** if $\tau = \text{Alt}(\tau)$ and we denote it by $\tau \in \mathcal{T}_p^{(k,0)a}(M) \equiv \Lambda_p^k(M)$.

A similar definition applies mutatis mutandis to elements lying in $\mathcal{T}_p^{(0,l)}(M)$.

We shall not discuss anti-symmetric tensor much in detail in the main body of these notes, but we will consider them in Appendix C. The reader is invited in particular to solve the following exercise:

Exercise 48. Under the hypotheses of Definition 47, prove that

1. the definition of antisymmetric tensor is equivalent to that of alternating tensor as in Definition 201,
2. $\dim(T_p(M)^{\otimes_s k}) = \binom{n+k-1}{k}$ for all $k = 0, \dots, n$.

Notation: We introduce a symbolic representation of the symmetrization and anti-symmetrization map which is widely used in general relativity. Given $\omega \in \mathcal{T}_p^{(0,l)}(M)$ and given a basis $\{dx^i\}_{i=1,\dots,n}$ of T_p^*M , we decompose $\omega = \omega_{i_1,\dots,i_l} dx^{i_1} \otimes \dots \otimes dx^{i_l}$. Then we denote by

$$\omega_{(i_1,\dots,i_l)} = \frac{1}{l!} \sum_{\sigma \in \mathfrak{S}_l} \omega_{i_{\sigma(1)} \dots i_{\sigma(l)}}, \quad (11)$$

while

$$\omega_{[i_1,\dots,i_l]} = \frac{1}{l!} \sum_{\sigma \in \mathfrak{S}_l} \epsilon(\sigma) \omega_{i_{\sigma(1)} \dots i_{\sigma(l)}}, \quad (12)$$

where both $\mathfrak{S}_l$ and $\epsilon(\sigma)$ are as per Definition 47.

Remark 14. The notion of symmetrization and anti-symmetrization map can be applied also to mixed tensors $\tau \in \mathcal{T}_p^{(k,l)}(M)$, though Sym and Alt can act only on the covariant and on the contravariant components of τ separately. It is also possible to symmetrize or antisymmetrize only a number of indices, e.g., given $\omega \in \mathcal{T}_p^{(0,3)}(M)$ and denoting as ω_{ijk} its components with respect to a given basis, we can write

$$\omega_{(ij)k} = \frac{1}{2}(\omega_{ijk} + \omega_{jik}), \quad \omega_{[ij]k} = \frac{1}{2}(\omega_{ijk} - \omega_{jik}). \quad i, j, k = 1, \dots, n$$

Having defined generic (k, l) -tensors at a point of a smooth manifold we can consider a global counterpart of these structures, much in the same spirit as in Section 2.1.

Definition 49. Let M be a smooth n -dimensional manifold, $n \geq 1$, as per Definition 9. We call

$$\mathcal{T}^{(k,l)}(M) = \bigsqcup_{p \in M} \mathcal{T}_p^{(k,l)}(M) \equiv \underbrace{TM \otimes \dots \otimes TM}_k \otimes \underbrace{T^*M \otimes \dots \otimes T^*M}_l.$$

A point in $\mathcal{T}^{(k,l)}(M)$ is a pair (p, τ) where $\tau \in \mathcal{T}_p^{(k,l)}(M)$ and we denote by $\pi_{(k,l)} : \mathcal{T}^{(k,l)}(M) \rightarrow M$ the canonical projection map such that $\pi(p, \tau) = p$. In addition we endow

$$\mathcal{T}^\bullet(M) = \bigoplus_{k,l=0}^n \mathcal{T}^{(k,l)}(M),$$

with the canonical algebra structure induced by the tensor product, that is

$$\otimes : \mathcal{T}^{(k,l)}(M) \times \mathcal{T}^{(k',l')}(M) \rightarrow \mathcal{T}^{(k+k',l+l')}(M), \quad (F, H) \mapsto \otimes(F, H) \doteq F \otimes H.$$

Remark 15. It is convenient to translate Definition 49 locally, namely, given any local chart (U, φ) of M and coordinates thereon $\{x^1, \dots, x^n\}$, $n = \dim M$, any $F \in \mathcal{T}^{(k,l)}(M)$ can be written for all $x \in U$ as

$$F(x) = F_{\mu_1 \dots \mu_l}^{\nu_1 \dots \nu_k}(x) \frac{\partial}{\partial x^{\nu_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{\nu_k}} \otimes dx^{\mu_1} \otimes \dots \otimes dx^{\mu_l},$$

while, considering $H \in \mathcal{T}^{(k',l')}(M)$, it can be written for all $x \in U$ as

$$H(x) = H_{\mu_1 \dots \mu_{l'}}^{\nu_1 \dots \nu_{k'}}(x) \frac{\partial}{\partial x^{\nu_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{\nu_{k'}}} \otimes dx^{\mu_1} \otimes \dots \otimes dx^{\mu_{l'}}.$$

Their tensor product reads

$$(F \otimes H)(x) = F_{\mu_1 \dots \mu_l}^{\nu_1 \dots \nu_k}(x) H_{\mu_{l+1} \dots \mu_{l+l'}}^{\nu_{k+1} \dots \nu_{k+k'}}(x) \frac{\partial}{\partial x^{\nu_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{\nu_{k+k'}}} \otimes dx^{\mu_1} \otimes \dots \otimes dx^{\mu_{l+l'}}.$$

A straightforward generalization of Theorem 31 shows that these are smooth manifolds of dimension n^{k+l} where $n = \dim M$. Similarly to Section 2.2, we can generalize the concepts of vector fields and differential forms by establishing the notion of a smooth assignment of a (k, l) -tensor at each point of the underlying manifold. This allows in turn to introduce the main kinematic and dynamical structure of general relativity, *the metric (tensor)*.

Definition 50. Let M be a smooth n -dimensional manifold, $n \geq 1$, as per Definition 9 and let $\mathcal{T}^{(k,l)}(M)$ be as per Definition 49. We call a **$(\mathbf{k}, \mathbf{l})$ -(smooth) tensor field** a map $\tau \in C^\infty(M; \mathcal{T}^{(k,l)}(M))$ such that $\pi_{(k,l)}\tau = \text{id}|_M$, where $\text{id}|_M$ is the identity map on M . The space of these maps is denoted by $\Gamma(\mathcal{T}^{(k,l)}(M))$. In addition, given $\omega \in \Gamma(\mathcal{T}^{(0,l)}(M))$, we say that

1. it is **symmetric** if, for all $p \in M$, $\omega(p)$ is symmetric as per Definition 47. We denote it by $\omega \in \Gamma(T^*M^{\otimes_s l})$.
2. it is **anti-symmetric** if, for all $p \in M$, $\omega(p)$ is anti-symmetric as per Definition 47. We denote it by $\omega \in \Omega^l(M)$.

We call **metric tensor** any $g \in \Gamma(T^*M^{\otimes_s 2})$ which is non-degenerate, that is, for all $p \in M$,

$$g|_p(v, v_1) = 0, \quad \forall v_1 \in T_p M \implies v = 0 \in T_p M.$$

Given two metric tensors $g, g' \in \Gamma(T^*M^{\otimes_s 2})$ we say that they are **conformal** to each other (or, related by a conformal transformation), if $\exists \Omega \in C^\infty(M; (0, \infty))$ such that $g' = \Omega^2 g$.

Henceforth we will be considering as main object of our investigation the metric tensor and we will use it as our main source of examples, together with other geometric structures constructed out of it. We leave to our beloved readers the task of working out more general examples.

Notation: We introduce a notation which is at first glance weird, but it is very convenient and it is widely used both in physics and in mathematics despite being not completely rigorous. More

precisely, given a metric tensor g as per Definition 50 and given a coordinate patch (U, φ) of M as well as a coordinate system thereon, $dx^1, \dots, dx^n$, following the discussion after Definition 46, we can write

$$g|_U = g_{\mu\nu} dx^\mu \otimes dx^\nu, \quad \mu, \nu = 1, \dots, n,$$

where $g_{\mu\nu} \equiv g_{\mu\nu}(x) \in C^\infty(U)$. Observe that we use Greek indices when working with geometric tensors and henceforth we will stick to this convention. In addition, it is convenient to recollect all the information of the metric in a single object, called **line element**:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (13)$$

The reason for this name will become clear once we will be studying geodesics. Observe that the line element is a mere formal object and the superscript 2 should not lead you to think that we are dealing with a positive definite object, although this becomes somehow true in Riemannian geometry.

Example 51. In order to make the above concepts more concrete, let us discuss a rather simple, yet instructive example. *We go once through all the steps, so to make the reader acquainted with the calculations.* Let us consider $\mathbb{R}^3$ endowed with the standard Cartesian coordinates x, y, z and let us consider the standard Euclidean metric $\delta \in \Gamma((T^*\mathbb{R}^3)^{\otimes 2})$, which reads

$$\delta = \delta_{\mu\nu} dx^\mu \otimes dx^\nu = dx \otimes dx + dy \otimes dy + dz \otimes dz. \quad (14)$$

The associated line element reads $ds^2 = \delta_{\mu\nu} dx^\mu dx^\nu = dx^2 + dy^2 + dz^2$ and it is also convenient to observe that this allows us to associate in the given coordinate system a matrix to δ whose entries are nothing but $\delta_{\mu\nu}$, i.e.,

$$\delta_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

This can be brought one step further by considering the *unit 2-sphere* $\mathbb{S}^2$ defined as

$$\mathbb{S}^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}.$$

In order to induce a metric thereon, it is convenient to introduce the polar coordinates

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta,$$

where $r = 1$, while $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$. These can be inverted as

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \arccos \frac{z}{\sqrt{x^2 + y^2 + z^2}}, \quad \varphi = \arctan \frac{y}{x}. \quad (15)$$

By considering the submanifold $\mathbb{R}^3 \setminus \{0\}$ and the restriction of the Euclidean metric thereon, we can switch from the Cartesian to the polar coordinates. First of all we can write the differentials, namely

$$dx = \sin \theta \cos \varphi dr - r \sin \theta \sin \varphi d\varphi - r \cos \theta \sin \varphi d\theta, \quad (16a)$$

$$dy = \sin \theta \sin \varphi dr + r \cos \theta \sin \varphi d\theta + r \sin \theta \cos \varphi d\varphi, \quad (16b)$$

$$dz = \cos \theta dr - r \sin \theta d\theta, \quad (16c)$$

which can be replaced in Equation (14) to obtain

$$\delta|_{\mathbb{R}^3 \setminus \{0\}} = dr \otimes dr + r^2(d\theta \otimes d\theta + \sin^2 \theta d\varphi \otimes d\varphi).$$

This is nothing but the flat metric though in a different coordinate system and we can observe that the line element as well as the matrix form of δ become

$$ds^2 = \delta_{\mu\nu}|_{\mathbb{R}^3 \setminus \{0\}} dx^\mu dx^\nu = dr^2 + r^2(d\theta^2 + \sin^2 \theta d\varphi^2),$$

and

$$\delta_{\mu\nu}|_{\mathbb{R}^3 \setminus \{0\}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}.$$

Observe that we have used on purpose the same symbols, namely $\delta_{\mu\nu}$ and $\delta_{\mu\nu} dx^\mu dx^\nu$ as these carry no specific intrinsic significance and they are tied to the choice of coordinate system. The reader should learn to recognize that, while computationally very efficient, every expression written in terms of components is intrinsically local and its expression coordinate-dependent.

Switching back to the unit 2-sphere, it suffices to notice that, in polar coordinates this can be also realized as the embedded submanifold $\mathbb{S}^2 \doteq \{(r, \theta, \varphi) \in \mathbb{R}^3 \setminus \{0\} \mid r = 1\}$. We denote the embedding as the map

$$\iota : \mathbb{S}^2 \rightarrow \mathbb{R}^3 \setminus \{0\}, \quad (\theta, \varphi) \mapsto \iota(\theta, \varphi) = (1, \theta, \varphi).$$

Generalizing to symmetric 2-tensors, that is elements of $(T^*M)^{\otimes s^2}$ Equation (10), we obtain that

$$g_{\mathbb{S}^2} = g_{\mu\nu} dx^\mu dx^\nu = \iota^* \delta = \iota^* (\delta_{\mu\nu}|_{\mathbb{R}^3 \setminus \{0\}}(x) dx^\mu dx^\nu) = \delta_{\mu\nu}|_{\mathbb{R}^3 \setminus \{0\}}(\iota(x)) d(x^\mu \circ \iota) d(x^\nu \circ \iota).$$

Since the generic point is $x \equiv (r, \theta, \varphi)$, $\iota(x) = (1, \theta, \varphi) \equiv (\theta, \varphi)$, while $d(x^\mu \circ \iota)$ is obtained setting $r = 1$ in Equation (16). This entails that

$$g_{\mathbb{S}^2} = g_{\mu\nu} dx^\mu dx^\nu = d\theta \otimes d\theta + \sin^2 \theta d\varphi \otimes d\varphi,$$

whose associated line element reads $ds_{\mathbb{S}^2}^2 = d\theta^2 + \sin^2 \theta d\varphi^2$, while the metric components can be written as the matrix

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{bmatrix}.$$

Remark 16 (Conceptual). *While, at first glance, the metric is just one among the many possible tensor fields on a smooth manifold, it is of paramount relevance even beyond the realm of general relativity. The etymology of the word itself suggests that it is related to the concept*

of measurement and this is indeed true since, given M a smooth n -dimensional manifold as per Definition 9, we recall that a metric $g \in \Gamma(T^*M^{\otimes_s 2})$. This entails that for any $p \in M$

$$g(p) \in T^*M \otimes_s T^*M \implies g(p) : T_pM \times T_pM \rightarrow \mathbb{R},$$

that is the metric is, per definition, a bilinear and non degenerate map on the space of vectors at each point on the underlying manifold. Heuristically speaking, one is tempted to say that a metric measures the length of a vector. This is indeed true, provided that one is ready to extend the standard notion of length by allowing it to be even vanishing or negative.

This idea can be clarified as follows: Given $p \in M$, consider any local chart (U, φ) such that $p \in U$ and any coordinate system thereon $\{x^1, \dots, x^n\}$. This induces a basis of T_pM , namely $\partial_1, \dots, \partial_n$ and we can therefore consider the associated matrix $g(p) \in \mathcal{M}(n, \mathbb{R})$ where $n = \dim M$. The entries are $g(p)_{\mu\nu} = g(p)(\partial_\mu, \partial_\nu)$, $\mu, \nu = 1, \dots, n$. On account of the assumptions on g , see Definition 50, $g(p)$ is a symmetric matrix and, therefore, using the spectral theorem, it can be diagonalized. Hence there exists an orthogonal matrix $O(p) \in O(n, \mathbb{R})$ such that

$$(O^T(p)g(p)O(p))_{\mu\nu} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}.$$

On top of this result, being g non degenerate we can conclude that $\lambda_i \neq 0$ for all $i = 1, \dots, n$. This entails that we can construct a basis $\{v_i(p)\}_{i=1, \dots, n}$ of $T_p(M)$ in terms of eigenvectors of $g(p)$. Since $g(p)(v_i(p), v_i(p)) = \lambda_i$, we can define an orthonormal basis $\{e_i(p)\}$ with $e_i(p) \doteq |\lambda_i|^{-\frac{1}{2}} v_i(p)$, which is called an **n-bein** at p . If $n = 3, 4$ then one usually calls it a **dreibein**/**vierbein** at p . Observe that, since we have no guarantee that the eigenvalues are positive, it turns out that $g(p)(e_i(p), e_i(p)) = \pm 1$, while λ_i can never vanish since the metric is per assumption non degenerate. The number of vectors of positive and negative length is an invariant at p since it measures the sign of the eigenvalues of $g(p)$ which does not depend on the choice of the basis. Therefore we can say that a smooth manifold M endowed with a metric tensor g is of **signature** (n_-, n_+) at $p \in M$ if $g(p)$ has n_- eigenvalues of negative sign and n_+ of positive sign.

In view of this remark we can classify smooth manifolds endowed with a metric tensor in terms of its signature. This leads to introducing one of the main concepts of general relativity whose definition is often never made precise.

Definition 52. Let M be an n -dimensional smooth manifold as per Definition 9 with $n \geq 1$ and let g be a metric tensor thereon as per Definition 50. We say that (M, g) is a **Riemannian manifold** if g is of signature $(0, n)$ for all $p \in M$, while we call it a **Lorentzian manifold** if g is of signature $(1, n-1)$ or $(n-1, 1)$ at all points $p \in M$. We call **spacetime** a connected, Lorentzian manifold (M, g) .

Observe that, in the definition of Lorentzian manifold, while both signatures $(1, n-1)$ and $(n-1, 1)$ are admitted, this is only a matter of convention and each manifold in this class abides

by one of the two possibilities globally. Hence a swap between the two options is not possible. In addition, the case $n = 1$ is degenerate since, in this instance, the notion of Riemannian and Lorentzian manifolds are equivalent. At last, we stress that many textbooks require implicitly or sometimes even explicitly that spacetimes are paracompact, namely, for every open cover of a smooth manifold M , there exists a locally finite refinement. Recall that a cover of M is a collection $\mathcal{O}$ of open subset U_i , $i \in I$, I being an index set, such that $M = \bigcup_{i \in I} U_i$. A refinement is a second cover $\mathcal{O}'$ such that, for any open set $V \in \mathcal{O}'$, there exists $U \in \mathcal{O}$ such that $V \subseteq U$. Locally finite means instead that for every point $p \in M$, there exists an open set which intersects at most a finite number of elements of the cover $\mathcal{O}'$. Yet this property is a consequence of all the assumptions we made, see in particular [Lee13, Thm. 1.15].

2.4 Connections and Covariant Derivatives

In the previous section, we have introduced the kinematic structure of general relativity, namely all pairs (M, g) where M is a connected, smooth differentiable structure while g is a metric. In this section this can be either Riemannian or Lorentzian, see Definition 52, though only the latter carries physical significance in general relativity. In the following we start introducing additional structures which will allow us to set up a dynamical problem from metric tensors.

Notation and Nomenclature [Important]: Henceforth, when considering a smooth manifold M and tensors thereon, it will be often convenient and, to a certain extent, necessary to work locally in a given chart and with a choice of coordinates assigned. Therefore, whenever we will be writing *in coordinates* or *locally*, we will be implicitly meaning that we are selecting an arbitrary but fixed $p \in M$, a local chart (U, φ) such that $p \in U$ and coordinates $\{x^1, \dots, x^n\}$, $n = \dim M$. With respect to these choices, every tensor $T \in \mathcal{T}^{(k,l)}(M)$ can be written as

$$T(x) = T_{\mu_1 \dots \mu_l}^{\nu_1 \dots \nu_k}(x) \frac{\partial}{\partial x^{\nu_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{\nu_k}} \otimes dx^{\mu_1} \otimes \dots \otimes dx^{\mu_l}. \quad x \in U$$

In this respect we shall identify T with its components $T_{\mu_1 \dots \mu_l}^{\nu_1 \dots \nu_k} \equiv T_{\mu_1 \dots \mu_l}^{\nu_1 \dots \nu_k}(x)$. It goes without saying that all results need and will be independent from all these implicit choices and this notation is adopted since it makes explicit calculations more transparent.

Let us focus for the time being on the tangent space TM . According to Definition 30 this can be seen as the disjoint union of the tangent spaces at all point of the underlying smooth manifold M . Given any pair $p, q \in M$, $T_p M$ is isomorphic to $T_q M$ and to $\mathbb{R}^n$, $n = \dim M$. Yet, from the mathematical viewpoint, such isomorphism is certainly not unique or canonical. From a more physical viewpoint, we can translate this remark as follows: Consider an observer sitting at p and interpret the elements of $T_p M$ as its internal degrees of freedom. Pictorially you can think of a capybara sitting peacefully at p with an arrow in his mouth, which is a vector $v \in T_p M$. Suppose that the capybara, still peacefully, wants to move from p to q along a curve γ . While γ constrains the path, it says nothing about what happens to the arrow in his/her mouth along the curve. Where does it point once we get to q ? Is it rotated? Is it left unchanged? We lack a

rule telling us how to compare vectors in $T_p M$ and in $T_q M$. While the capybara can cozily live without answering this question, the same cannot be said about us. Hence, in the following we shall introduce the notion of *connection* or *covariant derivative*. The first term is used more often in the theory of vector bundles to which this question pertains in its full generality. We shall not delve here in this topic, leaving the interested readers to the course of Metodi Matematici per la Fisica Teorica. On the contrary we shall use more often the second term, which is more common in the theory of Riemannian and Lorentzian manifolds, see [Lee11] for more technical details. As we shall see, in this scenario, the existence of a metric tensor allows to identify the Levi-Civita connection, which is a canonical structure. This is a sharp difference with respect to the theory of vector bundles where this is not possible under normal circumstances. The following definitions and results are here specialized to the case of the tangent space, but the reader should be warned that they have a wider range of applicability.

Definition 53. Let M be an n -dimensional smooth manifold, $n \geq 1$, as per Definition 9 and let TM be the tangent space as per Definition 30. We call **linear (or Koszul) connection** on TM a linear map $\nabla_L : \Gamma(TM) \rightarrow \Gamma(TM \otimes T^*M)$ such that, for all $\sigma \in \Gamma(TM)$ and for all $f \in C^\infty(M)$, it holds

$$\nabla_L(f\sigma) = f\nabla_L\sigma + df \otimes \sigma,$$

where $TM \otimes T^*M$ is as per Definition 49.

As it happens often for primary concepts, the above is just one among the many equivalent definitions of connections present in the literature. More importantly the concept of linear connection, given in Definition 53 seems a little bit detached from the heuristic notion given in the preceding discussion. We could have avoided giving this definition but it is often seen in more mathematical textbooks. The following proposition should dispel any doubt, since it specifies another, yet equivalent, characterization of a connection.

Proposition 54. Let M be an n -dimensional smooth manifold, $n \geq 1$, as per Definition 9 and let TM be the tangent space as per Definition 30. It is equivalent either to assigning a Koszul connection as per Definition 53 or a bilinear map

$$\nabla : \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TM), \quad (17)$$

such that, calling $\nabla_X : \Gamma(TM) \rightarrow \Gamma(TM)$ the map $\Gamma(TM) \ni X \mapsto \nabla_X \doteq \nabla(X, \cdot)$, it holds that, for all $f \in C^\infty(M)$ and for all $\sigma \in \Gamma(TM)$

$$\nabla_{fX}(\sigma) = f\nabla_X(\sigma), \quad \text{and} \quad \nabla_X(f\sigma) = f\nabla_X\sigma + X(f)\sigma. \quad (18)$$

The map ∇_X is called the **covariant derivative (relative to X)**.

Proof. Suppose a connection as per Equation (17) has been assigned. For any $\sigma \in \Gamma(TM)$, consider $\nabla(\cdot, \sigma) : \Gamma(TM) \rightarrow \Gamma(TM)$. This is a linear map which identifies therefore an element of $\Gamma(TM \otimes T^*M)$ which we denote as $\tilde{\nabla}(\sigma)$. Consider now $f \in C^\infty(M)$ and $X \in \Gamma(TM)$. It holds that

$$\nabla(X, f\sigma) = f\nabla(X, \sigma) + X(f)\sigma = f\tilde{\nabla}(\sigma)(X) + df(X)\sigma.$$

By the arbitrariness of X , it descends that

$$\tilde{\nabla}(f\sigma) = f\tilde{\nabla}\sigma + df \otimes \sigma,$$

namely we have identified a Koszul connection. Conversely, suppose that a linear connection ∇_L has been assigned in agreement to Definition 53. Define

$$\nabla : \Gamma^\infty(TM) \times \Gamma(TM) \rightarrow \Gamma(TM), \quad (X, \sigma) \mapsto \nabla(X, \sigma) = \nabla_X(\sigma) \doteq \nabla_L(\sigma)(X).$$

For any $f \in C^\infty(M)$, $X \in \Gamma^\infty(TM)$ and $\sigma \in \Gamma(TM)$, it holds

$$\nabla_X(f\sigma) = \nabla(X, f\sigma) = \nabla_L(f\sigma)(X) = f\nabla_L(\sigma)(X) + df(X)\sigma = f\nabla_X(\sigma) + X(f)\sigma,$$

and similarly

$$\nabla_{fX}(\sigma) = \nabla(fX, \sigma) = \nabla_L(\sigma)(fX) = f\nabla_L(\sigma)(X) = f\nabla_X(\sigma),$$

where we used the defining properties of a Koszul connection in the before last equality. $\square$

Disclaimer: In view of Definition 53 and of Proposition 54 we shall use indifferently the notion of Koszul connection and Equation (17) and with a slight abuse of notation we shall always use the symbol ∇ .

The first question that one should ask at this stage is whether connections do exist on TM , regardless of the assignment of a metric tensor. Luckily enough, though not surprisingly, the answer to our question is affirmative. As a matter of fact, consider a cover $\{U_\alpha\}_{\alpha \in I}$, I being an index set, of M such that each U_α identifies a chart of M . Choosing $\{\psi_\alpha\}_{\alpha \in I}$ a partition of unity subordinated to this cover, see Section A, it holds per linearity that, for any connection ∇ , as per equation (17), and for any $X \in \Gamma^\infty(TM)$,

$$\nabla_X \sigma = \sum_{\alpha \in I} \psi_\alpha \nabla_X \sigma = \sum_{\alpha \in I} \nabla_X(\psi_\alpha \sigma),$$

where we used implicitly that $\sum_{\alpha \in I} X(\psi_\alpha) \sigma = X(\sum_{\alpha \in I} \psi_\alpha) \sigma = 0$ since $\sum_{\alpha \in I} \psi_\alpha = 1$. This formula guarantees that, in constructing a connection ∇ , we can focus our attention to each open neighbourhood U_α . Since this is a chart, then $TM|_{U_\alpha} \simeq U_\alpha \times \mathbb{R}^n$. Thereon, consider a collection of linearly independent vector fields, $e_j : U_\alpha \rightarrow U_\alpha \times \mathbb{R}^n$, $j = 1, \dots, n$. For any $Y \in \Gamma^\infty(TU_\alpha)$ and for all $j = 1, \dots, n$ set $\nabla_{\alpha, Y}(e_j) = 0$. This identifies uniquely the map

$$\nabla_\alpha : \Gamma^\infty(TU_\alpha) \times \Gamma^\infty(TM|_{U_\alpha}) \rightarrow \Gamma^\infty(TM|_{U_\alpha}), \quad (Y, \eta) \mapsto \nabla_\alpha(Y, \eta) \doteq \sum_{j=1}^n Y(\eta_j) e_j,$$

where we have decomposed $\eta = \sum_{j=1}^n \eta_j e_j$, $\eta_j \in C^\infty(U_\alpha)$ for all $j = 1, \dots, n$. It is immediate to verify that we have defined a connection on $TM|_{U_\alpha}$ as per Proposition 54. By gluing together all these data, it turns out that, for all $X \in \Gamma^\infty(TM)$ and for all $\sigma \in \Gamma(TM)$, we can write

$$\nabla_X \sigma = \sum_{\alpha \in I} \nabla_{\alpha, X|_{U_\alpha}}(\psi_\alpha \sigma). \quad (19)$$

This is known as the **trivial connection on TM**.

Proposition 54 translates our heuristic idea of a connection since ∇ is a map which is applied to two vector fields. The first codifies the tangent vectors of all possible curves joining two points on the underlying manifold M , while the second is nothing but the vector that we want to transport between two points. Let us now assume that a linear connection ∇ as per Equation (17) has been assigned. Working in components in a chart (U, φ) , every $X, Y \in \Gamma(TM|_U)$ can be written as $X = X^i \partial_i$ and $Y = Y^j \partial_j$ with $X^i, Y^j \in C^\infty(U)$ for all $i, j = 1, \dots, n$. Hence, using Equation (18),

$$\nabla(X, Y) \equiv \nabla_X(Y) = \nabla_{X^i \partial_i}(Y^j \partial_j) = X^i Y^j \nabla_{\partial_i}(\partial_j) + X^i \partial_i(Y^j) \partial_j, \quad (20)$$

from which we infer that the only unknown quantity in this expression is one written in bold faced characters. Yet, since it identifies an element of $\Gamma(TM|_U)$,

$$\nabla_{\partial_i}(\partial_j) = \Gamma_{ij}^k \partial_k, \quad (21)$$

where $\Gamma_{ij}^k \in C^\infty(U)$ are known as **Christoffel symbols** of ∇ . Hence, when a connection ∇ is assigned, for every $Y \in \Gamma(TM)$ its action in components reads

$$\nabla_i Y^j = \partial_i Y^j + \Gamma_{ik}^j Y^k. \quad (22)$$

Having established the existence and the local characterization of a connection on a smooth manifold, our next step consists of extending its definition to an arbitrary tensor field. This can be done uniquely provided suitable constraints are imposed. Among these, one is subordinated to compatibility with the counterpart at the level of tensor fields of a notable operation on matrices: *the trace*. Let us first make precise this notion.

Definition 55. Let M be an n -dimensional smooth manifold, $n \geq 1$, as per Definition 9 and let $T \in \mathcal{T}^{(1,1)}(M)$, see Definition 49. We call **trace** the map

$$Tr : \mathcal{T}^{(1,1)}(M) \rightarrow C^\infty(M), \quad T \mapsto Tr(T),$$

such that, in every local chart and, given any coordinate system $\{x^1, \dots, x^n\}$,

$$Tr(T) = T_i^i, \quad (23)$$

where, $T = T_j^i \frac{\partial}{\partial x^i} \otimes dx^j$.

Observe that, in this definition, we have not claimed that we are working locally or, in coordinates, since the right hand side of Equation (23) does not depend on the choice of the coordinates, being Tr , pointwisely, nothing but the trace of a matrix which is a basis independent quantity. In addition Equation (23) can be readily extended to higher order tensor fields, although not in a unique way since we have the freedom the contract/trace out any two lower

and upper indices. With a slight abuse of notation, for any $k, l \geq 1$, we shall therefore call more generally **trace** the map

$$\text{Tr}_{ab} : \mathcal{T}^{(k,l)}(M) \rightarrow \mathcal{T}^{(k-1,l-1)}(M), \quad T \mapsto \text{Tr}_{ab}(T),$$

such that, in every local chart and, given any coordinate system $\{x^1, \dots, x^n\}$,

$$\text{Tr}_{ab}(T) = \delta_{i_b}^{j_a} T_{j_1 \dots j_a \dots j_l}^{i_1, \dots, i_b, \dots, i_k} \partial_{i_1} \otimes \dots \otimes \widehat{\partial_{i_b}} \otimes \dots \otimes \partial_{i_k} \otimes dx^i \otimes \dots \widehat{dx^{j_a}} \otimes \dots \otimes dx^{j_l}, \quad (24)$$

where a is any but fixed index running from 1 to l , while b span 1 to k . The hat symbol entails that the basis element does not appear. As in Equation (23) this coincides pointwisely with the trace of a matrix for any choice of the contracted indices and, therefore, this is a basis independent notion.

Remark 17. *The reader is warned that this operation is metric-independent and, when seeing in the literature, the components of a tensor written in the form $T_{i_1, \dots, m, \dots, i_k}^{j_1 \dots m \dots j_l}$, this entails that the indices m are summed from 1 to $n = \dim M$. Hence the trace is an operation which does not depend on the metric structure, say the least on its signature and it should not be confused with the action of raising or lowering indices in a tensor which is instead metric-dependent, as we shall see in the following.*

Proposition 56. *Let M be an n -dimensional smooth manifold, $n \geq 1$, as per Definition 9 and let ∇ be a linear connection as per Proposition 54. Then there exists a unique extension, still denoted by the same symbol*

$$\nabla : \Gamma(T(M)) \times \mathcal{T}^{(k,l)}(M) \rightarrow \mathcal{T}^{(k,l)}(M),$$

which is a multilinear map such that

1. if $k = 1$ and $l = 0$ it coincides with the given connection,
2. if $k = l = 0$, then, for all $X \in \Gamma(TM)$ and for all $f \in C^\infty(M)$, $\nabla(X, f) = \nabla_X(f) \doteq X(f)$,
3. it is an algebra homomorphism, that is, for all $F \in \mathcal{T}^{(k,l)}(M)$, $G \in \mathcal{T}^{(k',l')}(M)$ and for all $X \in \Gamma(TM)$

$$\nabla(X, F \otimes G) = \nabla_X(F \otimes G) = \nabla_X(F) \otimes G + F \otimes \nabla_X(G),$$

4. it commutes with the trace operation in Equation (24), namely, for all $F \in \mathcal{T}^{(k,l)}(M)$ and for all $X \in \Gamma(TM)$

$$\nabla(X, \text{Tr}_{ab}(T)) = \nabla_X(\text{Tr}_{ab}(T)) = \text{Tr}_{ab}(\nabla_X(T)),$$

where a is any but fixed index running from 1 to k , while b span 1 to l .

Proof. First of all we prove that the covariant derivative acts locally. Suppose that a connection exists and we consider $F, H \in \mathcal{T}^{(k,l)}(M)$ such that $F - H = 0$ for all $p \in V$, where V is an open subset of U , a local chart of M . Consider a bump function $\psi \in C_0^\infty(M)$ such that $\psi \geq 0$ and $\psi|_V = 1$, see [BT82, Prop. 2.25]. Calling $K = F - H$, it holds that, for all $X \in \Gamma(TM)$, $\nabla_X(\psi K) = X(\psi)K + \psi \nabla_X K = 0$ since $\psi K = 0$. In addition since $X(\psi) = 0$ for all $p \in \text{supp}(K)$, we can conclude that $\psi \nabla_X K = \nabla_X K = 0$ where we used that $\psi = 1$ on V . This entails that ∇ acts locally and it suffices to specify its action on a cover of M in local charts to reconstruct it globally.

Having established this fact, we can proceed by induction on the indices (k, l) and, where necessary, we work locally in a given chart and with respect to arbitrary but fixed coordinates. More precisely we observe that, if $l = 0$ and $k = 0, 1$, then uniqueness is guaranteed by items 1. and 2.. To prove the result on $\mathcal{T}^{(0,1)}(M)$, let $\omega \in \Gamma(T^*M)$ and let $X, Y \in \Gamma(TM)$. It turns out that items 3. and 4. entail that

$$\nabla_X(\text{Tr}(Y \otimes \omega)) = X(Y^i \omega_i) = X(Y^i) \omega_i + Y^i X(\omega_i),$$

while

$$\text{Tr}(\nabla_X(Y \otimes \omega)) = \text{Tr}(\nabla_X(Y) \otimes \omega + Y \otimes \nabla_X \omega).$$

Using Equation (20)

$$\text{Tr}(\nabla_X(Y) \otimes \omega) + \text{Tr}(Y \otimes \nabla_X \omega) = \omega_k (X^i Y^j \Gamma_{ij}^k + X^i \partial_i (Y^k)) + Y^k (\nabla_X \omega)_k^j \partial_j.$$

while

$$X(Y^i \omega_i) = X(Y^i) \omega_i + Y^i X(\omega_i) = X^j \partial_j (Y^i) \omega_i + Y^i X^j \partial_j \omega_i.$$

Comparing these formulae and renaming where necessary dummy indices, it holds that

$$\nabla_X \omega = X^j \partial_j (\omega_i) dx^i - X^j \omega_k \Gamma_{ji}^k dx^i,$$

or, in components

$$\nabla_i \omega_j = \partial_i \omega_j - \Gamma_{ij}^k \omega_k. \quad (25)$$

This identifies the action of the connection unambiguously of $\Gamma(T^*M)$. We can now proceed per induction assuming that ∇ has been defined on $\mathcal{T}^{(k,l)}(M)$ and we prove that it extends uniquely to $\mathcal{T}^{(k+1,l)}(M)$ and to $\mathcal{T}^{(k,l+1)}(M)$ separately. We consider only the first case, the second one following suit. Yet $T \in \mathcal{T}^{(k+1,l)}(M)$ can be written locally as

$$T = T_{j_1 \dots j_l}^{i_1 \dots i_k i_{k+1}} \partial_{i_1} \otimes \dots \otimes \partial_{i_k} \otimes \partial_{i_{k+1}} \otimes dx^1 \otimes \dots \otimes dx^{j_l} = \bar{T}^{i_{k+1}} \otimes \partial_{i_{k+1}},$$

where, in the last equality, we have written T as a linear combination of tensor product between elements $\bar{T}^{i_{k+1}} \in \mathcal{T}^{(k,l)}(TM|_U)$ and $\partial_{i_{k+1}} \in \Gamma(TM|_U)$. Yet the hypothesis of linearity and item 3. entail that, for all $X \in \Gamma(TM)$,

$$\nabla_X T = \nabla_X (\bar{T}^{i_{k+1}} \otimes \partial_{i_{k+1}}) = \nabla_X (\bar{T}^{i_{k+1}}) \otimes \partial_{i_{k+1}} + \bar{T}^{i_{k+1}} \otimes \nabla_X (\partial_{i_{k+1}}).$$

Yet the induction hypothesis guarantees the uniqueness of the right hand side. $\square$

We can infer two straightforward properties of a connection, the first of which is a direct consequence of condition 3. and 4. in Proposition 56.

Corollary 57. *Let M be an n -dimensional smooth manifold, $n \geq 1$, as per Definition 9 and let ∇ be a linear connection as per Proposition 54. Let $(,) : \Gamma(T^*M) \times \Gamma(TM) \rightarrow C^\infty(M)$ denote the standard pairing between 1-forms and vector fields. It holds that*

1. *for all $X, Y \in \Gamma(TM)$ and for all $\omega \in \Gamma(T^*M)$,*

$$\nabla_X(\omega, Y) = (\nabla_X \omega, Y) + (\omega, \nabla_X Y),$$

where, on the right hand side $(,)$ denotes the pairing between the same tensor entries as on the left hand side.

Proof. Since, as shown in the proof of Proposition 56, the action of ∇ is local, it suffices to prove the statement in an arbitrary but fixed chart. Hence, working in coordinates and using Equations (22) and (25)

$$X^i \nabla_i(\omega_j Y^j) = X^i \partial_i(\omega_j Y^j) = X^i(\partial_i \omega_j) Y^j + X^i \omega_j \partial_i Y^j = X^i(\partial_i \omega_j) Y^j + X^i \omega_j \partial_i Y^j \pm X^i \Gamma_{ij}^k Y^j \omega_k,$$

where, in the first equality we used item 2. of Proposition 56. Barring a redefinition of dummy indices the right hand side can be read as the local form of $(\nabla_X \omega, Y) + (\omega, \nabla_X Y)$. $\square$

Action in components of ∇ : Having established in Proposition 56 that the assignment of a linear connection as per Proposition 54 automatically induces a counterpart on all tensor fields, we are left with the operational question how to apply ∇ on a generic $T \in \mathcal{T}^{(k,l)}(M)$ when working with coordinates. The generalization of Equations (22) and (25) reads

$$\nabla_a T_{j_1 \dots j_l}^{i_1 \dots i_k} = \partial_a T_{j_1 \dots j_l}^{i_1 \dots i_k} - \sum_{b=1}^l \Gamma_{a j_b}^c T_{j_1 \dots c \dots j_l}^{i_1 \dots i_k} + \sum_{d=1}^k \Gamma_{ad}^{i_d} T_{j_1 \dots j_l}^{i_1 \dots d \dots i_k}, \quad (26)$$

where the indices c and d are places respectively in the j_b -th and in the i_d -th entry of the components of T . Hands on, this means that, barring the action of the partial derivative, ∇ acts on every lower index with $-\Gamma$ and on each upper one with $+\Gamma$.

Metric induced Connection: Until this point, we have only established that it suffices to assign a connection on vector fields to have it defined on all tensor counterparts, but the whole construction is purely subordinated to the differential structure of the manifold M and there is no reference to additional structures such as the metric. Henceforth suppose instead that a non-degenerate $g \in \Gamma((T^*M)^{\otimes_s 2})$ has been assigned and we make no assumption in this section on the signature, see Definition 50. Before delving into the details, we need to introduce some operations which are well-known in physics as the action of *raising and lowering indices*. Recall that, besides being symmetric, a metric tensor is degenerate and therefore

$$\forall \chi \in \Gamma(TM), \quad g(\chi, \cdot) : \Gamma(TM) \rightarrow C^\infty(M).$$

In other words $g(\chi, \cdot) \in \Gamma(TM)^* \simeq \Gamma(T^*M)$. Being non degenerate, this map is injective and, since at each point $p \in M$, $\dim T_p M = \dim T_p^* M = \dim M$, the dimension theorem from linear algebra entails that the map

$$\flat : \Gamma(TM) \rightarrow \Gamma(T^*M), \quad \chi \mapsto \chi^\flat \doteq g(\chi, \cdot), \quad (27)$$

is an isomorphism. Since the action of $\flat$ is pointwise, this also identifies a bijection between TM and T^*M , since, for all $v \in T_p M$, $v^\flat \doteq g(p)(v, \cdot) \in T_p^* M$. The map $\flat$ is called **musical isomorphism**. In components

$$(\chi^\flat)_\mu = g_{\mu\nu} \chi^\nu,$$

where we have restored the convention of using Greek indices for the components of geometric-related tensors.

Remark 18 (Notational). *Observe that, being non-degenerate, the metric tensor g admits an inverse $g^{-1} \in \Gamma(TM^{\otimes_s 2})$. In components $(g^{-1})^{\mu\nu}$ is such that*

$$(g^{-1})^{\mu\nu} g_{\nu\gamma} = \delta_\gamma^\mu, \quad g_{\gamma\mu} (g^{-1})^{\mu\nu} = \delta_\gamma^\nu.$$

*Yet it is customary in the literature to omit the superscript -1 when working in components and therefore we shall be writing $g^{\mu\nu} g_{\nu\gamma} = \delta_\gamma^\mu$. The superscript will be restored whenever we will be dealing with global sections. In addition observe that g^{-1} induces a second **musical isomorphism** between $\Gamma(T^*M)$ and $\Gamma(TM)$ which is the inverse of Equation (27), namely*

$$\sharp : \Gamma(T^*M) \rightarrow \Gamma(TM), \quad \omega \mapsto \omega^\sharp \doteq g^{-1}(\omega, \cdot), \quad (28)$$

*where, formally, on the right hand side we are interpreting g^{-1} as an element of $\Gamma(T^{**}M)$. In components this reads*

$$(\omega^\sharp)^i = g^{ij} \omega_j.$$

These are the well-known operations of raising and lowering indices and, from now on, we shall keep both musical isomorphisms implicit. We can combine these maps with the trace map in Definition 55 and in Equation (24) to give meaning to another well-known procedure used in many computation in theoretical physics. This is often referred to as the *index contraction* or *metric trace*. This is the first instance where one might wish to keep track of the position of the indices when dealing with the components of a tensor field. In principle this should not matter and we could still avoid it at this stage, but in later discussions, it will become very convenient doing so. The author strongly recommends the `LATEX`package `tensor` to implement it. That said, considering any $T \in \mathcal{T}^{(k,0)}(M)$, $k \geq 2$ and its components $T_{\mu_1, \dots, \mu_k}$ we can define for $a \neq b \in \{1, \dots, k\}$

$$\text{Tr}_{g,ab} : \mathcal{T}^{(k,0)}(M) \rightarrow \mathcal{T}^{(k-2,0)}(M), \quad T \mapsto \text{Tr}_{g,ab}(T), \quad (29)$$

which, written in coordinates reads

$$\text{Tr}_{g,ab}(T)^{\mu_1 \dots \hat{\mu}_a \dots \hat{\mu}_b \dots \mu_k} = g_{\mu_a \mu_b} T^{\mu_1 \dots \mu_a \dots \mu_b \dots \mu_k} \doteq \delta_{\mu_b}^\nu g_{\mu_a \nu} T^{\mu_1 \dots \mu_a \dots \mu_b \dots \mu_k},$$

where the hat symbol still entails that the index below it does not appear. In other words $\text{Tr}_{g,ab}$ is the composition of the musical isomorphism as in Equation (28) with the contraction map as per Equation (24).

Exercise 58. Construct the generalization of Equation (29) to a map

$$\text{Tr}_{g,ab} : \mathcal{T}^{(k,l)}(M) \rightarrow \mathcal{T}^{(k',l')}(M), \quad T \mapsto \text{Tr}_{g,ab}(T),$$

such that $k + l \geq 2$ and $k' + l' = k + l - 2$.

Remark 19 (Notational). *It is often convenient not to denote explicitly the action of the metric tensor when tracing out indices. Yet, to do it avoiding confusion, it is important to keep track of the position of indices as already mentioned. We shall therefore write*

$$g^{\mu_a \mu_b} T_{\mu_1 \dots \mu_a \dots \mu_b \dots \mu_k} = T_{\mu_1 \dots \mu_{a-1} \quad \mu_a \quad \mu_{a+1} \dots \mu_{b-1} \mu_a \mu_{b+1} \dots \mu_k}.$$

We have set up all the ingredients necessary to investigate whether a manifold (M, g) with a smooth metric comes endowed with a unique notion of connection. The answer is negative unless we impose further constraints. The first is motivated by Corollary 57, which highlights a rather straightforward interplay between a connection and the pointwise duality pairing between TM and T^*M . On the one hand we have remarked that the metric induces a canonical pairing between TM and T^*M by means of the musical isomorphism in Equation (27). On the other hand the pairing employed in Corollary 57 can be read as being the musical isomorphism induced by standard Euclidean metric on $\mathbb{R}^n$. Therefore we can look for a connection which allows us to replace pointwisely the Euclidean pairing with that induced by the metric. Heuristically, one can expect this to be possible, since, as mentioned in Remark 16, at each point $p \in M$, it is possible to find a coordinate system which makes the metric diagonal and that point. Furthermore, a reader can convince him/herself that a change from the Riemannian signature does not affect the content of Corollary 57. This motivates the following definition.

Definition 59. *Let (M, g) be an n -dimensional smooth manifold with a metric tensor g , $n \geq 1$, as per Definitions 9 and 50. In addition let ∇ be a linear connection as per Proposition 54. We call it **metric compatible** if, for all $X, Y, Z \in \Gamma(TM)$,*

$$\nabla_X \langle Y, Z \rangle_g = \langle \nabla_X Y, Z \rangle_g + \langle Y, \nabla_X Z \rangle_g, \quad (30)$$

where $\langle Y, Z \rangle_g \in C^\infty(M)$ reads locally $g^{\mu\nu} Y_\mu Z_\nu$, while $\langle \nabla_X Y, Z \rangle_g$ can be written as $g^{\mu\nu} X^\lambda (\nabla_\lambda Y_\mu) Z_\nu$ and similarly $\langle Y, \nabla_X Z \rangle_g$.

We can establish a very useful necessary and sufficient characterization for metric compatibility of a connection

Proposition 60. *Let (M, g) be an n -dimensional smooth manifold with a metric tensor g , $n \geq 1$, as per Definitions 9 and 50. In addition let ∇ be a linear connection as per Proposition 54. We say that ∇ is metric compatible as per Definition 59 if and only if, for all $X \in \Gamma(TM)$, $\nabla_X g = 0$, or, equivalently, working in coordinates $\nabla_\mu g_{\nu\rho} = 0$ for all $\mu, \nu, \rho = 1, \dots, n$.*

Proof. Since ∇ acts only locally, see proof of Proposition 54, it suffices to prove the statement in a single, yet arbitrary, local chart. Hence, working in coordinates, suppose that a metric compatible connection ∇ has been assigned. For every $X, Y, Z \in \Gamma(TM)$, the left hand side of Equation (30) reads

$$\begin{aligned} X^\rho \nabla_\rho (g^{\mu\nu} Y_\mu Z_\nu) &= X^\rho \partial_\rho (g^{\mu\nu} Y_\mu Z_\nu) = X^\rho \partial_\rho (g^{\mu\nu}) Y_\mu Z_\nu + X^\rho g^{\mu\nu} (\partial_\rho Y_\mu) Z_\nu + X^\rho g^{\mu\nu} Y_\mu \partial_\rho Z_\nu = \\ &= X^\rho \partial_\rho (g^{\mu\nu}) Y_\mu Z_\nu + X^\rho g^{\mu\nu} (\nabla_\rho Y_\mu) Z_\nu + X^\rho g^{\mu\nu} \Gamma_{\rho\mu}^\lambda Y_\lambda Z_\nu + X^\rho g^{\mu\nu} Y_\mu \nabla_\rho Z_\nu + X^\rho g^{\mu\nu} Y_\mu \Gamma_{\rho\nu}^\delta Z_\delta = \\ &= X^\rho \nabla_\rho (g^{\mu\nu}) Y_\mu Z_\nu + X^\rho g^{\mu\nu} (\nabla_\rho Y_\mu) Z_\nu + X^\rho g^{\mu\nu} Y_\mu \nabla_\rho Z_\nu, \end{aligned}$$

where, in the first equality we used item 2. in Proposition 57 since $g^{\mu\nu} Y_\mu Z_\nu$ is a smooth function. In the second equality we applied the Leibnitz rule, while, in the third and fourth one, we employed Equation (22), (25) and (26). The last two terms are the sought ones and, therefore, metric compatibility entails that the first one must vanish. Since X, Y, Z are arbitrary, it must hold $\nabla_\rho g^{\mu\nu} = 0$. Since $g^{\mu\nu} g_{\nu\sigma} = \delta_\sigma^\mu$, it turns out that

$$0 = \nabla_\rho (g^{\mu\nu} g_{\nu\sigma}) = (\nabla_\rho g^{\mu\nu}) g_{\nu\sigma} + g^{\mu\nu} \nabla_\rho g_{\nu\sigma} = g^{\mu\nu} \nabla_\rho g_{\nu\sigma},$$

where we have repeated implicitly the same steps as in the equality above, which imply that the connection ∇_X abides by the Leibnitz rule. Since $g^{\mu\nu} \nabla_\rho g_{\nu\sigma} = 0$ it suffices to multiply by $g_{\alpha\mu}$ to obtain

$$0 = g_{\alpha\mu} g^{\mu\nu} \nabla_\rho g_{\nu\sigma} = \delta_\alpha^\nu \nabla_\rho g_{\nu\sigma} = \nabla_\rho g_{\alpha\sigma},$$

where we used implicitly that $\nabla_\rho \delta_\alpha^\nu = 0$. This is the sought conclusion. Furthermore, since we have used only chain of equalities, we can trace back the reasoning to infer the converse statement, hence concluding the proof. $\square$

Compatibility of the metric is not enough to pinpoint a unique connection. The reason can be ascribed to Equation (21), since, as we shall see, a metric tensor allows to construct uniquely the symmetric part of Γ , namely $\Gamma_{(\mu\nu)}^\rho$, but no information can be inferred on its antisymmetric part. This is one of the key assumptions in general relativity and also in Riemannian and Lorentzian geometry, namely such antisymmetric components must vanish. From a mathematical viewpoint this is justified on the ground that it leads to a unique metric-induced connection, but, on physical grounds, this amounts to cancel several degrees of freedom in describing the gravitational field. While theoretically the implant of general relativity is satisfactory and we can live a happy life (no we cannot... it is just an idiomatic expression, #maiunagioia, #ForzaInter), this assumption must be tested experimentally.

Definition 61. Let M be an n -dimensional smooth manifold and let ∇ be a linear connection as per Proposition 54. We call **torsion tensor** $\tau \in \mathcal{T}^{(1,2)}$ associated to ∇ the map

$$\tau : \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TM), \quad \tau(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y], \quad (31)$$

where $[X, Y]$ is the Lie bracket as per Definition 36. A connection is said to be **torsion-free** (or **torsionless**) if $\tau = 0$.

Lemma 62. *Let M be an n -dimensional smooth manifold and let ∇ be a torsion-free, linear connection as per Proposition 54 and Definition 61. Then, the Christoffel symbols associated to ∇ are symmetric, that is, working in coordinates $\Gamma_{[ij]}^k = 0$.*

Proof. Let $X, Y, Z \in \Gamma(TM)$. Since both the connection and the Lie bracket between vector fields are local operations, see also the proof of Proposition 56, we can work on an arbitrary but fixed local chart, endowing it with arbitrary coordinates $\{x^1, \dots, x^n\}$. Hence

$$\begin{aligned}\tau(X, Y) &= X^i \nabla_i Y^j - Y^i \nabla_i X^j - X^i \partial_i Y^j - Y^i \partial_i X^j = \\ &= X^i \partial_i Y^j + X^i \Gamma_{ik}^j Y^k - Y^i \partial_i X^j - Y^i \Gamma_{ik}^j X^k - X^i \partial_i Y^j - Y^i \partial_i X^j = \\ &= X^i \Gamma_{ik}^j Y^k - Y^i \Gamma_{ik}^j X^k = 2X^i \Gamma_{[ik]}^j Y^k = 0,\end{aligned}$$

which entails, due to the arbitrariness of X and Y that $\Gamma_{[ik]}^j = 0$, which is the sought conclusion. $\square$

At last we are ready to state the main result of this section and one of the most relevant building blocks of general relativity. Despite its name, it applies verbatim to metrics of any signature.

Theorem 63 (Fundamental Theorem of Riemannian Geometry). *Let (M, g) be an n -dimensional smooth manifold with a metric tensor g as per Definition 50. Then there exists a unique torsion-free, metric compatible, linear connection $\nabla_g \equiv \nabla$ as per Proposition 54 and per Definitions 59 and 61. This connection is also called **(metric induced) covariant derivative or Levi-Civita connection**.*

Proof. Uniqueness: We start from a collection of useful identities. Assume that a connection ∇ with the sought properties exists. Since it has to be compatible with the metric, Equation (30) entails that for any $X, Y, Z \in \Gamma(TM)$

$$\begin{cases} \nabla_X \langle Y, Z \rangle_g = \langle \nabla_X Y, Z \rangle_g + \langle Y, \nabla_X Z \rangle_g = \langle \nabla_X Y, Z \rangle_g + \langle Y, \nabla_Z X \rangle_g + \langle Y, [X, Z] \rangle_g \\ \nabla_Y \langle Z, X \rangle_g = \langle \nabla_Y Z, X \rangle_g + \langle Z, \nabla_Y X \rangle_g = \langle \nabla_Y Z, X \rangle_g + \langle Z, \nabla_X Y \rangle_g + \langle Z, [Y, X] \rangle_g \\ \nabla_Z \langle X, Y \rangle_g = \langle \nabla_Z X, Y \rangle_g + \langle X, \nabla_Z Y \rangle_g = \langle \nabla_Z X, Y \rangle_g + \langle X, \nabla_Y Z \rangle_g + \langle X, [Z, Y] \rangle_g \end{cases},$$

where, in each line, in the last equality we have used that ∇ is torsion-free. Adding the first two equations and subtracting the third, we obtain that

$$\nabla_X \langle Y, Z \rangle_g + \nabla_Y \langle Z, X \rangle_g - \nabla_Z \langle X, Y \rangle_g = 2\langle \nabla_X Y, Z \rangle_g + \langle Y, [X, Z] \rangle_g + \langle Z, [Y, X] \rangle_g - \langle X, [Z, Y] \rangle_g,$$

where we have used implicitly that the pairing $\langle \cdot, \cdot \rangle_g$ is symmetric. It descends that

$$2\langle \nabla_X Y, Z \rangle_g = \nabla_X \langle Y, Z \rangle_g + \nabla_Y \langle Z, X \rangle_g - \nabla_Z \langle X, Y \rangle_g - \langle Y, [X, Z] \rangle_g - \langle Z, [Y, X] \rangle_g + \langle X, [Z, Y] \rangle_g. \quad (32)$$

Suppose now that there exists a second metric compatible, torsion free, linear connection ∇' . It has also to abide by this last equality. Yet a close inspection of the right hand side unveils that

the connection is acting only on scalar functions. Thereon its action is unique as per item 2. of Proposition 54. This entails that

$$\langle \nabla_X Y, Z \rangle_g - \langle \nabla'_X Y, Z \rangle_g = 0.$$

Since X, Y, Z are arbitrary and the metric pairing is not degenerate this entails that $\nabla_X Y = \nabla'_X Y$. Since Proposition 54 entails that the action of a connection on tensor fields is completely determined by its action on scalars and on $\Gamma(TM)$, uniqueness is proven.

Existence: As shown in the proof of Proposition 54, the action of a connection is local and therefore it suffices to determine it on an arbitrary, but fixed local chart (U, φ) endowed with coordinates $\{x^1, \dots, x^n\}$. In addition Equations (20) and (21) entail that it suffices to determine its Christoffel symbols or, rather the vector fields $\nabla_{\partial_\mu} \partial_\nu$ where we switched to Greek indices since we are constructing a metric-induced structure. In particular, setting $X = \partial_\mu$, $Y = \partial_\nu$ and $Z = \partial_\rho$, Equation (32) reads

$$\begin{aligned} & 2\langle \nabla_{\partial_\mu} \partial_\nu, \partial_\rho \rangle_g = \\ & \nabla_{\partial_\mu} \langle \partial_\nu, \partial_\rho \rangle_g + \nabla_{\partial_\nu} \langle \partial_\rho, \partial_\mu \rangle_g - \nabla_{\partial_\rho} \langle \partial_\mu, \partial_\nu \rangle_g - \langle \partial_\nu, [\partial_\mu, \partial_\rho] \rangle_g - \langle \partial_\rho, [\partial_\nu, \partial_\mu] \rangle_g + \langle \partial_\mu, [\partial_\rho, \partial_\nu] \rangle_g. \end{aligned}$$

Observing that the Lie bracket $[\partial_\mu, \partial_\nu] = 0$ for all $\mu, \nu = 1, \dots, n$ and that $\langle \partial_\mu, \partial_\nu \rangle_g = g_{\mu\nu}$, the last identity translates to

$$2\Gamma_{\mu\nu}^\lambda g_{\lambda\rho} = \partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu},$$

which, multiplying each side by $g^{\rho\kappa}$ yields

$$\boxed{\Gamma_{\mu\nu}^\kappa = \frac{1}{2} g^{\rho\kappa} (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu})}, \quad (33)$$

which is the sought result. **Engrave this formula in your soul.** $\square$

First and foremost, observe that, on an n -dimensional manifold, the number of metric-induced Christoffel symbols is equal to $\frac{n^2(n+1)}{2}$ since they inherit the symmetry property of the metric in any coordinate system, namely

$$\Gamma_{\mu\nu}^\rho = \Gamma_{(\mu\nu)}^\rho, \quad (34)$$

as a consistency check with respect to Lemma 62. This is the beginning of the end? No! This is just the true beginning. We have proven that a connection exists, it is unique and it can be constructed unambiguously from the metric with explicit formulae. Time to use it to cook up a physical theory.

Example 64. It is worth emphasizing that, given a smooth manifold (M, g) with a metric tensor, the Levi-Civita connection is unique but its local expression as per Equation (33) depends heavily on the underlying coordinates. We analyze a simple, yet very instructive example. Let us consider Minkowski spacetime that is $(\mathbb{R}^4, \eta)$ where η is a metric whose associated line element

in Cartesian coordinates (t, x, y, z) reads $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$. Writing $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$ we can infer that the matrix representation associated to it is diagonal, namely

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Since there is no explicit dependence on the coordinates in the entries of the matrix representation of η , it turns out that $\partial_\rho \eta_{\mu\nu} = 0$ for all $\mu, \nu, \rho = t, x, y, z$. Applying Equation (33), we infer that

$$\Gamma_{\mu\nu}^\rho = 0, \quad \forall \mu, \nu, \rho = t, x, y, z$$

which entails, that, in standard Cartesian coordinates $\nabla_\mu = \partial_\mu$ at all points in $\mathbb{R}^4$. Yet, we can consider in place of the spatial coordinates (x, y, z) their spherical counterparts (r, θ, φ) , see Example 51. This is only a diffeomorphism if we restrict the attention to $\mathbb{R}^4 \setminus \mathbb{R}_t$ where $\mathbb{R}_t \doteq \{(t, 0, 0, 0) \in \mathbb{R}^4\}$. Still using Example (51), we can infer that the line element reads

$$ds^2 = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2 \theta d\varphi^2).$$

Writing $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$ we can infer that the matrix representation in these coordinates reads

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}. \quad (35)$$

In this case, there is an explicit dependence on the coordinates in the entries of the matrix representation of η . Hence we have no reason to expect all Christoffel symbols to vanish. We have to apply Equation (33) to compute them. This is an algorithm and the first step consists of computing the inverse metric, namely η^{-1} whose components $(\eta^{-1})^{\mu\nu} \equiv \eta^{\mu\nu}$ are obtained by inverting Equation (35), namely

$$\eta^{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^{-2} & 0 \\ 0 & 0 & 0 & r^{-2} \sin^{-2} \theta \end{bmatrix}. \quad (36)$$

The second step consists in despairing since the a priori number of Christoffel symbols one needs to compute is 40 since the underlying background is four-dimensional. In general this cannot be helped, but, in the case under scrutiny we observe that the metric coefficients do not depend on t and on φ . Therefore all derivatives involving such coordinates do vanish automatically. This observation, in combination with the metric and its inverse being diagonal in the chosen coordinates, helps us concluding that, when computing $\Gamma_{\mu\nu}^\rho$, $\rho, \mu, \nu = t, r, \theta, \varphi$ at least one among the indices ρ, μ, ν must be equal either to r or to θ , while none of them can be equal to t . A

quick computation shows us that we have reduced the number of possibilities to 15. In addition, since the metric is diagonal, a direct inspection of Equation (21) unveils that at least two indices must coincide. This further reduces the number of possibilities to 12. This is not a guarantee that all of these remaining symbols are non vanishing and actually, it is not the case. Hence we need to compute them and we shall do one example explicitly. Let us fix $\mu = r$ and Equation (21) becomes

$$\Gamma_{r\nu}^\rho = \frac{1}{2}\eta^{\rho\kappa} (\partial_r\eta_{\nu\kappa} + \partial_\nu\eta_{r\rho} - \partial_\rho\eta_{\mu r}) = \frac{1}{2}\eta^{\rho\kappa}\partial_r\eta_{\nu\kappa},$$

where we used that all the components of the form $\eta_{r\nu}$ are constant. Since only $\eta_{\theta\theta}$ and $\eta_{\varphi\varphi}$ have an explicit dependence on r , we can conclude that $\Gamma_{rr}^r = \Gamma_{r\theta}^r = \Gamma_{r\varphi}^r = \Gamma_{rr}^\varphi = \Gamma_{r\theta}^\varphi = \Gamma_{r\varphi}^\varphi = 0$, while we already knew by an abstract reasoning that

$$\Gamma_{rt}^t = \Gamma_{rr}^t = \Gamma_{r\theta}^t = \Gamma_{r\varphi}^t = \Gamma_{rt}^\theta = \Gamma_{r\varphi}^\theta = \Gamma_{rt}^\varphi = \Gamma_{r\theta}^\varphi = \Gamma_{r\varphi}^\varphi = \Gamma_{rt}^r = 0.$$

The remaining two symbols read

$$\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{2}r^{-2}\partial_r(r^2) = \frac{1}{r}, \quad \text{and} \quad \Gamma_{r\varphi}^\varphi = \Gamma_{\varphi r}^\varphi = \frac{1}{2}(r\sin\theta)^{-2}\partial_r(r^2\sin^2\theta) = \frac{1}{r}.$$

Proceeding similarly one can find that the only additional non-vanishing Christoffel symbols are

$$\Gamma_{\theta\theta}^r = -r, \quad \Gamma_{\varphi\varphi}^r = -r\sin^2\theta, \quad \Gamma_{\varphi\varphi}^\theta = -\frac{1}{2}\sin(2\theta), \quad \Gamma_{\theta\varphi}^\varphi = \Gamma_{\varphi\theta}^\varphi = \cot\theta.$$

This shows, not only that the explicit form of the Christoffel symbols heavily depends on the chosen coordinates, but also that the number of these which are not vanishing is highly varying, in this case hopping from zero to 7.

The reader should reflect on the fact that Example 64 shows, on the one hand, that the computation of Christoffel symbols is for all practical purposes an algorithm, but also, that, depending on the choice of coordinates, it can become long and tedious. For this reason it is fair to say that it is often convenient and wise to rely on programs such as *Mathematica* which are based on symbolic calculus. Creating your own package to compute geometric quantities or use existing ones is certainly an option.

2.5 Lie Derivatives and Killing Fields

In the previous section, particularly in Example 64, we have realized that the computation of geometric structures such as the metric coefficients or the Christoffel symbols is heavily dependent on the coordinates chosen. In addition, one can observe that the lack of an explicit dependence of the metric on a given coordinate entails a drastic simplification of the computation of Equation (21). This is reminiscent to a similar situation which occurs in classical mechanics when studying the motion of a collection of point particles or of a rigid body. In this case when the Lagrangian is independent from a Lagrangian coordinate, hence the latter is cyclic, then we have an underlying symmetry which reflects itself at the level of dynamics with the existence

of a conserved quantity. Such symmetries have a geometric interpretation since the presence of a cyclic coordinate entails either translation or rotation invariance in a given direction in the underlying configuration space. Focusing for simplicity on a single point particle moving on the Euclidean space $(\mathbb{R}^3, \delta)$, δ being the standard Euclidean metric, at a physical level, this is interpreted by saying that, at the kinematic level, the configuration space is both *homogeneous and isotropic*. In other words there is an invariance under rotations and translations, a fact, which does not surprise if we think of the line element associated to $(\mathbb{R}^3, \delta)$ written in Cartesian coordinates x, y, z , namely

$$ds^2 = dx^2 + dy^2 + dz^2.$$

One can swiftly realize that any change of coordinates of the form $(x, y, z) \mapsto (x', y', z')$ with $x' = x + a$, $y' = y + b$ or $z' = z + c$ with $a, b, c \in \mathbb{R}$ or with $(x', y', z')^T = O(x, y, z)^T$, with $O \in O(3, \mathbb{R})$, the orthogonal group, the superscript T denoting the transpose, yields

$$ds^2 = d(x')^2 + d(y')^2 + d(z')^2.$$

In other words translations and rotations leave the line element invariant in form. Cyclic coordinates can therefore be seen as a statement that certain kinematic symmetries are preserved at a dynamical level. Going back to Example 64, this detour prompts the question what is the counterpart of the kinematic symmetries of the Euclidean space when we consider a generic smooth manifold (M, g) with a metric tensor and, whether, this might give a hint on the existence of preferred coordinates which simplify the computation of the Christoffel symbols.

We shall delve in a topic which is often marginalized in mathematics textbooks since, from the mere perspective of differential geometry, it is not especially interesting, whereas its applications to general relativity are countless. Hence we refer a reader to [Jost08, Chap. 1.9] or to [Wal84, App. C].

Bearing in mind the idea of symmetry from classical mechanics, we seek first and foremost maps from an underlying smooth manifold M into itself preserving the differential structure. This is nothing but the *diffeomorphism group* $\text{Diff}(M)$ introduced in Definition 40. Since we are interested in those diffeomorphism which leave the metric unchanged (or, using coordinates, invariant in form), we want to make precise what does it mean for a diffeomorphism to act on a generic tensor field $T \in \Gamma(\mathcal{T}^{(k,l)}(M))$, $k, l \geq 0$. Recalling that, if $k = 0$, one can pull-back elements lying in $\mathcal{T}^{(0,l)}(M)$ by a direct generalization of Equation (10), while, if $l = 0$, one can push-forward tensor field in $\mathcal{T}^{(k,0)}(M)$ by an extension of Definition 4, a diffeomorphism allows us to mix them by exploiting that, for all $\Phi \in \text{Diff}(M)$, also $\Phi^{-1} \in \text{Diff}(M)$.

Definition 65. Let M be an n -dimensional smooth manifold as per Definition 9. For any $\Phi \in \text{Diff}(M)$, see Definition 40, and for any $T \in \Gamma(\mathcal{T}^{(k,l)}(M))$, $k + l > 0$, see Definition 49, we call **pull-back** of T under Φ , the tensor field $\Phi^*T \in \Gamma(\mathcal{T}^{(k,l)}(M))$, such that, for all $\omega_1, \dots, \omega_k \in \Gamma(T^*M)$ and for all $\chi_1, \dots, \chi_l \in \Gamma(TM)$

$$\Phi^*(T)(\omega_1, \dots, \omega_k, \chi_1 \dots \chi_l) \doteq T((\Phi^{-1})^*\omega_1, \dots, (\Phi^{-1})^*\omega_k, \Phi_*\chi_1, \Phi_*\chi_l). \quad (37)$$

Observe that, on the right hand side of Equation (37), it appears $(\Phi^{-1})^*$ in place of Φ_* , since we are using implicitly that, for every $\Phi \in \text{Diff}(M)$, the push-forward of a k -form $k \geq 1$ under the action of Φ is defined as its pull-back under the action of Φ^{-1} .

Exercise 66. Under the assumptions of Definition 65, find the expression in coordinates of Equation (37).

If we endow the underlying smooth manifold with a metric tensor g , the discussion at the beginning of the section, suggests that, among all diffeomorphisms, distinguished are those leaving invariant g since they represent in this generalized framework the counterpart of translations and rotations on $(\mathbb{R}^3, \delta)$, namely the kinematic symmetries.

Definition 67. Let (M, g) be an n -dimensional smooth manifold as per Definition 9 with a metric tensor as per Definition 50. We call **isometry group** $\text{Iso}(M, g)$ the subgroup of all diffeomorphisms $\Phi \in \text{Diff}(M)$ such that $\Phi^*g = g$, where the pull-back is defined according to Definition 65.

Remark 20. In the spirit of our comparison to classical mechanics, isometries as per Definition 67 are close to what we are looking for, but they are not the whole story. To be precise, suppose that we consider on $\mathbb{R}^2 \ni (q_1, q_2)$ the following Lagrangian $L = \frac{m}{2} (\dot{q}_1^2 + \dot{q}_2^2) - \frac{\kappa}{2} q_1^2$, $\kappa > 0$. We would say that q_2 is a cyclic coordinate since L does not depend explicitly on it. Yet, if we would have to translate this statement in a mathematical language, we would write $\frac{\partial L}{\partial q_2} = 0$. This is a local statement since it involves derivatives, which is nonetheless valid at all points on $\mathbb{R}^2$. A global counterpart of this assertion is instead $L \equiv L[q_1, q_2] = L[q_1, q_2 + a]$ for all $a \in I \subseteq \mathbb{R}$ depending on the domain of q_2 , i.e., if it is a translation or rotation coordinate. Isometries are a generalization at the level of metric tensors of this last viewpoint since the action of a diffeomorphism is a global statement. We seek therefore also a counterpart for the first viewpoint.

Integral Curves: In order to answer this question, we need to go via a short detour. More precisely, in Section 2.1, we have introduced tangent vectors at a point $p \in M$ as directional derivatives, which is a rather analytic approach to the construction of $T_p M$. A more traditional viewpoint instead consists of looking at all curved passing through p and at their tangent vectors. This seems to suggest that elements in $T_p M$ and smooth curved passing through p are closely related concept. We will make precise this heuristic idea, starting from a key concept in the theory of dynamical systems.

Definition 68. Let M be an n -dimensional smooth manifold as per Definition 9 and let $X \in \Gamma(TM)$. A map $\gamma \in C^\infty(I; M)$ with $I \subseteq \mathbb{R}$ is called an **integral curve** of X , if $\dot{\gamma}(t) \equiv d\gamma\left(\frac{d}{dt'}\bigg|_{t'=t}\right) = X(\gamma(t))$.

Observe that the writing $\dot{\gamma}(t) \in T_{\gamma(t)}M$ is a merely and more common form for $d\gamma\left(\frac{d}{dt'}\bigg|_{t'=t}\right) \in T_{\gamma(t)}M$. Assuming henceforth and without loss of generality that $0 \in I$, if we work locally around

$\gamma(0) \in M$, we can find a local chart (U, φ) and a coordinate system $\{x^1, \dots, x^n\}$ such that

$$d\gamma \left(\frac{d}{dt} \Big|_{t'=t} \right) = X(\gamma(t)) \implies \sum_{i=1}^n \dot{\gamma}^i(t) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)} = \sum_{i=1}^n X^i(\gamma(t)) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}, \quad (38)$$

Since all vectors $\frac{\partial}{\partial x^i}$ are linearly independent, this reduces to the following system of n -coupled ODEs:

$$\dot{\gamma}^i(t) = X^i(\gamma^1(t), \dots, \gamma^n(t)), \quad \forall i = 1, \dots, n \quad (39)$$

to which we can assign an initial condition, $\gamma^i(0) = a^i \in \mathbb{R}$, $i = 1, \dots, n$. In particular, if we require the integral curve to pass through the point $p \in M$ at $t = 0$, we can set

$$a^i = x^i(p). \quad i = 1, \dots, n$$

Since $X \in \Gamma(TM)$, it turns out that also the right hand side of Equation (39) is smooth and, hence, we can apply the Picard-Lindelöf theorem to conclude that there exists an interval $I \subseteq \mathbb{R}$ centered at $t = 0$ where a smooth solution exists and it is unique. As mentioned several times in these notes, it is important to understand how to translate abstract mathematical concepts in an operational and computationally efficient tool. Hence we discuss a few examples to highlight how to construct in concrete cases integral curves and how it is not so uncommon that the domain of existence I does not coincide with the whole $\mathbb{R}$. Before proceeding in this task, we state a lemma which teaches us, how we can exploit that the equation defining integral curves is a system of autonomous ODEs.

Lemma 69. *Let M be an n -dimensional smooth manifold as per Definition 9 and let $X \in \Gamma(TM)$. Given $J \subseteq \mathbb{R}$ such that $0 \in J$ and such that $\gamma \in C^\infty(J; M)$ is an integral curve of X going through $\gamma(0) = p \in M$, we call $\tilde{J} \doteq \{t \in \mathbb{R} \mid t + a \in J\}$ with $a \in J$. Then $\tilde{\gamma}(t) \doteq \gamma(t + a) \in C^\infty(\tilde{J}; M)$ is an integral curve of X going through $\gamma(a)$ at $t = 0$.*

Proof. Let (U, φ) be any local chart of M such that $\tilde{\gamma}(t_0)$ lies therein where $t_0 \in \tilde{J}$. For any $f \in C^\infty(U)$ Equation (38) entails that

$$\begin{aligned} \dot{\tilde{\gamma}}(t_0)(f) &= \frac{d((f \circ \tilde{\gamma})(t))}{dt} \Big|_{t=t_0} = \frac{d[f(\gamma(t+a))]}{dt} \Big|_{t=t_0} = \frac{d(f \circ \gamma)(t)}{dt} \Big|_{t=t_0+a} = \\ &= \dot{\gamma}(t_0 + a)(f) = X(\gamma(t_0 + a))(f) = X(\tilde{\gamma}(t_0))(f), \end{aligned}$$

from which the sought conclusion descends. $\square$

Example 70. Let $M = \mathbb{R}^2$ and let $X \in \Gamma(\mathbb{R}^2)$ be in standard Cartesian coordinates $X = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}$. We seek its integral curve $\gamma \in C^\infty(I; \mathbb{R}^2)$. With a slight abuse of notation we write $\dot{\gamma}(t) = \dot{x}(t) \frac{\partial}{\partial x} + \dot{y}(t) \frac{\partial}{\partial y}$ from which we can write $\dot{\gamma}(t) = X(\gamma(t))$ as the system of ODEs

$$\begin{cases} \dot{x}(t) = x \\ \dot{y}(t) = y \end{cases}.$$

Given an initial condition $\gamma(0) = (x_0, y_0) \in \mathbb{R}^2$, the unique solution is

$$x(t) = x_0 e^t, \quad y(t) = y_0 e^t,$$

which exists for all $t \in I = \mathbb{R}$.

Example 71. Let $M = \mathbb{R}^2$ and let $X \in \Gamma(\mathbb{R}^2)$ be such that, in Cartesian coordinates, it reads $X = x^2 \frac{\partial}{\partial x}$. Calling $\gamma \in C^\infty(I; \mathbb{R}^2)$ the integral curve of X , as in the previous example, we can write $\dot{\gamma}(t) = \dot{x}(t) \frac{\partial}{\partial x} + \dot{y}(t) \frac{\partial}{\partial y}$. The equation $\dot{\gamma}(t) = X(\gamma(t))$ becomes the system of ODEs

$$\begin{cases} \dot{x}(t) = x^2 \\ \dot{y}(t) = 0 \end{cases}.$$

Assigning the initial condition $\gamma(0) = (x_0, y_0) \in \mathbb{R}^2$, the unique solution reads

$$x(t) = \frac{x_0}{1 - x_0 t}, \quad y(t) = y_0,$$

which is defined in $I = \left(-\infty, \frac{1}{x_0}\right)$ if $x_0 > 0$ or if $\left(\frac{1}{x_0}, \infty\right)$ if $x_0 < 0$. In this case the solution does not exist for all values of t on account of the form of the vector field, in particular due to the coefficient x^2 .

Example 72. Let $M = \mathbb{R}^2 \setminus \{(x, y) \in \mathbb{R}^2 \mid x \geq 0\}$ and let $X \in \Gamma(TM)$ be such that, in standard Cartesian coordinates restricted to M , $X = \frac{\partial}{\partial x}$. Once more we look for $\gamma \in C^\infty(I; M)$ integral curve of X . Writing once more $\dot{\gamma}(t) = \dot{x}(t) \frac{\partial}{\partial x} + \dot{y}(t) \frac{\partial}{\partial y}$ the equation $\dot{\gamma}(t) = X(\gamma(t))$ can be rewritten as the following system of ODEs:

$$\begin{cases} \dot{x}(t) = 1 \\ \dot{y}(t) = 0 \end{cases}.$$

Assigning the initial condition $\gamma(0) = (x_0, y_0) \in M$ with $x_0 < 0$, there exists the unique solution

$$x(t) = x_0 + t, \quad y(t) = y_0,$$

which is defined only on the interval $I = (-\infty, -x_0)$. In this case the restriction to the domain of existence of the integral curve is due to the structure of the underlying manifold, since, the simplest among all vector fields, namely the generator of the translations along the x -direction yields an integral curve which reaches in a finite amount of proper time the “end” of the manifold itself.

Let us now consider a smooth manifold M and a vector field $X \in \Gamma(TM)$ such that its integral curve, see Equation (38) exists for all $t \in \mathbb{R}$ and for all $\gamma(0) = p \in M$. This allows us to identify two distinguished maps:

$$\gamma_p : \mathbb{R} \rightarrow M \quad \text{such that} \quad \begin{cases} \dot{\gamma}_p(t) = X(\gamma_p(t)) \\ \gamma_p(0) = p \end{cases} \quad (40)$$

and, for any but fixed $t \in \mathbb{R}$

$$\tilde{\gamma}_t : M \rightarrow M \quad \text{such that} \quad \tilde{\gamma}_t(p) \doteq \gamma_p(t). \quad (41)$$

The advantage of introducing these two maps is that, if, for $\bar{t} \in \mathbb{R}$, if $q = \gamma_p(\bar{t})$, then Lemma 69 entail that $\gamma_q(t) = \gamma_p(t + \bar{t})$. In other words $\tilde{\gamma}$ abides by

$$\tilde{\gamma}_t \circ \tilde{\gamma}_{\bar{t}} = \tilde{\gamma}_{t+\bar{t}}, \quad \tilde{\gamma}_0 = \text{id}|_M,$$

where $\text{id}|_M$ is the identity map on M . We have constructed an action of the Abelian group $\mathbb{R}$ on M , namely

$$\gamma : \mathbb{R} \times M \rightarrow M, \quad (t, p) \mapsto \gamma(t, p) = \gamma_p(t) = \tilde{\gamma}_t(p).$$

The following definition generalizes these remarks.

Definition 73. Let M be an n -dimensional smooth manifold as per Definition 9. We call **global flux** on M any map $\theta \in C^\infty(\mathbb{R} \times M; M)$ such that $\forall s, t \in \mathbb{R}$ and $\forall p \in M$

$$\theta(t, \theta(s, p)) = \theta(s + t, p), \quad \theta(0, p) = p.$$

Furthermore, for every $t \in \mathbb{R}$, the map

$$\tilde{\theta}_t : M \rightarrow M, \quad p \mapsto \tilde{\theta}_t(p) = \theta(t, p), \quad (42)$$

lies in $\text{Diff}(M)$. It is called **one-parameter group of diffeomorphisms** since $\forall s, t \in \mathbb{R}$,

$$\tilde{\theta}_t \circ \tilde{\theta}_s = \tilde{\theta}_{t+s}, \quad \tilde{\theta}_0 = \text{id}|_M, \quad (43)$$

where $\text{id}|_M$ is the identity map on M . For any $p \in M$, the set $\tilde{\theta}_t(p) \subset M$, $t \in \mathbb{R}$, is referred to as an **orbit** of $\tilde{\theta}$. We call **local flux** on M any map $\theta \in C^\infty(I \times M; M)$ with $I \subset \mathbb{R}$ such that $0 \in I$, provided that it abides by the above conditions whenever they are meaningful.

We investigate how global fluxes are related to vector fields, a feature which will be relevant in the following discussions.

Lemma 74. Let M be an n -dimensional smooth manifold as per Definition 9 and let $\theta \in C^\infty(I \times M; M)$ be a global or local flux as per Definition 73. Then, $\forall p \in M$, $X_p \doteq \dot{\theta}_p(0) \doteq \left. \frac{d\theta(t, p)}{dt} \right|_{t=0} \in \Gamma(TM)$ and $\theta_p(t)$ is its integral curve.

Proof. We start by showing that $X \in \Gamma(TM)$. On account of Theorem 35, this is equivalent to proving that, for all $f \in C^\infty(M)$, $X(f) \in C^\infty(M)$. Yet, the following chain of equalities holds true:

$$X_p(f) = \dot{\theta}_p(0)f = \left. \frac{d(f \circ \theta_p)(t)}{dt} \right|_{t=0} = \left. \frac{df[\theta(t, p)]}{dt} \right|_{t=0}, \quad (44)$$

from which it descends that, being both f and θ smooth, $X_p \in \Gamma(TM)$. To conclude we show that $\theta_p(t)$ is an integral curve of X . Let $t_0 \in \mathbb{R}$ and $q = \theta_p(t_0)$. On account of Definition 73 it holds that, whenever $t + t_0 \in I$,

$$\theta_q(t) = \tilde{\theta}_t(q) = \tilde{\theta}_t(\tilde{\theta}_{t_0}(p)) = \tilde{\theta}_{t+t_0}(p) = \theta_p(t + t_0),$$

from which it descends that, for all $f \in C^\infty(M)$,

$$X_q(f) = \dot{\theta}_q(0)f = \left. \frac{df(\theta_q(t))}{dt} \right|_{t=0} = \left. \frac{df(\theta_p(t + t_0))}{dt} \right|_{t=0} = \dot{\theta}_p(t_0)f,$$

namely $\theta_p(t)$ is an integral curve of X . $\blacksquare$

□

Observe that the Examples 71 e 72 teach us that a converse statement to that in Lemma 74 cannot be true. There exists smooth vector fields to which no global flux is associated. This motivates the following definition.

Definition 75. *Let M be an n -dimensional smooth manifold as per Definition 9 and let $X \in \Gamma(TM)$. If X generates a global flux as per Definition 73 or, equivalently, if all its integral curves are defined for all $t \in \mathbb{R}$, then X is called a **complete vector field**.*

Lie Derivative: We can now move back to the main topic of this section and in particular we can look for a local counterpart of the action of a diffeomorphism on a tensor field. We start from the following building block:

Definition 76. *Let M be an n -dimensional smooth manifold as per Definition 9 and let us consider therefore a global flux as per Definition 73 and let us denote by $\tilde{\Phi}_t$, $t \in \mathbb{R}$ the one-parameter family of diffeomorphisms generated by it as per Equation (42). Let us denote by $X \in \Gamma(TM)$ the vector field generating $\tilde{\Phi}_t$ as per Lemma 74. For any $T \in \mathcal{T}^{(k,l)}(M)$, $k + l > 0$, we call **Lie derivative** of T along X*

$$\mathcal{L}_X : T \in \mathcal{T}^{(k,l)}(M) \rightarrow T \in \mathcal{T}^{(k,l)}(M), \quad T \mapsto \mathcal{L}_X(T) \doteq \left. \frac{d}{dt} \tilde{\Phi}_t^* T \right|_{t=0}, \quad (45)$$

where $\tilde{\Phi}_t^* T$ is as per Definition 65.

The Lie derivative is a key tool in general relativity and the following proposition aims at characterizing its main properties.

Remark 21. *We have given the definition of Lie derivative on all tensor fields barring those lying in $C^\infty(M) \equiv \mathcal{T}^{(0,0)}(M)$. In this instance, one can realize that Equation (45) is still meaningful, but it boils down to the standard action of a vector field on a smooth function, namely, for all $X \in \Gamma(TM)$ and for all $f \in C^\infty(M)$,*

$$\mathcal{L}_X(f) = X(f).$$

Remark 22. Observe that, for all $X \in \Gamma(TM)$, Equation (37) entails that, for all $t \in \mathbb{R}$, $\tilde{\Phi}_t^*(X) = (\tilde{\Phi}_t^{-1})_*(X)$. Indeed, Equation (43) entails that, for all $t \in \mathbb{R}$ $\tilde{\Phi}_t \circ \tilde{\Phi}_{-t} = \text{id}|_M$, $\text{id}|_M$ being the identity map on M . Yet, at the same time, it holds that $\tilde{\Phi}_t \circ \tilde{\Phi}_t^{-1} = \text{id}|_M$, which implies that $\tilde{\Phi}_t^{-1} = \tilde{\Phi}_{-t}$ and, therefore, $(\tilde{\Phi}_t^{-1})_*(X) = (\tilde{\Phi}_{-t})_*(X)$. Consequently Equation (45) reads in the case under scrutiny

$$\mathcal{L}_X(Y) = \left. \frac{d}{dt} \tilde{\Phi}_t^*(Y) \right|_{t=0} = \left. \frac{d}{dt} (\tilde{\Phi}_{-t})_*(Y) \right|_{t=0}.$$

Proposition 77. Let M be an n -dimensional smooth manifold as per Definition 9 and let $X \in \Gamma(TM)$. It holds that

1. for every $Y \in \Gamma(TM)$, $\mathcal{L}_X(Y) = [X, Y]$, where $[\cdot, \cdot]$ is the Lie bracket as per Definition (36),
2. for every $\omega \in \Gamma(T^*M)$, $\mathcal{L}_X(\omega)$ is the differential 1-form, which reads locally

$$\mathcal{L}_X \omega = (X^i \partial_i \omega_j + (\partial_j X^i) \omega_i) dx^j.$$

Proof. Let us denote by $\tilde{\Phi}_t$, $t \in \mathbb{R}$ the one-parameter family of diffeomorphisms generated by X as per Equation (42).

1. For any $f \in C^\infty(M)$ and for every $Y \in \Gamma(TM)$, on account of Remark 21 and of Equation (45), it holds that

$$X(Y(f)) = \left. \frac{d}{dt} \Phi_t^*(Y(f)) \right|_{t=0} = \left. \frac{d}{dt} (\Phi_t^*(Y)(\Phi_t^*(f))) \right|_{t=0}. \quad (46)$$

Similarly it descends that

$$Y(X(f)) = Y \left(\left. \frac{d}{dt} \Phi_t^*(f) \right|_{t=0} \right). \quad (47)$$

Considering the difference between Equation (46) and Equation (47), it descends

$$\begin{aligned} X(Y(f)) - Y(X(f)) &= \left. \frac{d}{dt} (\Phi_t^*(Y)(\Phi_t^*(f))) \right|_{t=0} - Y \left(\left. \frac{d}{dt} \Phi_t^*(f) \right|_{t=0} \right) = \\ &= Y \left(\left. \frac{d}{dt} \Phi_t^*(f) \right|_{t=0} \right) + \left(\left. \frac{d}{dt} \Phi_t^*(Y) \right|_{t=0} \right) (f) - Y \left(\left. \frac{d}{dt} \Phi_t^*(f) \right|_{t=0} \right) = \\ &= \left(\left. \frac{d}{dt} \Phi_t^*(Y) \right|_{t=0} \right) (f) = (\mathcal{L}_X(Y))(f), \end{aligned}$$

which is the sought result. Observe that, in the second equality we have used the Leibnitz rule and the property that Φ_t^* reduces to the identity operator at $t = 0$.

2. Considering $\omega \in \Gamma(T^*M)$ and $\chi \in \Gamma(TM)$, working in coordinates, it holds that

$$\mathcal{L}_X(\omega_i \chi^i) = X(\omega_i \chi^i) = X^j (\partial_j \omega_i) \chi^i + \omega_i X^j \partial_j \chi^i.$$

At the same time, using the Leibnitz rule, it holds that

$$\mathcal{L}_X(\omega_i \chi^i) = \mathcal{L}_X(\omega_i) \chi^i + \omega_i \mathcal{L}_X(\chi^i).$$

Comparing these two equations and recalling item 1., we obtain

$$\begin{aligned} \mathcal{L}_X(\omega_i) \chi^i &= X^j (\partial_j \omega_i) \chi^i + \omega_i X^j \partial_j \chi^i - \omega_i \mathcal{L}_X(\chi^i) = \\ X^j (\partial_j \omega_i) \chi^i + \omega_i X^j \partial_j \chi^i - \omega_i [X, \chi]^i &= X^j (\partial_j \omega_i) \chi^i + \omega_i X^j \partial_j \chi^i - \omega_i (X^j \partial_j \chi^i - \chi^j \partial_j X^i) = \\ (X^i \partial_i \omega_j + (\partial_j X^i) \omega_i) \chi^j, \end{aligned}$$

which entails the sought result. $\square$

Exercise 78. Under the same hypotheses of Proposition 77 prove item 2. therein using directly Definition 76.

Exercise 79. Let M be an n -dimensional smooth manifold as per Definition 9 and let $X \in \Gamma(TM)$. Prove that the Lie derivative satisfies the following version of the Leibnitz rule: for all $T \in \mathcal{T}^{(k,l)}(M)$ and for all $S \in \mathcal{T}^{(k',l')}(M)$,

$$\mathcal{L}_X(S \otimes T) = \mathcal{L}_X(S) \otimes T + S \otimes \mathcal{L}_X(T).$$

A rather interesting application of Lie derivatives arises when applying it to a metric tensor g on an underlying smooth manifold M . In this case, for any $X \in \Gamma(TM)$, it descends that, working in coordinates and using Exercise 79

$$\mathcal{L}_X(g) = \mathcal{L}_X(g_{\mu\nu} dx^\mu \otimes dx^\nu) = \mathcal{L}_X(g_{\mu\nu}) dx^\mu \otimes dx^\nu + g_{\mu\nu} \mathcal{L}_X(dx^\mu) \otimes dx^\nu + g_{\mu\nu} dx^\mu \otimes \mathcal{L}_X(dx^\nu).$$

On account of Remark 21 $\mathcal{L}_X g_{\mu\nu} = X(g_{\mu\nu}) = X^\rho \partial_\rho g_{\mu\nu}$, while item 2. of Proposition 77 entails that $\mathcal{L}_X(dx^\mu) = (\partial_\rho X^\mu) dx^\rho$. Up to a renaming of dummy indices, these identities allow us to conclude that, in every local chart,

$$\mathcal{L}_X(g_{\mu\nu} dx^\mu \otimes dx^\nu) = (X^\rho \partial_\rho g_{\mu\nu} + g_{\rho\nu} \partial_\mu X^\rho + g_{\mu\rho} \partial_\nu X^\rho) dx^\mu \otimes dx^\nu. \quad (48)$$

Using Equation (26) we can rewrite this last identity as

$$\begin{aligned} \mathcal{L}_X(g_{\mu\nu} dx^\mu \otimes dx^\nu) &= (X^\rho \nabla_\rho g_{\mu\nu} + g_{\rho\nu} \nabla_\mu X^\rho + g_{\mu\rho} \nabla_\nu X^\rho) dx^\mu \otimes dx^\nu = \\ &= (g_{\rho\nu} \nabla_\mu X^\rho + g_{\mu\rho} \nabla_\nu X^\rho) dx^\mu \otimes dx^\nu. \end{aligned}$$

Lowering the indices of X , this leads to the celebrated (almost on par of Halloween) **Killing Equation**

$$\boxed{(\mathcal{L}_X g)_{\mu\nu} = \nabla_\mu X_\nu + \nabla_\nu X_\mu = 2\nabla_{(\mu} X_{\nu)}} \quad (49)$$

Engrave this formula in your soul.

Definition 80. Let (M, g) be an n -dimensional smooth manifold as per Definition 9 with a metric tensor as per Definition 50. We call $X \in \Gamma(TM)$ a **Killing field** (or infinitesimal isometry) if

$$\mathcal{L}_X g = 0.$$

This allows us to establish one of the main structural results in differential geometry, since it allows to characterize locally the symmetries of a metric tensor. The application in general relativity will be countless.

Lemma 81. Let (M, g) be an n -dimensional smooth manifold as per Definition 9 with a metric tensor as per Definition 50. A vector field X is a Killing field as per Definition 80 if it is the generator of a one-parameter group of isometries as per Definition 67.

Proof. If $\tilde{\Phi}_t$ is a one parameter group of isometries, it entails that $\tilde{\Phi}_t^* g = g$ for all $t \in \mathbb{R}$. It turns this implies that, for all $t \in \mathbb{R}$,

$$\left. \frac{d}{dt} \tilde{\Phi}_t^* g \right|_{t=0} = 0 \implies \mathcal{L}_X(g) = 0,$$

where X is the generator of $\tilde{\Phi}_t$ as per Definition 73. □

Remark 23 (Conceptual). The relation between global isometries and Killing fields is very subtle since one is prone to believe that they are substantially carrying the same information. This is not true, but this attitude is understandable since in many instances this is sort of true. The source of this reasoning lies first of all in the fact that Killing fields form a sub-algebra of the algebra of vector fields: if $X, Y \in \Gamma(TM)$ are such that $\mathcal{L}_X g = \mathcal{L}_Y g = 0$, then also $\mathcal{L}_{[X,Y]} g = 0$, see [Jost08, Lem. 1.9.8]. This might lead to the idea that $\text{Iso}(M, g)$ as per Definition 67 can be reconstructed from the Killing fields via a form of exponentiation as much as all (Lie) groups of matrices can be recovered from their (Lie) algebra by means of the exponential function – Ask Bisio for more details!

Unfortunately the reconstruction of a group from its algebra, also known as Lie’s third theorem, requires that the group, seen as a differentiable manifold, is finite dimensional. The diffeomorphism group associated to a smooth manifold does not fall in this category. Hence the reconstruction of $\text{Iso}(M, g)$ from its algebra via a form of exponential map is not always possible. We shall not delve in these very technical topics (yeah, sure, these notes are ages from being technical in the real world out there! The world is full of monster and of perils, usually all of them with a Fields medal at their neck), but the message is that isometries carry information of local symmetries of the metric, which often, but not necessarily admit a global counterpart.

At the same time, we should not expect that all isometries yield a Killing field, since we need to be able to consider a one-parameter group of isometries to construct it. In the language of theoretical physics, we need to be able to allow for small deformations of an isometry. This is similar to the construction of infinitesimal rotations in classical mechanics which required the rotation, an isometry of $(\mathbb{R}^3, \delta)$, to admit a continuous deformation to the identity. In other

words Killing fields will never be able to encode information on discrete symmetries. As an example consider the parity inversion map

$$P : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad (x, y, z) \mapsto P(x, y, z) \doteq (-x, y, z).$$

This is a smooth and invertible map which preserves the Euclidean metric δ , hence $P \in \text{Iso}(\mathbb{R}^3, \delta)$, but it cannot be associated to any one-parameter group of isometries and thus to a Killing field.

To conclude we stress an analogy with classical mechanics, mentioned at the beginning of the section. When considering a Lagrangian function $L : \mathbb{R}^n \rightarrow \mathbb{R}$, we say that a coordinate q^i is cyclic if L does not depend explicitly on it. In other words we require $\frac{\partial L}{\partial q^i} = 0$, or, in a geometric language $X(L) = 0$ where $X \in \Gamma(T\mathbb{R}^n)$ is the vector field $X = \frac{\partial}{\partial q^i}$. It should not come as a surprise that $\mathcal{L}_X \delta = 0$, in other words, X is a Killing field for the Euclidean metric. In addition, in classical mechanics we are used to consider just $\mathbb{R}^n$ as the lone possible underlying manifold, but, suppose that, you want to describe a free particle moving on the half line, that is $L = \frac{m}{2} \dot{q}^2$ where $q \in (0, \infty)$. We can easily see that q is a cyclic coordinate and that, denoting by $X = \frac{\partial}{\partial q}$, $\mathcal{L}_X L = 0$ and $\mathcal{L}_X \delta = 0$. Yet, there does not exist a global flux associated to X , since this would correspond to the translations, namely $\theta : \mathbb{R} \times (0, \infty) \rightarrow (0, \infty)$ such that $\theta(t, q_0) = q_0 + t$. It cannot exist for all values of t since if $t < -q_0$, $\theta(t, q_0) < 0$ which is not admissible. Hence the coordinate q is cyclic, the Lagrangian does not depend on it, $\frac{\partial}{\partial q}$ is a Killing field, but it represents only a local isometry and not a global one.

What about conserved quantities? Is this concept lost? In the next section we shall give an answer to this question and we shall elaborate more on this analogy and comparison with Lagrangian mechanics.

Before proceeding with discussing a few notable examples, it is useful to generalize Definition 80.

Definition 82. Let (M, g) be an n -dimensional smooth manifold as per Definition 9 with a metric tensor as per Definition 50. We call $T \in \Gamma(TM^{\otimes sk})$, $k \geq 1$, a **Killing tensor** if, working in coordinates,

$$\nabla_\mu T^{\nu_1, \dots, \nu_k} = \nabla_\mu T^{(\nu_1, \dots, \nu_k)} = 0.$$

We conclude the section with a few examples aimed at showing how to transform the notion of Killing fields discussed above in a computationally efficient tool. The first one could be transformed in an actual proposition and it translates to the language of differential geometry the well-known property in Lagrangian mechanics that, if the Lagrange function does not depend explicitly on a coordinate, then the derivative along such direction is the generator of a symmetry.

Example 83. Let M be an n -dimensional smooth manifold endowed with a metric tensor g . Assume that, in a local chart (U, φ) , there exists a coordinate system $x^1, \dots, x^n$ such that the metric does not depend explicitly on x^k where k is a fixed index. We show that $\xi = \frac{\partial}{\partial x^k}$ is an infinitesimal isometry on U seen as a submanifold of M . In other words we know that $g_{\mu\nu} \neq g_{\mu\nu}(x^k)$ and, barring a raise of indices, Equation (49) reads

$$\nabla_\nu \xi_\mu + \nabla_\mu \xi_\nu = \nabla_\nu (\xi^\rho g_{\rho\mu}) + \nabla_\mu (\xi^\rho g_{\rho\nu}) = \Gamma_{\nu\lambda}^\rho \xi^\lambda g_{\rho\mu} + \Gamma_{\mu\lambda}^\rho \xi^\lambda g_{\rho\nu},$$

where we used metric compatibility of the connection. Since $\xi^\lambda = \delta^{\lambda k}$, δ being here the Kronecker delta, it turns out that

$$\nabla_\nu \xi_\mu + \nabla_\mu \xi_\nu = \Gamma_{\nu k}^\rho g_{\rho\mu} + \Gamma_{\mu k}^\rho g_{\nu\rho}.$$

Using Equation (21) it holds that

$$\Gamma_{\nu k}^\rho = \frac{1}{2} g^{\alpha\rho} (\partial_\nu g_{k\alpha} + \partial_k g_{\mu\alpha} - \partial_\alpha g_{k\nu}) = \frac{1}{2} g^{\alpha\rho} (\partial_\nu g_{k\alpha} - \partial_\alpha g_{k\nu}),$$

where we used the hypothesis that the metric coefficients do not depend on x^k . It turns out that

$$\begin{aligned} \nabla_\nu \xi_\mu + \nabla_\mu \xi_\nu &= \frac{1}{2} g^{\alpha\rho} (\partial_\nu g_{k\alpha} - \partial_\alpha g_{k\nu}) g_{\rho\mu} + \frac{1}{2} g^{\alpha\rho} (\partial_\mu g_{k\alpha} - \partial_\alpha g_{k\mu}) g_{\nu\rho} = \\ &= \frac{1}{2} \delta_\mu^\alpha (\partial_\nu g_{k\alpha} - \partial_\alpha g_{k\nu}) + \frac{1}{2} \delta_\nu^\alpha (\partial_\mu g_{k\alpha} - \partial_\alpha g_{k\mu}) = \frac{1}{2} (\partial_\nu g_{k\mu} - \partial_\mu g_{k\nu} + \partial_\mu g_{k\nu} - \partial_\nu g_{k\mu}) = 0. \end{aligned}$$

Example 84. In this example, we show how to infer that certain vector fields are Killing. Consider Minkowski spacetime $(\mathbb{R}^4, \eta)$ and the standard Cartesian coordinates, so that the metric reads as in Example 64. Consider the vector field which is called in special relativity a *generator of the Lorentz boosts*, namely $\xi = x \frac{\partial}{\partial t} + t \frac{\partial}{\partial x}$. This is a Killing field since Equation (49) becomes

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu,$$

where we used that all Christoffel symbols are vanishing. Using the metric tensor we can infer that $\xi_\mu = \eta_{\mu\rho} \xi^\rho$ and therefore $\xi_t = -x$, $\xi_x = t$, while $\xi_y = \xi_z = 0$. Hence, by direct inspection

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = 0,$$

and therefore ξ is a Killing vector field. Observe that this is one instance where infinitesimal isometries are associated to a global counterpart, namely the Lorentz boosts. Yet, one should also keep in mind that, had we elected to consider in place of $\mathbb{R}^4$ a submanifold say $\mathbb{R}_+^4 \doteq \{(t, x, y, z) \in \mathbb{R}^4 \mid x > 0\}$, then ξ would still be an infinitesimal isometry but the constraint on the domain of x would have hindered the existence of a global flux.

Example 85. In this last example, we show how to use Equation (49) to construct Killing fields once a smooth manifold M and a metric tensor have been assigned. In this case the higher the dimension of M , the more complicated becomes the procedure. Therefore, let us set $M = \mathbb{S}^2$, so that $\dim M = 2$ and we consider the round metric whose associated line element is in angular coordinates

$$ds^2 = d\theta^2 + \sin^2 \theta d\varphi^2.$$

First of all we need to compute all 6 independent Christoffel symbols by means of Equation (21). A direct computation yields

$$\Gamma_{\theta\theta}^\theta = \Gamma_{\theta\varphi}^\theta = \Gamma_{\theta\theta}^\varphi = \Gamma_{\varphi\varphi}^\varphi = 0,$$

and

$$\Gamma_{\theta\varphi}^\varphi = \cot \theta, \quad \Gamma_{\varphi\varphi}^\theta = -\sin \theta \cos \theta.$$

Looking at Equation (49), this is written with respect to the components of a 1-form, which we denote by $\xi = \xi_\theta(\theta, \varphi)d\theta + \xi_\varphi(\theta, \varphi)d\varphi$. Hence the Killing equation becomes the following set of PDEs:

$$\nabla_\theta \xi_\theta = \partial_\theta \xi_\theta - \Gamma_{\theta\theta}^\rho \xi_\rho = \partial_\theta \xi_\theta = 0, \quad (50a)$$

$$\nabla_\theta \xi_\varphi + \nabla_\varphi \xi_\theta = \partial_\theta \xi_\varphi + \partial_\varphi \xi_\theta - 2\Gamma_{\theta\varphi}^\rho \xi_\rho = \partial_\theta \xi_\varphi + \partial_\varphi \xi_\theta - 2\cot \theta \xi_\varphi = 0, \quad (50b)$$

$$\nabla_\varphi \xi_\varphi = \partial_\varphi \xi_\varphi - \Gamma_{\varphi\varphi}^\rho \xi_\rho = \partial_\varphi \xi_\varphi + \sin \theta \cos \theta \xi_\theta = 0. \quad (50c)$$

Equation (50a) entails that $\xi_\theta = \xi_\theta(\varphi)$. Hence we can consider Equations (50b) and (50c) and the derivative with respect to φ of the first and with respect to θ of the second to obtain

$$\begin{cases} \partial_\varphi \partial_\theta \xi_\varphi + \frac{d^2 \xi_\theta}{d\varphi^2} - 2\cot \theta \partial_\varphi \xi_\varphi = 0 \\ \partial_\theta \partial_\varphi \xi_\varphi + \cos(2\theta) \xi_\theta = 0 \end{cases}.$$

Replacing Equation (50c) and the second identity in the first one, we obtain

$$\frac{d^2 \xi_\theta}{d\varphi^2} + (2\cos^2 \theta - \cos(2\theta)) \xi_\theta = 0 \implies \frac{d^2 \xi_\theta}{d\varphi^2} + \xi_\theta = 0.$$

This is the equation of an harmonic oscillator hence

$$\xi_\theta = a \sin \varphi + b \cos \varphi,$$

where $a, b \in \mathbb{R}$ are integration constants. Plugging in this result in Equation (50c) yields

$$\partial_\varphi \xi_\varphi = -\frac{\sin(2\theta)}{2}(a \sin \varphi + b \cos \varphi),$$

which can be integrated as

$$\xi_\varphi = \frac{\sin(2\theta)}{2}(a \cos \varphi - b \sin \varphi) + K(\theta),$$

where K is an integration function. At last we replace this identity in Equation (50b) to obtain out of a few trigonometric identities

$$\frac{dK}{d\theta} = 2\cot \theta K \implies K(\theta) = c \sin^2 \theta,$$

where $c \in \mathbb{R}$ is an integration constant. In other words

$$\xi_\varphi = \frac{\sin(2\theta)}{2}(a \cos \varphi - b \sin \varphi) + c \sin^2 \theta.$$

Yet, to conclude we need to raise the indices with the metric tensor, which is diagonal and therefore the most general Killing fields on $\mathbb{S}^2$ reads

$$\xi = \xi^\theta \frac{\partial}{\partial \theta} + \xi^\varphi \frac{\partial}{\partial \varphi} = (a \sin \varphi + b \cos \varphi) \frac{\partial}{\partial \theta} + (\cot \theta (a \cos \varphi - b \sin \varphi) + c) \frac{\partial}{\partial \varphi}. \quad (51)$$

As a consistency check, we observe that, if we set $a = b = 0$, then $\frac{\partial}{\partial \varphi}$ is a Killing field in full agreement with Example 83 since the coefficients of the metric tensor do not depend explicitly on φ . In other words, starting from a point $(\theta_0, \varphi_0) \in \mathbb{S}^2$, the curve $(\theta_0, \varphi_0 + t)$ is a global isometry provided we take into account the periodicity of both θ and φ . At the same time, if we set $c = 0$ and we choose a, b so that $a \cos \varphi_0 = b \sin \varphi_0$, it turns out that $\xi = \kappa \frac{\partial}{\partial \theta}$ is not a Killing field since the vector field generating translation along the θ -direction depends on the values of a, b as well as on that of φ_0 . This entails that also the curve $(\theta_0 + t, \varphi_0)$ is not a global isometry even if we take into account the periodicity of both θ and φ .

3 The Geodesic Equation

Although it seems apparently completely out of context, let us remember for a moment the first two laws of Newtonian mechanics:

[First Law of Newtonian mechanics]: *Every object perseveres in its state of rest, or of uniform motion in a right line, except insofar as it is compelled to change that state by forces impressed thereon*

[Second Law of Newtonian mechanics]: *The change of motion of an object is proportional to the force impressed; and is made in the direction of the straight line in which the force is impressed*

This is a literal translation from their Latin formulation and it is not my goal to comment on their interpretation or applications, but I want to highlight the rôle of straight lines in the statements. If one thinks about the reason why one refers to them, besides phenomenological motivations, it is clear that, at a theoretical level, one can read this as the implicit assumption that we are considering Euclidean geometry. In other words our model for the configuration space of a point particle or of an idealized observer is the Euclidean space $\mathbb{E}^3 \equiv (\mathbb{R}^3, \delta)$. Herein straight lines are known to be geometrically privileged since the shortest path connecting two points in $\mathbb{E}^3$ is indeed a straight line. On top of that Newton's second law is consistent with this viewpoint since it entails, that, in absence of external forces, a point particle will move along a path of vanishing acceleration, that is a straight line. If we adopt the paradigm shift that the Euclidean space or, more generally even Minkowski space $\mathbb{M}^4 \equiv (\mathbb{R}^4, \eta)$ is a mere local approximation of the true configuration space, then we are a little bit in trouble of making sense of the above two laws of mechanics, since it not clear what should replace the words *straight line*. On top of this, even the concepts of *acceleration* or of *shortest path* are somehow blurred. In this section, we shall address the question whether, given a smooth manifold M endowed with a metric tensor g , there exists a natural replacement of the notion of straight line. This

shall lead to the concept of **geodesic**, or, in a more physical language, of *free falling observer*. Most notably, it turns out that, to establish a notion of acceleration, it will be sufficient to assign a linear connection on the underlying manifold and no metric tensor is necessary, while this becomes a key ingredient when making precise the idea of a shortest path.

Geodesics and Acceleration: Let us consider henceforth a smooth n -dimensional manifold M as per Definition 9 and a linear connection ∇ as per Proposition 54. Let us consider a smooth curve $\gamma : I \subseteq \mathbb{R} \rightarrow M$. As already mentioned in Definition 68 this comes naturally accompanied by the notion of tangent vector to the curve, *i.e.*

$$\dot{\gamma}(t) = d\gamma \left(\left. \frac{d}{dt'} \right|_{t'=t} \right) \in T_{\gamma(t)}M,$$

which, in a physical language, can be read as the *velocity vector*. This motivated by the fact that, if we consider a local chart (U, φ) centered at $\gamma(t)$ and coordinates $\{x^1, \dots, x^n\}$ thereon, we can write

$$\dot{\gamma}(t) = \dot{\gamma}^i(t) \frac{\partial}{\partial x^i}.$$

We are interested in defining a notion of acceleration, although, in this context, we cannot start from $\frac{d^2}{dt^2}$ defined on I since there is no notion of push-forward for higher derivatives. To this end we want to focus our attention to vector fields living on $\gamma(t)$.

Definition 86. Let $\gamma : I \subseteq M$ be a smooth curve. We call vector fields along γ the set

$$T_\gamma(M) \doteq \{V \in C^\infty(I; TM) \mid V(t) \in T_{\gamma(t)}M, \forall t \in I\}. \quad (52)$$

One can realize that, for each $X \in \Gamma(TM)$, this induces a counterpart on $T_\gamma(M)$ by restriction. Yet not all elements in $T_\gamma(M)$ can be obtained in this way, for example whenever there exist $t_1, t_2 \in I$ such that $t_1 \neq t_2$, $\gamma(t_1) = \gamma(t_2)$ but $\dot{\gamma}(t_1) \neq \dot{\gamma}(t_2)$. Our next step is to prove that ∇ induces a distinguished operator on $T_\gamma(M)$.

Having endowed M with a linear connection, we can show that it induces a counterpart on $T_\gamma(M)$. Notice that this is not a mere pullback along γ since there are vector fields which are not extensible to $\Gamma(TM)$ and, hence the mantra “extend, act with ∇ and then restrict again on $\gamma(t)$ ” cannot work. Yet, when such operation is admissible, we expect this induced linear connection to work in this way

Proposition 87. Let M be an n -dimensional smooth manifold as per Definition 9 and let ∇ be a linear connection as per Proposition 54. Let $\gamma \in C^\infty(I; M)$ with $I \subseteq M$. Then there exists a unique linear operator, called **covariant derivative (or induced connection)**

$$D_t : T_\gamma(M) \rightarrow T_\gamma(M), \quad (53)$$

such that the following properties hold true:

1. for every $f \in C^\infty(I)$ and for all $V \in T_\gamma(M)$,

$$D_t(fV) = \dot{f}V + fD_t(V), \quad (54)$$

2. Given $V \in T_\gamma(M)$, if there exists $X, Y \in \Gamma(TM)$ such that $X|_{\gamma(t)} = V$ and $Y|_{\gamma(t)} = \dot{\gamma}(t)$, then

$$D_t(V) = \nabla_{\dot{\gamma}(t)} X|_{\gamma(t)}. \quad (55)$$

Proof. Suppose D_t exists and let $t_0 \in I$. Using an argument similar to that of Proposition 56, we can conclude that D_t is local, namely its action on V at $t = t_0$ depends only on the values of V in an arbitrary small neighbourhood of t_0 . Hence, we can consider a local chart (U, φ) in M such that $\gamma(t) \in U$ and we can select a coordinate system $\{x^1, \dots, x^n\}$. Accordingly $V(t) = V^i(t) \frac{\partial}{\partial x^i}$. Observe that $\frac{\partial}{\partial x^i}$ is per construction defined also outside $\gamma(t)$ and therefore, we can apply Conditions 1. and 2. to infer that

$$D_t(V)(\gamma(t_0)) = D_t \left(V^i(t) \frac{\partial}{\partial x^i} \right) \Big|_{t=t_0} = \dot{V}^i(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} + V^i(t_0) \nabla_{\dot{\gamma}(t)} \left(\frac{\partial}{\partial x^i} \right) \Big|_{\gamma(t_0)}.$$

Writing $Y = Y^j \frac{\partial}{\partial x^j}$ we can use both Equation (18) and (21) to infer that

$$\begin{aligned} D_t(V)(\gamma(t_0)) &= \dot{V}^i(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} + V^i(t_0) Y^j(\gamma(t_0)) \Gamma_{ij}^k(\gamma(t_0)) \frac{\partial}{\partial x^k} \Big|_{\gamma(t_0)} = \\ &= \dot{V}^i(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} + V^i(t_0) \dot{\gamma}^j(t_0) \Gamma_{ij}^k(\gamma(t_0)) \frac{\partial}{\partial x^k} \Big|_{\gamma(t_0)}. \end{aligned} \quad (56)$$

This shows that there is no ambiguity in defining D_t which is therefore unique. In addition, having established that it suffices to define D_t locally around any point of I , we can use Equation (56) as its definition. This entails existence and the sought conclusion.  $\square$

Remark 24 (Conceptual). *The reader might be grumbling that this yet another and new mathematical concept. Though the Watcher might have disappeared because he has intervened in the Multiverse, here I am grinning. Seriously? New? And what do you think is the material derivative in fluid dynamics?!*

We have all the ingredients to answer our original question. Suppose that a curve $\gamma \in C^\infty(I; M)$ has been assigned with $I \subseteq \mathbb{R}$. If we interpret it as the trajectory on M of a point particle, we can define its velocity at any instant of time as the tangent vector to the curve, namely $\dot{\gamma}(t)$, $t \in I$. Yet, if we endow M with a linear connection, we can now define the **acceleration** at any instant of time as

$$a_\gamma(t) \doteq D_t \dot{\gamma}(t), \quad t \in I. \quad (57)$$

where D_t is the covariant derivative along γ induced by ∇ . We can now give our first modification of Newton's law since we have a mean to define a path with vanishing acceleration.

Definition 88. Let M be an n -dimensional smooth manifold as per Definition 9 and let ∇ be a linear connection as per Proposition 54. Let $\gamma \in C^\infty(I; M)$ with $I \subseteq \mathbb{R}$. We call it a **geodesic** if it has vanishing acceleration, namely

$$a_\gamma(t) \doteq D_t \dot{\gamma}(t) = 0, \quad t \in I. \quad (58)$$

where D_t is the covariant derivative along γ induced by ∇ as per Proposition 87. We denote by $\gamma_V(t)$ a geodesic such that $d\gamma_V(0) = V$.

Observe that, if we consider a local chart (U, φ) in M such that $\gamma(t) \in U$ and we can select a coordinate system $\{x^1, \dots, x^n\}$, then Equation (58) in combination with Equation (56) becomes the celebrated **geodesic equation**

$$\dot{\gamma}^i(t) \nabla_i \dot{\gamma}^k(t) = 0, \quad (59a)$$

$$\boxed{\frac{d^2 x^k(t)}{dt^2} + \Gamma_{ij}^k(x(t)) \frac{dx^i(t)}{dt} \frac{dx^j(t)}{dt} = 0}. \quad (59b)$$

Engrave this formula in your soul.

Remark 25 (conceptual). Equation (59b) meets all our expectations since, on the one hand, if we consider a flat connection, that is $\Gamma_{ij}^k = 0$, such as when we work with the metric induced covariant derivative on $(\mathbb{R}^3, \delta)$ it reduces to $\frac{d^2 x^k(t)}{dt^2} = 0$. This is the standard equation for vanishing acceleration in Newtonian mechanics. On the other hand, since the Christoffel symbols are always assumed to be smooth, we can read Equation (59b) as a coupled system on non-linear ODEs with smooth coefficients and, once more, the Picard-Lindelöf theorem entails existence of the solution in a small interval around value $t_0 \in I$ where initial data are assigned.

Exercise 89. Prove the rescaling lemma, namely, show that, under the assumptions of Definition 88, if $\gamma_V : I \rightarrow M$ is a geodesic, then for all $c \in \mathbb{R}$, whenever $ct \in I$,

$$\gamma_{cV}(t) = \gamma_V(ct).$$

(Riemannian) Geodesics: We have established the notion of a geodesic as a curve with vanishing acceleration using only a linear connection. We will now specialize to the case where we endow the underlying smooth manifold with a metric tensor and therefore ∇ can be chose as the covariant derivative, see Theorem 63. While Equation (59b) is left untouched, the only difference being that the Christoffel symbols are defined out of Equation (33), the physical significance is greatly improved.

To start with, we introduce a new notion which has a marginal application in general relativity but finds its main relevance in the geometric formulation of gauge theories and in quantum field theory on curved spacetime: *parallel transport*.

Definition 90. Let (M, g) be an n -dimensional smooth manifold as per Definition 9 with a metric tensor as per Definition 50. Let ∇ be the associated Levi-Civita connection as per Theorem 63. Given a curve $\gamma \in C^\infty(I; M)$, $I \subseteq \mathbb{R}$, we say that $V \in T_\gamma(M)$ is **parallel** along γ if

$$D_t V = 0,$$

where D_t is as per Proposition 87.

On the one hand we observe that, given a geodesic γ as per Definition 88, the tangent vector to the curve, $\dot{\gamma}(t)$ is per definition parallel to it. On the other hand a more general question is whether, given a vector $V_0 \in T_{\gamma(t_0)}M$, we can construct $V(t) \in T_{\gamma(t)}(M)$ parallel to γ and such that $V(t_0) = V_0$. The answer is affirmative as this proposition shows.

Proposition 91. Under the assumptions of Definition 90, for any $V_0 \in T_{\gamma(t_0)}M$, there exists a unique $V(t) \in T_{\gamma(t)}(M)$ parallel to γ and such that $V(t_0) = V_0$

Proof. Equation (56) entails that $D_t V = 0$ with $V(t) = V_0$ translates into the following system of ODEs:

$$\begin{cases} \dot{V}^i + \Gamma_{jk}^i V^j \dot{\gamma}^k(t) = 0 \\ V^i(t_0) = V_0^i \end{cases} . \quad (60)$$

This is a linear system of ODEs and, once an initial condition has been assigned, the solution exists for all $t \in I$ (ve lo avevo detto che il corso di Vitali va seguito!) $\square$

The term parallel transport might seem a little bit obscure at this stage, but the following lemma should clarify that the nomenclature has been chosen on purpose.

Lemma 92. Under the assumptions of Definition 90, given $V, W \in T_\gamma(M)$, it holds that

1. $\frac{d}{dt}g(V, W) = g(D_t V, W) + g(V, D_t W)$,
2. If V, W are parallel along γ then $g(V, W)$ is constant along $\gamma(t)$.

Proof. We need to prove only item 1. since the second statement descends immediately from it once we recall that, if a vector field, *e.g.* V , is parallel along γ , then $D_t V = 0$, see Definition 90. Let us therefore consider $V, W \in T_\gamma(M)$. Working locally, Equation (60) entails that

$$g(D_t V, W) = g_{\mu\nu}(\dot{V}^\mu + \Gamma_{\rho\lambda}^\mu V^\rho \dot{\gamma}^\lambda(t))W^\nu,$$

which, in turn, gives

$$g(D_t V, W) + g(V, D_t W) = g_{\mu\nu}(\dot{V}^\mu + \Gamma_{\rho\lambda}^\mu V^\rho \dot{\gamma}^\lambda(t))W^\nu + g_{\mu\nu}V^\mu(\dot{W}^\nu + \Gamma_{\rho\lambda}^\nu W^\lambda \dot{\gamma}^\rho(t)) =$$

At the same time, on account of the Leibnitz rule,

$$\frac{d}{dt}(g_{\mu\nu}V^\mu W^\nu) = \left(\frac{d}{dt}g_{\mu\nu}\right)V^\mu W^\nu + g_{\mu\nu}\dot{V}^\mu W^\nu + g_{\mu\nu}V^\mu \dot{W}^\nu.$$

Using metric compatibility, we can infer that $\nabla_{\dot{\gamma}(t)}g = 0$ which reads in coordinates $\frac{d}{dt}g_{\rho\nu} - g_{\mu\rho}\Gamma_{\nu\lambda}^\rho \dot{\gamma}^\lambda(t) - g_{\rho\nu}\Gamma_{\mu\lambda}^\rho \dot{\gamma}^\lambda(t) = 0$. This gives the sought identity. $\square$

We can now prove one of the most underestimated, yet very informative corollary which highlights the interplay between geodesics and Killing fields.

Corollary 93. *Let (M, g) be an n -dimensional smooth manifold as per Definition 9 with a metric tensor as per Definition 50. Let ∇ be the associated Levi-Civita connection as per Theorem 63. Given a geodesic $\gamma \in C^\infty(I; M)$, $I \subseteq \mathbb{R}$ as per Definition 88*

1. *the velocity vector associated to γ has constant length, namely, for all $t \in I$,*

$$\frac{d}{dt} \left(g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t)) \right) = 0,$$

2. *if ξ is a Killing field, then, along $\gamma(t)$,*

$$\frac{d}{dt} \left(g_{\gamma(t)}(\xi(\gamma(t)), \dot{\gamma}(t)) \right) = 0,$$

which reads in coordinates

$$g_{\mu\nu}(\gamma(t))\xi^\mu \dot{\gamma}^\nu = \text{const.} \quad (61)$$

Proof. The first statement is a consequence of item 2. of Lemma 92 once we observe that, per definition, if γ is a geodesic, its velocity vector is parallel along it. Let us focus on the second statement. Locally it holds that

$$\begin{aligned} \frac{d}{dt} \left(g_{\gamma(t)}(\xi(\gamma(t)), \dot{\gamma}(t)) \right) &= \dot{\gamma}^k(t) \nabla_k (g_{ij} \xi^i \dot{\gamma}^j) = \\ \dot{\gamma}^k \left(\nabla_k (g_{ij}) \xi^i \dot{\gamma}^j + g_{ij} (\nabla_k \xi^i) \dot{\gamma}^j + g_{ij} \xi^i \nabla_k \dot{\gamma}^j \right) &= (\nabla_k \xi_j) \dot{\gamma}^k \dot{\gamma}^j + \xi_j \dot{\gamma}^k \nabla_k \dot{\gamma}^j = 0, \end{aligned}$$

where, in the second equality, we used metric compatibility, see Proposition 60. In the third one we use instead that, being the indices dummy, $(\nabla_k \xi_j) \dot{\gamma}^k \dot{\gamma}^j = (\nabla_{(k} \xi_{j)}) \dot{\gamma}^k \dot{\gamma}^j$ which vanishes due to the Killing equation (49), while $\dot{\gamma}^k \nabla_k \dot{\gamma}^j = 0$ due to the geodesic equation (59a). $\square$

One can generalize Corollary 93 using Killing tensors as introduced in Definition 82. The following exercise serves this purpose.

Exercise 94. Let (M, g) be an n -dimensional smooth manifold as per Definition 9 with a metric tensor as per Definition 50. Let ∇ be the associated Levi-Civita connection as per Theorem 63. Given a geodesic $\gamma \in C^\infty(I; M)$, $I \subseteq \mathbb{R}$ as per Definition 88. If T is a Killing tensor as per Definition 82, then along $\gamma(t)$,

$$g_{\mu_1 \nu_1}(\gamma(t)) \dots g_{\mu_k \nu_k}(\gamma(t)) \dot{\gamma}^{\mu_1} \dots \dot{\gamma}^{\mu_k} T^{(\nu_1, \dots, \nu_k)} = \text{const.} \quad (62)$$

Remark 26 (Riemannian vs Lorentzian). *Item 1. of Corollary 93 guarantees us that the velocity vector of a geodesic is constant and, barring a rescaling of the coordinate t along I , it can also be normalized. Yet here is a substantial difference between the scenario where M is endowed with a Riemannian or with a Lorentzian metric, see Definition 50. In the first case*

we can always choose t so that the length of the velocity vector of a geodesic γ is normalized to 1, while this is not possible in the second case. In a Lorentzian geometry, we can recognize three different options, the velocity vector $\dot{\gamma}(t)$ is such that $g(\dot{\gamma}(t), \dot{\gamma}(t)) > 0$, $g(\dot{\gamma}(t), \dot{\gamma}(t)) = 0$ or $g(\dot{\gamma}(t), \dot{\gamma}(t)) < 0$ to which correspond three different possibility of normalization, 1, 0, -1 . When dealing with black hole spacetimes in Section 7, we will be exploiting considerably this peculiarity of Lorentzian metrics to study the motion of a massless (photon) or of a massive point particle around a Schwarzschild black hole.

The Lagrangian and Hamiltonian Formulation of Geodesics: In this section, we have established two important results concerning smooth manifolds endowed with a metric tensor. The first is the existence of curves of vanishing acceleration, see Definition 88 which abide by a system of ODEs as per Equation (59b). These are called geodesics and the length of their tangent vector is constant. The second result concerns the existence of a non trivial interplay between the velocity vector of a geodesic and Killing fields, that is infinitesimal isometries of the underlying metric. Most notably their scalar product is constant along the geodesic. This allows for a novel and more physically inspired interpretation and derivation of geodesics. In the following we assume that (M, g) is an n -dimensional smooth Riemannian manifold, see Definition 50, although the reasoning can be extended to arbitrary signatures, barring some aspects of the physical interpretation.

We shall work locally and, hence we assume to have chosen a local chart (U, φ) and coordinates $\{x^1, \dots, x^n\}$ thereon. Considering a point particle of mass m , if its trajectory is a smooth curve $\gamma \in C^\infty(I; M)$ the velocity vector is $\dot{\gamma}(t) \equiv \dot{x}^\mu(t) \frac{\partial}{\partial x^\mu}$ where we have denoted $\gamma(t) = (x^1(t), \dots, x^n(t))$. In absence of external forces, the dynamics of the point particle m can be constructed out of its Lagrangian which coincides with the kinetic energy, namely

$$L[x^1, \dots, \dot{x}^n, x^1, \dots, x^n] = \frac{m}{2} g_{\mu\nu} \dot{x}^\mu(t) \dot{x}^\nu(t) \equiv \frac{m}{2} g(\dot{\gamma}(t), \dot{\gamma}(t)), \quad (63)$$

where the right hand side is positive definite since g is Riemannian. Therefore the particle shall follow a trajectory determined from the variation of L , that is the associated Euler-Lagrange Equation reads

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial \dot{x}^\rho} - \frac{\partial L}{\partial x^\rho} &= \frac{m}{2} \frac{d}{dt} (g_{\mu\rho} \dot{x}^\mu + g_{\rho\nu} \dot{x}^\nu) - \frac{m}{2} \frac{\partial g_{\mu\nu}}{\partial x^\rho} \dot{x}^\mu \dot{x}^\nu = \\ \frac{m}{2} (g_{\mu\rho} \ddot{x}^\mu + g_{\rho\nu} \ddot{x}^\nu) + \frac{m}{2} \left(\frac{\partial g_{\mu\rho}}{\partial x^\nu} + \frac{\partial g_{\rho\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\rho} \right) \dot{x}^\mu \dot{x}^\nu &= \\ = m g_{\mu\rho} \ddot{x}^\mu + \frac{m}{2} \left(\frac{\partial g_{\mu\rho}}{\partial x^\nu} + \frac{\partial g_{\rho\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\rho} \right) \dot{x}^\mu \dot{x}^\nu &= 0, \end{aligned} \quad (64)$$

where we have renamed suitably all dummy indices. By multiplying the last identity by $g^{\rho\lambda}$ and comparing with Equation (33), we end up with

$$\ddot{x}^\lambda + \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0. \quad (65)$$

In other words we have obtained once more the geodesic equation as the Euler-Lagrange equation associate to the Lagrangian of a free particle moving along a trajectory in (M, g) . In addition, since there are no external forces L coincides with the *total energy* $E(t)$ at time t . In turn, since g is Riemannian, the Lagrangian is a convex function and, therefore one can associate to it its conjugate momenta:

$$p^\mu = \frac{\partial L}{\partial \dot{x}^\mu} = mg_{\mu\nu} \dot{x}^\nu \quad (66)$$

On the one hand, the musical isomorphisms in Equation (27) guarantee us that momenta can be interpreted as the components of an element lying in $T^*M|_U$. On other hand the Hamiltonian's function associated to Equation (63) reads

$$H[x^1, \dots, x^n, p_1, \dots, p_n] = \frac{1}{2m} g^{\mu\nu} p_\mu p_\nu, \quad (67)$$

whose Hamilton's equations read

$$\dot{q}^\mu(t) = \frac{1}{m} g^{\mu\nu} p_\nu, \quad \dot{p}_\mu(t) = -\frac{1}{2m} \frac{\partial}{\partial x^\mu} g^{\rho\nu} p_\rho p_\nu. \quad (68)$$

In addition, Corollary 93 entails two important bits of physical information. Since Equation (65) guarantees that the dynamics of a free particle on (M, g) takes place on a geodesic, this means that, along the motion trajectories, the total energy $E(t) \equiv L(t)$ is constant. In addition, we have also recovered the classical notion of constant of motions since, still Corollary 93 entails that, given a Killing field, that is an infinitesimal isometry of the metric, its scalar product with the tangent vector to a geodesics is a constant of motion. In particular, suppose that the Lagrangian admits a cyclic coordinate, namely there exists x^λ such that $L \neq L[x^\lambda]$. This entails that $g_{\mu\nu} \neq g_{\mu\nu}(x^\lambda)$ and Example 83 tells us that $\frac{\partial}{\partial x^\lambda}$ is a Killing field. Hence $g(\frac{\partial}{\partial x^\lambda}, \dot{\gamma}(t))$ is constant of motion:

$$g_{\lambda\mu} \dot{x}^\mu = p_\lambda = \text{const.},$$

which is nothing but *momentum conservation*. To conclude, observe that, in the Riemannian setting, conservation of the energy comes from the constancy of the length of geodesics, whereas momentum conservation from Killing vector fields. This is another signature that, in classical mechanics, one should not ascribe energy conservation to Noether's theorem. The situation changes in a Lorentzian setting where the generator of time translations is a genuine Killing vector field. It is also noteworthy remarking that the conservation laws are associated to local isometries and there is no requirement that they originate from a global isometry.

Geodesics from minimizing lengths: In order to convince ourselves that a Riemannian or a Lorentzian manifold are a natural generalization of the space of configurations of Newtonian dynamics, we are left with one open question, namely on $(\mathbb{R}^3, \delta)$ straight lines are a priori distinguished as being the shortest path connecting two points. This does not require any notion of acceleration, nor of energy. We shall explain that geodesics meet also this criterion.

As before we shall assume for simplicity and to facilitate the physical interpretation that (M, g) is a Riemannian manifold as per Definition 50, although we could consider any signature. Given a curve $\gamma \in C^\infty(I; M)$ with $I \subseteq \mathbb{R}$ and $a, b \in I$, we call **length of γ from a to b**

$$\mathsf{L}[\gamma; a, b] \equiv \mathsf{L}[\gamma] = \int_a^b dt g(\dot{\gamma}(t), \dot{\gamma}(t))^{\frac{1}{2}} = \int_a^b dt \sqrt{g_{\mu\nu}(\gamma(t)) \dot{x}^\mu(t) \dot{x}^\nu(t)}, \quad (69)$$

where the last identity is valid if and only if the whole curve connecting a to b lies within a single local chart (U, φ) where we have chosen coordinates $\{x^1, \dots, x^n\}$.

Remark 27. Observe that Equation (69) is a natural generalization of the Euclidean counterpart for the length of a path. As a matter of fact, if $(M, g) = (\mathbb{R}^3, \delta)$ and if employ the standard Cartesian coordinates (x, y, z) , the right hand side of Equation (69) reads as expected:

$$\int_a^b dt \sqrt{\dot{x}^2(t) + \dot{y}^2(t) + \dot{z}^2(t)}.$$

Remark 28. Observe that the integrand on the right hand side of Equation (69) is positive since we have assumed g to be Riemannian. Should we have considered a Lorentzian signature then, we need to replace the argument of the square root with its absolute value:

$$\mathsf{L}[\gamma; a, b] \equiv \mathsf{L}[\gamma] = \int_a^b dt \sqrt{|g_{\mu\nu}(x(\gamma(t))) \dot{x}^\mu(t) \dot{x}^\nu(t)|}, \quad (70)$$

Having established the notion of length of a curve, we seek the one which minimizes L , namely the distance between a and b .

Definition 95. Let (M, g) be a smooth n -dimensional Riemannian manifold as per Definition 50. Given $p, q \in M$, we call **distance** between p and q

$$d(p, q) \doteq \inf\{\mathsf{L}[\gamma; p, q] \mid \gamma : [a, b] \rightarrow M \text{ is a piecewise smooth curve with } \gamma(a) = p, \text{ and } \gamma(b) = q\}. \quad (71)$$

Observe that we have chosen the interval $[a, b]$ to be the same for all curves since we can always exploit the freedom in redefining the coordinate along γ to hard code this requirement. For this reason, in many instances, one chooses $I = [0, 1]$. In addition we have allowed the curve to be piecewise smooth since any two points on a Riemannian manifold can be joined by such a curve. The term *piecewise* refers to the property that γ has to be smooth everywhere except for an at most countable set of points where it needs only to be continuous. Such departure from the smooth structure does not affect Equation (69) since it involves a set of zero measure and we can use the additive property of the integral to bypass it.

Exercise 96. Under the assumptions of Definition 95 prove that the function d abides by the defining properties of a distance, namely, for all $p, q, r \in M$

1. $d(p, q) \geq 0$ for all p, q and $d(p, q) = 0$ if and only if $p = q$,
2. $d(p, q) = d(q, p)$,
3. the triangle inequality holds: $d(p, r) \leq d(p, q) + d(q, r)$.

Observe that, on a Lorentzian manifold, these properties would not hold true, in particular $d(p, q)$ can vanish even if $p \neq q$. For this reason, in such scenario, the function d is called distance with an abuse of nomenclature. In addition to the length of a curve, inspired by Equation (67), we can also associate to any (piecewise) smooth curve $\gamma : I \rightarrow M$ with $I \subseteq \mathbb{R}$ an **action functional** seen as the integral of the (kinetic) energy along γ connecting a to b :

$$S[\gamma] \doteq \frac{1}{2} \int_a^b dt g(\dot{\gamma}(t), \dot{\gamma}(t)) = \frac{1}{2} \int_a^b dt g_{\mu\nu}(x(\gamma(t))) \dot{x}^\mu(t) \dot{x}^\nu(t), \quad (72)$$

where the last identity is valid if and only if the whole curve connecting a to b lies within a single local chart (U, φ) where we have chosen coordinates $\{x^1, \dots, x^n\}$. In this expression no mass parameter appears, or, if you prefer, we set $m = 1$. The most striking property is that Equation (72) controls from above Equation (69).

Lemma 97. *Let (M, g) be a smooth n -dimensional Riemannian manifold as per Definition 50. For any (piecewise) smooth curve $\gamma : I \rightarrow M$ with $I = [a, b]$, it hold that*

$$L[\gamma]^2 \leq 2(b-a)S[\gamma], \quad (73)$$

where L is defined in Equation (69), while S in Equation (72). The equality holds if and only if, for all $t, t' \in I$, $g(\dot{\gamma}(t), \dot{\gamma}(t)) = g(\dot{\gamma}(t'), \dot{\gamma}(t'))$.

Proof. Since the interval I is bounded and since the curve is (piecewise) smooth, $f(t) \doteq g(\dot{\gamma}(t), \dot{\gamma}(t))^{\frac{1}{2}} \in L^p(I)$ for all $p \in [1, \infty]$. Hence we can apply the Cauchy-Schwarz (or the Hölder) inequality in combination with the characteristic function χ on I to infer that $\|f\chi\|_{L^1} \leq \|\chi\|_{L^2} \|f\|_{L^2}$. This reads also

$$L[\gamma] \leq (b-a)^{\frac{1}{2}} S[\gamma]^{\frac{1}{2}},$$

which, upon taking the square of each side, is the sought result. In addition, still the Cauchy-Schwarz inequality entails that the equality is true if and only if $g(\dot{\gamma}(t), \dot{\gamma}(t))$ is proportional to the characteristic function. In other words it is constant along γ . $\square$

As a notable consequence of this lemma, it turns out that a curve which minimizes $S[\gamma]$ is also such for $L[\gamma]$. Yet, minimizing the action functional is tantamount to considering the Euler-Lagrange equation for Equation (63). These have been computed in Equation (65) and therefore we can conclude that *geodesics* are the shortest path connecting two points p, q in a Riemannian manifold.

Exercise 98. Consider the functional $L[\gamma]$ as per Equation (69) and compute its minimum directly. [Hint: You can consider the integrand as a Lagrangian density on its own right or compute the variation from scratch – if, in trouble, see [Kre91, §51].]

Remark 29. *It is important to stress that, in deriving the geodesic equation as the minimizing curve both for the energy and for the length functional, we have implicitly assumed that the curve is parameterized by its arc length. If one chooses a different option, the outcome changes. We shall not delve into the details leaving an interested reader to [Lee11, Jost08]. A second issue concerns the fact that, when computing the extrema of a functional, a priori we might find both its minima and its maxima. Here we have implicitly assumed that the latter option does not occur. We will not prove it and an interested reader can refer to [Lee11, Chap. 6].*

As the surviving readers should have understood, there is no real theory without accompanying examples, showing that all these concepts can be translated in a computationally efficient framework. Since we will use geodesics extensively also in Section 7, we shall discuss just one example, highlighting two different, albeit equivalent ways to approach the computation of a geodesic. We shall discuss only local existence, hence working in a coordinate patch. We shall not address here the question of global existence.

Example 99. Let us consider once more $\mathbb{S}^2$ endowed with the standard angular coordinates (θ, φ) as well as the round metric whose associated line element is in angular coordinates

$$ds^2 = d\theta^2 + \sin^2 \theta d\varphi^2.$$

Recalling that the associated Christoffel symbols have been computed in Example (85), we show two ways to compute geodesics in this background.

Method 1: We start from the defining identity, Equation (59b), which yields the following system of second order non-linear ODEs:

$$\ddot{\theta} - \frac{\sin(2\theta)}{2} \dot{\varphi}^2 = 0, \tag{74a}$$

$$\ddot{\varphi} + 2 \cot \theta \dot{\theta} \dot{\varphi} = 0. \tag{74b}$$

This is a rather complicated set of equations and it is quite hard to solve it without further information. One can realize that Equation (74b) can be rewritten as

$$\ddot{\varphi} + 2 \cot \theta \dot{\theta} \dot{\varphi} = 0 \implies \frac{1}{\sin^2 \theta} \frac{d}{dt} (\sin^2 \theta \dot{\varphi}) = 0.$$

Yet, it is not at all straightforward how to repeat the procedure for Equation (74a) and therefore we develop a second approach which helps us solving this quandary.

Method 2: We recall that Corollary 93 tells us that the velocity vector along a geodesic has constant length, which, up to a reparameterization of the time variable, we set to 1 and that every Killing vector field generates a conserved quantity via Equation (61). Recalling that

$\xi = \frac{\partial}{\partial \varphi}$ is Killing vector field, see Examples (83) and (85), these two pieces of data translate in the following set of non-linear, first order ODEs:

$$g(\dot{\gamma}(t), \dot{\gamma}(t)) = 1 \implies \dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2 = 1, \quad (75a)$$

$$g(\dot{\gamma}(t), \dot{\gamma}(t)) = M \implies \sin^2 \theta \dot{\varphi} = M, \quad (75b)$$

where M is a constant of motion smaller than 1 due to the chosen parameterization of the length of the velocity vector. Assigning an initial condition of the form $(\theta_0, \dot{\theta}_0, \varphi_0, \dot{\varphi}_0) \in \mathbb{S}^2$, we can infer that $M = \sin^2 \theta_0 \dot{\varphi}_0$. The above set of two equations can be rewritten as

$$\begin{cases} \dot{\theta} = \pm \sqrt{1 - \sin^2 \theta \dot{\varphi}^2} \\ \dot{\varphi} = \frac{M}{\sin^2 \theta} \end{cases}. \quad (76)$$

We select the plus sign since the other case can be dealt with slavishly. The phase portrait of this set of coupled ODEs abides by the ordinary differential equation

$$\frac{d\theta}{d\varphi} = \sqrt{1 - \sin^2 \theta \dot{\varphi}^2} \frac{\sin^2 \theta}{M} = \frac{\sin \theta \sqrt{\sin^2 \theta - M^2}}{M},$$

where, in the second equality, we used Equation (75b). This entails that

$$\varphi - \varphi_0 = \int_{\theta_0}^{\theta} dx \frac{M}{\sin(x) \sqrt{\sin^2(x) - M^2}} = \int_{\theta_0}^{\theta} dx \frac{M}{\sin^2(x) \sqrt{1 - \frac{M^2}{\sin^2(x)}}}.$$

By observing that $\frac{dx}{\sin^2(x)} = -d(\cot(x))$ and that $\sin^{-2}(x) = 1 + \cot^2(x)$, we can set $y = \cot(x)$ to obtain

$$\varphi - \varphi_0 = - \int_{\cot(\theta_0)}^{\cot \theta} dy \frac{M}{\sqrt{1 - M^2(1 + y^2)}} = - \arcsin \left(\frac{My}{\sqrt{1 - M^2}} \right) \Big|_{y=\cot \theta_0}^{y=\cot \theta}. \quad (77)$$

This yields a relation between φ and θ which is commonly written in the literature as

$$\sin(\tilde{\varphi}_0 - \varphi) = \frac{M \cot \theta}{\sqrt{1 - M^2}} \implies \tilde{\varphi}_0 = \varphi_0 + \arcsin \left(\frac{M \cot \theta_0}{\sqrt{1 - M^2}} \right). \quad (78)$$

In other words if we can find the curve $\theta(t)$, we have found the sought geodesics using Equation (77). Consider the first of the identities in Equation (76) and replacing therein the second one we end up with the ordinary differential equation

$$\dot{\theta} = \sqrt{1 - \frac{M^2}{\sin^2 \theta}} \implies t - t_0 = \int_{\theta_0}^{\theta} dx \frac{\sin(x)}{\sqrt{\sin^2(x) - M^2}}, \quad (79)$$

where, once more, we have chosen the plus sign, the other option following suit. By introducing the new variable $y = \cos(x)$, we can rewrite Equation (79) as

$$t - t_0 = \int_{\cos \theta_0}^{\cos \theta} dy \frac{1}{\sqrt{(1 - M^2) - y^2}} = \arcsin \left(\frac{y}{\sqrt{1 - M^2}} \right) \Big|_{\cos \theta_0}^{\cos \theta},$$

which can be inverted as

$$\theta(t) = \sqrt{1 - M^2} \arccos(\sin(t - \tilde{t}_0)), \quad \tilde{t}_0 = t_0 - \arcsin \left(\frac{\cos \theta_0}{\sqrt{1 - M^2}} \right). \quad (80)$$

Exercise 100. Give a physical interpretation to the result of Example (99). In particular show that the geodesics represent an arc of a **great circle or orthodrome**. This is the circular intersection of the unit sphere and a plane passing through the sphere's center point. [Hint: Consider Equation (78), argue by spherical symmetry that it is possible to work with $\theta_0 = \frac{\pi}{2}$ and switch to Cartesian coordinates once $\mathbb{S}^2$ is realized in $\mathbb{R}^3$.]

Remark 30. A reader should be warned of two important aspects which we have not really discussed in detail in Example (99). The first concerns the presence in several formulae of integrals with square roots in the integrand which we have always assumed implicitly to be positive definite. A priori this might not be true, but, if one checks the constraints on the domain of definition on the angular variables as well as the limitations on the admissible values for M , this turns out to be the case. In addition, if we compare Equation (75b) with Equation (51), we can realize that if we have employed just one of the three possible independent Killing fields, the one obtained setting $a = b = 0$ and $c = 1$. One could repeat the whole procedure also including the conserved quantities obtained from the vector fields constructed setting $a = 1$, $b = c = 0$ and $b = 1$, $a = c = 0$. Yet, as one can expect, the outcome is not altered and they do not carry additional information.

3.1 Exponential Map and Normal Coordinates

We need one last bit of geometric information before getting to the core of these lecture notes. Let us consider for the moment the Euclidean space $(\mathbb{R}^n, \delta)$. If we adopt Cartesian coordinates, the Christoffel symbols vanish and, therefore Equation (59b) can be solved as

$$x^i = x_0^i + \dot{x}_0^i t, \quad \forall t \in \mathbb{R}$$

which tells us that, as expected, geodesics are straight lines. Yet, if we fix $x_0 = 0$, we can infer that every point $y = (y^1, \dots, y^n) \in \mathbb{R}^n$ can be reached from the origin by means of a geodesic setting $t = 1$ and $\dot{x}^i = y^i$. Since Equation (60) applies to geodesics, this entails that they exist for all $t \in \mathbb{R}$. Setting $t = 1$ can be done without loss of generality since, choosing $t = \bar{t} \neq 1$ yields the same point in $\mathbb{R}^n$ provided that $\dot{x}_0^i = \frac{y^i}{\bar{t}}$. Hence this unveils that the geodesics allow to establish a distinguished smooth bijection between $T_0 \mathbb{R}^n$ and $\mathbb{R}^n$ which is the mathematical

transliteration of drawing arrows between the origin as a source and a target point. A natural question is how much of this reasoning survives when we drop the assumption that the underlying manifold is a Euclidean space. This section gives an answer and as a by product it allows us to establish the existence of a distinguished class of coordinates, known as *normal*, which are the closest relatives of the Cartesian ones in a generic scenario.

In the following we shall always consider (M, g) to be an n -dimensional, $n \geq 1$, Riemannian manifold as per Definition 50. This is done for mere convenience, but all results, unless explicitly stated otherwise, can be translated verbatim to every signature.

Definition 101. Let (M, g) be an n -dimensional, $n \geq 1$, Riemannian manifold as per Definition 50. We call **exponential map**

$$\exp : \mathcal{E} \rightarrow C^\infty(M), \quad V \mapsto \exp(V) = \gamma_V(1),$$

where

$$\mathcal{E} \doteq \{V \in \Gamma(TM) \mid \exists \gamma_V(t) \forall t \in [0, 1]\},$$

where γ_V is the integral curve associated to V as per Definition 68. In addition we denote by $\exp_p \doteq \exp|_{T_p M} : T_p M \rightarrow M$, $p \in M$.

The exponential map is a cornerstone of differential geometry since it enjoys the following properties.

Proposition 102. Under the assumption of Definition 101, it holds that

1. $\mathcal{E}$ is an open subset of $\Gamma(TM)$ containing the points of the form $\{(p, 0), p \in M\}$, while $\mathcal{E}_p = \mathcal{E} \cap T_p M$ is star-shaped,
2. For every $V \in \Gamma(TM)$, the associated geodesic abides by $\gamma_V(t) = \exp(tV)$ provided both sides of the identity are well-defined,
3. $\exp \in C^\infty(\mathcal{E}; M)$.

We avoid giving the proof since it long and every student who takes one among the courses of Dynamical Systems, Group Theory or Mathematical Methods for Theoretical Physics, shall see it. In any case, if interested, check [Lee11, Prop 5.7].

Lemma 103. Under the assumption of Definition 101, for every $p \in M$, there exists V an open subset of $T_p M$ containing the origin as well as $U \subseteq M$ with $p \in U$ such that $\exp_p$ is a diffeomorphism from U to V . The set U is called **normal (or geodesic) neighbourhood of M centered at p** .

Proof. Since Proposition 102 guarantees us that $\exp$ is smooth, the statement follows if we can prove that, for all $p \in M$, the map $d\exp_p$ is invertible at the origin of $T_p M$. This entails that we can apply the inverse function theorem and the sought conclusion descends. Recall that, on account of Definition 28, $d\exp_p : T_0(T_p M) \rightarrow T_p M$, where we have used implicitly that TM is a

smooth manifold, see Theorem 31. In addition, since $T_p M$ is diffeomorphic to $\mathbb{R}^n$, $n = \dim M$, we can conclude that $T_0(T_p M)$ is also diffeomorphic to $T_p M$. Bearing in mind this observation, for every $V \in T_p M$, we can use Definition 101 to infer that $d \exp_p(V)$ abides by

$$d \exp_p(V) = \left. \frac{d}{dt} (\exp_p \circ \tau)(t) \right|_{t=0},$$

where $\tau(t)$ is the integral curve of V in $T_p M$ such that $\tau(0) = 0$. In view of the discussion at the beginning of the section, up to the identification of $T_p M$ with $\mathbb{R}^n$, $\tau(t) = Vt$. In other words, using item 2. of Proposition 102,

$$d \exp_p(V) = \left. \frac{d}{dt} \exp_p(Vt) \right|_{t=0} = \left. \frac{d}{dt} \gamma_V(t) \right|_{t=0} = V,$$

where γ_V is the integral curve of V such that $\gamma_V(0) = 0$. This identity entail that, up to the diffeomorphic identifications of the tangent spaces with $\mathbb{R}^n$, $d \exp_p$ is the identity map and, hence, it is invertible. $\square$

Lemma 103 constitutes one of the operational building blocks in differential geometry and in general relativity. The reason lies in the fact that many quantities of interest shall turn out to be invariant under the action of the diffeomorphism group of the underlying smooth Riemannian or Lorentzian manifold (M, g) . This is often proven using abstract nonsense arguments and, if one is interested in establishing their actual functional form, diffeomorphism invariance entails that, at least locally, we can compute them in any coordinate system. As in classical mechanics, there exists infinite many possible such systems, but the vast majority of them is not a wise choice to do explicit calculations. Using Lemma 103 we can individuate a distinguished choice of a coordinate system at any point $p \in M$, which generalizes the notion on Cartesian coordinates.

Let $p \in M$ and let $\{e_i(p), \dots, e_n(p)\}$ be an orthonormal basis of $T_p M$. This establishes an isomorphism

$$E : \mathbb{R}^n \rightarrow T_p M, \quad v \equiv (v^1, \dots, v^n) \mapsto E(v) = v^i e_i(p).$$

Considering $U \subseteq M$ to be a geodesic neighbourhood of p as per Lemma 103 and the projections

$$\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}, \quad (v^1, \dots, v^n) \mapsto \pi_i(v^1, \dots, v^n) = v^i,$$

then the maps

$$x^i \doteq \pi_i \circ E^{-1} \circ \exp_p^{-1} : U \rightarrow \mathbb{R}, \tag{81}$$

individuate **normal coordinates** on U . As one can infer from Equation (81), the set of normal coordinates is in 1:1 correspondence with orthonormal bases of $T_p M$. A reader might complaint that the maps of $\mathbb{R}^n$ are implicitly selecting Cartesian coordinates thereon. This is true and we would be free to make other choices, but this would be tantamount to considering a different orthonormal basis on $T_p M$. In other words we are not missing possible sets of normal coordinates, but actually we are avoiding over-counting them. In Equation (81) there is just one freedom, either in selecting E or π_i , not both. That said, the following proposition makes clear why normal coordinates are thought to be so-special.

Proposition 104. *Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50. Let $p \in M$ and let $U \subseteq M$ be geodesic neighbourhood centered thereat as per Lemma 103. Let $\{x^1, \dots, x^n\}$ be a set of normal coordinates. Then*

1. *for any $V \in T_p M$, the geodesic $\gamma_V(t)$ as per Definition 88 such that $\gamma_V(0) = p$ can be written as*

$$\gamma_V(t) = (tV^1, \dots, tV^n), \quad V = V^\mu \frac{\partial}{\partial x^\mu}$$

for all $t \in \mathbb{R}$ such that $\gamma_V(t) \in U$.

2. *at p $\partial_\rho g_{\mu\nu}(p) = 0$ and $\Gamma_{\mu\nu}^\rho(p) = 0$ where $g_{\mu\nu}$ are the matrix components of the metric tensor in the normal coordinates, where $\Gamma_{\mu\nu}^\rho$ are the Christoffel symbols built out of Equation (33).*

Proof. 1.: Consider a geodesic $\gamma_V(t)$ whose tangent vector at $t = 0$ is $V \in T_p M$. Reading its components in normal coordinates entails applying Equation (81), that is

$$x^i(t) = x^i[\gamma(t)] = \pi_i \circ E^{-1} \circ \exp_p^{-1}[\gamma_V(t)].$$

Using Exercise (89), it holds that $\gamma_V(t) = \gamma_{tV}(1)$ from which it descends

$$x^i(t) = \pi_i \circ E^{-1} \circ \exp_p^{-1}[\gamma_{tV}(1)] = \pi_i \circ E^{-1} \circ \exp_p^{-1}[\exp_p(tV)] = \pi_i \circ E^{-1}(tV) = tV^i,$$

where we used Definition 101.

2. Since $\gamma_V(t)$ is a geodesic, it must abide by Equation (59b). Along γ , it holds that $x^i(t) = V^i t$ and hence

$$\Gamma_{\mu\nu}^\rho V^\mu V^\nu = 0.$$

Since, at $p \in M$, this must hold true for all geodesics and therefore for all vectors V^μ , this identity entails that $\Gamma_{\mu\nu}^\rho(p) = 0$. Using this result in combination with metric compatibility of the Levi-Civita connection, see Proposition 60, it holds that, for all $\mu, \nu, \rho = 1, \dots, n$,

$$\nabla_\rho g_{\mu\nu}(p) = \partial_\rho g_{\mu\nu}(p) = 0,$$

which concludes the proof. □

3.2 Riemann Tensor

At last we can introduce the basic geometric structures which are at the heart of general relativity. For the last time in these notes, we consider an n -dimensional smooth manifold M with a metric tensor of arbitrary signature g . From a geometric view point, the question that one might ask at this stage is whether, given two smooth manifolds (M, g) and (N, h) with metric tensors g and h , then, if $\dim M = \dim N$, they are locally isometric. It is clear that globally this is not possible since it would entail in addition that M and N are diffeomorphic and the examples of $\mathbb{S}^n$ and $\mathbb{R}^n$ suffice to give a negative answer. Yet, locally this is less foregone. One still expects the statement not to be true, but one cannot rely on topological arguments as in

the global case. Hence, a possible way out consists in finding *non trivial local invariants*, namely tensors built out of the metric and invariant under the action of (local) isometries.

Although what follows can be defined for an arbitrary linear connection, henceforth we work only with the covariant derivative, see Theorem 63, which will be denoted simply by ∇ omitting the subscript g unless a disambiguation becomes necessary.

Definition 105. Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50. We call **Riemann curvature endomorphism** the tensor fields $\text{Rm} \in \mathcal{T}^{(1,3)}(M)$ such that

$$\begin{aligned} \text{Rm} : \Gamma(TM) \times \Gamma(TM) \times \Gamma(TM) &\rightarrow \Gamma(TM) \\ (X, Y, Z) &\mapsto \text{Rm}(X, Y, Z) \doteq \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, \end{aligned} \quad (82)$$

where $[\cdot, \cdot]$ denotes the Lie bracket between two vector fields as per Definition 36. At the same time we define the **Riemann curvature tensor** as the tensor field $\text{Riem} \in \mathcal{T}^{(0,4)}(M)$ such that

$$\begin{aligned} \text{Riem} : \Gamma(TM) \times \Gamma(TM) \times \Gamma(TM) \times \Gamma(TM) &\rightarrow C^\infty(M) \\ (X, Y, Z, W) &\mapsto \text{Riem}(X, Y, Z, W) \doteq g(\text{Rm}(X, Y, Z), W). \end{aligned} \quad (83)$$

Remark 31. In Definition 105 we individuate two tensor fields which are, for all practical purposes equivalent, to the point that their difference becomes completely moot when working in components, namely, locally we shall write

$$\text{Rm} = \text{Rm}^\lambda_{\mu\nu\rho} dx^\mu \otimes dx^\nu \otimes dx^\rho \otimes \frac{\partial}{\partial x^\lambda} \equiv R^\lambda_{\mu\nu\rho} dx^\mu \otimes dx^\nu \otimes dx^\rho \otimes \frac{\partial}{\partial x^\lambda},$$

whereas, when considering the Riemann curvature tensor, we obtain

$$\text{Riem} = \text{Riem}_{\mu\nu\rho\lambda} dx^\mu \otimes dx^\nu \otimes dx^\rho \otimes dx^\lambda \equiv R_{\mu\nu\rho\lambda} dx^\mu \otimes dx^\nu \otimes dx^\rho \otimes dx^\lambda.$$

The two expressions are related via a musical isomorphism, see Equations (83) and (27), namely

$$\text{Riem}_{\mu\nu\rho\lambda} = \text{Rm}^\delta_{\mu\nu\rho} g_{\delta\lambda}. \quad (84)$$

This is the reason why, in the literature the difference is dropped and only the symbols $R_{\mu\nu\rho\lambda}$ or $R^\lambda_{\mu\nu\rho}$ are employed. We shall stick to this (good and healthy) habit. In addition, it is worth noting that the definition of the Riemann tensor is subject to several sign conventions, second in number only to those appearing in the definition of the Fourier transform. We shall not even try listing all of them, but we warn the reader that it is fairly common in general relativity and in quantum field theory on curved spacetimes to find global sign discrepancies between different textbooks and research papers. One source lies in the convention of the metric signature and a second one in the definition of the Riemann tensor. We emphasize the importance to pay attention to this detail... did I ever tell you the story about the Dirac equation on Lorentzian manifolds? No, oh well... maybe I should.

Remark 32 (On the physical significance of the Riemann tensor). *It is worth highlighting that, in many physics textbooks, the Riemann tensor is also dubbed curvature tensor with reference to the motivation why it is introduced. More precisely, given an observer sitting at a point $p \in M$, one might wonder what happens if a vector $V \in T_p M$ is moved around an infinitesimal loop. Does it go back to its original self or does it get transformed? To make more concrete this question, one considers two infinitesimal vectors $A^\mu \frac{\partial}{\partial x^\mu}$ and $B^\nu \frac{\partial}{\partial x^\nu}$ so that both A^μ and B^ν are constant, $\mu, \nu = 1, \dots, n = \dim M$ and so that we can work in a local patch where a coordinate system $\{x^1, \dots, x^n\}$ has been defined. Suppose to transport parallelly V first along the integral curve generated by A and then along that by B . Subsequently we carry back the vector transporting it first along the curve generated by A and then along that associated to B . We obtain a new vector $V' \in T_p M$ and denoting by $\delta V = V' - V$, we expect that there exists a tensor lying in $T_p^* M \times T_p^* M \times T_p^* M \times TM$ such that*

$$(\delta V)^\rho = R_{\mu\nu\lambda}^\rho V^\mu A^\nu B^\lambda.$$

The Riemann tensor enjoys several notable properties. We split them in two. The first is a verification that Equation (82) identifies actually a tensor field. This is based on the *tensor characterization lemma*, see [Lee11, Thm. 2.4] which states that a map

$$T : \underbrace{\Gamma(TM) \times \dots \times \Gamma(TM)}_k \times \underbrace{\Gamma(T^*M) \times \dots \times \Gamma(T^*M)}_l \rightarrow C^\infty(M),$$

identifies an element of $\mathcal{T}^{(k,l)}(M)$ if and only if it is multilinear over $C^\infty(M)$. A direct inspection of Equation (82) unveils that the only potential difficulty can arise from the compatibility with respect to the product of smooth function

Proposition 106. *Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50. Then for all $X, Y, Z \in \Gamma(TM)$ and for all $f, f', \tilde{f} \in C^\infty(M)$, it holds that*

$$\text{Rm}(fX, f'Y, \tilde{f}Z) = ff'\tilde{f}\text{Rm}(X, Y, Z).$$

Proof. The identity can be proven in two steps. Let us start by assuming that $f = f' = 1$. It holds that for all $\tilde{f} \in C^\infty(M)$ and for all $Z \in \Gamma(TM)$,

$$\nabla_X \nabla_Y (\tilde{f}Z) = \nabla_X (\tilde{f} \nabla_Y Z) + Y(\tilde{f})Z = \tilde{f} \nabla_X \nabla_Y Z + X(\tilde{f}) \nabla_Y Z + Y(\tilde{f}) \nabla_X Z + X(Y(\tilde{f}))Z,$$

where we applied iteratively the second identity in Equation (18). Similarly exchanging X and Y

$$\nabla_Y \nabla_X (\tilde{f}Z) = \nabla_Y (\tilde{f} \nabla_X Z) + X(\tilde{f})Z = \tilde{f} \nabla_Y \nabla_X Z + Y(\tilde{f}) \nabla_X Z + X(\tilde{f}) \nabla_Y Z + Y(X(\tilde{f}))Z,$$

while, recalling that $[X, Y] \in \Gamma^\infty(TM)$

$$\nabla_{[X,Y]}(\tilde{f}Z) = \tilde{f} \nabla_{[X,Y]}Z + [X, Y](\tilde{f})Z.$$

Putting together these three last identities, we obtain

$$\text{Rm}(X, Y, \tilde{f}Z) = \tilde{f}\text{Rm}(X, Y, Z).$$

The second step of the proof consists of assuming that $\tilde{f} = 1$. It holds that for all $f, f' \in C^\infty(M)$ and for all $Z \in \Gamma(TM)$, the first identity in Equation (18) entails

$$\nabla_{fX} \nabla_{f'Y} Z = f \nabla_X (f' \nabla_Y Z) = f f' \nabla_X \nabla_Y Z + f X(f') \nabla_Y Z.$$

At the same time exchanging X and Y we obtain

$$\nabla_{f'Y} \nabla_{fX} Z = f' \nabla_Y (f \nabla_X Z) = f f' \nabla_Y \nabla_X Z + f' Y(f) \nabla_X Z.$$

Finally, if we recall the structural properties of the Lie brackets, it holds that $[fX, f'Y] = f f' [X, Y] + f X(f')Y - f' Y(f)X$. This entails that

$$\nabla_{[fX, f'Y]} Z = \nabla_{f f' [X, Y] + f X(f')Y - f' Y(f)X} Z = f f' \nabla_{[X, Y]} Z + f X(f') \nabla_Y Z - f' Y(f) \nabla_X Z,$$

where we used that ∇ is a bilinear map, cf. Equation (17). Putting together all these identities we obtain

$$R(fX, f'Y, Z) = f f' R(X, Y, Z),$$

which concludes the proof. $\square$

The second set of properties is merely a collection of algebraic identities, whose relevance has not to be underestimated since they are the door which opens the path to constructing the Einstein tensor.

Proposition 107. *Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50. Then for all $X, Y, Z, W \in \Gamma(TM)$, the **first (algebraic) Bianchi identities** hold true:*

1. $\text{Riem}(X, Y, Z, W) = -\text{Riem}(Y, X, Z, W)$,
2. $\text{Riem}(X, Y, Z, W) = -\text{Riem}(X, Y, W, Z)$,
3. $\text{Riem}(X, Y, Z, W) + \text{Riem}(Y, Z, X, W) + \text{Riem}(Z, X, Y, W) = 0$,
4. $\text{Riem}(X, Y, Z, W) = \text{Riem}(Z, W, X, Y)$.

Proof. Statement 1. is a direct consequence of Equation (82). We focus therefore on 2. observing that the statement is equivalent to proving that

$$\text{Riem}(X, Y, Z, Z) = 0, \quad \forall X, Y, Z \in \Gamma(TM),$$

since, being Riem multilinear, it descends that, for any $W \in \Gamma(TM)$

$$\begin{aligned} 0 &= \text{Riem}(X, Y, Z + W, Z + W) = \\ &= \text{Riem}(X, Y, Z, Z) + \text{Riem}(X, Y, W, W) + \text{Riem}(X, Y, Z, W) + \text{Riem}(X, Y, W, Z) \implies \\ &\implies 0 = \text{Riem}(X, Y, Z, W) + \text{Riem}(X, Y, W, Z). \end{aligned}$$

Using repeatedly Equation (30) we can infer that

$$\begin{cases} X(Y(g(Z, Z))) = 2X(g(\nabla_Y Z, Z)) = 2g(\nabla_X \nabla_Y Z, Z) + 2g(\nabla_Y Z, \nabla_X Z) \\ Y(X(g(Z, Z))) = 2Y(g(\nabla_X Z, Z)) = 2g(\nabla_Y \nabla_X Z, Z) + 2g(\nabla_X Z, \nabla_Y Z) \\ [X, Y](g(Z, Z)) = 2g(\nabla_{[X, Y]} Z, Z) \end{cases}.$$

Subtracting from the first equation the second and third ones, we obtain applying Definition 36

$$0 = g((\nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]} Z, Z) = \text{Riem}(X, Y, Z, Z).$$

3. First we observe that, on account of Equation (83), the sought identity descends if we prove that

$$\text{Rm}(X, Y, Z) + \text{Rm}(Y, Z, X) + \text{Rm}(Z, X, Y) = 0.$$

This statement descends from a direct computation applying Equation (82), namely

$$\begin{aligned} & \text{Rm}(X, Y, Z) + \text{Rm}(Y, Z, X) + \text{Rm}(Z, X, Y) = \\ & \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z + \nabla_Y \nabla_Z X - \nabla_Z \nabla_Y X - \nabla_{[Y, Z]} X + \\ & \nabla_Z \nabla_X Y - \nabla_X \nabla_Z Y - \nabla_{[Z, X]} Y = \nabla_X (\nabla_Y Z - \nabla_Z Y) + \nabla_Y (\nabla_Z X - \nabla_X Z) + \\ & + \nabla_Z (\nabla_X Y - \nabla_Y X) - \nabla_{[X, Y]} Z - \nabla_{[Y, Z]} X - \nabla_{[Z, X]} Y, \end{aligned}$$

where we rearranged all terms. Applying now Equation (20), it descends that

$$\nabla_X(Y) - \nabla_Y(X) = [X, Y], \quad \forall X, Y \in \Gamma(TM).$$

This implies

$$\begin{aligned} & \text{Rm}(X, Y, Z) + \text{Rm}(Y, Z, X) + \text{Rm}(Z, X, Y) = \\ & \nabla_X([Y, Z]) - \nabla_{[Y, Z]} X + \nabla_Y([Z, X]) - \nabla_{[Z, X]} Y + \nabla_Z([X, Y]) - \nabla_{[X, Y]} Z = \\ & [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0, \end{aligned}$$

where we applied the Jacobi identity to conclude, see Corollary 37.

4. Applying the identity at item 3. repeatedly we obtain for all $X, Y, Z, W \in \Gamma(TM)$

$$\begin{cases} \text{Riem}(X, Y, Z, W) + \text{Riem}(Y, Z, X, W) + \text{Riem}(Z, X, Y, W) = 0 \\ \text{Riem}(Y, Z, W, X) + \text{Riem}(Z, W, Y, X) + \text{Riem}(W, Y, Z, X) = 0 \\ \text{Riem}(Z, W, X, Y) + \text{Riem}(W, X, Z, Y) + \text{Riem}(X, Z, W, Y) = 0 \\ \text{Riem}(W, X, Y, Z) + \text{Riem}(X, Y, W, Z) + \text{Riem}(Y, W, X, Z) = 0 \end{cases}$$

Adding up all four equations, while using the identity at items 1. and 2. we end up with

$$2\text{Riem}(Z, X, Y, W) + 2\text{Riem}(W, Y, Z, X) = 0 \implies \text{Riem}(Z, X, Y, W) = \text{Riem}(Y, W, Z, X),$$

which concludes the proof. 

□

Remark 33 (Comparison with GR textbooks). Bearing in mind Remark 31, we highlight a curious fact, namely that in many GR textbooks, see e.g. [Wal84], the last term in Equation (82) seems not to appear. This is a glitch which arises when expressing the Riemann tensor locally. More precisely, fixing a local chart (U, φ) in M and a coordinate system $\{x^1, \dots, x^n\}$, it turns out that

$$R_{\mu\nu\rho}^\lambda \frac{\partial}{\partial x^\lambda} = \text{Rm}(\partial_\mu, \partial_\nu, \partial_\rho)^\lambda \frac{\partial}{\partial x^\lambda} = \nabla_{\partial_\mu} \nabla_{\partial_\nu} \partial_\rho - \nabla_{\partial_\nu} \nabla_{\partial_\mu} \partial_\rho - \nabla_{[\partial_\mu, \partial_\nu]} \partial_\rho,$$

where in the last expression, we have omitted for simplicity to expand the vector field and therefore the index λ is left implicit. Observe that, on account of Definition 36, $[\partial_\mu, \partial_\nu] = 0$. For this reason, in several GR textbooks, c.f. [Wal84, Eq. (3.2.11)], it is written that, for any $T \in \Gamma(TM)$, expanding $T = T^\rho \frac{\partial}{\partial x^\rho}$, it holds

$$R_{\mu\nu\rho}^\lambda T^\rho = \nabla_\mu \nabla_\nu T^\lambda - \nabla_\nu \nabla_\mu T^\lambda. \quad (85)$$

This expression allows to establish a computationally efficient formula for the components of the Riemann tensor in terms of the Christoffel symbols and hence of the metric tensor. More precisely using repeatedly Equation (22), it descends that

$$\begin{aligned} R_{\mu\nu\rho}^\lambda T^\rho &= \partial_\mu (\nabla_\nu T^\lambda) - \Gamma_{\mu\nu}^\alpha \nabla_\alpha T^\lambda + \Gamma_{\mu\beta}^\lambda \nabla_\nu T^\beta - \partial_\nu (\nabla_\mu T^\lambda) - \Gamma_{\nu\mu}^\gamma \nabla_\gamma T^\lambda + \Gamma_{\nu\delta}^\lambda \nabla_\mu T^\delta = \\ &= \partial_\mu \partial_\nu T^\lambda + \partial_\mu (\Gamma_{\nu\chi}^\lambda T^\chi) - \Gamma_{\mu\nu}^\alpha (\partial_\alpha T^\lambda + \Gamma_{\alpha\epsilon}^\lambda T^\epsilon) + \Gamma_{\mu\beta}^\lambda (\partial_\nu T^\beta + \Gamma_{\nu\theta}^\beta T^\theta) + \\ &= -(\partial_\nu \partial_\mu T^\lambda + \partial_\nu (\Gamma_{\mu\omega}^\lambda T^\omega) - \Gamma_{\nu\mu}^\gamma (\partial_\gamma T^\lambda + \Gamma_{\gamma\varphi}^\lambda T^\varphi) + \Gamma_{\nu\delta}^\lambda (\partial_\mu T^\delta + \Gamma_{\mu\Xi}^\delta T^\Xi)) = \\ &= \partial_\mu (\Gamma_{\nu\chi}^\lambda T^\chi) - \partial_\nu (\Gamma_{\mu\omega}^\lambda T^\omega) + \Gamma_{\mu\beta}^\lambda (\partial_\nu T^\beta + \Gamma_{\nu\theta}^\beta T^\theta) - \Gamma_{\nu\delta}^\lambda (\partial_\mu T^\delta + \Gamma_{\mu\Xi}^\delta T^\Xi), \end{aligned}$$

where, in the last equality, we canceled out all terms which were identical up to a sign, renaming dummy indices where necessary and exploiting Equation (34). Applying once more the Leibnitz rule to the action of the partial derivatives, renaming dummy indices, we end up with

$$R_{\mu\nu\rho}^\lambda T^\rho = \partial_\mu (\Gamma_{\nu\chi}^\lambda T^\chi) - \partial_\nu (\Gamma_{\mu\omega}^\lambda T^\omega) + \Gamma_{\mu\beta}^\lambda \Gamma_{\nu\theta}^\beta T^\theta - \Gamma_{\nu\delta}^\lambda \Gamma_{\mu\Xi}^\delta T^\Xi,$$

from which, due to the arbitrariness of the vector field T , it descends the expression of the components of the Riemann tensor.

$$\boxed{R_{\mu\nu\rho}^\lambda = \partial_\mu \Gamma_{\nu\rho}^\lambda - \partial_\nu \Gamma_{\mu\rho}^\lambda + \Gamma_{\mu\alpha}^\lambda \Gamma_{\nu\rho}^\alpha - \Gamma_{\nu\beta}^\lambda \Gamma_{\mu\rho}^\beta, \quad R_{\mu\nu\rho\delta} = g_{\lambda\delta} R_{\mu\nu\rho}^\lambda.} \quad (86)$$

Engrave this formula in your soul. Ok, you can check it on Wikipedia if strictly necessary, and appreciate the value of learning how to type hieroglyphics in $\text{\LaTeX}$ since we were on the verge of running out of Greek letters. To conclude this Remark, we highlight that the local form of the first Bianchi identities, see Proposition 107 reads, for all $\mu, \nu, \rho, \delta = 1, \dots, n$

$$R_{\mu\nu\rho\delta} = -R_{\nu\mu\rho\delta}, \quad (87a)$$

$$R_{\mu\nu\rho\delta} = -R_{\mu\nu\delta\rho}, \quad (87b)$$

$$R_{\mu\nu\rho\delta} + R_{\nu\rho\mu\delta} + R_{\rho\mu\nu\delta} = 0, \quad (87c)$$

$$R_{\mu\nu\rho\delta} = R_{\delta\rho\mu\nu}, \quad (87d)$$

We should keep in mind that we have introduced the Riemann tensor as a tool to compare locally different geometries. In order to claim that this is exactly what we need, we need to establish its invariance under the action of local isometries.

Proposition 108. *Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50. Then both Rm and Riem as in Definition 105 are local isometric invariants, namely, for any local chart (U, φ) and for every $f \in \text{Diff}(U)$ such that $f^*g|_U = g|_U$, then*

$$f^* \text{Rm}|_U = \text{Rm}|_U, \quad f^* \text{Riem}|_U = \text{Riem}|_U. \quad (88)$$

Proof. In view of Equation (83), it suffices to prove the statement either for Rm or for Riem . We shall focus on the former and, dropping the subscript U for simplicity of the notation, Equation (88) is equivalent to showing that for all $X, Y, Z \in \Gamma(TM|_U)$

$$(f^* \text{Rm})(X, Y, Z) \doteq \text{Rm}(f_*X, f_*Y, f_*Z) = f_*(\text{Rm}(X, Y, Z)).$$

First of all we observe that, we can define a linear connection ∇' as per Proposition 54 by setting for all $X, Y \in \Gamma(TM|_U)$

$$\nabla'_X(Y) \doteq f_*^{-1} \nabla_{f_*X}(f_*Y),$$

where f_* denotes the push-forward of f , see Definition 28. Since f is a diffeomorphism on U , f_*^{-1} is also well-defined. The map ∇' is linear since it is a composition of linear maps. On account of Equation (18), for any $h \in C^\infty(U)$, using that ∇ is a linear connection per construction, it holds that

$$\begin{aligned} \nabla'_{hX}(Y) &= f_*^{-1} \nabla_{f_*(hX)} f_*Y = f_*^{-1} \nabla_{h(f(x))f_*X} f_*Y = \\ &= f_*^{-1} h(f(x)) \nabla_{f_*X}(f_*Y) = h(x) f_*^{-1} \nabla_{f_*X}(f_*Y) = h \nabla'_X(Y). \end{aligned}$$

For the same reasons

$$\begin{aligned} \nabla'_X(hY) &= f_*^{-1} \nabla_{f_*X} f_*hY = f_*^{-1} \nabla_{f_*X}(h(f(x))f_*Y) = \\ &= f_*^{-1} \left(h(f(x)) \nabla_{f_*X}(f_*Y) + \underbrace{f_*(X)(h(f(x)))}_{f_*(X(h))} f_*Y \right) = h \nabla'_X(Y) + X(h)Y. \end{aligned}$$

In addition ∇' is metric compatible. To this end, we must verify that it abides by Equation (30). Hence, for all $X, Y, Z \in \Gamma(M|_U)$, it holds that

$$\begin{aligned} g(\nabla'_X Y, Z) &= g(f_*^{-1} \nabla_{f_*X}(f_*Y), Z) = g(f_*^{-1} \nabla_{f_*X}(f_*Y), f_*^{-1} f_*(Z)) = \\ &= ((f^{-1})^* g)(\nabla_{f_*X} f_*(Y), f_*Z) = g(\nabla_{f_*X} f_*(Y), f_*Z), \end{aligned}$$

where we used that, if f is an isometry so it is f^{-1} . Repeating the same calculation for $g(Y, \nabla'_X Z)$, we end up with

$$\begin{aligned} g(\nabla'_X Y, Z) + g(Y, \nabla'_X Z) &= g(\nabla_{f_*X} f_*(Y), f_*Z) + g(f_*(Y), \nabla_{f_*X}(f_*Z)) = \nabla_{f_*X}(g(f_*Y, f_*Z)) = \\ &= f_*(X)(g(f_*(Y), f_*Z)) = f_*(X)(f^*g(Y, Z)) = X(g(Y, Z)) = \nabla'_X(g(Y, Z)), \end{aligned}$$

where the last identity descends from the fact that, on scalars, linear connections act uniquely, see Proposition 56. At last we need to check that also the torsion tensor vanishes, namely we consider for all $X, Y \in \Gamma(TM|_U)$

$$\tau'(X, Y) = \nabla'_X Y - \nabla'_Y X - [X, Y] = f_*^{-1} \nabla_{f_*(X)} f_*(Y) - f_*^{-1} \nabla_{f_*(Y)} f_*(X) - [X, Y].$$

Since $[X, Y] = (f_*^{-1} \circ f_*)[X, Y] = f_*^{-1}[f_*X, f_*Y]$, it holds that

$$\tau'(X, Y) = f_*^{-1} (\nabla_{f_*(X)} f_*(Y) - \nabla_{f_*(Y)} f_*(X) - [f_*X, f_*Y]) = f_*^{-1} \tau(f_*X, f_*Y) = 0,$$

where τ is the torsion tensor built out of ∇ . In the first identity we used Exercise (41). In other words ∇' is a torsion free, metric compatible linear connection and by the uniqueness property proven in Theorem 63, it descends that $\nabla' = \nabla$. Having established this property, for all $X, Y, Z \in \Gamma(TM)$, using Equation (82), it holds that

$$\begin{aligned} f^* \text{Rm}(X, Y, Z) &= \text{Rm}(f_*X, f_*Y, f_*Z) = \nabla_{f_*X} \nabla_{f_*Y} (f_*Z) - \nabla_{f_*Y} \nabla_{f_*X} (f_*Z) - \nabla_{[f_*X, f_*Y]} f_*Z = \\ &= \nabla_{f_*X} f_* \circ \underbrace{f_*^{-1} \nabla_{f_*Y} (f_*Z)}_{\nabla'_Y(Z)} - \nabla_{f_*Y} f_* \circ \underbrace{f_*^{-1} \nabla_{f_*X} (f_*Z)}_{\nabla'_X(Z)} - f_* \circ \underbrace{f_*^{-1} \nabla_{[f_*X, f_*Y]} f_*Z}_{\nabla'_{[X, Y]}(Z)} = \\ &= \nabla_{f_*X} f_* (\nabla'_Y(Z)) - \nabla_{f_*Y} f_* (\nabla'_X(Z)) - f_* (\nabla'_{[X, Y]}(Z)) = \\ &= f_* \left(\nabla'_X \nabla'_Y(Z) - \nabla'_Y \nabla'_X(Z) - \nabla'_{[X, Y]}(Z) \right) = \\ &= f_* (\nabla_X \nabla_Y(Z) - \nabla_Y \nabla_X(Z) - \nabla_{[X, Y]}(Z)) = f_* (\text{Rm}(X, Y, Z)), \end{aligned}$$

where we used in the before last equality that $\nabla = \nabla'$. □

Example 109. In the following chapters we shall plenty of Riemann tensor and hence we reckon that it is sufficient to discuss only one example of Riemann tensor. Observe that, if $\dim M = n$, then the number of independent components of Riem that we need to compute is equal to $\frac{n^2(n^2-1)}{12}$. These are tamed by the first Bianchi identities, see Proposition 107, which entail the existence of several symmetries. Besides observing that, since the times of when you computed the inertia tensor of a 1D homogeneous rod, a factor $\frac{1}{12}$ never appeared again (well, yes, for once, it is a coincidence), it turns out that if $n = 2$, we have just one single independent component. Denoting the coordinates by x^1, x^2 , the only unknown corresponds to R_{1212} .

Focusing on the case of S^2 , we can use Equation (86) to evaluate it using the Christoffel symbols computed in Example 85. It turns out that, identifying $x^1 = \theta$ and $x^2 = \varphi$

$$R_{\theta\varphi\theta\varphi} = g_{\varphi\varphi} R_{\theta\varphi\theta}^\varphi = g_{\varphi\varphi} [\partial_\theta \Gamma_{\varphi\theta}^\varphi - \partial_\varphi \Gamma_{\theta\theta}^\varphi + \Gamma_{\theta\alpha}^\varphi \Gamma_{\varphi\theta}^\alpha - \Gamma_{\varphi\beta}^\varphi \Gamma_{\theta\varphi}^\beta] = -\sin^2 \theta.$$

As a by-product of this straightforward computation, we can realize that, as expected, all manifolds are not locally isometric since, if we would have considered $(\mathbb{R}^2, \delta)$, working in Cartesian coordinates, all Christoffel symbols are identically vanishing and therefore $\text{Riem} = 0$.

The Ricci Tensor and the Scalar Curvature: Having established the relevant properties of the Riemann tensor we can proceed in unveiling a second set of identities which they abide by. These have far reaching consequences especially in general relativity as we shall see by the end of this section.

Proposition 110. *Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50. Then for all $X, Y, V, Z, W \in \Gamma(TM)$, the **second (or differential) Bianchi identity** holds true:*

$$\nabla_W \text{Riem}(X, Y, Z, V) + \nabla_Z \text{Riem}(X, Y, V, W) + \nabla_V \text{Riem}(X, Y, W, Z) = 0 \quad (89)$$

Proof. First of all, we observe that, using the first Bianchi identities, see Proposition 107, Equation (89) is equivalent to

$$\nabla_W \text{Riem}(Z, V, X, Y) + \nabla_Z \text{Riem}(V, W, X, Y) + \nabla_V \text{Riem}(W, Z, X, Y) = 0.$$

Since using generic vector fields entails a long, tedious and painful series of computations, we use shrewdness. Exploiting that the tensor ∇Riem is multilinear, we prove the statement only locally and for a basis of the tangent space, namely $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ where we employ normal coordinates $\{x^1, \dots, x^n\}$ centered at $p \in M$, see Equation (81). It turns out that the first term of Equation (89), evaluated at $p \in M$ yields

$$\nabla_W \text{Riem}(X, Y, Z, V) = \nabla_W g(\text{Rm}(X, Y, Z), V) = g(\nabla_W \nabla_X \nabla_Y Z - \nabla_W \nabla_Y \nabla_X Z, V).$$

Repeating this computation for all three terms in Equation (89), we obtain

$$\begin{aligned} & \nabla_W \text{Riem}(Z, V, X, Y) + \nabla_Z \text{Riem}(V, W, X, Y) + \nabla_V \text{Riem}(W, Z, X, Y) = \\ & g(\nabla_W \nabla_Z \nabla_V X - \nabla_W \nabla_V \nabla_Z X + \nabla_Z \nabla_V \nabla_W X - \nabla_Z \nabla_W \nabla_V X + \nabla_V \nabla_W \nabla_Z X - \nabla_V \nabla_Z \nabla_W X, Y). \end{aligned}$$

Rearranging the terms this becomes

$$g(\text{Rm}(W, Z, \nabla_V X) + \text{Rm}(Z, V, \nabla_W X) + \text{Rm}(V, W, \nabla_Z X), Y) = 0,$$

since all the terms $\nabla_V X = \nabla_W X = \nabla_Z X = 0$ due to the Christoffel symbols vanishing at p and due to the fact the all vector fields that we consider are constant. $\square$

It is convenient to rewrite Equation (89) in coordinates obtaining

$$R_{\mu\nu\rho\delta;\lambda} + R_{\mu\nu\delta\lambda;\rho} + R_{\mu\nu\lambda\rho;\delta} = 0 \iff R_{\mu\nu(\delta\lambda;\rho)} = 0, \quad (90)$$

where we introduced the notation $;$ to denote the covariant derivative, namely

$$R_{\mu\nu\rho\delta;\lambda} = \nabla_\lambda R_{\mu\nu\rho\delta}.$$

We can at last construct additional geometric tensors starting from the Riemann tensor. These are obtained using the metric trace introduced in Equation (29).

Definition 111. Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50 and let Riem be the Riemann curvature tensor as per Equation (83). We call **Ricci tensor** the tensor field $\text{Ric} \in \mathcal{T}^{(0,2)}(M)$ such that

$$\text{Ric} \doteq \text{Tr}_{g,14}(\text{Riem}). \quad (91)$$

Similarly we call **Ricci scalar** the function $R \in C^\infty(M)$ such that

$$R \doteq \text{Tr}_{g,12}(\text{Ric}), \quad (92)$$

where, in both instances, $\text{Tr}_{g,14}$ and $\text{Tr}_{g,12}$ are defined according to Equation (29).

The Ricci tensor and the Ricci scalar are among the building blocks of general relativity and it is very convenient to express them locally. Combining Equation (91) with Equation (84) we obtain

$$R_{\mu\nu} = g^{\rho\lambda} R_{\rho\mu\nu\lambda} = g^{\rho\lambda} R_{\rho\mu\nu}^\alpha g_{\alpha\lambda} = \delta_\alpha^\rho R_{\rho\mu\nu}^\alpha = R_{\rho\mu\nu}^\rho, \quad (93)$$

which combined with Equation (86) yields

$$R_{\mu\nu} = R_{(\mu\nu)} = \partial_\rho \Gamma_{\mu\nu}^\rho - \partial_\mu \Gamma_{\rho\nu}^\rho + \Gamma_{\rho\alpha}^\rho \Gamma_{\mu\nu}^\alpha - \Gamma_{\mu\beta}^\rho \Gamma_{\rho\nu}^\beta. \quad (94)$$

At the same time the Ricci scalar admits a rather straightforward local representation, namely

$$R = g^{\mu\nu} R_{\mu\nu}. \quad (95)$$

Remark 34 (Notation and Convention). There are several possible ambiguities in defining the Ricci tensor, namely in Equation (91), we have chosen to contract the indices (1, 4). We could have equivalently chosen the pair (2, 3) and the first Bianchi identities, see Proposition 107, entail that the result is left unchanged. On the contrary, contracting the pairs (1, 2) or (3, 4) would have given zero due to the antisymmetry property of the Riemann tensor. If one chooses instead the indices (1, 3) or (2, 4) the overall result would change by a global minus sign! It is important to keep track of these conventions. **Observe** that, in many general relativity textbooks such as [Wal84], the Riemann tensor is defined with a global minus sign compared to the one used in these notes. In order to reabsorb this mismatch with the physics literature, we have used a different convention in comparison to [Lee11] for defining the Ricci tensor.

Example 112. We complete our survey of the unit 2-sphere by computing their Ricci and scalar curvature. This is a direct application of Equations (94) and (95) using the results of Example 109. In particular it descends that

$$R_{\theta\theta} = g^{\varphi\varphi} R_{\varphi\theta\theta\varphi} = 1, \quad R_{\varphi\varphi} = g^{\varphi\varphi} R_{\varphi\theta\theta\varphi} = \sin^2 \theta, \quad (96)$$

while

$$R = g^{\mu\nu} R_{\mu\nu} = g^{\theta\theta} R_{\theta\theta} + g^{\varphi\varphi} R_{\varphi\varphi} = 2, \quad (97)$$

namely the sphere has constant (scalar) curvature. This can be seen as a particular instance of a very important result in differential geometry, known as *Gauss-Bonnet* theorem (Bonnet, not bonèt like the Piedmontese custard dessert), see [Lee11, Chap. 9].

A notable result can be obtained combining the definition of Ricci tensor and of scalar curvature with the differential Bianchi identities. Most notably, if we consider Equation (90) and we contract it with $g^{\mu\delta}$, on account of metric compatibility, we obtain

$$0 = g^{\mu\delta} (R_{\mu\nu\rho\delta;\lambda} + R_{\mu\nu\delta\lambda;\rho} + R_{\mu\nu\lambda\rho;\delta}) = R_{\nu\rho;\lambda} - R_{\nu\lambda;\rho} + g^{\mu\delta} R_{\mu\nu\lambda\rho;\delta}.$$

Contracting once more with $g^{\nu\rho}$ we obtain

$$R_{;\lambda} - g^{\nu\rho} R_{\nu\lambda;\rho} - g^{\mu\delta} R_{\mu\lambda;\delta} = 0.$$

Renaming the indices we obtain

$$\nabla^\mu G_{\mu\nu} = 0, \quad G_{\mu\nu} = R_{\mu\nu} - \frac{R}{2} g_{\mu\nu}. \quad (98)$$

Definition 113. Let (M, g) be an n -dimensional smooth manifold, endowed with a metric tensor as per Definition 50. We call **Einstein tensor** the divergence free tensor field $G \in \mathcal{T}^{(0,2)}(M)$ such that

$$G = \text{Ric} - \frac{R}{2} g, \quad (99)$$

where Ric and R are respectively the Ricci tensor as in Equation (91) and the scalar curvature as in Equation (92). In components it reads

$$G_{\mu\nu} = R_{\mu\nu} - \frac{R}{2} g_{\mu\nu}. \quad (100)$$

3.3 Jacobi Fields

Before delving into the deep realm of General Relativity we can address one last question concerning the motion of point particles on a given manifold (M, g) where g is a metric tensor of arbitrary signature as per Definition 50. More precisely, one of the postulate of Euclidean geometry is that parallel straight line remain such indefinitely and never intersect. Heuristically, one might expect this statement not to hold true on (M, g) , but making it precise can be quite hard since even the notion of parallel itself becomes somehow blurry.

Let us therefore consider an *admissible family* of smooth curves that is

$$\Upsilon : (-\epsilon, \epsilon) \times [a, b] \rightarrow M, \quad (s, t) \mapsto \Upsilon(s, t), \quad (101)$$

such that $\epsilon > 0$. In addition we require that, for each but fixed value of s , $\Upsilon_s(t) = \Upsilon(s, t)$ is a geodesic. If we introduce $T(s, t) \doteq \partial_t \Upsilon(s, t)$ and $S(s, t) \doteq \partial_s \Upsilon(s, t)$, in view of equation (58), it holds that

$$D_t T(s, t) = 0 \implies D_s D_t T(s, t) = 0,$$

where D_t and D_s are as per Equation (53). We can infer the following technical result.

Lemma 114. *Let Υ be any admissible family of smooth curves as per Equation (101) and let $V \in C^\infty((-\epsilon, \epsilon) \times [a, b]; TM)$ be such that, for all $(s, t) \in (-\epsilon, \epsilon) \times [a, b]$, $V(s, t) \in T_{\Upsilon(s, t)}M$. Given D_t as per Equation (53), it holds that*

$$D_s D_t V - D_t D_s V = \text{Rm}(S, T, V),$$

where $S = \frac{\partial}{\partial s}$, while $T = \frac{\partial}{\partial t}$.

Proof. It suffices to prove the statement locally. Hence, considering a chart (U, φ) as well as a coordinate system $\{x^1, \dots, x^n\}$ thereon, it descends that $V = V^i \frac{\partial}{\partial x^i}$. Using Equation (54) it holds that

$$D_t V = D_t(V^i \partial_i) = \dot{V}^i \partial_i + V^i D_t(\partial_i),$$

which entails, in turn, that

$$D_s D_t V = \frac{\partial^2 V^i}{\partial s \partial t} \partial_i + \dot{V}^i D_s(\partial_i) + \frac{\partial V^i}{\partial s} D_t(\partial_i) + V^i D_s D_t(\partial_i).$$

The left hand side of Equation (114) can be written as

$$D_s D_t V - D_t D_s V = V^i (D_s D_t - D_t D_s)(\partial_i).$$

Writing $S = \frac{\partial x^k}{\partial s} \partial_k$ and similarly $T = \frac{\partial x^j}{\partial t} \partial_j$, Equation (55) yields

$$D_t(\partial_i) = \nabla_T(\partial_i) = \frac{\partial x^j}{\partial t} \nabla_{\partial_j}(\partial_i),$$

where we used Equation (18) and where, with a slight abuse of notation, we have omitted to write the restrictions to the underlying curves. Consequently Equation (54) entails that

$$\begin{aligned} D_s D_t(\partial_i) &= D_s \left(\frac{\partial x^j}{\partial t} \nabla_{\partial_j}(\partial_i) \right) = \frac{\partial^2 x^j}{\partial s \partial t} \nabla_{\partial_j}(\partial_i) + \left(\frac{\partial x^j}{\partial t} D_s(\nabla_{\partial_j}(\partial_i)) \right) = \\ &= \frac{\partial^2 x^j}{\partial s \partial t} \nabla_{\partial_j}(\partial_i) + \frac{\partial x^j}{\partial t} \nabla_S(\nabla_{\partial_j}(\partial_i)) = \\ &= \frac{\partial^2 x^j}{\partial s \partial t} \nabla_{\partial_j}(\partial_i) + \frac{\partial x^j}{\partial t} \frac{\partial x^k}{\partial s} \nabla_{\partial_k}(\nabla_{\partial_j}(\partial_i)). \end{aligned}$$

This identity, in combination with Proposition 106 entails that

$$V^i D_s D_t(\partial_i) = V^i \frac{\partial x^j}{\partial t} \frac{\partial x^k}{\partial s} (\nabla_{\partial_k}(\nabla_{\partial_j}(\partial_i)) - \nabla_{\partial_j}(\nabla_{\partial_k}(\partial_i))) = \text{Rm}(S, T, V),$$

which concludes the proof. $\square$

If we now apply this result to the case where $\Upsilon_s(t)$ is a geodesic for all $s \in (-\epsilon, \epsilon)$ and where $V = \nabla_T S$, we obtain a remarkable specialization of Lemma 114.

Proposition 115. *Under the assumptions of Lemma 114, whenever $\Upsilon_s(t)$ is a geodesic for all $s \in (-\epsilon, \epsilon)$ and $V = \nabla_T S$, the **Jacobi equation (geodesic deviation equation)** holds:*

$$D_t^2 V + \text{Rm}(V, \dot{\gamma}, \dot{\gamma}) = 0,$$

where $\dot{\gamma}$ is the velocity vector of $\Upsilon_s(t)$. Under these circumstances V is called a **Jacobi field**.

Proof. We apply Lemma 114 setting $V = T$, obtaining

$$D_s D_t T - D_t D_s T = \text{Rm}(S, T, T) \implies D_s D_t T = D_t D_s T + \text{Rm}(S, T, T).$$

Since a direct computation yields that $D_s T = D_t S$ and, since, being $\Upsilon_s(t)$ a geodesic, $D_t T = 0$, we obtain

$$D_t^2 S + \text{Rm}(S, T, T) = 0,$$

which entails the sought result once we evaluate it at $s = 0$ and we observe that $S(0, t) = V$, while $T(0, t) = \dot{\gamma}(t)$. $\square$

It is worth translating locally Equation (115) to obtain

$$\boxed{A^\mu \doteq D_t^2 S^\mu = -R_{\nu\rho\sigma}^\mu V^\nu T^\rho T^\sigma}, \quad (102)$$

whose interpretation is to characterize the *relative acceleration* between two neighboring geodesics. Observe that, as expected, it vanishes whenever the Riemann tensor is identically 0 as in Euclidean space.

3.4 Causal Structures

Until this point, it was not necessary to disambiguate between manifold (M, g) endowed with a smooth metric on the basis of their signature, see Definition 52. Yet, one cannot forget that relativity, be it special or general, is closely tied to the Lorentzian signature. This entails the existence of special features which need to be discussed. A reader familiar with special relativity will recognize them as being a natural generalization of what occurs on Minkowski spacetime. Hence, for definiteness, in this section (M, g) is an n -dimensional spacetime with $\dim M \geq 2$ and signature $(-, \underbrace{+, \dots, +}_n)$. This entails the possibility of separating the elements of $T_p M$ for any $p \in M$ in three distinguished classes.

Definition 116. *Let (M, g) be an n -dimensional spacetime with $\dim M \geq 2$ and signature $(-, \underbrace{+, \dots, +}_n)$, see Definition 52. For any $p \in M$, we say that a vector $v \in T_p M$ is*

- **timelike** if $g|_p(v, v) < 0$,
- **lightlike** if $g|_p(v, v) = 0$,
- **spacelike** if $g|_p(v, v) > 0$.

Vectors $v \in T_p M$ such that $g|_p(v, v) \leq 0$ are called **causal** and their collection is denoted by V_p .

Observe that, at each point $p \in M$, V_p can be realized as the union of two (light) cones which share the origin as their tip. One can arbitrarily denote them as $V_p^\pm$ to indicate that there is a freedom to designate which one is the *future* or the *past* light cone. While Definition 116 suffices in the analysis of models on Minkowski spacetime $(\mathbb{R}^n, \eta)$, we need to boost it up on a generic Lorentzian manifold.

Definition 117. Let (M, g) be an n -dimensional spacetime with $\dim M \geq 2$ and signature $(-, \underbrace{+, \dots, +}_{n-1})$, see Definition 52. Let $\gamma \in C^\infty(I; M)$ with $I \subseteq \mathbb{R}$. We say that this curve is

timelike (resp. **lightlike**, **spacelike**) if its tangent vector at any point is timelike (resp. lightlike, spacelike). Similarly, a vector field $\chi \in \Gamma(TM)$ is timelike (resp. lightlike, spacelike) if, for all $p \in M$, $\chi(p)$ is timelike (resp. lightlike, spacelike).

Definition 118. Let (M, g) be an n -dimensional spacetime with $\dim M \geq 2$ and signature $(-, \underbrace{+, \dots, +}_{n-1})$, see Definition 52. Let $\iota : \Sigma \rightarrow M$ identify an embedded submanifold as per

Definition 40. We say that Σ is **timelike** (resp. **lightlike**, **spacelike**) if, for every $p \in \Sigma$ and for every $v \in T_p \Sigma$ $\iota_*(v) = d\iota(v)$ is a timelike (resp. lightlike, spacelike) vector.

These notions are rather natural and they allow to characterize what is often known as the *causal structure* of a spacetime (M, g) . As we mentioned in Section C.2, in order for a smooth manifold to admit a well-defined integration theory for differential forms, it is necessary that it is orientable, see Definition 213. Henceforth we shall assume that an orientation $\mathcal{O}$ has been fixed. Similarly, on a Lorentzian spacetime (M, g) , we want to be able to assign a coherent notion of what does it mean to lie in the future or in the past of an arbitrary point $p \in M$. Definition 117 entails that this notion can be established arbitrarily at each point $p \in m$, but we want to determine a criterion which guarantees us that such assignment is coherent with a change of p itself. The following definition solves this quandary

Definition 119. Let (M, g) be an n -dimensional spacetime with $\dim M \geq 2$ and signature $(-, \underbrace{+, \dots, +}_{n-1})$, see Definition 52. We call it **time-orientable** if $\exists \chi \in \Gamma(TM)$ which is timelike and nowhere vanishing.

The existence of a timelike vector field X as per Definition 119 ensures the sought notion of fixing a time orientation, since, given any causal vector $v \in T_p M$ as per Definition 116 we can say that it is *future directed* at $p \in M$ if $g|_p(X(p), v) < 0$, while, otherwise, it is *past directed*. Similarly we can say that a curve $\gamma \in C^\infty(I; M)$, $I \subseteq \mathbb{R}$, is future or past directed if such is its tangent vector at any point $t \in I$.

Exercise 120. Let $(\mathbb{R}^n, \eta)$ be the n -dimensional Minkowski spacetime endowed with the standard Cartesian coordinates $(t, x_1, \dots, x_{n-1})$. Show that the vector field $X \in \Gamma(T\mathbb{R}^n)$ such that $X(p) = \frac{\partial}{\partial t}$ at each $p \in \mathbb{R}^n$ induces the canonical time orientation of Minkowski spacetime.

Exercise 121. Let (M, g) be an n -dimensional spacetime with $\dim M \geq 2$. Prove that, if it is time orientable as per Definition 119, there exists only two possible choices of time orientations.

Remark 35. *In the following we will update our notion of spacetime in comparison to Definition 52 by requiring always implicitly that the underlying manifold is both orientable and time-orientable. In addition, we also assume that one among the possible choices has been made. In other words denoting by $\mathcal{O}$ the spacetime orientation and by $\mathfrak{t}$ the time one, to be precise we should always indicate the choice of a spacetime with the quadruple*

$$(M, g, \mathcal{O}, \mathfrak{t}).$$

For simplicity of the notation, we shall always leave $\mathcal{O}$ and $\mathfrak{t}$ understood keeping only (M, g) .

Under the above assumptions it is possible to extend the notion of causality which is tied to tangent vectors as per Definition 116. The reason lies in our desire to extend as much as possible to a generic spacetime the very convenient notion in Minkowski which allows us to establish a causal relation between two arbitrary points therein

Definition 122. *Let (M, g) be an n -dimensional spacetime with $\dim M \geq 2$ and signature $(-, \underbrace{+, \dots, +}_{n-1})$, see Definition 52. We call*

- **chronological future (+) or past (-) of a point $p \in M$**

$$I^\pm(p) \doteq \{q \in M \mid \exists \text{ a future (+) or past (-) directed timelike curve } \gamma \text{ connecting } p \text{ to } q\},$$

- **causal future (+) or past (-) of a point $p \in M$**

$$J^\pm(p) \doteq \{q \in M \mid \exists \text{ a future (+) or past (-) directed causal curve } \gamma \text{ connecting } p \text{ to } q\},$$

where causality is as per Definition 117. Similarly, given $\Omega \subseteq M$, we call

$$I^\pm(\Omega) \doteq \bigcup_{p \in \Omega} I^\pm(p) \quad \text{and} \quad J^\pm(\Omega) \doteq \bigcup_{p \in \Omega} J^\pm(p). \quad (103)$$

Remark 36. *It is worth observing that, given two spacetimes (M, g) and (M, g') , if they are related by a conformal transformation, see Definition 50, their causal structure is the same. More precisely if $v \in T_p M$ is timelike/lightlike/spacelike with respect to g , so is also with respect to g' . consequently also the notions of causal or chronological future and past do not depend on a conformal rescaling of the metric. In other words conformal structures are in 1:1 correspondences with equivalence classes $[(M, g)]$ where $(M, g) \sim (M, g')$ if and only if g is conformally related to g' as per Definition 50.*

Establishing a causal structure is just the first step in studying the causal properties of a manifold. In these notes we shall refrain from delving more into this topic, unless the author decides that he has enough time to write more sections. Yet, we urge the reader to observe that, given a spacetime (M, g) , nothing prevents the existence of pathological scenarios, such as closed timelike or causal curve. A reader might be prone to think that, once we will have introduced Einstein's equations, these will prohibit the occurrence of these pathologies. Yet this is not the case as the somehow famous Gödel Universes have shown. For this reason it is often customary to impose stronger requirements on physically admissible spacetimes, the most important being that of **global hyperbolicity**. A reader interested in this topic should refer to [Wal84, Chap. 8] or to the monograph [BEE96], which is very complete though lacking some very important result proven in the past 25 years.

4 Einstein's Equations

It is time to recap what we have understood up to this point. As we have argued in Section 3, one can consider the laws of Newtonian mechanics and, by generalizing the idea that, in absence of external forces, a point particle must follow a straight line, we have come to accept the option that the underlying configuration space can be generalized from $(\mathbb{R}^3, \delta)$ to a Riemannian manifold (M, g) . If we add to this picture special relativity, we should hop from Minkowski spacetime $(\mathbb{R}^4, \eta)$ to a Lorentzian manifold (M, g) . As a consequence of Remark 16, we have understood that, depending on the case under scrutiny, δ and η can be recovered at any point $p \in M$ choosing a suitable coordinate system. The curvature tensor Riem is a measure of our deviation from the idealized regime of a *flat spacetime*, that is a manifold where the Riemann tensor vanishes identically at every point. From a mere mathematical viewpoint, one might argue that quantifying such departure from the flat regime depends on the choice of a notion of a (semi-)norm, which is true. Here, let us reason only from a physical viewpoint and let us postpone these technical aspects to a later stage. In this novel, extended picture, the concept of *straight line* is replaced by that of *geodesic*, see Definition 88. This is a trajectory of vanishing acceleration and observe that this concept has been defined independently of the internal structure of the moving test object.

As always, every generalization necessitates paying a due price: from a unique choice of the configuration space, say $(\mathbb{R}^4, \eta)$, we have now infinitely many possibilities, de facto all Lorentzian spacetimes (M, g) . Additional considerations might restrict our options, but never to a single one. This leeway might be seen as a hurdle, but actually it constitutes an opportunity, since we have still a few unsolved issues, hidden under the rug. Let us briefly review them:

1. The first law of Newtonian mechanics establishes the classical principle of *inertia*, which states that all inertia observers are equivalent and none is preferred. Yet, this entails that, once we have elected one reference system to be inertial, all those related to it by a transformation subordinated either to the Galilei group (classical mechanics) or to the Poincaré group (special relativity) is equivalent to it. Yet, if we consider a generic Riemannian or Lorentzian spacetime, we have a problem since it is easy to see that Poincaré transforma-

tions are the global isometries of $(\mathbb{R}^4, \eta)$, see Definition 67. On a generic spacetime (M, g) , see Definition 52, there is no guarantee even that isometries exist. Hence, if we translate the choice of a reference system as fixing $p \in M$ and an orthonormal basis of $T_p M$ as per Remark 16 and even if we declare arbitrarily one of these as being inertial, how do we construct all others? The only set of transformations available regardless of the choice of (M, g) is encoded in the diffeomorphism group, $\text{Diff}(M)$, see Definition 40. Yet, if we accept to work with $\text{Diff}(M)$, it is manifest that, in the special case of $(M, g) \equiv (\mathbb{R}^4, \eta)$, we would be able to construct way more reference systems than if we were using the Poincaré group and many of these would not be considered inertial under the standard notion.

2. In classical Newtonian mechanics the behaviour of gravity is ruled by the **weak equivalence principles** which was stated originally as

Mass (measured with a balance) and weight (measured with a scale) are locally in identical ratio for all bodies.

From a more modern perspective this entails that freely falling test point particles follow the same trajectories, once an initial position and velocity have been prescribed. Hence their acceleration is the same and it leads to the equivalence between the gravitational and the inertial masses. Yet, already in the realm of special relativity this formulation needs to be revised since it entails the exertion of instantaneous forces and it is manifestly incompatible with the existence of a limit speed, that of light.

Taken separately, these two problems might lead to the impression that they can be solved independently, especially if one seeks a modification of Newtonian gravity which is compatible with the dictates of special relativity. Yet one can observe that the weak equivalence principle is stating that, contrary to other forces, *e.g.*, the electromagnetic, gravity acts independently from the nature of the matter constituents and its effect is to alter the trajectory of test particles. Yet, this is exactly what happens if one changes the metric of a smooth manifold. If we start from (M, g) and we consider subsequently (M, g') , it turns out that, from the view point of a test particle, the net effect is that of changing its trajectory in M , since, in the first case, it shall follow the geodesics subordinated to g and, in the second one, those of g' . This is the core of Einstein's idea concerning the existence of strict interplay between geometry and gravity. Yet, in view of the remarks above this also entails that, in order to accept this connection, we have to give up all together on the possibility of establishing the existence of a class of inertial, hence privileged, reference systems. All of them are equally admissible. This novel viewpoint can be codified in the *Einstein's equivalence principle* which is often heuristically summarized as stating that, in arbitrary small regions of spacetime it is impossible to detect the existence of a gravitational field by means of local experiments. A more complete statement is given in [HL01], to which we refer also for the discussion of the experimental validations:

Einstein Equivalence Principle

The outcome of any local, non-gravitational test experiment is independent of the experimental apparatus' velocity relative to the gravitational field and is independent of where and when in the gravitational field the experiment is performed.

Having established the core principle that gravity is a manifestation of spacetime geometry, we can infer the basic kinematic building blocks of general relativity are 1) a *four-dimensional Lorentzian spacetime* (M, g) as per Definition of 52 and 2) the diffeomorphism group $\text{Diff}(M)$ as per Definition 40. Our next step consists of translating the above ideas in a dynamical theory.

The Stress-Energy Tensor: Having accepted the idea that geometry is a manifestation of gravity, we need to establish how this occurs quantitatively. Let us start by observing that, in classical mechanics, the gravitational potential Φ abides by the Poisson equation

$$\Delta\Phi = \rho,$$

where Δ is the D'Alembert operator on $(\mathbb{R}^n, \delta)$, while ρ is the underlying mass density. The first step consists of observing that considering ρ alone is already not compatible with the dictates of special relativity. If we consider $(\mathbb{R}^4, \eta)$ and ρ is a constant function, we can introduce the 4-vector $j^\mu = (\rho, 0, 0, 0)$ in the reference system comoving with the mass density and use the Lorentz transformations to construct the ensuing current in any other reference system. The outcome is the celebrated energy-momentum density ρu^μ where u^μ is to be interpreted as the tangent vector to the (worldline) curve representing the motion of the underlying body in spacetime and such that $\eta_{\mu\nu} u^\mu u^\nu = -1$. Following classical, continuum mechanics [MH94], it turns out that, since $\partial_\mu j^\mu = 0$, there must exist a symmetric tensor field $T \in \mathcal{T}^{(0,2)}(\mathbb{R}^4)$ such that, working in Cartesian coordinates, $j^\mu = \eta^{\rho\mu} T_{\rho\nu} u^\nu$. This is called **stress-energy tensor** for the underlying matter distribution and considering that the indices $\mu, \nu, \rho = 0, 1, 2, 3$, the component T_{00} codifies the energy density. On top of that $T_{\mu\nu}$ is divergence free, namely

$$\partial^\mu T_{\mu\nu} = 0.$$

If, given an arbitrary n -dimensional spacetime (M, g) and a suitable matter distribution with an associated divergence free stress-energy tensor $T \in \mathcal{T}^{(0,2)}(M)$, it turns out that we have found the last building block for our theory. On the matter side and working locally $T_{\mu\nu}$ encodes in a manifestly covariant form the energy content. On the gravity side, we have already individuated a candidate for encoding dynamically the content of the metric tensor, namely $G_{\mu\nu}$, see Equation (99). Since the differential Bianchi identity entails that G is divergence free, see (98), this suggests that their relation can be codified in the celebrated **Einstein's equations**

$$G_{\mu\nu} = \kappa T_{\mu\nu}, \tag{104}$$

where κ is a proportionality constant to be determined. Observe that, even accepting the rationale which has lead us to Equation (104), several open cogent questions are left open:

1. The stress energy-tensor is a well-established concept for physical models on Minkowski spacetime. Yet, it is unclear how to make sense of the right hand side of Equation (104) on a generic Lorentzian manifold (M, g) . We shall address this issue in Section 4.1.
2. Equation (104) expresses that the Einstein tensor and therefore the metric are modified by the matter content but there is no a priori reason to set the value of the proportionality constant κ to any specific value. We shall investigate this issue in Section 5.1.
3. If we accept the postulate that the matter content of a physical model is codified by the stress-energy tensor, then, one might wonder how general is the left hand side of Equation (104). By mere consistency we need it to be a tensor field of rank $(0, 2)$ which is divergence free and symmetric. The Einstein tensor, see Equation (99) abides by these requirements and it depends only on the second derivatives of the underlying metric tensor g . In addition we need to make sure that, in a suitable limiting procedure, we can recover Newtonian gravity, so to guarantee that this new theory is an extension on the classical one. As we shall see in Section 5.1 this entails that on the left hand side of Equation (104) G must be present, but this does not guarantee that no additional term can be present. This shall lead to renown **cosmological constant**. We shall briefly comment on it at the end of Section 5.1.

4.1 The Einstein-Hilbert Action

Henceforth, we shall only consider (M, g) to be a spacetime as per Definition 52 with $\dim M = n \geq 2$, although the physically relevant scenario is undoubtedly $n = 4$ (in these notes, we do not refer nor we bully string theory. We ignore it). If we consider the Einstein's equations (104) and we want to promote them to be the fundamental law ruling the gravitational field, we cannot disregard that, in modern theoretical physics as well as in continuum mechanics, the equations of motion ruling the dynamics of a model are always constructed from a variational principle, that is as the Euler-Lagrange equations of a suitable Lagrange density.

Following this premise, in this chapter, we assume that all matter content on the spacetime (M, g) can be described in terms of a Lagrangian density $\mathcal{L}[\Phi, g]$, where Φ is a shortcut notation to denote the whole collection of underlying fields, *e.g.*, scalar, spinor, electromagnetic and so on and so forth. We can associate to it a corresponding action by integrating over the whole manifold. Following Theorem 216, we can write

$$S[\Phi, g] \doteq \int_M \omega_g \mathcal{L}[\Phi, g] \equiv \int_M d^n x \sqrt{|g|} \mathcal{L}[\Phi, g], \quad (105)$$

where ω_g is the metric induced volume form, see Theorem 216 while the rightmost expression is a translation in local coordinates of the integral form, see Corollary 217. Observe that writing

$$\int_M d^n x \sqrt{|g|},$$

is formally incorrect, unless M is covered by a single local chart, which is rarely the case. Yet we write this expression since it appears in every single physics textbooks as well as in many research papers on quantum field theory on curved backgrounds. More elusive is instead the formulation of the left-hand side of Equation (104) in terms of an action principle. In the early days of general relativity this has been one of the recurring and main nightmares for Einstein, who managed to overcome this hurdle only thanks to the help of one of the most famous mathematicians at the time, David Hilbert. This has brought to the formulation of the **Einstein-Hilbert action**

$$S_{EH}[g] = \int_M \omega_g R \equiv \int_M d^n x \sqrt{|g|} R, \quad (106)$$

where R is the Ricci scalar as per Equation (92). Putting together Equations (105) and (106) we obtain the full action

$$S_{TOT}[\Phi, g] = \int_M d^n x \sqrt{|g|} \frac{R}{2\kappa} + \mathcal{L}[\Phi, g],$$

where we opted to use the improper notation for the volume form since our next task will be to compute the underlying stationary points. More precisely we wish to consider the variation of S_{TOT} with respect to the metric tensor and to show that this yields Equation (104). It is probable that many readers have been already exposed to variational problems, but, to be on the safe side, we recall some basic facts. We assume the reader to be familiar with the notion of Fréchet spaces, else one can refer to [Tre06].

Definition 123. *Let X, Y be two Fréchet spaces and let $U \subseteq X$. Given $F : U \rightarrow Y$, we call **directional derivative** of F at $u \in U$ along the direction $v \in X$*

$$DF(u)v \doteq \lim_{t \rightarrow 0} \frac{F(u + tv) - F(u)}{t} \in Y,$$

provided that the limit exist. We say that $F \in C^1(U; Y)$ or, equivalently, that it is continuously differentiable if the limit exists for all $(u, v) \in U \times X$ and if the function

$$DF : U \times X \rightarrow Y, \quad (u, v) \mapsto DF(u)v,$$

is continuous.

Under the assumptions of Definition 123, given $u \in U$ and $h \in X$ such that $u + h \in U$, we can write

$$F(u + h) = F(u) + DF(u)h + \mathcal{R}[u, h],$$

where $\mathcal{R}[u, h]$ is remainder which tends to 0 as $h \rightarrow 0$. In the scenario of our interest, the role of X is played by $\Gamma(TM^{\otimes 2})$, that is the space of smooth metrics, while $Y = \mathbb{R}$. We shall also employ the notation

$$h \longleftrightarrow \delta g.$$

Observe in addition that we can work in components since we can consider a subset of $\Gamma(T^*M^{\otimes_s 2})$ which corresponds to the space of all smooth metric in a local coordinate patch of M . This leads to considering

$$\delta S_{TOT}[\Phi, g] = S_{TOT}[\Phi, g + \delta g] - S_{TOT}[\Phi, g] = \int_M \delta(\sqrt{|g|} \frac{R}{2\kappa} + \sqrt{|g|} \mathcal{L}[\Phi, g]). \quad (107)$$

In addition observe that, as a byproduct of Definition 123, the variation δ abides by the Leibnitz rule. Hence it holds the following chain of identities:

$$0 = \delta(\underbrace{g_{\mu\nu} g^{\nu\rho}}_{\delta_\mu^\rho}) = \delta(g_{\mu\nu}) g^{\nu\rho} + g_{\mu\nu} \delta g^{\nu\rho} \implies \delta g_{\mu\nu} = -g_{\mu\lambda} g_{\nu\rho} \delta g^{\lambda\rho}. \quad (108)$$

Having these ingredients, we can rewrite Equation (107) to write

$$\delta S_{TOT}[\Phi, g] = \int_M d^n x \frac{1}{2\kappa} \left(\underbrace{\delta(\sqrt{|g|}) R}_1 + \underbrace{\sqrt{|g|} \delta g^{\mu\nu} R_{\mu\nu}}_2 + \underbrace{\sqrt{|g|} g^{\mu\nu} \delta R_{\mu\nu}}_3 + 2\kappa \underbrace{\delta(\sqrt{|g|} \mathcal{L}[\Phi, g])}_4 \right). \quad (109)$$

We analyze all four terms separately although the second one is already in the desired form.

Term 1. in Equation(109): In order to manipulate this expression we use the Jacobi formula which states that, given $A \equiv A(t) \in \mathcal{C}^\infty(\mathbb{R}; \mathcal{M}(n, \mathbb{R}))$, then

$$\frac{d \det(A(t))}{dt} = \det(A(t)) \text{Tr} \left(A^{-1}(t) \frac{dA(t)}{dt} \right). \quad (110)$$

Replacing A with g we end up with

$$\frac{1}{\det(|g|)} \delta(\det |g|) = g^{\mu\nu} \delta g_{\mu\nu} \implies \delta(\det |g|) = -\det(|g|) g_{\mu\nu} \delta g^{\mu\nu}.$$

This entails in turn

$$\delta(\sqrt{\det |g|}) R = -\frac{1}{2\sqrt{\det |g|}} \delta(\det |g|) R = \frac{1}{2\sqrt{\det |g|}} g_{\mu\nu} R \delta g^{\mu\nu}. \quad (111)$$

Term 3. in Equation(109): In order to evaluate $\delta R_{\mu\nu}$ we start from Equation (94) and we write

$$\begin{aligned} \delta R_{\mu\nu} &= \delta(\partial_\rho \Gamma_{\mu\nu}^\rho - \partial_\mu \Gamma_{\rho\nu}^\rho + \Gamma_{\rho\alpha}^\rho \Gamma_{\mu\nu}^\alpha - \Gamma_{\mu\beta}^\rho \Gamma_{\rho\nu}^\beta) = \\ &= \partial_\rho \delta \Gamma_{\mu\nu}^\rho - \partial_\mu \delta \Gamma_{\rho\nu}^\rho + \delta(\Gamma_{\rho\alpha}^\rho) \Gamma_{\mu\nu}^\alpha + \Gamma_{\rho\alpha}^\rho \delta(\Gamma_{\mu\nu}^\alpha) - \delta(\Gamma_{\mu\beta}^\rho) \Gamma_{\rho\nu}^\beta - \Gamma_{\mu\beta}^\rho \delta(\Gamma_{\rho\nu}^\beta), \end{aligned} \quad (112)$$

where we exploited the Leibnitz rule and we switched the action of the variation and of the partial derivative since this does not depend on the metric tensor. Therefore we have reduced our task to that of computing the variation of the Christoffel symbols. Yet, at this stage we

can rely on a barba-trick. Since $\delta\Gamma_{\mu\nu}^\rho$ is a tensor of rank $(1,2)$ we can compute its covariant derivative:

$$\nabla_\lambda \delta\Gamma_{\mu\nu}^\rho = \partial_\lambda \delta\Gamma_{\mu\nu}^\rho - \Gamma_{\lambda\mu}^\alpha \delta\Gamma_{\alpha\nu}^\rho - \Gamma_{\lambda\nu}^\beta \delta\Gamma_{\mu\beta}^\rho + \Gamma_{\epsilon\lambda}^\rho \delta\Gamma_{\mu\nu}^\epsilon.$$

This expression can be replaced in Equation (112) yielding after a long and tedious calculation

$$\delta R_{\mu\nu} = \nabla_\rho \delta\Gamma_{\mu\nu}^\rho - \nabla_\nu \delta\Gamma_{\rho\mu}^\rho. \quad (113)$$

Exercise 124. Given a spacetime (M, g) , prove that

$$\delta\Gamma_{\mu\nu}^\rho = -\frac{1}{2} \left[g_{\lambda\mu} \nabla_\nu (\delta g^{\lambda\rho}) + g_{\lambda\nu} \nabla_\mu (\delta g^{\lambda\rho}) - g_{\mu\alpha} g_{\nu\beta} g^{\rho\chi} \nabla_\chi (\delta g^{\alpha\beta}) \right]. \quad (114)$$

Replacing this result in term 3. in Equation(109), we end up with

$$\begin{aligned} \int_M d^n x \sqrt{|g|} g^{\mu\nu} (\nabla_\rho \delta\Gamma_{\mu\nu}^\rho - \nabla_\nu \delta\Gamma_{\mu\rho}^\rho) &= \int_M d^n x \sqrt{|g|} \nabla_\rho (g^{\mu\nu} \delta\Gamma_{\mu\nu}^\rho) - \nabla_\nu (g^{\mu\nu} \delta\Gamma_{\mu\rho}^\rho) = \\ &= \int_M d^n x \sqrt{|g|} \nabla_\mu (g^{\rho\nu} \delta\Gamma_{\rho\nu}^\mu - g^{\mu\nu} \delta\Gamma_{\rho\nu}^\rho) = \\ &= \int_M d^n x \sqrt{|g|} \nabla_\mu [\nabla_\alpha (\delta g^{\mu\alpha}) - g^{\mu\nu} g_{\rho\lambda} \nabla_\nu (\delta g^{\rho\lambda})] = 0, \end{aligned} \quad (115)$$

where, in the last line, we used first Exercise 124 and subsequently the divergence theorem 235

Remark 37. When we apply the divergence theorem we are implicitly assuming that we are considering only variations of the metric which are compactly supported, that is $\delta g \in \Gamma_0(TM^{\otimes_s 2})$. It is an intrinsic feature of variational principle that the outcome depends strictly on the space with respect to which variations are allowed (remember, remember functional analysis). We shall not discuss here other options which lead to the possibility of adding boundary terms and we content ourselves with considering this standard class of variations.

Putting together all the results obtained we are finally in the position of rewriting Equation (109) as

$$\delta S_{TOT}[\Phi, g] = \int_M d^n x \sqrt{|g|} \frac{1}{2\kappa} \left(R_{\mu\nu} - \frac{R}{2} g_{\mu\nu} + \frac{2\kappa}{\sqrt{|g|}} \frac{\delta(\sqrt{|g|} \mathcal{L}[\Phi, g])}{\delta g^{\mu\nu}} \right) \delta g^{\mu\nu}, \quad (116)$$

from which we obtain that the variation vanishes for all possible $\delta g^{\mu\nu}$ if and only if

$$G_{\mu\nu} = -\frac{2\kappa}{\sqrt{|g|}} \frac{\delta(\sqrt{|g|} \mathcal{L}[\Phi, g])}{\delta g^{\mu\nu}},$$

where we inserted the Einstein tensor as per Equation (99). This is nothing but Einstein's equations (104) provided that we identify the right hand side with the stress-energy tensor of the underlying theory.

Definition 125. Let (M, g) be an n -dimensional spacetime with $n \geq 2$ and let $\mathcal{L}[\Phi, g]$ be the Lagrangian density of a collection of classical fields on (M, g) . We call **stress-energy tensor** associated to Φ the symmetric rank $(0, 2)$ tensor field whose components read

$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\delta(\sqrt{|g|}\mathcal{L}[\Phi, g])}{\delta g^{\mu\nu}}. \quad (117)$$

In order to confirm that Equation (117) is a legitimate candidate for the definition of stress-energy tensor when we consider matter fields whose dynamics descends from a Lagrangian density, we need to establish that it is divergence free. The following corollary does the job

Corollary 126. Let (M, g) be an n -dimensional spacetime with $n \geq 2$ and let $T_{\mu\nu}$ be as per Equation (117). Then it is divergence free, namely

$$\nabla^\mu T_{\mu\nu} = g^{\mu\lambda} \nabla_\lambda T_{\mu\nu} = 0.$$

Proof. Consider Equation (116) and, for every $X \in \Gamma_0(TM)$, consider the variation $\delta g_{\mu\nu} = (\mathcal{L}_X g)_{\mu\nu}$ where $\mathcal{L}_X$ is the Lie derivative as per Definition 76. Using the Killing equation (49) as well as Equation (108), it descends that

$$\delta g_{\mu\nu} = \nabla_{(\mu} X_{\nu)} \implies \delta g^{\mu\nu} = -g^{\mu\rho} g^{\nu\lambda} \delta g_{\rho\lambda} = -g^{\mu\rho} g^{\nu\lambda} \nabla_{(\rho} X_{\lambda)}.$$

Inserting this result in the variation of the total action, we obtain

$$\delta S_{TOT}[\Phi, g] = - \int_M d^n x \sqrt{|g|} \frac{1}{2\kappa} \left(R_{\mu\nu} - \frac{R}{2} g_{\mu\nu} - \kappa T_{\mu\nu} \right) g^{\mu\rho} g^{\nu\lambda} \nabla_{(\rho} X_{\lambda)}.$$

Since the integrand is symmetric under exchange of $\mu \leftarrow \nu$ and therefore of $\rho \leftrightarrow \lambda$, we can rewrite this last expression as

$$\delta S_{TOT}[\Phi, g] = - \int_M d^n x \sqrt{|g|} \frac{1}{2\kappa} \left(R_{\mu\nu} - \frac{R}{2} g_{\mu\nu} - \kappa T_{\mu\nu} \right) g^{\mu\rho} g^{\nu\lambda} \nabla_\rho X_\lambda.$$

Using metric compatibility as per Definition 59, integrating by parts and dropping all boundary terms since X is compactly supported, we obtain

$$\delta S_{TOT}[\Phi, g] = \int_M d^n x \sqrt{|g|} \frac{1}{2\kappa} \nabla^\mu \left(R_{\mu\nu} - \frac{R}{2} g_{\mu\nu} - \kappa T_{\mu\nu} \right) X^\nu = \frac{1}{2} \int_M d^n x \sqrt{|g|} (\nabla^\mu T_{\mu\nu}) X^\nu.$$

Since X is an arbitrary, compactly supported vector field on M , $\delta S_{TOT}[\Phi, g]$ vanishes if and only if $\nabla^\mu T_{\mu\nu} = 0$, which concludes the proof. $\square$

Remark 38. The identification of the stress-energy tensor as the variation of the Lagrangian density in Equation (117) calls for the natural question whether the outcome is equivalent to the canonical stress-energy tensor or, more generally, to the Belinfante-Rosenfeld stress-energy tensor, see [MH94]. We shall not enter into the details of this discussion, but the rationale is that, when both can be constructed, then they coincide, barring mild technical assumptions. A nice survey on all the possible definition and comparison can be found here, see [FR03].

Remark 39. *A mathematically oriented reader (why are you smirking? I know, you know that I am talking to you!) might be displeased about Definition 125 since the assumptions are not written in the same formal language used in the preceding chapters. There is no excuse and this is true. The reason is that a formulation of classical field theories on a smooth manifold requires the notion of vector and, possibly, even of principal bundles, see [Lee13, Chap. 10] and [Mor01]. This goes beyond the strict scopes of these notes and, should you be interested, you are encouraged to take the course of Mathematical Methods for Theoretical Physics, a.k.a. Metodi III.*

We conclude this chapter emphasizing that one should not be prone to believe that all stress-energy tensors are of the form of Equation (117). This is a definition which is tuned to models subordinated to a Lagrangian density encoding information on the dynamics. Although the vast majority of physical systems of interest falls in this category, this is not always the case. In these scenarios a different approach to the definition of $T_{\mu\nu}$ should be taken into account. We shall see a sample of this possibility when discussing cosmological models in Section 6.

5 Linearized Gravity and Gravitational Waves

In the formulation of Einstein's equation (104) we have postulated the existence of a proportionality relation between the Einstein tensor and the stress-energy counterpart. Yet no values for the constant κ has been given. In this section we shall discuss how it is possible to fix it, the rationale being to look for a suitable limit in which it is possible to recover Newtonian gravity.

5.1 Linearized Gravity

In order to investigate whether the Einstein's equations encode in a suitable limit Newtonian gravity, we should consider a scenario where the effect on the metric of the stress-energy tensor is negligible. To make this idea precise we should start from the case where no gravitational field is present at all, namely Minkowski spacetime $(\mathbb{R}^n, \eta)$, adopting here for definiteness the convention $(-, \underbrace{+, \dots, +}_{n-1})$ for the signature. In order to encode the idea of looking for a small gravitational field, we should work at the level of perturbation theory.

Notation: Henceforth we shall often denote by a superscript $^{(0)}$, 0-th order quantities and by $^{(1)}$ their first order counterpart. The only exception shall be Equation (30) and the inverse of this expression. Furthermore, we shall write generically $O(n)$ with $n \geq 1$ to denote that the expression is valid up to the n -th order in perturbation theory.

We consider a metric tensor whose components read

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2), \quad (118)$$

where $h_{\mu\nu}$ is interpreted as a perturbation of the Minkowski metric, which is therefore meant to be small in comparison to it. Furthermore we assume to work with Cartesian coordinates.

Remark 40. A more precise formulation of Equation (118) requires to give a mathematical meaning to the statement that h is small in comparison to η . This requires to consider a suitable norm which acts as a quantifier. The choice is highly non-unique, but, for the purposes of this section, we can require $h \in \Gamma(T^*(\mathbb{R}^n)^{\otimes s^2})$ to be bounded. Working, at the level of coordinates, this entails that, for every $\mu, \nu = 1, \dots, n$, $\|h_{\mu\nu}\|_{L^\infty(\mathbb{R}^n)} \ll 1$ and Equation (118) should be written as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(\|h_{\mu\nu}\|_{L^\infty(\mathbb{R}^n)}^2).$$

Our next goal is to expand the Einstein tensor $G_{\mu\nu}$ as in Equation (99) to first order in perturbation theory. This is not difficult but it is painfully long and mechanical. Hence brace yourself for the long computation. Let us start from the Einstein tensor observing that

$$G_{\mu\nu} = G_{\mu\nu}^{(0)} + G_{\mu\nu}^{(1)} + O(2) = G_{\mu\nu}^{(1)} + O(2), \quad (119)$$

where we used that, at 0-th order we have the Einstein tensor built out of the Minkowski metric, which is vanishing. Equation (119) entails that we need to consider

$$G_{\mu\nu}^{(1)} = R_{\mu\nu}^{(1)} - \frac{\eta_{\mu\nu}}{2} R^{(1)} - \frac{h_{\mu\nu}}{2} R^{(0)} = R_{\mu\nu}^{(1)} - \frac{\eta_{\mu\nu}}{2} R^{(1)}, \quad (120)$$

where we used that the Ricci scalar built out of the Minkowski metric vanishes. Recalling that

$$R_{\mu\nu} = g^{\rho\lambda} R_{\rho\mu\nu\lambda} \quad R = g^{\mu\nu} R_{\mu\nu},$$

we can infer that

$$R_{\mu\nu}^{(1)} = \eta^{\rho\lambda} R_{\rho\mu\nu\lambda}^{(1)} + O(2), \quad R^{(1)} = \eta^{\mu\nu} R_{\mu\nu}^{(1)} + O(2), \quad (121)$$

where 0-th order terms are not present since the Ricci tensor and the scalar curvature built out of Minkowski metric do vanish. This boils down to expanding the Riemann tensor to first order in perturbation theory. Using Equation (86) we can infer that

$$R_{\mu\nu\rho\delta}^{(1)} = \eta_{\lambda\delta} (R^{(1)})_{\mu\nu\rho}^\lambda + O(2), \quad (R^{(1)})_{\mu\nu\rho}^\lambda = \partial_\mu (\Gamma^{(1)})_{\nu\rho}^\lambda - \partial_\nu (\Gamma^{(1)})_{\mu\rho}^\lambda + O(2). \quad (122)$$

Observe that all terms involving products of Christoffel symbols do not contribute since, omitting all indices for simplicity, it turns out that $(\Gamma\Gamma)^{(1)} = \Gamma^{(0)}\Gamma^{(1)} + \Gamma^{(1)}\Gamma^{(0)} = 0$ since all Christoffel symbols built out of $\eta_{\mu\nu}$ vanish. If we consider at last Equation (33), we obtain

$$(\Gamma^{(1)})_{\mu\nu}^\rho = \frac{1}{2} \eta^{\rho\sigma} (\partial_\mu h_{\nu\sigma} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu}). \quad (123)$$

Replacing this expression in Equation (122), we end up with

$$\begin{aligned} R_{\mu\nu\rho\alpha}^{(1)} &= \frac{1}{2} \underbrace{\eta_{\lambda\alpha} \eta^{\lambda\sigma}}_{\delta_\alpha^\sigma} (\partial_\mu \partial_\nu h_{\rho\sigma} + \partial_\mu \partial_\rho h_{\nu\sigma} - \partial_\mu \partial_\sigma h_{\rho\nu} - \partial_\nu \partial_\mu h_{\rho\sigma} - \partial_\nu \partial_\rho h_{\mu\sigma} + \partial_\nu \partial_\sigma h_{\rho\mu}) = \\ &= \frac{1}{2} (\partial_\mu \partial_\rho h_{\nu\alpha} - \partial_\mu \partial_\alpha h_{\rho\nu} - \partial_\nu \partial_\rho h_{\mu\alpha} + \partial_\nu \partial_\alpha h_{\rho\mu}), \end{aligned} \quad (124)$$

from which it descends

$$R_{\mu\nu}^{(1)} = \frac{1}{2} (\partial^\alpha \partial_\mu h_{\nu\alpha} - \square_\eta h_{\mu\nu} - \partial_\nu \partial_\mu h + \partial_\nu \partial^\alpha h_{\mu\alpha}) \text{ and } R^{(1)} = -\square_\eta h + \partial^\mu \partial^\alpha h_{\mu\alpha}, \quad (125)$$

where we adopt the convention that $\partial^\alpha \doteq \eta^{\alpha\lambda} \partial_\lambda$, $h \doteq \eta^{\mu\nu} h_{\mu\nu}$ is the trace of the perturbation, while we define $\square_\eta = \eta^{\mu\nu} \partial_\mu \partial_\nu = -\frac{1}{c^2} \partial_t^2 + \sum_{i=1}^3 \partial_{x_i}^2$. We do not set $c = 1$ since our ultimate goal is to determine the dimensionful constant κ in Equation (104). After achieving our goal, we shall revert to our good old habit setting $c = 1$. Using Equation (120) and (125), we end up with

$$G_{\mu\nu}^{(1)} = \frac{1}{2} \left(\partial^\alpha \partial_\mu h_{\nu\alpha} - \square_\eta h_{\mu\nu} - \partial_\nu \partial_\mu h + \partial_\nu \partial^\alpha h_{\mu\alpha} + \eta_{\mu\nu} \square_\eta h - \eta_{\mu\nu} \partial^\lambda \partial^\alpha h_{\lambda\alpha} \right). \quad (126)$$

Gauge Transformations: Regardless of the explicit form of the stress-energy tensor, Equation (126) is rather unwieldy. Yet, there is still one bit of information that we have not made use of, *the action of the diffeomorphism group* $\text{Diff}(M)$. More precisely one of the cornerstone of general relativity is the statement that there does not exist a distinguished class of reference systems.

This is tantamount to stating that, given a spacetime (M, g) and $\phi \in \text{Diff}(M)$ it is equivalent working on $(M, \phi^* g)$, where ϕ^* denotes the pull-back as per Definition 29. This is known as a *gauge transformation*. In the physics literature this correspondence is always referred with a slight abuse of terminology as *the intrinsic freedom in general relativity (GR) to choose any underlying coordinate system*. A rather interesting consequence of this principle arises when we consider a family of diffeomorphisms $\Phi_\epsilon \in C^\infty(I; \text{Diff}(M))$ where $0 \in I \subseteq \mathbb{R}$, which is generated in terms of integral curves of a vector field $X \in \Gamma(TM)$, see Lemma 74. Here we use ϵ as the parameter generating the diffeomorphism since we do not want to confuse it with a time coordinate, being the background Lorentzian. In addition ϵ and its powers can be seen as a book keeping device to keep track of the perturbative order of an expression, much in the same spirit as the superscripts (n) used above. In this case, when we work in perturbation theory, we can consider either

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2) \quad \text{or} \quad (\phi_\epsilon^* g)_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}^{(\epsilon)} + O(2),$$

from which it descends that

$$h_{\mu\nu}^{(\epsilon)} = (\phi_\epsilon^* g)_{\mu\nu} - \eta_{\mu\nu} + O(2) = (\phi_\epsilon^* \eta)_{\mu\nu} - \eta_{\mu\nu} + (\phi_\epsilon^* h)_{\mu\nu} + O(2).$$

This identity can be approximated as

$$h_{\mu\nu}^{(\epsilon)} = (\phi_\epsilon^* h)_{\mu\nu} + \epsilon \left. \frac{d(\phi_\epsilon^* \eta)_{\mu\nu}}{d\epsilon} \right|_{\epsilon=0} + O(2) = h_{\mu\nu} + \epsilon (\mathcal{L}_X \eta)_{\mu\nu} + O(2),$$

where we used Definition 76. Observe that we have approximated $(\phi_\epsilon^* h)_{\mu\nu} \equiv (\phi_0^* h)_{\mu\nu} = h_{\mu\nu}$ since the following term in the expansion, namely $\epsilon \left. \frac{d(\phi_\epsilon^* h)_{\mu\nu}}{d\epsilon} \right|_{\epsilon=0}$ would be of order $O(2)$ since

h is already a contribution to g at first order in perturbation theory. The last expression can be further refined using the Killing equation (49), inferring that an *infinitesimal gauge transformation* acting on a metric perturbation yields

$$h_{\mu\nu} \mapsto h_{\mu\nu} + 2\partial_{(\mu}X_{\nu)}. \quad (127)$$

Parameterization of the degrees of freedom – A convenient way to exploit Equation (127) comes from a clever reparameterization of the degrees of freedom of the metric perturbation h :

$$h_{00} = -2\Phi, \quad (128a)$$

$$h_{0i} = A_i, \quad (128b)$$

$$h_{ij} = -2s_{ij} - 2\Psi\delta_{ij}, \quad (128c)$$

where δ is the Krönecker delta and where we employ Latin indices since they refer to the spatial components of the metric tensor or of its perturbations. In addition

$$\Psi = -\frac{1}{2(n-1)}\delta^{ij}h_{ij}. \quad (129)$$

This entails that Equation (128c) yields

$$s_{ij} = \frac{1}{2} \left(h_{ij} - \frac{1}{n-1} \delta^{kl} h_{kl} \delta_{ij} \right), \quad (130)$$

which is known as **strain**. Observe that it is traceless, that is $\delta^{ij}s_{ij} = 0$. The line element associated to $\eta + h$ reads

$$ds^2 = -(1 + 2\Phi)dt^2 + 2A_i dt dx^i + [(1 - 2\Psi)\delta_{ij} + s_{ij}]dx^i dx^j + O(2). \quad (131)$$

Exercise 127. Compute the components of the Einstein tensor associated to Equation (131) proving that it reads

$$\begin{cases} G_{00} = 2\Delta_\delta \Psi + \partial_k \partial_l s^{kl} + O(2) \\ G_{0j} = -\frac{1}{2}\Delta_\delta A_j + \frac{1}{2}\partial_j \partial_k A^k + 2\partial_0 \partial_j \Psi + \partial_0 \partial_k s_j^k + O(2) \\ G_{ij} = (\delta_{ij}\Delta_\delta - \partial_i \partial_j)(\Phi - \Psi) + \delta_{ij}\partial_0 \partial_k A^k - \partial_0 \partial_{(i} A_{j)} + 2\delta_{ij}\partial_0^2 \Psi \\ \quad - \square_\eta s_{ij} + 2\partial_k \partial_{(i} s_{j)}^k - \delta_{ij}\partial_k \partial_l s^{kl} + O(2). \end{cases} \quad (132)$$

where indices run from 0 to $n-1$ if $\dim M = n$. Furthermore all indices are raised and lowered by means of the flat Minkowski metric. Furthermore $\Delta_\delta = \delta^{ij}\partial_i \partial_j$ is the Laplace operator on $(\mathbb{R}^{n-1}, \delta)$ while $\square = \frac{1}{c^2}\partial_0^2 - \Delta$ is the wave operator on $(\mathbb{R}^n, \eta)$.

Transverse Gauge – In order to study the Newtonian limit of general relativity, it is convenient to exploit the existing gauge freedom, that is Equation (127) to further reduce the complexity of Equations (128a), (128b) and (128c). This procedure is very similar in spirit and in practice to the one employed in the study of free Maxwell equations when introducing the vector potential and its dynamics. Gauge fixing does not follow a set of a priori rules but it must be adapted to each scenario... it is more of an art than of an algorithm. We shall not discuss what is known as **transverse gauge**:

Step 1 - Assume that the strain s_{ij} is not divergence free. Exploiting Equation (127) we can look for a vector field $\xi \in \Gamma(T\mathbb{R}^n)$ such that

$$\Delta_\delta \xi_j + \frac{1}{3} \partial_j \partial^i \xi_i = -2 \partial^i s_{ij}, \quad (133)$$

where we have used Equation (130) and where, once more, $i, j = 1, \dots, n-1$, $n = \dim M$ and Δ_δ is the Laplace operator on $(\mathbb{R}^{n-1}, \delta)$.

Exercise 128 (Very Technical). Prove that Equation (133) is an elliptic PDE and, hence it admits a solution. Characterize the residual gauge freedom.

Step 2 - Assume that the A_i as per Equation (128b) is not divergence free. Once more using Equation (127) we can observe that the time component of $\xi \in \Gamma(T\mathbb{R}^n)$ has not been fixed nor constrained by Equation (133). Hence we can still look for

$$\tilde{A}_i = A_i + \partial_{(0} \xi_i),$$

which is divergence free, namely

$$\partial^i \tilde{A}_i = 0 \implies \Delta_\delta \xi_0 = -\partial_0 \partial^i \xi_i - \partial^i A_i.$$

In this case, this is manifestly a Poisson equation for ξ_0 with a non vanishing source and hence a solution certainly exists.

Having fixed the transverse gauge greatly simplifies the Einstein tensor in Equation (132) which reads

$$\begin{cases} G_{00} = 2\Delta_\delta \Psi + O(2) \\ G_{0j} = -\frac{1}{2} \Delta_\delta A_j + 2\partial_0 \partial_j \Psi + O(2) \\ G_{ij} = (\delta_{ij} \Delta_\delta - \partial_i \partial_j) (\Phi - \Psi) - \partial_0 \partial_{(i} A_{j)} + 2\delta_{ij} \partial_0^2 \Psi - \square_\eta s_{ij} + O(2). \end{cases} \quad (134)$$

Newtonian Limit – In order to discuss the Newtonian limit, we have to move our attention to the stress-energy tensor. The discussion is very similar to that of Equation (119), namely we can expect to expand $T_{\mu\nu} = T_{\mu\nu}^{(0)} + T_{\mu\nu}^{(1)} + O(2)$, but compatibility with Einstein's equations order by order in perturbation theory entails that, since the background metric is that of Minkowski and therefore $G_{\mu\nu}^{(0)} = 0$, then $T_{\mu\nu}^{(0)} = 0$. Hence the lowest non-vanishing contribution of the stress-energy tensor to the Einstein's equations is automatically that of the metric perturbation.

From a physical viewpoint, this is to be expected since the stress-energy tensor sources the deviation of the metric from the Minkowski counterpart and, therefore they need to be of the same order as the deviation itself. In what follows, we shall not need to consider not to assume the existence of higher order contribution in perturbation theory from $T_{\mu\nu}$ and, hence, without loss of generality, we can set

$$T_{\mu\nu} \equiv T_{\mu\nu}^{(1)}. \quad (135)$$

In addition, to make contact with Newtonian gravity, we assume that the matter content in the underlying spacetime is that of mass density ρ such that the internal pressure forces are negligible. As we shall see in Section 6, this is a special case of a perfect fluid, called *dust*. Still working on Minkowski spacetime and in Cartesian coordinates, this entails that

$$T_{\mu\nu} = \text{diag}(c^2\rho, \underbrace{0, \dots, 0}_{n-1}),$$

which implies, in turn, via Corollary 126 that

$$\partial^\mu T_{\mu\nu} = c^2 \partial_0 \rho = 0.$$

In other words we are considering a static mass density $\rho \neq \rho(t)$. Observe in addition that we have restored the explicit dependence on the speed of light, discarding for a second the convention $c = 1$. Consequently we can look for time-independent solutions of Einstein's equations using Equation (134), that is, setting $\partial_0 \Psi = 0$,

$$\begin{cases} 2\Delta_\delta \Psi = \kappa c^2 \rho \\ \Delta_\delta A_j = 0 \\ (\delta_{ij} \Delta_\delta - \partial_i \partial_j) \left(\frac{\Phi}{c^2} - \Psi \right) - \Delta_\delta s_{ij} = 0. \end{cases}, \quad (136)$$

where the factor c^{-2} multiplying Φ arises restoring the explicit dependence on the speed of light in Equation (131), namely

$$-(1 + 2\Phi)dt^2 \mapsto -c^2 \left(1 + \frac{2\Phi}{c^2} \right) dt^2.$$

Recalling that the trace of s_{ij} vanishes per construction, contracting the third equation via δ^{ij} , yields

$$(n-2)\Delta_\delta(c^{-2}\Phi - \Psi) = 0 \implies \frac{\Phi}{c^2} = \Psi,$$

provided that we look for harmonic functions vanishing at infinity. This entails in turn that

$$\Delta_\delta s_{ij} = 0 \implies s_{ij} = 0.$$

For the same reason the second equation entails that $A_j = 0$, which leaves us with the equation

$$\Delta_\delta \Phi = \frac{\kappa c^4}{2} \rho, \quad (137)$$

which coincides with the scalar Poisson equation for gravity if $\kappa = \frac{8\pi G}{c^4} \simeq 2.07665 \times 10^{-43} N$, where G is the Newton's constant and c is the speed of light. A reader might be wondering why we expect Equation (137) to be closely related to Newtonian gravity. The following exercise should dispel any doubt.

Exercise 129. Suppose that $\mathbb{R}^4$ is endowed with the line element

$$ds^2 = - \left(1 - 2\frac{\Phi}{c^2}\right) c^2 dt^2 + \left(1 - 2\frac{\Phi}{c^2}\right) (dx^2 + dy^2 + dz^2),$$

where $\Phi \neq \Phi(t)$ and it abides by Equation (137). Compute the geodesic equation for a point particle of mass $m > 0$.

To summarize we have established the final form of Einstein's equations

$$\boxed{G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.} \quad (138)$$

Engrave this formula in your soul. Yet, it is customary to set $c = 1$ and even $G = 1$ and we shall stick to this convention unless stated otherwise, namely we work with natural units (no, alas, no one sets $8\pi = 1$).

5.2 Gravitational Waves

We shall present one of the most famous consequences of the weak field approximation of general relativity: gravitational waves. These have been for decades one of the cornerstones of the theoretical predictions of Einstein's theory, which lacked direct experimental evidences, although indirect ones were available since the mid seventies. The observation of a signal compatible with gravitational waves in September 2015 by the Ligo experiment opened a new era, validating the last among the key predictions of Einstein's theory which still lacked a confirmation. In the following we shall not delve in the complicated and rather interesting experimental aspects of gravitational waves, but we will outline the basic theoretical concepts.

Our starting point is once more a perturbative expansion at first order of Einstein's equations, but we shall assume in addition that there is no matter content, namely the stress-energy tensor vanishes. Already at the level of the full Einstein's equation (138), the hypotheses $T_{\mu\nu} = 0$ entails by tracing out with respect to the metric

$$G_{\mu\nu} = 0 \implies g^{\mu\nu} G_{\mu\nu} = \frac{2-n}{2} R = 0,$$

which in turn guarantees us that, unless $\dim M = n = 2$

$$G_{\mu\nu} = 0 \iff R_{\mu\nu} = 0. \quad (139)$$

Remark 41. *One might wonder why Equation (139) fails to be true if $n = 2$. In order to grasp the significance of this constraint, we should be counting how many degrees of freedom are involved in solving Einstein's equations. Let us start from the basic underlying unknown, the metric. Working locally, this is nothing but a function with values in the symmetric matrices having $\frac{n(n+1)}{2}$ independent entries.*

If we look instead at the Einstein's tensor, being it symmetric we have the same amount of independent entries, although the differential Bianchi identities entail that $\nabla^\mu \mathbf{G}_{\mu\nu} = 0$ gives n -additional constraints. These can be ascribed to the underlying freedom in choosing a coordinate system. At the level of the metric, this is encoded by Equation (127), which is ultimately telling us that the number of independent entries goes down to $\frac{n(n-1)}{2}$. In other words, if we consider the physically interesting scenario $n = 4$, we are left with 6 independent equations.

Yet, if set $n = 2$, apparently there is only one equation to be solved. Yet, one has to account for another bit of information, namely

$$\int_M d^2x \sqrt{|g|} R = 4\pi\chi(M),$$

where $\chi(M) = 2 - 2h$ is the Euler characteristic of the manifold, while h is its genus. If M is compact and Riemannian this is known also as the Gauss-Bonnet theorem, see [Lee11]. Yet, for our current purposes, it is relevant to observe that the right hand side is a topological invariant. Hence, any variation of the Hilbert-Einstein action is automatically vanishing, namely $\mathbf{G}_{\mu\nu} = 0$ for any metric. In other words, in two-dimensions gravity is known to be a topological theory and no propagating degrees of freedom exist. A similar statement holds true in $2+1$ -dimensions, but the interpretation is different and we shall not discuss it here, rather we refer to the monograph [Car10].

In other words, if impose a vanishing stress-energy tensor, and we look for small deviations from Minkowski metric, Equation (139) entails that we are free to consider Equation (125), namely

$$\mathbf{R}_{\mu\nu}^{(1)} = \frac{1}{2} (\partial^\alpha \partial_\mu h_{\nu\alpha} - \square_\eta h_{\mu\nu} - \partial_\nu \partial_\mu h + \partial_\nu \partial^\alpha h_{\mu\alpha}) = 0. \quad (140)$$

Contrary to the preceding section, we opt to study this equation still exploiting the freedom coming from Equation (127), but opting for a different gauge fixing. This is a two step procedure, the first relying on the following result.

Proposition 130. *Let (M, g) be a spacetime, $\dim M > 2$, such that, working in coordinates, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2)$. Then, in every local chart (U, φ) , there exists $X \in \Gamma(TM|_U)$ such that*

$$\partial^\mu \left(\tilde{h}_{\mu\nu} - \eta_{\mu\nu} \frac{\tilde{h}}{2} \right) = 0, \quad (141)$$

*where $\tilde{h}_{\mu\nu} = h_{\mu\nu} + 2\partial_{(\mu} X_{\nu)}$. Imposing Equation (141) is known as **De Donder gauge**.*

Proof. It suffices to start from $\tilde{h}_{\mu\nu} = h_{\mu\nu} + 2\partial_{(\mu}X_{\nu)}$ and to compute

$$\partial^\mu \left(\tilde{h}_{\mu\nu} - \eta_{\mu\nu} \frac{\tilde{h}}{2} \right) = \partial^\mu \left(h_{\mu\nu} - \eta_{\mu\nu} \frac{h}{2} + 2\partial_{(\mu}X_{\nu)} \right) = \partial^\mu \left(h_{\mu\nu} - \eta_{\mu\nu} \frac{h}{2} \right) + \frac{\square_\eta X_\nu}{2}.$$

This implies that we need to find X such that

$$\square_\eta X_\nu = -2\partial^\mu \left(h_{\mu\nu} - \eta_{\mu\nu} \frac{h}{2} \right), \quad (142)$$

which admits always a solution being a vector valued, second order hyperbolic differential equation. $\square$

The second step consists in exploiting that Equation (142) does not imply that we are selecting a unique vector field X and therefore a residual gauge freedom is present. This can be exploited as follows:

Proposition 131. *Let (M, g) be a spacetime, $\dim M > 2$ such that, working in coordinates, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2)$ where h abides by the de Donder gauge as per Proposition 130. Then, in every local chart (U, φ) , there exist a residual gauge freedom, namely $Y \in \Gamma(TM|_U)$ such that $\tilde{h} = 0$ where $\tilde{h}_{\mu\nu} = h_{\mu\nu} + 2\partial_{(\mu}Y_{\nu)}$. Imposing this constraint in addition to Equation (141) is known as **Transverse-Traceless (TT) gauge**.*

Proof. On account of Equation (142), we have a residual freedom of looking for $Y \in \Gamma(TM|_U)$ such that $\square_\eta Y_\mu = 0$. The condition of vanishing trace entails that, in addition

$$\tilde{h} = h + 2\partial^\mu Y_\mu = 0.$$

This system of equations admits a solution as a reader can verify in [Hunt12, Sec. 4.5.2]. $\square$

Denoting for simplicity $h_{\mu\nu}^{TT}$ a metric perturbation in the TT gauge, Equation (140) becomes

$$\square_\eta h_{\mu\nu}^{TT} = 0. \quad (143)$$

This result entails that, up to a gauge transformation, metric perturbations abide by a wave equation, which justifies the name of *gravitational waves* for linear perturbations of Einstein's equations. These enjoy several notable physical properties which distinguish them from other fields whose dynamics is ruled by a wave-like operator, such as the vector potential in electromagnetism or the Higgs field and the inflaton in particle physics and cosmology respectively. In the following, to make contact with the more interesting physical scenarios, we work with 4 spacetime dimensions.

Proposition 132. *Let (M, g) be a spacetime, $\dim M = 4$ such that, working in coordinates, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2)$ where h abides by the de TT gauge as per Proposition 131. Then, in every local chart (U, φ) , there exist a residual gauge freedom, namely $Z \in \Gamma(TM|_U)$ such that $\tilde{h}_{0i} = 0$ where $\tilde{h}_{\mu\nu} = h_{\mu\nu} + 2\partial_{(\mu}Z_{\nu)}$. Imposing this constraint in addition to those accounted by Proposition 131 is known as **radiation gauge**.*

Proof. Our first step consists of observing that Proposition 131 does not entail that we have performed a complete gauge fixing since, we are still free to consider $Z \in \Gamma(TM|_U)$ so that

$$\square_\eta Z_\mu = 0 \quad \text{and} \quad \partial^\mu Z_\mu = 0.$$

If we consider $h_{\mu\nu} \mapsto \tilde{h}_{\mu\nu} \doteq h_{\mu\nu} + \partial_{(\mu} Z_{\nu)}$, then $\tilde{h}$ still abides by the TT gauge. Hence, if we start from $h_{0i} \neq 0$, we are looking for Z such that

$$\partial_t Z_i + \partial_i Z_0 = -h_{0i},$$

which needs to be combined with the additional constraints

$$-\partial_t Z_0 + \partial^i Z_i = 0, \tag{144a}$$

$$-\partial_t^2 Z_0 + \partial_t \partial^i Z_i = -\Delta Z_0 + \partial^i \partial_t Z_i = 0, \tag{144b}$$

where Equation (144a) corresponds to $\partial^\mu Z_\mu = 0$. At the same time Equation (144b) is nothing but $\partial_t(\partial^\mu Z_\mu) = 0$ where we have exploited that $\square_\eta Z_\mu = 0$ entails that $\partial_t^2 Z_0 = \Delta Z_0$. All these identities can be read as constraints on the initial data and therefore we are left with the following systems of hyperbolic partial differential equations:

$$\begin{cases} \square_\eta Z_\mu = 0 \\ \partial_t Z_i + \partial_i Z_0 = -h_{0i} \\ -\partial_t Z_0 + \partial^i Z_i = 0 \quad \text{and} \quad \Delta Z_0 + \partial^i \partial_t Z_i = 0 \end{cases}. \tag{145}$$

Observe that since $[\square, \partial_\mu] = 0$ for all $\mu = t, i$ with $i = x, y, z$ and since $\square h_{0i} = 0$ for all $i = x, y, z$, then we can conclude that the constraints on the initial data if imposed at any but fixed time t_0 are preserved at all $t \in \mathbb{R}$. Using standard arguments in the theory of PDEs, we can conclude that this system admits a solution and this concludes the proof. $\square$

In addition to Proposition 132 the icing on the cake lies in the following result.

Corollary 133. *Let (M, g) be a spacetime, $\dim M = 4$ such that, working in coordinates, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2)$ where h abides by the de radiation gauge as per Proposition 132. Then also $h_{00} = 0$.*

Proof. Since $\partial^\mu h_{\mu\nu} = 0$, setting $\nu = 0$ and recalling that $h_{i0} = 0$, we obtain that $\partial_t h_{00} = 0$, which entails that h_{00} is constant in time. Yet, since also $h = 0$, Equation (140) for the component 00 boils down to

$$\square_\eta h_{00} = \Delta h_{00} = 0,$$

which entails that $h_{00} = 0$. $\square$

Physical Properties of Gravitational waves: The first most notable feature of gravitational waves, comes, from the observation that Equation (143) is the translation in the TT gauge of the linearized Einstein's equation. This entails that, considering the radiation gauge which is a specialization of the TT gauge and using the parameterization in Equations (128a)–(128c), Equation (132) becomes

$$\begin{cases} G_{00} = 2\Delta_\delta \Psi^{TT} = 0 \\ G_{0j} = -\frac{1}{2}\Delta_\delta A_j^{TT} = 0 \\ G_{ij} = \partial_i \partial_j \Psi^{TT} - \partial_0 \partial_{(i} A_{j)}^{TT} + 2\delta_{ij} \partial_0^2 \Psi^{TT} - \square_\eta s_{ij}^{TT} = 0 \end{cases}, \quad (146)$$

where, the traceless condition $h^{TT} = 0$ entails that $\Psi = 0$, while $\partial^\mu h_{\mu\nu}^{TT} = 0$ entail

$$\partial^i s_{ij}^{TT} = 0, \quad \text{and} \quad \partial^i A_i^{TT} = 0.$$

Yet, we observe that the equations for G_{00} and G_{0j} entail that

$$\Psi^{TT} = 0 \quad \text{and} \quad A_j^{TT} = 0,$$

provided that we impose a vanishing condition at infinity, which is justified on physical grounds since we are considering small perturbations with respect to Minkowski spacetime. This entails in turn that

$$G_{ij} = 0 \iff \square_\eta s_{ij}^{TT} = 0. \quad (147)$$

This result entails that, barring for s_{ij}^{TT} all other components of $h_{\mu\nu}^{TT}$ vanish. A priori, being s_{ij}^{TT} symmetric and traceless, it has 5 independent degrees of freedom, but, if we add the condition $\partial^i s_{ij}^{TT} = 0$ inherited from the de Donder gauge, we boil down to only 2 degrees left. It is important also to realize how we can parameterize them and, to this end, it is convenient to work in Fourier space.

For the mathematically oriented reader, since Equation (143) is a partial differential equation with constant coefficients, the Hörmander-Łojasiewicz theorem [Hör58, Los59] guarantees that there exists a solution lying in $\mathcal{S}'(\mathbb{R}^4)$. This guarantees that it is legitimate to consider the Fourier transform, that is, for all $\mu, \nu = 0, \dots, 3$,

$$(\omega^2 - |k|^2) \hat{s}_{\mu\nu}^{TT} = 0,$$

where ω is the dual variable to the time coordinate, while (k_x, k_y, k_z) are dual to the spatial ones. In addition $|k|^2 = k_x^2 + k_y^2 + k_z^2$ and the condition $\partial^i s_{ij}^{TT} = 0$ becomes

$$k^i \hat{s}_{ij}^{TT} = 0.$$

These two constraints are rather familiar in electromagnetism and they can be dealt with similarly. We can consider a coordinate systems such that the gravitational wave is propagating in the z -direction, which entails therefore that

$$k^z \hat{s}_{3j}^{TT} = 0 \implies \hat{s}_{3j}^{TT} = 0, \quad (148)$$

where we identified z with the subscript 3. Hence we end up with the following form for the Fourier transform of $h_{\mu\nu}^{TT}$ and of s_{ij}^{TT} :

$$\hat{h}_{\mu\nu}^{TT} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & & & \\ 0 & 2\hat{s}_{ij}^{TT} & & \\ 0 & & & \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2\hat{s}_{11}^{TT} & 2\hat{s}_{12}^{TT} & 0 \\ 0 & 2\hat{s}_{12}^{TT} & -2\hat{s}_{11}^{TT} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (149)$$

where, in the last equality, we have accounted for Equation (148) as well as for the traceless gauge condition $h^{TT} = 0$ which entails $\hat{h}^{TT} = 0$. In other words a gravitational wave has only 2 independent degrees of freedom.

Remark 42. [Physical Interpretation] *The preceding analysis guarantees us that a linear perturbation of the gravitational field on top of Minkowski spacetime behaves, after a gauge fixing, as a propagating wave with two degrees of freedom which are transversal to the direction of propagation of the wave itself. This is reminiscent of electrodynamics, although, at this level we are working at the level of perturbation theory. A rather interesting re-interpretation of Equations (143) and (149) comes from relativistic field theory. As a matter of fact h^{TT} can be read as a covariant field on Minkowski spacetime which transforms under a unitary and irreducible, massless representation of the component connected to the identity of the Poincaré group of helicity $\lambda = 2$. For this reason, depending on the scenario of interest and often on the community addressed, gravitational waves can also be called **gravitons**. Their behaviour, as fields, is very much similar to that of the photon both at the classical and at a quantum level, although the former has helicity equal to 1. In a heuristic language one says that photons are associated to vectorial propagating modes, while gravitons to tensorial ones. In the following we shall make more precise this last statement. It is instead incorrect to say that gravitons are spin 2 particles since this term should be reserved only to massive fields. A fully comprehensive analysis of these topics requires tools proper of the theory of Lie groups and of their representation. A reader is encouraged to refer to [BaRa86, §21].*

Effect of a Gravitational Wave: An historically controversial and hotly debated topic has been the analysis of the effect of a passing gravitational wave on objects with their own mass and energy. Nowadays this is well-understood and tested at an experimental level, but, in the early days of general relativity, what follows was often not accepted.

Let us therefore consider on four-dimensional Minkowski spacetime two idealized observers, namely two point particles of unit mass $m = 1$, which are at rest. In addition they have no initial spatial velocity nor external forces are present. In view of the analysis in Section 3, they both move in spacetime along a geodesic γ whose tangent vector at $t = 0$ is $\dot{\gamma}^\mu = (1, 0, 0, 0)$ where we set $c = 1$. We have not inserted a subscript discriminating between the two particles since, on Minkowski spacetime, by translation invariance, one can deduce that the two geodesics will run along parallel trajectories. This is codified by the Jacobi equation, see Proposition 115 and Equation (102) which reads

$$A^\mu = 0,$$

since the Riemann tensor is vanishing. In other words there is no relative acceleration between two neighboring geodesics. Yet, if a gravitational wave abiding by Equation (143) passes by, we can expect this conclusion not to hold true since there is a displacement of the Riemann tensor as one can infer from Equation (124). More precisely this equation entails that, up to a lowering of the indices by means of the Minkowski metric, the two geodesics experience a relative acceleration equal to

$$A_\mu = (R^{(1)})_{\nu 00\mu} V^\nu = \frac{1}{2} (\partial_\nu \partial_0 h_{0\mu}^{TT} - \partial_\nu \partial_\mu h_{00}^{TT} - \partial_0^2 h_{\mu\nu}^{TT} + \partial_0 \partial_\mu h_{0\nu}^{TT}) V^\nu = -\frac{1}{2} \partial_0^2 h_{\mu\nu}^{TT} V^\nu, \quad (150)$$

where V^ν are the components of the vector field codifying the separation between the geodesics and where we used that $h_{\mu 0}^{TT} = 0$ for all $\mu = 0, \dots, 3$, see Equation (149). Under our assumptions and comparing with Equation (102), this expression can be rewritten as

$$\frac{d^2 S_\mu}{dt^2} = -\frac{1}{2} \partial_0^2 h_{\mu\nu}^{TT} S^\nu.$$

A comparison with Equation (149) entails that if $\mu = 0, 3$ the right hand side vanishes and therefore, assuming vanishing initial velocity, S^0 and S^3 are constant in time. On the contrary S^1 and S^2 change in time and, to analyze their behaviour, we consider separately the two building blocks:

1. a monochromatic perturbation $h_{\mu\nu}^{TT} = 2\hat{s}_{11}^{TT} \cos(\omega t - k_3 z)$ such that $\hat{s}_{12}^{TT} = 0$ while $2\hat{s}_{11}^{TT} \equiv h_+$ is constant.
2. a monochromatic perturbation $h_{\mu\nu}^{TT} = 2\hat{s}_{12}^{TT} \cos(\omega t - k_3 z)$ such that $\hat{s}_{11}^{TT} = 0$ while $2\hat{s}_{12}^{TT} \equiv h_-$ is constant.

[*Disclaimer*] In what follows one should work with the theory of distributions and of convolutions, but we shall refrain from doing it since we want to make manifest the physical consequences of Equation (150).

In the first case,

$$\begin{cases} \frac{d^2 S_1}{dt^2} = -\frac{1}{2} S_1 \frac{d^2}{dt^2} h_+ \cos(\omega t - k_3 z) = -\frac{\omega^2}{2} S_1 h_+ \cos(\omega t - k_3 z) \\ \frac{d^2 S_2}{dt^2} = \frac{1}{2} S_2 \frac{d^2}{dt^2} h_+ \cos(\omega t - k_3 z) = \frac{\omega^2}{2} S_2 h_+ \cos(\omega t - k_3 z) \end{cases}.$$

Imposing as initial conditions $S_i(0) = S_i$ and $\dot{S}_i(0) = 0$, $i = 1, 2$ these two equations can be solved as

$$S_1(t) = -S_1 \frac{\left(ce'_0 \left(-h_+, \frac{k_3 z}{2} \right) se_0 \left(-h_+, \frac{1}{2}(k_3 z - \omega t) \right) - se'_0 \left(-h_+, \frac{k_3 z}{2} \right) ce_0 \left(-h_+, \frac{1}{2}(k_3 z - \omega t) \right) \right)}{\left(ce'_0 \left(-h_+, \frac{k_3 z}{2} \right) se_0 \left(-h_+, \frac{1}{2}(k_3 z) \right) - se'_0 \left(-h_+, \frac{k_3 z}{2} \right) ce_0 \left(-h_+, \frac{1}{2}(k_3 z) \right) \right)}, \quad (151a)$$

$$S_2(t) = S_2 \frac{\left(ce'_0 \left(-h_+, \frac{k_3 z}{2} \right) se_0 \left(-h_+, \frac{1}{2}(k_3 z - \omega t) \right) - se'_0 \left(-h_+, \frac{k_3 z}{2} \right) ce_0 \left(-h_+, \frac{1}{2}(k_3 z - \omega t) \right) \right)}{\left(ce'_0 \left(-h_+, \frac{k_3 z}{2} \right) se_0 \left(-1, \frac{1}{2}(k_3 z) \right) - se'_0 \left(-h_+, \frac{k_3 z}{2} \right) ce_0 \left(-h_+, \frac{1}{2}(k_3 z) \right) \right)}, \quad (151b)$$

where $se_n(q, z)$ and $ce_n(q, z)$ are the odd and even Mathieu functions with characteristic value n and parameter q , see [Nist, §28.2], while $se'_n(q, z) = \frac{d se_n(q, z)}{dz}$ and $ce'_n(q, z) = \frac{d ce_n(q, z)}{dz}$. At first glance Equations (151a) and (151b) might appear not be very informative, but, for small time intervals a good approximation reads

$$S_1(t) \simeq -S_1 \left(1 + \frac{1}{2} h_+ \cos(k_3 z - \omega t) \right), \quad S_2(t) \simeq S_2 \left(1 - \frac{1}{2} h_+ \cos(k_3 z - \omega t) \right),$$

which tells us that the separation along the spatial $x-y$ plane, orthogonal to that of propagation of the gravitational wave, oscillates with opposite sign.

In the second case instead, the two equations are coupled, namely

$$\begin{cases} \frac{d^2 S_1}{dt^2} = -\frac{1}{2} S_2 \frac{d^2}{dt^2} h_- \cos(\omega t - k_3 z) = -\frac{\omega^2}{2} S_2 h_- \cos(\omega t - k_3 z) \\ \frac{d^2 S_2}{dt^2} = \frac{1}{2} S_1 \frac{d^2}{dt^2} h_- \cos(\omega t - k_3 z) = \frac{\omega^2}{2} S_1 h_- \cos(\omega t - k_3 z) \end{cases}.$$

Imposing as initial conditions $S_i(0) = S_i$ and $\dot{S}_i(0) = 0$, $i = 1, 2$ these two equations can still be solved but the procedure is less straightforward. We content ourselves with observing that for small time intervals a good approximation reads

$$S_1(t) \simeq S_1 + \frac{S_2}{2} h_- \cos(k_3 z - \omega t), \quad S_2(t) \simeq S_2 + \frac{S_1}{2} h_- \cos(k_3 z - \omega t).$$

This tells us that the separation along the spatial $x-y$ plane, orthogonal to that of propagation of the gravitational wave, oscillates with an “X-pattern”. This means that, if we consider a collection of point particles sitting initially on a circle in the $x-y$ plane, their relative position is distorted so to form ellipses oscillating in such a way that their pattern looks like a big X in the plane.

Production of Gravitational Waves: The last question that we want to address concerning gravitational waves is their production mechanism. For this reason we cannot assume a vanishing stress-energy tensor, as we did until now, but we need to retain it. Our main goal is to understand the folklore statement that gravitational waves are generated by the quadrupole moment of a mass distribution. In order to make our analysis precise, we need to consider a third, convenient gauge fixing. Let us start introducing the **trace-reversed metric perturbation**

$$\tilde{h}_{\mu\nu} \doteq h_{\mu\nu} - \frac{h}{2} \eta_{\mu\nu}, \quad (152)$$

where $h = \eta^{\mu\nu} h_{\mu\nu}$. Observe that, since we are working in four spacetime dimensions, $\tilde{h} \doteq \eta^{\mu\nu} \tilde{h}_{\mu\nu} = -h$.

Proposition 134. *Let (M, g) be a four-dimensional spacetime such that, working in coordinates, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2)$. Then, in every local chart (U, φ) , there exists $X \in \Gamma(TM|_U)$ such that $\partial^\mu \tilde{h}_{\mu\nu} = 0$ where $\tilde{h}_{\mu\nu} = \tilde{h}_{\mu\nu} + 2\partial_{(\mu} X_{\nu)} - \eta_{\mu\nu} \partial^\lambda X_\lambda$. Imposing this constraint is known as **Lorenz gauge**.*

Proof. Let $h_{\mu\nu}$ be a metric perturbation. Equation (152) entails that Equation (127) becomes for the trace-reversed counterpart $\tilde{h}_{\mu\nu} = \tilde{h}_{\mu\nu} + 2\partial_{(\mu} X_{\nu)} - \eta_{\mu\nu} \partial^\lambda X_\lambda$. Imposing the condition $\partial^\mu \tilde{h}_{\mu\nu} = 0$, yields

$$\square_\eta X_\nu = -\partial^\mu \tilde{h}_{\mu\nu},$$

which is a wave equation with smooth source. This always admits a solution, hence proving the sought statement. $\square$

Working with the Lorenz gauge, greatly simplifies the first order correction to the Einstein tensor which reads, according to Equation (126)

$$G_{\mu\nu}^{(1)} = -\frac{1}{2} \square_\eta \tilde{h}_{\mu\nu}, \quad (153)$$

where $\tilde{h}_{\mu\nu}$ is the trace-reversed metric perturbation as per Equation (152) abiding by $\partial^\mu \tilde{h}_{\mu\nu} = 0$. Following the same reasoning which has lead to Equation (135), assuming that the matter content is codified by the stress-energy tensor $T_{\mu\nu}$, the dynamics of the metric perturbation is ruled by

$$\square_\eta \tilde{h}_{\mu\nu} = -16\pi G T_{\mu\nu}, \quad (154)$$

where we have set $c = 1$. The most general solution to this equation reads

$$\tilde{h}_{\mu\nu} = -16\pi G (E_{ret} \star T)_{\mu\nu} + \tilde{h}_{\mu\nu,0},$$

where $\tilde{h}_{\mu\nu,0}$ is a solution of the homogeneous equation

$$\square_\eta \tilde{h}_{\mu\nu,0} = 0,$$

while E_{ret} is the retarded fundamental solution of $\square_\eta$ acting on $\Gamma((T^*\mathbb{R}^4)^{\otimes s^2})$ by means of the convolution $\star$. We focus on this term which encodes the contribution of the source term and we observe that, since the wave operator acts independently on each component of $\tilde{h}_{\mu\nu}$, it turns out that

$$(E_{ret} \star T)_{\mu\nu} = E_{ret} \star T_{\mu\nu},$$

where, with a slight abuse of notation, we still use the symbol E_{ret} which denotes now the retarded fundamental solution of the wave operator on the $(\mathbb{R}^4, \eta)$ acting on scalars. Working at the level of integral kernels

$$E_{ret}(t, \underline{x}, t', \underline{x}') = -\frac{\Theta(t - t')}{4\pi} \delta(-(t - t')^2 + |\underline{x} - \underline{x}'|^2), \quad (155)$$

where $\underline{x} = (x, y, z) \in \mathbb{R}^3$ and, similarly, $\underline{x}' = (x', y', z') \in \mathbb{R}^3$. Fixing for simplicity and without loss of generality $t' = 0$, we can use the distributional identity

$$\delta(f(t)) = \sum_{i=1}^n \frac{\delta(t - t_i)}{|f'(t_i)|},$$

where $f \in C^\infty(\mathbb{R})$ and where $\{t_i\}_{i=1, \dots, n}$ is the collection of points such that $f(t_i) = 0$ but $f'(t_i) \neq 0$. Hence letting $t^2 - |\underline{x}|^2$ play the rôle of $f(t)$, we can rewrite Equation (155) as

$$E_{ret}(t, \underline{x}, 0, 0) = -\frac{\Theta(t)}{4\pi|\underline{x} - \underline{x}'|} (\delta(t - |\underline{x} - \underline{x}'|) + \delta(t + |\underline{x} - \underline{x}'|)) = -\frac{\Theta(t)}{4\pi|\underline{x} - \underline{x}'|} \delta(t - |\underline{x} - \underline{x}'|), \quad (156)$$

which entails that

$$\tilde{h}_{\mu\nu}(t, \underline{x}) = -4G \int_{\mathbb{R}^3} d^3 \underline{x}' \frac{T_{\mu\nu}(t - |\underline{x} - \underline{x}'|, \underline{x}')}{|\underline{x} - \underline{x}'|}. \quad (157)$$

It is convenient to consider the Fourier transform with respect to the time variable of Equation (157), which we will still denote by the symbol $\widehat{}$ with a slight abuse of notation:

$$\widehat{h}_{\mu\nu}(\omega, \underline{x}) = -4G \int_{\mathbb{R}^3} d^3 \underline{x}' e^{i\omega |\underline{x} - \underline{x}'|} \frac{\widehat{T}_{\mu\nu}(\omega, \underline{x}')}{|\underline{x} - \underline{x}'|}.$$

Let us now assume that the matter source in Equation (154) is localized in a small region of spacetime and that the observe lies sufficiently far from it, at a distance r so that we can approximate the factor $\frac{e^{i\omega |\underline{x} - \underline{x}'|}}{|\underline{x} - \underline{x}'|}$ with $\frac{e^{i\omega r}}{r}$. In other words, denoting by δr the size of the matter distribution, we require that $\delta r \ll r$ and also that $\omega \delta r \ll 1$. This entails

$$\widehat{h}_{\mu\nu}(\omega, \underline{x}) = -4G \frac{e^{i\omega r}}{r} \int_{\mathbb{R}^3} d^3 \underline{x}' \widehat{T}_{\mu\nu}(\omega, \underline{x}'). \quad (158)$$

First of all we observe that, upon Fourier transform of the time variable, the Lorenz gauge condition entails that, if we raise all indices,

$$\widehat{h}^{0\nu}(\omega, \underline{x}) = \frac{i}{\omega} \partial_j \widehat{h}^{j\nu}(\omega, \underline{x}),$$

which entails that, first setting $\nu = 1, 2, 3$ and subsequently $\nu = 0$, we can reconstruct the left hand side if we know the spatial components of the metric perturbation tensor. In addition observe that, for $i, j = 1, 2, 3$,

$$\begin{aligned} \int_{\mathbb{R}^3} d^3 \underline{x}' \widehat{T}^{ij}(\omega, \underline{x}') &= \int_{\mathbb{R}^3} d^3 \underline{x}' \partial_k \left((x')^i \widehat{T}^{kj}(\omega, \underline{x}') \right) - (x')^i \partial_k \widehat{T}^{kj}(\omega, \underline{x}') = \\ &= - \int_{\mathbb{R}^3} d^3 \underline{x}' (x')^i \partial_k \widehat{T}^{kj}(\omega, \underline{x}') = i\omega \int_{\mathbb{R}^3} d^3 \underline{x}' (x')^i \widehat{T}^{0j}(\omega, \underline{x}'), \end{aligned}$$

where, in the second equality, we used that the first integral vanishes being a total derivative. In the third equality we exploited that the conservation of the stress-energy tensor, *i.e.*, $\partial_\mu T^{\mu\nu} = 0$ entails in Fourier space that

$$-\partial_j \hat{T}^{j\mu}(\omega, \underline{x}') = i\omega \hat{T}^{0\mu}(\omega, \underline{x}').$$

Repeating one more the same procedure we end up with

$$\hat{h}_{\mu\nu}(\omega, \underline{x}) = -4G \frac{e^{i\omega r}}{r} \int_{\mathbb{R}^3} d^3 \underline{x}' \hat{T}_{\mu\nu}(\omega, \underline{x}') = 2G\omega^2 \frac{e^{i\omega r}}{r} \int_{\mathbb{R}^3} d^3 \underline{x}' (x')^i (x')^j \hat{T}^{00}(\omega, \underline{x}'). \quad (159)$$

If we introduce the **quadrupole moment tensor**

$$I^{ij}(t) \doteq \int_{\mathbb{R}^3} d^3 \underline{x}' (x')^i (x')^j \hat{T}^{00}(t, \underline{x}'), \quad (160)$$

we end up with the following formula for the spatial components of the trace-reversed metric perturbation tensor [**quadrupole formula**]

$$\hat{h}_{jk}(\omega, \underline{x}) = 2G\omega^2 \frac{e^{i\omega r}}{r} \hat{I}_{jk}(\omega). \quad (161)$$

Remark 43. Equations (161) and (160) give an explanation to the folklore statement that gravitational waves are produced only by mass distributions with a non-vanishing quadrupole or higher moment. Observe in particular that higher momenta do not appear in the calculations above due to the approximation leading to Equation (158). Yet, regardless of this assumption, it remains valid the statement that mass distributions which are either spherically symmetric or with a lone dipole moment do not act as sources of gravitational waves.

6 Cosmological Spacetimes

In the previous section we have discussed the weak field limit of Einstein's equations, but we have not yet constructed or discussed exact solutions. Before tackling this issue, it is important to highlight the following structural features of Einstein's equations which mark the complexity of their analysis:

- Equation (138) does not necessarily form a closed system of partial differential equations. The reason lies in the stress-energy tensor which is canonically interpreted as a source term. If this were the whole story, this would require us to know $T_{\mu\nu}$ globally. Yet, in many instances, the stress-energy tensor is built from Equation (117) as the variation with respect to the metric tensor of a Lagrangian encoding the dynamics of all matter fields, which we denote collectively by Φ . In other words $T_{\mu\nu} \equiv T_{\mu\nu}[\Phi, g]$ and the condition $\nabla^\mu T_{\mu\nu} = 0$ is realized only for dynamically allowed configurations of Φ , namely the underlying fields must be in turn solutions of the Euler-Lagrange equations of motion derived from their

Lagrangian density. Hence, in presence of classical fields, the Einstein's equations need to be studied together with the dynamics of the matter content, else it is an ill-defined system. There are exceptions, one of which will be discussed in this section. The most notable one consists of assuming that $T_{\mu\nu} = 0$, namely there is no matter in spacetime. This special scenario is usually referred to as *vacuum Einstein's Equations*

- Even under the assumption that $T_{\mu\nu} = 0$, Equation (138) yields for $\dim M > 2$

$$G_{\mu\nu} = 0 \implies R_{\mu\nu} = 0.$$

A direct inspection of Equation (94) combined with Equation (33) unveils that the vacuum Einstein's equations are a system of non-linear, second order partial differential equations for the metric coefficients. In addition the linear component does not fall in the standard classification which distinguishes among elliptic, hyperbolic and parabolic second order partial differential equations. We shall not enter into the details of such classifications, but, suffices to say that elliptic PDEs are associated to boundary value problems, that is, existence and uniqueness of the solutions can be discussed only when boundary (or asymptotic) conditions on the solutions are assigned. On the contrary hyperbolic and parabolic PDEs are known also as evolution equations, which means that existence and uniqueness of the solutions can be discussed only when initial data on a codimension 1 hypersurface are assigned. In this respect, Einstein's equations are an unicum since they do not fall in the standard scheme of classification of PDEs. Yet, at the same time, this should not surprise, since boundary value problems are naturally associated to the description of static systems, which is not what we expect as one can infer from the discussion of gravitational radiation in Section 5.2. At the same time, initial value problems refer implicitly to the existence of a distinguished direction codifying the evolution of the underlying system. This is interpreted usually as the time variable, but Einstein's theory rests on diffeomorphism invariance being a building block. This entails that there should not exist a distinguished coordinate system, all being equivalent and, in turn, there cannot exist a preferred time direction.

These comments point towards the direction that studying the solutions of Einstein's equations is not a topic which can be treated systematically and in full generality. We shall discuss it on a case by case basis.

Homogeneous and Isotropic Backgrounds: In this section we discuss one of the first applications of general relativity, aimed at the description of the observed Universe at large scales. This problem has been studied originally and independently by Friedmann, Lemaître, Robertson and Walker in the early thirties of the former century and it represents one of the biggest successes of Einstein's theory since it has been the starting point of what is now known as *cosmology*. Our current understanding of the large scale structures of spacetimes has changed considerably since the beginning of the eighties, also thanks to a large influx of reliable experimental data. Our goal is not to survey the current status of this field of research, but to explain the building

blocks of what is often called *standard (model of) cosmology*. A reader is encouraged to refer to the specialized and recent literature for more information on the more advanced topics, see *e.g.* [Bau22]. Henceforth we shall work for definiteness with $\dim M = 4$, although a generalization to any dimension is possible.

A precise description of the metric of the observed Universe is certainly not feasible since, even restricting the attention to macroscopic matter, we would not be able to account for the distribution of all matter components, ranging from cluster of galaxies to single stars. On the one hand, we look for a coarse grained description of the Universe at such large scales that one is not sensitive to the local variations of matter densities. On the other hand, as customary in these scenarios, think for example at statistical mechanics, we need to rely on some defining postulates. This is sometimes referred to as **Copernican principle** and it can be heuristically translated as follows:

At each instant of time there is neither a distinguished point nor a preferred direction in the observed Universe.

As a first step we translate this heuristic idea in a concrete, mathematical statement. We start from the lack of a distinguished point, which corresponds to the notion of *homogeneity*.

Definition 135. *Let (M, g) be a Lorentzian spacetime such that $\dim M = 4$. We say that it is **homogeneous** if there exists a diffeomorphism $\Phi : \mathbb{R} \times \Sigma \rightarrow M$ as per Definition 40 such that*

- *for each $t \in \mathbb{R}$, $\iota_t : \Sigma \rightarrow \{t\} \times \Sigma$ identifies a spacelike hypersurface as per Definition 118 with respect to the induced metric Φ^*g ,*
- *for every $p, q \in \{t\} \times \Sigma$, $t \in \mathbb{R}$, $\exists$ an isometry $f \in \text{Iso}(\mathbb{R} \times \Sigma, \Phi^*g)$, see Definition 67, such that $f(p) = q$.*

Henceforth we will be leaving the action of Φ understood and we shall identify g with Φ^*g . It is also customary to refer to $\mathbb{R} \times \Sigma$ as a **foliation** of the spacetime M .

Our next step consists of translating the idea that there does not exist a preferred direction in the Universe.

Definition 136. *Let (M, g) be a Lorentzian spacetime such that $\dim M = 4$. We say that it is **isotropic** if*

- *we can individuate a family of timelike curves such that for all $p \in M$, there exists one and only one among these curves passing through p (congruence),*
- *for every $p \in M$, denoting by $u \in T_p M$ the vector tangent to the timelike curve lying in the congruence and for every pair $v, w \in T_p M$ such that $g|_p(v, u) = g|_p(w, u) = 0$, there exists $f \in \text{Iso}(M, g)$ as per Definition 67 such that $f(p) = p$, $df(u) = u$ and $df(v) = w$, where df denotes the differential of f as per Definition 4.*

Imposing that a spacetime (M, g) is both homogeneous and isotropic has far reaching consequences, mainly in terms of existing constraints on the form of the underlying metric. Starting from Definition 135, we can identify (M, g) with $(\mathbb{R} \times \Sigma, g)$ and via pull-back we can induce on Σ a time-dependent Riemannian metric $h(t) \doteq \iota_t^*(g)$, where we interpret henceforth the coordinate along $\mathbb{R}$ as time. Denoting by ${}^{(3)}R_{abc}{}^d[h]$ the associated Riemann tensor via Equation (84), we can observe that it induces a map

$$R_2 : \Omega^2(\Sigma_t) \rightarrow \Omega^2(\Sigma_t), \quad F \mapsto R_2(F), \quad (162)$$

where $\Omega^2(\Sigma_t)$ is the space of 2-forms on Σ_t , see Definition 198 and where, working with coordinates, if we decompose $F = F_{ab}dx^a \wedge dx^b$,

$$R_2(F) = R_2(F)_{ab}dx^a \wedge dx^b \doteq {}^{(3)}R_{ab}{}^{cd}F_{cd}dx^a \wedge dx^b.$$

If we consider an arbitrary but fixed $t \in \mathbb{R}$ and $p \in \Sigma_t$, it holds that, choosing a coordinate system in a neighbourhood of p , the collection of all smooth 2-forms on Σ_t evaluated at p is isomorphic to $\mathcal{M}_a(\mathbb{R}^3)$, the vector space of real valued, antisymmetric matrices. Using the Riemannian metric h_t we can endow this space with the following positive inner product:

$$(\cdot, \cdot)_h : \mathcal{M}_a(\mathbb{R}^3) \times \mathcal{M}_a(\mathbb{R}^3) \rightarrow \mathbb{R}, \quad (F, F') \mapsto (F, F')_h \doteq h_t(F, F') = h_t^{ac}(p)h_t^{bd}(p)F_{ab}(p)F'_{cd}(p). \quad (163)$$

Hence $(\mathcal{M}_a(\mathbb{R}^3), (\cdot, \cdot)_h)$ is a finite-dimensional Hilbert space and, using Equation (162), we can induce on $\mathcal{M}_a(\mathbb{R}^3)$ a linear operator which we denote still by R_2 with a slight abuse of notation:

$$(F, R_2(F'))_h = \overline{F}_{ab} \underbrace{h^{ac}h^{bd}{}^{(3)}R_{cd}{}^{ef}}_{h^{ac}h^{bd}{}^{(3)}R_{cdlm}h^{el}h^{fm}} F'_{ef} = (R_2(F), F')_h.$$

Here we used implicitly Equation (87d). Hence R_2 is a symmetric operator on $(\mathcal{M}_a(\mathbb{R}^3), (\cdot, \cdot)_h)$ and, being the Hilbert space finite dimensional, this entails that R_2 is a bounded, self-adjoint operator [Mor17]. Hence R_2 has only point spectrum $\sigma_P(R_2)$ and it can be decomposed as $R_2 = \sum_{i=1}^{N < \infty} \lambda_i P_{\lambda_i}$ where $\lambda_i \in \mathbb{R}$ is an eigenvalue and P_{λ_i} the orthogonal projector on the associated, finite-dimensional eigenspace. Suppose that there exist $\lambda_1, \lambda_2 \in \sigma_P(R_2)$ such that $\lambda_1 \neq \lambda_2$ and without loss of generality we consider $\lambda_1 \neq 0$. Consequently, we can find $F_1, F_2 \in \mathcal{M}_a(\mathbb{R}^3)$ such that $R_2(F_i) = \lambda_i(F_i)$, $i = 1, 2$. Yet, recalling that $\mathcal{M}_a(\mathbb{R}^3)$ is isomorphic to the space of smooth 2-forms evaluated at $p \in M$, by applying the Hodge operator $\star$, see Definition 229 and Exercise 231, we can construct two distinguished vector fields, namely $(\star F_i)^\sharp$, $i = 1, 2$, where we used the musical isomorphism as per Equation (28). Working at the level of coordinates and recalling that we have fixed $t \in \mathbb{R}$ and $p \in \Sigma_t$,

$$[(\star F_1)^\sharp]^i = h_t^{ij}(\star F_1)_j = \frac{\sqrt{h_t}}{2} h_t^{ij}(F_1)_{lm} h_t^{lk} h_t^{mq} \epsilon_{mkj},$$

where ϵ is the totally antisymmetric Levi-Civita symbol. An identical formula is valid for F_2 . Yet Definition 136 guarantees us that there must exist an isometry $f \in \text{Iso}(\mathbb{R} \times \Sigma, g)$ which

leaves the base point fixed and it maps via push-forward $d\iota_t((\star F_1)^\sharp)$ in $d\iota_t((\star F_2)^\sharp)$. Yet, since at least one among λ_1 and λ_2 must be different from 0, say λ_1

$$\begin{aligned} df \circ d\iota_t((\star F_1)^\sharp) &= \frac{1}{\lambda_1} df \circ d\iota_t((\star R_2(F_1))^\sharp) = \frac{1}{\lambda_1} d\iota_t((\star R_2(F_2))^\sharp) = \\ &= \frac{\lambda_2}{\lambda_1} d\iota_t((\star F_2)^\sharp) = d\iota_t((\star F_2)^\sharp). \end{aligned}$$

In this chain of identities we used that the operator R_2 is geometric, since it is constructed out of the action of the Riemann tensor on Σ_t . Hence the last identity holds true if and only if $\lambda_1 = \lambda_2$. In other words all eigenvalues must coincide. One can realize that the assumption of working with the point spectrum is not necessary and the statement holds true if one assumes to consider the continuous spectrum. At the end of the day all eigenvalues must coincide, which entails that

$$R_2 = \lambda \mathbb{I},$$

where $\mathbb{I}$ is the identity operator on $\mathcal{M}_a(\mathbb{R}^3)$, while $\lambda \in \mathbb{R}$. Working in coordinates, this entails that at $(t, p) \in \mathbb{R} \times \Sigma$

$$R_2 = \lambda \mathbb{I} \implies {}^{(3)}R_{ab}{}^{cd} = 2\lambda \delta_{[a}^c \delta_{b]}^d. \quad (164)$$

We can now let (t, p) vary and Equation (164) is left unchanged, barring that $\lambda \in C^\infty(\mathbb{R} \times \Sigma; \mathbb{R})$. Lowering the indices with the metric h_t yields

$${}^{(3)}R_{abcd} = 2\lambda h_{c[a} h_{b]d}. \quad (165)$$

Yet, we can apply the differential Bianchi identity in Equation (90) to infer that

$$\nabla_{[e} {}^{(3)}R_{ab]cd} = 2\nabla_{[e} (\lambda h_{|c|a} h_{b]d}) = 2(\nabla_{[e} \lambda) h_{|c|a} h_{b]d} = 0,$$

where we used metric compatibility of the connection in the last identity, see Definition 59 and where the symbol $|c|$ entails that the index must be excluded from the antisymmetrization procedure. Since this last identity holds true for all indices, it must hold that

$$\nabla_e \lambda = 0 \implies \lambda = \text{const..}$$

We can conclude that an **isotropic and homogeneous four-dimensional spacetime** (M, g) must

1. be isometric to $(\mathbb{R} \times \Sigma, g)$,
2. the hypersurfaces $\iota_t : \Sigma \rightarrow M$, where $\iota_t(\Sigma) = \Sigma_t = \{t\} \times \Sigma$, $t \in \mathbb{R}$ are Riemannian manifolds with respect to the induced metric $h \equiv h_t = \iota_t^* g$ whose Riemann tensor reads

$${}^{(3)}R_{abcd} = K h_{c[a} h_{b]d}. \quad K \in \mathbb{R} \quad (166)$$

A Riemannian manifold whose Riemann tensor abides by Equation (166) is known in the literature as being of **constant sectional curvature**, see [Lee11, Prop. 8.8]. In addition observe that, by contracting with respect to the metric, see Equations (94) and (95) we obtain

$${}^{(3)}R_{bc} = K \left(h_c^d h_{bd} - 3h_{bc} \right) \quad \text{and} \quad {}^{(3)}R = -6K. \quad K \in \mathbb{R} \quad (167)$$

We can now classify all three-dimensional Riemannian manifolds of constant sectional curvature using the celebrated **Killing-Hopf** theorem which we report here verbatim, leaving an interested reader to [Lee11, Thm 11.12 & Cor. 11.13] for a proof. Here we specialize it to three-dimensions.

Theorem 137. *Let (M, h) be a 3-dimensional, connected and complete Riemannian manifold. Then it is isometric to $\tilde{M}/\Gamma$, where $\tilde{M}$ is a constant curvature model space, namely one among*

1. *the Euclidean space $\mathbb{R}^3$, whose line element reads in Cartesian coordinates (x, y, z)*

$$ds^2 = dx^2 + dy^2 + dz^2.$$

2. *the 3-sphere $\mathbb{S}^3$, whose line element reads in spherical coordinates $(\chi, \theta) \in (0, \pi)$ and $\varphi \in (0, 2\pi)$*

$$ds^2 = d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2).$$

3. *the hyperbolic 3-plane $\mathbb{H}^3$, whose line elements reads in hyperbolic coordinates $\chi \in (0, \infty)$, $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$*

$$ds^2 = d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2).$$

In addition Γ is a discrete subgroup of the first homotopy group $\pi_1(M)$ which acts freely and properly discontinuously on $\tilde{M}$.

Remark 44. *The Killing-Hopf Theorem has rather important consequences. If one ignores the action of Γ or, equivalently, if one considers only simply connected manifolds, this is telling us that homogeneity and isotropy entail that, in our spacetime, there exists a distinguished time direction, leading to the foliation of M as $\mathbb{R} \times \Sigma$ and that, at constant values of time, only three options are possible: a plane, a sphere or an hyperbolic plane. Yet, we are making a topological, hence global assumption on M , namely that $\pi_1(M) = \{0\}$. If we drop it, then one needs to allow for other options coming from the action of discrete subgroups of $\pi_1(M)$. In cosmology, usually when one refers to Friedmann-Lemaître-Robertson-Walker (FRLW) model, this possibility is discarded. Yet, from an experimental viewpoint, it is very hard to test a topological feature like simple-connectedness and in the literature one can often find works on **Lens Spaces** which are referring to some among the alternative options offered by the Killing-Hopf theorem. Observe also that, when working with local coordinates, this feature becomes irrelevant since coordinate patches are insensible to global topological properties.*

Theorem 137 guarantees us that, at least locally, if we consider an homogeneous and isotropic manifold $M \simeq \mathbb{R} \times \Sigma$, on each hypersurface the induced metric can be built from one the constant curvature model spaces. Our only leeway stems from the fact that their scalar curvature has been normalized to be ${}^{(3)}R = 0$ (Euclidean space), ${}^{(3)}R = 1$ (3-sphere) or ${}^{(3)}R = -1$ (hyperbolic space), while Equation (167) entails that ${}^{(3)}R$ can take a priori any real value. In other words the line element associated to h_t , $t \in \mathbb{R}$ is

$$ds_{h_t}^2 = a^2(t) \begin{cases} d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2) & \text{if } {}^{(3)}R = -1 \\ dx^2 + dy^2 + dz^2 & \text{if } {}^{(3)}R = 0 \\ d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2) & \text{if } {}^{(3)}R = 1 \end{cases}, \quad (168)$$

where $a \in C^\infty(\mathbb{R}; \mathbb{R})$ is a time-dependent function which varies smoothly the value of the scalar curvature on Σ_t . Observe that we are still working locally in a local chart and no changes of topology are allowed. Hence, if, at a given instant of time, Σ_t is (locally) spherical, flat or hyperbolic, such property is preserved for all $t \in \mathbb{R}$. In the literature it is often considered convenient to use an alternative and more compact form, which is constructed as follows:

- if ${}^{(3)}R = -1$ we replace χ with $r \doteq \sinh \chi \in \mathbb{R}$,
- if ${}^{(3)}R = 0$ we switch to spherical coordinates, $r \in (0, \infty)$, $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$,
- if ${}^{(3)}R = 1$ we replace χ with $r \doteq \sin \chi \in (0, 1]$.

Accordingly Equation (168) becomes

$$ds_{h_t}^2 = a^2(t) \left[\frac{dr^2}{1 - \kappa r^2} + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2) \right], \quad \kappa = 0, \pm 1 \quad (169)$$

where $\kappa = 0$ corresponds to **flat** spatial sections, namely $\mathbb{R}^3$, $\kappa = -1$ to **hyperbolic** spatial sections, namely $\mathbb{H}^3$, while $\kappa = 1$ to **spherical spatial sections**, namely $\mathbb{S}^3$. Observe that, in this last case, the coordinate change covers only half of the 3-sphere since the map $r = \sin \chi$ on the interval $(0, \pi)$ is 2:1.

The underlying hypotheses of isotropy and homogeneity allows also to establish a form for the full four-dimensional metric g associated to $\mathbb{R} \times \Sigma$. More precisely, Definition 136 entails the existence of a congruence of timelike vectors in the underlying four-dimensional spacetime (M, g) . Let $q \in M$ and let $u \in T_q M$ be the tangent vector at q lying in the congruence. According instead to Definition 135, up to a diffeomorphism which we leave implicit, M can be foliated in spacelike hypersurfaces, that is $M \simeq \mathbb{R} \times \Sigma$. Hence q must lie on one of such hypersurfaces, say Σ_t . In addition, for any $v \in T_q \Sigma_t$, denoting by $\iota_t : \Sigma_t \rightarrow M$ the embedding map, isotropy entails that $g(u, d\iota_t(v)) = 0$. Hence the congruence of timelike vectors is orthogonal to the hypersurfaces Σ_t . In view of these considerations, for any $q \in M$, we can consider a local patch with coordinates $(t, \underline{x})$ where t is the affine parameter along the integral lines of u , promoted to coordinate by setting $t(q) = 0$. We employ the symbol t , because, being u orthogonal to the foliation of hypersurfaces, existing due to the hypothesis of homogeneity, we can use the

affine parameter of the integral lines as the label of these hypersurfaces. The only difference with the preceding discussion is that we have no guarantee that $t \in \mathbb{R}$, but it could be defined only on a subset $I \subset \mathbb{R}$. We shall denote this choice as **cosmological time**. Combining these considerations with Equation (168) we can conclude that the line element of an *homogeneous and isotropic, four-dimensional, spacetime* (M, g) reads

$$ds^2 = -dt^2 + a^2(t) \begin{cases} d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2) & \text{if } {}^{(3)}R = -1 \\ dx^2 + dy^2 + dz^2 & \text{if } {}^{(3)}R = 0 \\ d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2) & \text{if } {}^{(3)}R = 1 \end{cases}, \quad (170)$$

or equivalently

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - \kappa r^2} + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2) \right], \quad \kappa = 0, \pm 1 \quad (171)$$

Behold the Friedmann-Lemaître-Robertson-Walker (FLRW) metric.

Remark 45. *It is often convenient to introduce a different coordinate, called **conformal time** τ and defined in such a way that $d\tau = \frac{dt}{a(t)}$. The line element (170) becomes*

$$ds^2 = a^2(\tau) \left[-d\tau^2 + \begin{cases} d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2) & \text{if } {}^{(3)}R = -1 \\ dx^2 + dy^2 + dz^2 & \text{if } {}^{(3)}R = 0 \\ d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2) & \text{if } {}^{(3)}R = 1 \end{cases} \right] \quad (172)$$

In particular, if we consider flat spatial sections, we realize that the FLRW metric is conformal to the Minkowski metric, see Definition 50.

The preceding analysis tells us that the hypothesis of homogeneity and isotropy greatly restrict the number of degrees of freedom of the metric tensor, namely, once we have a priori designed which constant time hypersurface we wish to consider among the three possibilities, there is only one unknown function left in Equation (170), namely $a(t)$ which is commonly called the **scale factor**. The reason for this name is clear if we consider as an example flat spatial sections and the locus $B_R(0) = \{(0, x, y, z) \mid x^2 + y^2 + z^2 \leq R^2\}$, $R > 0$. The volume of this ball induced by the Riemannian metric h_0 on Σ_0 is $\text{Vol}[B_R(0)] = \frac{4a^3(0)}{3\pi} R^3$. If we consider instead $B_R(\bar{t}) = \{(\bar{t}, x, y, z) \mid x^2 + y^2 + z^2 \leq R^2\}$, $0 \neq \bar{t} \in \mathbb{R}$, the volume of this ball induced by the Riemannian metric $h_{\bar{t}}$ on $\Sigma_{\bar{t}}$ is $\text{Vol}[B_R(\bar{t})] = \frac{4a^3(\bar{t})}{3\pi} R^3$. This yields the ratio

$$\frac{\text{Vol}[B_R(\bar{t})]}{\text{Vol}[B_R(0)]} = \left(\frac{a(\bar{t})}{a(0)} \right)^3,$$

namely a rules how volumes are scaling in time on a FLRW spacetime.

In order to determine the scale factor, we need to resort to Einstein's equations. To this end, a stress-energy tensor must be assigned, but, as we shall see, homogeneity and isotropy greatly

constraint our options. To realize it, first of all we write down the Einstein tensor starting from the Christoffel symbols. Using Equation (33) in combination with Equation (171) we obtain that the only non vanishing symbols are the following:

$$\begin{aligned}
\Gamma_{11}^0 &= \frac{a\dot{a}}{1-\kappa r^2}, & \Gamma_{22}^0 &= a\dot{a}r^2, & \Gamma_{33}^0 &= a\dot{a}r^2 \sin^2 \theta, \\
\Gamma_{01}^1 &= \Gamma_{10}^1 = \frac{\dot{a}}{a}, & \Gamma_{11}^1 &= \frac{\kappa r}{1-\kappa r^2}, & \Gamma_{22}^1 &= -r(1-\kappa r^2), & \Gamma_{33}^1 &= -r(1-\kappa r^2) \sin^2 \theta, \\
\Gamma_{02}^2 &= \Gamma_{20}^2 = \frac{\dot{a}}{a}, & \Gamma_{12}^2 &= \Gamma_{21}^2 = r^{-1}, & \Gamma_{33}^2 &= -\sin \theta \cos \theta, \\
\Gamma_{03}^3 &= \Gamma_{30}^3 = \frac{\dot{a}}{a}, & \Gamma_{23}^3 &= \Gamma_{32}^3 = \cot \theta.
\end{aligned} \tag{173}$$

where we employed the convention $0 \longleftrightarrow t$, $1 \longleftrightarrow r$, $2 \longleftrightarrow \theta$, $3 \longleftrightarrow \varphi$ and where $\dot{a} \doteq \frac{da}{dt}$. Consequently the components of the Ricci tensor and the scalar curvature can be computed from this chain of identities using Equations (94) and (95):

$$\begin{aligned}
R_{00} &= -3\frac{\ddot{a}}{a}, & R_{11} &= \frac{a\ddot{a} + 2\dot{a}^2 + 2\kappa}{1-\kappa r^2}, \\
R_{22} &= r^2(a\ddot{a} + 2\dot{a}^2 + 2\kappa), & R_{33} &= r^2 \sin^2 \theta (a\ddot{a} + 2\dot{a}^2 + 2\kappa),
\end{aligned} \tag{174}$$

while

$$R = 6 \left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{\kappa}{a^2} \right]. \tag{175}$$

At last we can infer that the Einstein tensor is diagonal and its non-vanishing components read:

$$\begin{cases} G_{00} = 3 \left[\left(\frac{\dot{a}}{a} \right)^2 + \frac{\kappa}{a^2} \right] \\ G_{11} = -\frac{2a\ddot{a} + \dot{a}^2 + \kappa}{1-\kappa r^2} \\ G_{22} = -r^2 (2a\ddot{a} + \dot{a}^2 + \kappa) \\ G_{33} = -r^2 \sin^2 \theta (2a\ddot{a} + \dot{a}^2 + \kappa) \end{cases}. \tag{176}$$

This calculation entails that, in order for a stress-energy tensor to be compatible with Einstein's equations, it must be both divergence free and diagonal. A priori there are infinitely many possibilities, but, we should not forget that we are trying to construct a model for the observed Universe at very large scales. In this respect, at a first approximation, we should not consider classical or quantum fields, but ordinary matter. Hence, one could think of galaxies as the basic building blocks and, at cosmic scales, they can be approximated as point particles which are not interacting among themselves since the gravitational force is negligible over such large distances. In classical models we are describing what is known as *dust*, a special case of a **perfect fluid**. This is a model for matter which is characterized by two functions, ρ , the *mass density* and p , the *pressure* with an associated stress-energy tensor which reads in components

$$T_{\mu\nu} = \rho u_\mu u_\nu + P(g_{\mu\nu} + u_\mu u_\nu), \tag{177}$$

where $u \equiv \frac{\partial}{\partial t}$. Observe that compatibility with the Einstein's equations, see Equation (176) entails that $\rho = \rho(t)$ and $p = p(t)$, while, in addition

$$\nabla^\mu T_{\mu\nu} = 0 \implies \dot{\rho} = -3\frac{\dot{a}}{a}(\rho + P). \quad (178)$$

This is an evolution equation for the mass density, but there is no dynamical law for the pressure. Hence Einstein's equations together with Equation (178) is an undetermined system. To bypass this hurdle we shall consider an *equation of state* relating P to ρ .

The Friedmann Equations: Summarizing the content of our analysis up to this point, if we assume that the spacetime is four-dimensional, homogeneous and isotropic as well as that the matter content is described by a perfect fluid whose stress-energy tensor is that of Equation (178), the spatial, diagonal components in Equation (138) (setting $c = 1$) are identical and we end up with

$$\begin{cases} G_{00} = 8\pi G T_{00} \implies 3 \left[\left(\frac{\dot{a}}{a} \right)^2 + \frac{\kappa}{a^2} \right] = 8\pi G \rho \\ G_{ii} = 8\pi G T_{ii} \implies 2a\ddot{a} + \dot{a}^2 + \kappa = -8\pi G P \end{cases}.$$

We can rewrite these identities in a different, albeit equivalent, form obtaining the celebrated **Friedmann Equations**:

$$\boxed{\begin{cases} \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{\kappa}{a^2} \\ \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) \\ \dot{\rho} = -3\frac{\dot{a}}{a}(\rho + P) \end{cases}}, \quad (179)$$

where we have also included the conservation law for the stress-energy tensor, although, in many textbooks, only the first two are referred as Friedmann equations. Before proceeding in the analysis of Equation (179), two comments are in due course:

1. One might think that there might exist a potential clash between the first and the second Friedmann equations since they are both ODEs for the scale factor $a(t)$. Yet, by taking the time derivative of the first and using the third one, we can obtain the second, hence proving that the dynamics of $a(t)$ is ruled by the second equation. This entails that a unique solution can be found provided that $\dot{a}$ and a are assigned at an initial value of time, say $t = 0$. Yet these two data are not completely independent since the first Friedmann equation represents a constraint they have to abide by.
2. It is convenient, especially if we consider flat spatial sections $\kappa = 0$, to introduce a different variable, namely the **Hubble parameter**

$$H \doteq \frac{\dot{a}}{a}, \quad (180)$$

which allows to rewrite Equation (179) with $\kappa = 0$ as

$$\boxed{\begin{cases} H^2 = \frac{8\pi G}{3}\rho \\ \dot{H} + H^2 = -\frac{4\pi G}{3}(\rho + 3p) \\ \dot{\rho} = -3H(\rho + P). \end{cases}} \quad (181)$$

The value of H measured now in the Universe is often referred as the *Hubble constant*, say H_0 , although it is now understood that one should interpret it as an initial datum. Its value is approximately 70Km/(s· Mpc). An additionally useful quantity is the **density parameter**

$$\Omega = \frac{8\pi G}{3H^2}\rho = \frac{\rho}{\rho_{crit}}, \quad (182)$$

where we introduce the critical density $\rho_{crit} = \frac{3H^2}{8\pi G}$. This is a function of time and, if we restore the possibility that $\kappa \neq 0$, its sign is determined by Ω itself, since we can rewrite the first Friedmann equation as

$$\Omega - 1 = \frac{\kappa}{H^2 a^2},$$

which entails that $\kappa > 0$ (*resp.* $\kappa < 0$ or $\kappa = 0$) if $\Omega > 1$ (*resp.* $\Omega < 1$ or $\Omega = 0$).

Equation of State: As already mentioned Equation (179) is an ill-defined system unless we assign the pressure globally in space and time or we relate it to ρ via an equation of state. Here, we follow this procedure and we assume henceforth that

$$P = \gamma\rho, \quad (183)$$

where we shall assume γ to be a constant real number. In addition, classical thermodynamics and matter suggest that $\gamma \geq 0$ so to have a positive or at most vanishing pressure, although, as we shall discuss, one might wish to relax this hypothesis. For the time being we accept this viewpoint and, replacing it in Equation (178), we obtain

$$\dot{\rho} = -3\frac{\dot{a}}{a}(1 + \gamma)\rho \implies \rho a^{3(1+\gamma)} = K, \quad (184)$$

where K is a positive constant which can also be written as $\rho(t_0)a^{3(1+\gamma)}(t_0)$.

Remark 46. Equation (184) is valid for all values of $\gamma \in \mathbb{R}$. Yet, we observe that, if $\gamma < -1$, it turns out that, whenever $a \rightarrow 0$, also $\rho \rightarrow 0$. While mathematically this is no contradiction, from a physical viewpoint, it is not acceptable. As explained above the scale factor measures the variation in time of the volume of a region in space, while ρ is the density of mass in a given region. If we consider that a total mass M is localized in a region V at time $t = t_0$, if a tends to 0 and the volume of V also tends to 0, then we expect the density to diverge, while, if $\gamma < -1$, it should vanish. For this reason, we discard completely all values $\gamma < -1$.

Flat Spatial Section $\kappa = 0$: We can solve by separation of variables the first Friedmann equation using Equation (184) since

$$\dot{a} = \sqrt{\frac{8\pi GK}{3}} a^{-\frac{3}{2}(1+\gamma)+1} \implies a^{\frac{3}{2}(1+\gamma)-1} da = \sqrt{\frac{8\pi GK}{3}} dt \implies d(a^{\frac{3}{2}(1+\gamma)}) = \sqrt{\frac{8\pi GK}{3}} dt. \quad (185)$$

If $\gamma > -1$ then we obtain

$$a(t) = \sqrt{\frac{8\pi GK}{3}} (t - t_0)^{\frac{2}{3(1+\gamma)}}, \quad (186)$$

where $a(t_0)$ is an integration constant. The case $\gamma = -1$ is special and we will investigate separately in Section 6.1.

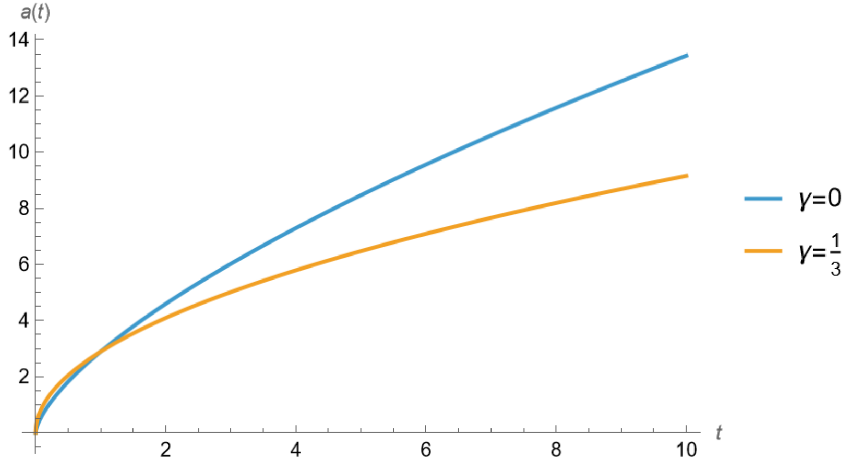


Figure 1: Plot of Equations (187) and (189) assuming $t_0 = 0$ and $GK = 1$.

Let us now specify two values of γ which have a distinguished physical interpretation when working with perfect fluids:

1. **Matter Dominated Universe $\gamma = 0$:** In this case we are considering a gas of non-colliding particles, yielding, therefore, a vanishing pressure. As mentioned above, at a cosmological level, this is a good approximation when we consider at very large scales galaxies which are far apart and whose volume is negligible with respect to the distances involved. In this case the dynamics is fully determined by the equations:

$$\rho a^3 = K \quad \text{and} \quad a(t) = \sqrt{\frac{8\pi GK}{3}} (t - t_0)^{\frac{2}{3}}. \quad (187)$$

Using the conformal time τ such that $d\tau = a^{-1}dt$ in place of the cosmological one, we obtain that

$$\rho a^3 = K \quad \text{and} \quad a(\tau) = \sqrt{12\pi GK} (\tau - \tau_0)^2. \quad (188)$$

2. **Radiation Dominated Universe $\gamma = \frac{1}{3}$:** In this case we are considering a gas of relativistic massless particles, for example photons. This is considered a reasonably good model in the early stage of the Universe where quantum effects have become negligible, that is post-inflation, baryogenesis and nucleosynthesis, but large scale structures have not been formed yet. Hence the Universe is filled only by highly energetic particles. In this case the dynamics is fully determined by the equations:

$$\rho a^4 = K \quad \text{and} \quad a(t) = \sqrt{\frac{8\pi GK}{3}}(t - t_0)^{\frac{1}{2}}. \quad (189)$$

Using the conformal time τ such that $d\tau = a^{-1}dt$ in place of the cosmological one, we obtain that

$$\rho a^4 = K \quad \text{and} \quad a(\tau) = \frac{4\pi GK}{3}(\tau - \tau_0). \quad (190)$$

Remark 47. A rather famous consequence of Equation (184) is that, for all $\gamma > -1$, then

$$a(\bar{t}) = 0 \quad \text{if} \quad \bar{t} = \left(-\sqrt{\frac{3}{8\pi GK}} a(t_0) \right)^{\frac{3(1+\gamma)}{2}}.$$

Hence at $\bar{t}$ the spatial coefficients of a FLRW metric vanish and, hence, it is not invertible. This can be inferred also from Figure 1. More importantly, using Equation (175), we can infer that the scalar curvature tends to diverge as the scale factor tends to vanish. This is the very famous **Big Bang Singularity**. Observe that, if $t \rightarrow \infty$, the scale factor is always growing and therefore the Universe is expanding eternally with a power law.

One might wonder whether the conclusions of Remark 47 depend critically on the assumption that $\kappa = 0$, hence we are considering flat spatial sections. Solving the Friedmann's equations for all cases contemporary is no longer possible and we need to analyze each single case separately.

Open (or hyperbolic) Spatial Section $\kappa = -1$: In this scenario Equation (184) is still valid, but it turns out that it is more convenient to work with the conformal time τ such that $d\tau = a^{-1}dt$, see Remark 45.

1. **Matter Dominated Universe $\gamma = 0$:** In this case, using Equation (179), the dynamics is fully determined by the equation

$$\rho a^3 = K \quad \text{and} \quad \dot{a} = \sqrt{\frac{8\pi GK}{3a} + 1} \implies \frac{da}{d\tau} = a \sqrt{\frac{8\pi GK}{3a} + 1},$$

whose solution, setting $C \doteq \frac{4\pi GK}{3}$ is

$$\tau - \tau_0 = \int_0^a \frac{dx}{\sqrt{2Cx + x^2}} = \ln \left(\frac{a + C + \sqrt{a(2C + a)}}{C} \right) = \cosh^{-1} \left(\frac{a}{C} + 1 \right).$$

$$a(\tau) = \frac{4\pi GK}{3} (\cosh(\tau - \tau_0) - 1). \quad (191)$$

2. **Radiation Dominated Universe $\gamma = \frac{1}{3}$:** In this case, using Equation (179), the dynamics is fully determined by the equation

$$\rho a^4 = K \quad \text{and} \quad \dot{a} = \sqrt{\frac{8\pi GK}{3a^2} + 1} \implies \frac{da}{d\tau} = a \sqrt{\frac{8\pi GK}{3a^2} + 1} = \sqrt{\frac{8\pi GK}{3} + a^2},$$

whose solution is

$$\rho a^4 = K \quad \text{and} \quad a(\tau) = \sqrt{\frac{8\pi GK}{3}} \sinh(\tau - \tau_0). \quad (192)$$

Remark 48. *The above calculation shows a behaviour very similar to that observed in the case $\kappa = 0$. As a matter of fact, in both scenarios as $\tau \rightarrow \tau_0$, the scale factor tends to 0 and we get yet another Big Bang scenario with a singularity occurring. Similarly, for large times the expansion does not stop and actually, it increases exponentially.*

Closed (or spherical) Spatial Section $\kappa = 1$: In this scenario Equation (184) is still valid, but, as in the previous case, it turns out that it is more convenient to work with the conformal time τ such that $d\tau = a^{-1}dt$, see Remark 45.

1. **Matter Dominated Universe $\gamma = 0$:** In this case, using Equation (179), the dynamics is fully determined by the equation

$$\rho a^3 = K \quad \text{and} \quad \dot{a} = \sqrt{\frac{8\pi GK}{3a} - 1} \implies \frac{da}{d\tau} = a \sqrt{\frac{8\pi GK}{3a} - 1},$$

whose solution, setting $C \doteq \frac{4\pi GK}{3}$ is

$$\tau - \tau_0 = \int_0^a \frac{dx}{\sqrt{2Cx - x^2}} = \cos^{-1} \left(1 - \frac{a}{C} \right).$$

$$a(\tau) = \frac{4\pi GK}{3} (1 - \cos(\tau - \tau_0)). \quad (193)$$

2. **Radiation Dominated Universe $\gamma = \frac{1}{3}$:** In this case, using Equation (179), the dynamics is fully determined by the equation

$$\rho a^4 = K \quad \text{and} \quad \dot{a} = \sqrt{\frac{8\pi GK}{3a^2} - 1} \implies \frac{da}{d\tau} = a \sqrt{\frac{8\pi GK}{3a^2} - 1} = \sqrt{\frac{8\pi GK}{3} - a^2},$$

whose solution is

$$\rho a^4 = K \quad \text{and} \quad a(\tau) = \sqrt{\frac{8\pi GK}{3}} \sin(\tau - \tau_0). \quad (194)$$

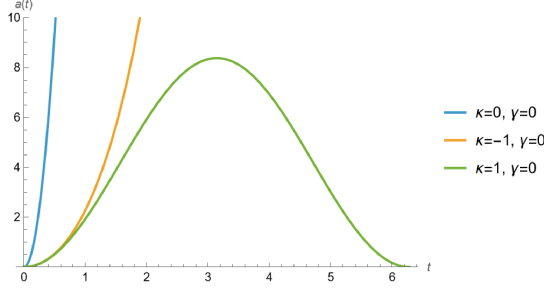


Figure 2: Profile of the scale factor for a Universe filled with dust with all three possible spatial sections setting $\tau_0 = 0$ and $KG = 1$.

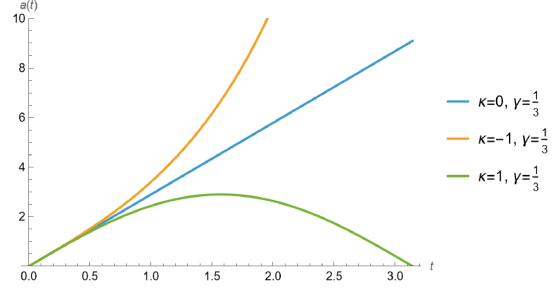


Figure 3: Profile of the scale factor for a Universe filled with dust with all three possible spatial sections setting $\tau_0 = 0$ and $KG = 1$.

Remark 49. Both Equations (193) and (194) display the usual behaviour as $\tau \rightarrow \tau_0$, namely the scale factor vanishes and there is a Big Bang singularity. Yet, since all the solutions are written in terms of trigonometric functions, we can infer that also when $\tau = \tau_0 + \pi$, $a(\tau) = 0$. In other words the Universe starts expanding and it collapses under its own gravitational force. This scenario is known as **Big Crunch**. Experimental data seem to suggest that the flat spatial section scenario $\kappa = 0$ seems favored by current observations... well, if a big Crunch would occur, what would happen to Cthulhu? Unbearable!

The Red Shift: As a last bit of information that we can extract from a FLRW spacetime, we introduce a notable variable which is commonly used in cosmology when measure distances and related properties. The starting point is the observation that, in addition to several Killing fields, the metric in Equation (171) come endowed also with a Killing tensor, see Definition 82:

$$K_{\mu\nu} = a^2(g_{\mu\nu} + U_\mu U_\nu), \quad (195)$$

where $a(t)$ is the scale factor appearing in the metric while $U \in \Gamma(TM)$ is such that $U = \frac{\partial}{\partial t}$.

Exercise 138. Prove that $\nabla_{(\rho} K_{\mu\nu)} = 0$.

Using Exercise 94, we can infer that, if we consider a point particle moving along a geodesic γ whose tangent vector is denoted by V then

$$a^2(V_\mu V^\mu + (U_\mu V^\mu)^2) = \text{const.} \quad (196)$$

Assuming that the geodesic is lightlike, *e.g.* considering a photon, then $g(V, V) = V^\mu V_\mu = 0$, which entails that, denoting by K the constant on the right hand side of Equation (196),

$$(U_\mu V^\mu)^2 = \frac{K}{a^2}.$$

Since $U = \frac{\partial}{\partial t}$ the left hand side is nothing but the component V^0 , which can be interpreted as the frequency ω (or $-\omega$ depending on the signature convention) of the photon. This entails that, if we emit a massless particle at time $t = t_i$ and observe it at time $t = t_f$, then

$$\frac{\omega_f}{\omega_i} = \frac{a(t_i)}{a(t_f)}.$$

Switching to wavelengths λ_f and λ_i , it is convenient to introduce the **redshift** between two events:

$$z_i = \frac{\lambda_f - \lambda_i}{\lambda_i}. \quad (197)$$

If we observe a photon or a massless particle now setting $a_f = 1$, then we can construct the scale factor at the time of its emission using Equation (197) via the relation:

$$a_i = \frac{1}{1 + z_i}.$$

6.1 The Cosmological Constant

At the end of Section 4 we enumerated three open questions concerning Einstein's equations. We have addressed the first two, while leaving open the third one, namely whether the form of Equation (104) is the most general one we can conceive. To be precise, the question concerns the left hand side, since the stress-energy tensor is not under discussion. Hence, we are wondering whether the Einstein tensor G , see Equation (99) can be replaced with a different or more general expression.

From a structural viewpoint, being the stress-energy tensor divergence free, so must by the geometric counterpart on the left hand side of Einstein's equation. This reduces our option, but not by much. A second, physically motivated constraint, lies in the interpretation of the metric tensor as the key kinematic and dynamical variable/field of the theory. In this respect, we look only for tensors which depend only on the metric coefficients and on its derivatives up to second order and such that we can recover Newtonian gravity in a suitable limit as discussed in Section 5.1.

This limits considerably our options but it does not oblige us to consider only the Einstein tensor. It is nowadays a rather interesting historical curiosity that this problem was investigated by Einstein himself, since, at that time, there was no experimental evidence about an expanding universe and the Friedmann Equations (179) seemed to suggest that such a behaviour was unavoidable – remember that QFT was still under development and much in its infancy. Starting from these premises, we observe that, if we consider symmetric, divergence free, $(0,2)$ -tensors which depend only on the metric coefficients up to the second derivative, we have only two options:

$$G_{\mu\nu} \quad \text{and} \quad g_{\mu\nu},$$

that is the Einstein tensor and the metric itself. This suggests the following modified form of the Einstein's equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (198)$$

where $\Lambda \in \mathbb{R}$ is known as **cosmological constant**. Before discussing whether there are experimental reasons to accept a non vanishing value of Λ , it is worth looking at the immediate consequence of this additional leeway. First and foremost, it is imperative that the structural results obtained with $\Lambda = 0$ are still valid when $\Lambda \neq 0$. This turns out to be the case, but it is important to check it and we leave it to our readers with the following two exercises.

Exercise 139. Let (M, g) be an n -dimensional spacetime as per Definition 52. Show that the left hand side of Equation (198) can be obtained as the critical point of the variation of the following modified Einstein-Hilbert action

$$S_{EH}[g] = \int_M \omega_g R \equiv \int_M d^n x \sqrt{|g|} (R + \Lambda), \quad (199)$$

Exercise 140. Discuss the Newtonian limit of Equation (198) repeating the analysis of Section 5.1.

A different issue that we shall discuss in these notes concerns instead the necessity of “replacing” Minkowski spacetime. More precisely, if one thinks at the analysis of Section 5.1, we considered an expansion of the metric of the form

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(2).$$

The reason for considering perturbation around Minkowski spacetime stems from the fact that η is a solution of Einstein’s vacuum equations, a condition which is reasonable when we study perturbation theory for Equation (138) since, as we commented, this entails that the stress-energy tensor contributes only starting from the first order. Yet, there are several solutions of vacuum Einstein’s equations and the choice of Minkowski is, strictly speaking, not necessary from a mathematical viewpoint. It is motivated by physics observing that this is the natural background in special relativity and, in addition it is *maximally symmetric*.

Given an n -dimensional spacetime (M, g) , $\text{Iso}(M, g)$ as per Definition 67 is a Lie group whose dimension cannot exceed the number of independent components of the metric, namely $\dim \text{Iso}(M, g) \leq \frac{n(n+1)}{2}$. We call maximally symmetric a spacetime where the dimension of the isometry group is exactly $\frac{n(n+1)}{2}$. This is the indeed the case for Minkowski spacetime if we look at the underlying Poincaré group. In addition, it is not difficult to realize that Poincaré covariance of the metric automatically determines it completely and unambiguously.

Exercise 141. Let (M, g) be a 4-dimensional spacetime. Prove that, if $\text{Iso}(M, g) = \text{Poin}_4$, where Poin_4 is the Poincaré group, then (M, g) must be isometric to $(\mathbb{R}^4, \eta)$. Generalize this result to n -dimensions with $n \geq 2$.

If we allow for a cosmological constant to be present, the situation changes drastically since we cannot consider any longer Minkowski spacetime. As a matter of fact the new vacuum Einstein’s equation become

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0 \implies \left(\frac{1}{2} - \frac{1}{n} \right) R = \Lambda,$$

where we have taken the metric-trace of both sides. Hence, for $n > 2$, this is incompatible with Minkowski spacetime, which has vanishing scalar curvature, unless $\Lambda = 0$. This prompts the question of what replaces Minkowski spacetime as the maximally symmetric background if $\Lambda \neq 0$. The result is different depending on the sign of Λ . In the following we shall discuss $\Lambda > 0$ commenting only at the end on the other case. In order to address this question, there are several possibilities and unfortunately the literature is not particularly rich in this respect despite the appearances. An old, yet still the most complete reference on this topic is [HaEl73, §5.2], but here we follow a slightly different approach.

De Sitter spacetime: Let us consider a spacetime (M, g) of dimension $\dim M > 2$ and a cosmological constant $\Lambda \neq 0$. We are interested in looking for vacuum solutions of Equation (198), namely

$$G_{\mu\nu} = -\Lambda g_{\mu\nu}.$$

By putting the contribution of the cosmological constant on the right hand side, we are suggesting the option to interpret this term as a geometric contribution to the stress-energy tensor, namely the vacuum Einstein's equations with a non vanishing cosmological constant are formally equivalent to Equation (138) with

$$T_{\mu\nu} = -\Lambda g_{\mu\nu}.$$

Yet this form of the stress-energy tensor is not new to us if we focus on $\dim M = 4$. A close inspection of Equation (177) unveils that, if we consider a perfect fluid,

$$\rho = -P \implies T_{\mu\nu} = -\rho g_{\mu\nu}. \quad (200)$$

In addition we need to take into account the conservation law in Equation (178) which entails

$$\dot{\rho} = 0 \implies \rho(t) = \rho(0).$$

In other words we have established that

A FRLW Universe with equation of state $P = -\rho$ is equivalent to the vacuum Einstein's equation with $\Lambda = \rho$.

On the one hand this correspondence does not guarantee a priori that solving the Friedmann equations will give us a maximally symmetric solution, but, on the other hand, it tells us that the cosmological constant behaves like a rather exotic matter component. If $\Lambda > 0$, it is like a perfect fluid with a negative pressure, that is, we expect that it facilitates the expansion of the Universe. If $\Lambda < 0$ this behaviour is reversed but we have to account for negative energy densities. In comparison to Section 6 we have to study the Friedmann equations both for flat, open and closed spatial sections for the only equation of state that we left out from the previous analysis. In the following we fill this gap.

Flat Spatial Sections, $\kappa = 0$: Let us consider Equation (181) setting $\rho(t) = \Lambda > 0$. It descends

$$H^2 = \frac{8\pi G\Lambda}{3} \implies \frac{d \ln(a(t))}{dt} = \pm \sqrt{\frac{8\pi G\Lambda}{3}} \implies a_{\pm}(t) = a(t_0)e^{\pm H_0(t-t_0)}, \quad (201)$$

where $H_0 = \sqrt{\frac{8\pi G\Lambda}{3}}$ is sometimes called Hubble constant and where $a(t_0)$ is an integration constant. As we can see we get two possible solutions, whose line element reads setting $t_0 = 0$ and $a(0) = a_0$

$$ds^2 = -dt^2 + a_0^2 e^{\pm 2H_0 t} (dx^2 + dy^2 + dz^2). \quad (202)$$

This describes either an exponentially fast, expanding (+) or a shrinking (−) Universe with no Big Bang singularity at finite time since the domain of the cosmological time variable is $t \in \mathbb{R}$. Both are solutions of the Friedmann equation but, in the literature, it is commonly used the one with the plus sign, which is also called **cosmological (four-dimensional) de Sitter spacetime**.

Open Spatial Sections, $\kappa = -1$: The second option that we consider are FLRW spacetimes with hyperbolic constant time hypersurfaces. Starting from Equation (179) setting $\rho(t) = \Lambda > 0$ and for simplicity an initial condition $a(t_0) = 0$ for $t_0 \in \mathbb{R}$, we obtain

$$\dot{a} = \pm \sqrt{\frac{8\pi G\Lambda a^2}{3} + 1} \implies a_{\pm}(t) = \pm \sqrt{\frac{3}{8\pi G\Lambda}} \sinh \left(\sqrt{\frac{8\pi G\Lambda}{3}} (t - t_0) \right). \quad (203)$$

In this case we obtain two solutions which are actually describing the same spacetime if we observe that the hyperbolic sine is an odd function. More precisely, since $a(t)$ must be positive, setting for simplicity $t_0 = 0$, we should consider a_+ if $t > 0$ and a_- if $t < 0$. The ensuing line element reads

$$ds^2 = -dt^2 + \frac{3}{8\pi G\Lambda} \sinh^2 \left(\sqrt{\frac{8\pi G\Lambda}{3}} (t - t_0) \right) (d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2)), \quad (204)$$

which displays a vanishing scale factor as $t \rightarrow t_0$. In this case the metric requires the cosmological time to be defined either on the interval $(-\infty, t_0)$ or on (t_0, ∞) . Yet, as we will clarify later, we should not call $t = t_0$ a Big Bang singularity. In the physics literature this second case is usually considered and it describes an exponentially expanding Universe.

Closed Spatial Sections, $\kappa = 1$: The third and last option that we consider are FLRW spacetimes with spherical constant time hypersurfaces. Starting from Equation (179) setting $\rho(t) = \Lambda > 0$ and for simplicity an initial condition $a(t_0) = 0$ for $t_0 \in \mathbb{R}$, we obtain

$$\dot{a} = \pm \sqrt{\frac{8\pi G\Lambda a^2}{3} - 1} \implies a(t) = \sqrt{\frac{3}{8\pi G\Lambda}} \cosh \left(\sqrt{\frac{8\pi G\Lambda}{3}} (t - t_0) \right). \quad (205)$$

In this case we obtain one single scale factor since we exploit that the hyperbolic cosine is an even function. The ensuing line element reads

$$ds^2 = -dt^2 + \frac{3}{8\pi G\Lambda} \cosh^2 \left(\sqrt{\frac{8\pi G\Lambda}{3}} (t - t_0) \right) (d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2)), \quad (206)$$

which is well-defined for all $t \in \mathbb{R}$ and it displays an exponential growth both at $t \rightarrow \pm\infty$. This metric is sometimes referred to as **(full) four-dimensional de Sitter metric dS_4** .

de Sitter as an homogeneous manifold: In the previous discussion, we have identified possible candidates for replacing the rôle of Minkowski spacetime in four dimensions, although it is not clear whether any of them is maximally symmetric and how they can be generalized to any dimension greater than 2. Equation (203) does not seem to be up to the task since it individuates two distinct solutions with a time coordinate defined only on a half line. Actually, it turns out that both this solution as well as Equation (201) and Equation (205) describe the same spacetime, though with different coordinate patches. In order to make this statement precise, one should discuss the theory of homogeneous manifolds, a rather advanced and interesting topic at the interface between differential geometry and the theory of Lie groups. A complete reference is [Hel01], but here we shall refrain going into the details, presenting only the geometric construction of de Sitter spacetime.

Definition 142. *Given the $(n+1)$ -dimensional Minkowski spacetime $(\mathbb{R}^{n+1}, \eta)$ with Cartesian coordinates $(X^0, \dots, X^n)$, $n \geq 2$, we call **(n-dimensional) de Sitter spacetime dS_n** , the hyperboloid*

$$-(X^0)^2 + \sum_{i=1}^n (X^i)^2 = \ell^2, \quad (207)$$

where $\ell > 0$ is called de Sitter radius, endowed with the induced metric.

Remark 50. *In the language of Lie groups we can recognize that dS_n is a homogeneous manifold, namely it is diffeomorphic to the quotient $dS_n \simeq \frac{SO(n,1)}{SO(n-1,1)}$ where $SO(n,1)$ is the proper Lorentz group on the $(n+1)$ -dimensional Minkowski spacetime. In this respect we can immediately realize that $SO(n,1)$ is acting transitively on dS_n . Consequently, being a subgroup of all isometries of $(\mathbb{R}^{n+1}, \eta)$ and being dS_n endowed with the induced metric, we can conclude that $SO(n,1) \subseteq \text{Iso}(dS_n, g|_{dS_n})$. Yet, since $\dim SO(n,1) = \frac{n(n+1)}{2}$ and since $\dim \text{Iso}(M, g) \leq \frac{n(n+1)}{2}$ for every n -dimensional spacetime, we can conclude that $SO(n,1)$ is the full isometry group of dS_n , which is, therefore, maximally symmetric.*

Consider now the following change of coordinates on the hyperboloid in Equation (207):

$$\begin{cases} X^0 = \ell \sinh\left(\frac{t}{\ell}\right) \\ X^1 = \ell \cosh\left(\frac{t}{\ell}\right) \cos \theta^1 \\ X^2 = \ell \cosh\left(\frac{t}{\ell}\right) \sin \theta^1 \cos \theta^2 \\ \dots \\ X^n = \ell \cosh\left(\frac{t}{\ell}\right) \sin \theta^1 \sin \theta^2 \dots \sin \theta^{n-1} \end{cases}, \quad (208)$$

where $(\theta_1, \dots, \theta_{n-1})$ are coordinates on the unit $(n-1)$ -sphere $\mathbb{S}^{n-1}$. If we use these coordinates on the line element

$$ds^2|_{\mathbb{R}^{n+1}} = -d(X^0)^2 + d(X^1)^2 + \dots + d(X^n)^2,$$

we end up with

$$ds^2 = -dt^2 + \ell^2 \cosh\left(\frac{t}{\ell}\right) (d\mathbb{S}_{n-1}^2), \quad (209)$$

where $d\mathbb{S}_{n-1}^2$ is the metric of the unit $(n-1)$ -sphere. Using the coordinates $\theta^1, \dots, \theta^{n-1}$ it reads

$$d\mathbb{S}_{n-1}^2 = d(\theta^1)^2 + \sin^2 \theta^1 (d(\theta^2)^2 + \sin^2(\theta^2) (\dots (d\theta^{n-2} + \sin^2(\theta^{n-2}) d(\theta^{n-2})^2))).$$

A direct comparison in $n = 4$ between Equation (209) and Equation (206) unveils that de Sitter spacetime is exactly a FLRW spacetime with closed spatial sections.

Exercise 143. Consider de Sitter spacetime as per Definition 142 and the following change of coordinates on the hyperboloid in Equation (207):

$$\begin{cases} X^0 = \ell \sinh\left(\frac{t}{\ell}\right) + \frac{r^2}{2\ell} e^{\frac{t}{\ell}} \\ X^1 = \ell \cosh\left(\frac{t}{\ell}\right) - \frac{r^2}{2\ell} e^{\frac{t}{\ell}} \\ X^2 = \ell \cosh\left(\frac{t}{\ell}\right) x^1 \\ \dots \\ X^n = \ell \cosh\left(\frac{t}{\ell}\right) x^{n-1} \end{cases}, \quad (210)$$

where $x^i \in \mathbb{R}$ for all $i = 1, \dots, n-1$ and where $r^2 = \sum_{i=1}^{n-1} (x^i)^2$. Prove that the metric induced on dS_n if $n = 4$ is equivalent to Equation (201) with the identification $\ell = \frac{3}{8\pi G\Lambda}$.

Exercise 144. Consider de Sitter spacetime as per Definition 142 and the following change of coordinates on the hyperboloid in Equation (207):

$$\begin{cases} X^0 = \ell \sinh\left(\frac{t}{\ell}\right) \cos \chi \\ X^1 = \ell \cosh\left(\frac{t}{\ell}\right) \\ X^2 = \ell \cosh\left(\frac{t}{\ell}\right) \sin \chi \cos \theta^1 \\ X^3 = \ell \cosh\left(\frac{t}{\ell}\right) \sin \chi \sin \theta^1 \cos \theta^2 \\ \dots \\ X^n = \ell \cosh\left(\frac{t}{\ell}\right) \sin \chi \sin \theta^1 \sin \theta^2 \dots \sin \theta^{n-2} \end{cases}, \quad (211)$$

where $\chi \in \mathbb{R}$ while $(\theta_1, \dots, \theta_{n-2})$ are coordinates on the unit $(n-2)$ -sphere $\mathbb{S}^{n-2}$. Prove that the metric induced on dS_n if $n = 4$ is equivalent to Equation (205) with the identification $\ell = \frac{3}{8\pi G\Lambda}$.

Experimental Evidence for $\Lambda > 0$: While claiming that our Universe is filled with matter and with radiation is based on straightforward evidences, introducing a cosmological constant seems rather unnatural. The original motivations which lead Einstein to considering it are long gone since he believed in a non-expanding Universe, to the point that he introduced in 1917 what is nowadays known as *Einstein Static Universe*. This is a four-dimensional spacetime (M, g) isometric to $\mathbb{R} \times \mathbb{S}^3$ and endowed with a metric whose associated line element is

$$ds^2 = -dt^2 + R^2 (d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2)), \quad (212)$$

where $t \in \mathbb{R}$ while (χ, θ, φ) are the standard angular coordinates on the unit 3-sphere $\mathbb{S}^3$. In addition $R > 0$ is a constant which can be interpreted as the radius of the Universe at each instant of time. Although we have not yet defined the notion of static spacetime, we can still infer from Example 83 that $\frac{\partial}{\partial t}$ is a Killing vector field and there is also invariance under time reversal $t \rightarrow -t$. In order for Equation (212) to be a solution of Einstein's equation with a positive energy density, it was necessary to introduce a cosmological constant $\Lambda = R^{-2}$ in the equations. Yet Hubble's observation in 1929 on the redshift of galaxies proved incontrovertibly that the Universe is expanding. This prompted the adoption of the FLRW models as the standard background in cosmology. At the same time it relegated the cosmological constant to a mere mathematical curiosity.

Under these premises, if we look at the first Friedmann equation (179), it is convenient to rewrite it in terms of the density parameter $\rho_{crit,0}$ where the subscript 0 refers to the fact that we replace in Equation (182) the value of H with the Hubble parameter H_0 measured at a fixed time $t = t_0$ (for example now):

$$\frac{H^2}{H_0^2} = \frac{\rho}{\rho_{crit,0}} - \frac{\kappa}{a^2 H_0^2} = \frac{\rho}{\rho_{crit,0}} + \Omega_\kappa a^{-2}, \quad (213)$$

where we have introduced

$$\Omega_\kappa \doteq -\frac{\kappa}{H_0^2}. \quad (214)$$

Let us now make the assumption that the density ρ in Our Universe is the sum of the contribution of three components: dust, radiation and the cosmological constant. Recalling Equations (187), (189) as well as the identification $\rho \equiv \Lambda$ if we work with the equation of state $\rho = -P$, then Equation (213) can be rewritten as

$$\frac{H^2}{H_0^2} = \Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \Omega_\kappa a^{-2}, \quad (215)$$

where, assuming for simplicity that at time t_0 , $a(t_0) = 1$, the density matter and radiation are respectively $\rho_{m,0}$ and $\rho_{crit,0}$ while

$$\Omega_m = \frac{\rho_{m,0}}{\rho_{crit,0}} \quad \text{and} \quad \Omega_r = \frac{\rho_{r,0}}{\rho_{crit,0}}.$$

If we evaluate Equation (215) at time $t = t_0$, we get a balance law:

$$1 = \Omega_m + \Omega_r + \Omega_\Lambda + \Omega_\kappa. \quad (216)$$

One of the most striking successes of modern day cosmology has been a precise measurement of these four parameters. We refer a reader interested in these aspects of gravitational physics to specialized textbooks, *e.g.* [Bau22] on which this paragraph is based. We summarize here the main facts:

1. in 1964 Penzias and Wilson (not the ball of Cast Away! Did you watch the movie?) discovered that the Universe is filled with a background radiation, that is very weakly interacting photons, at an average temperature of approximately $T=2.725\text{K}$. This is known as **Cosmic Microwave Background (CMB)** and it allows for very good estimates of $\Omega_r \sim 10^{-4}$. Hence the contribution of radiation to the present day density content of matter and energy in the Universe is very small,
2. in 1992 the COBE experiment (not to be misunderstood with Kobe beef) unveiled anisotropies in the angular distribution of the CMB although relative fluctuations of the temperature $\frac{\Delta T}{T}$ are of the order of 10^{-5} . The precise shape of the fluctuations and in particular their peaks depend on the underlying cosmological model. Using FLRW spacetimes yields a very interesting estimate, namely

$$|\Omega_\kappa| \leq 0.095.$$

On the one hand this tells us that the contribution of spatial curvature to Equation (216) is negligible, while, on the other hand, it points towards favoring the value $\kappa = 0$.

3. The observed distribution of visible galaxies, combined with the theory of nucleosynthesis and with the data of CMB, suggests that visible matter is mostly constituted by baryons and it amounts to

$$\Omega_b \sim 0.05,$$

that is at most 5% of the total energy density. We employed the subscript b since it turns out that, still the data of CMB, but also other experimental evidences ranging from the rotational curves of galaxies to the dynamics of entire clusters of galaxies suggest that

$$\Omega_m \sim 0.32.$$

In other words the Universe appears to be filled by weakly interacting matter, known as **(Cold) Dark Matter (CDM)** which constitutes 27% of the total energy density of the Universe and around 80% of the existing matter. There are dozens of models trying to explain dark matter, but, as of 2025, there is no evidence that such matter can be described in terms of standard particles as we understand them in quantum field theory. Attempts to modify general relativity to account for this effect are almost a daily routine, but they must face very precise constraints and experimental data coming from Einstein's theory. Among these very popular are MO(dified) N(ewtonian) D(ynamics) theories, since they provide an alternative explanation to dark matter. Yet no experimental evidence supporting MONDs is available.

4. Having constrained experimentally three out of the four parameters in Equation (216), we can infer that

$$\Omega_\Lambda \sim 0.68,$$

namely 68% of the total energy density of our Universe has to be ascribed to the cosmological constant which is also known as **Dark Energy**. We observe that, in an expanding

Universe, as the scale factor a grows, Equation (216) is telling us that the contribution from dark energy is becoming more and more dominant. Hence, according to the assumptions at the heart of FLRW models, we can conclude that, if $\kappa = 0$, then at late cosmological times, the best description of the Universe is given by the *cosmological de Sitter metric* as in Equation (202), which suggests a phase of exponential growth, implementing therefore a forced social distancing at a cosmological scale.

Remark 51. *One might wonder why the contribution from the cosmological constant is dubbed dark energy. From the viewpoint of Einstein's theory Λ is a fundamental constant of nature much in the same spirit of c , G or $\hbar$. Yet, one needs to take into account that, if one takes seriously, as he or she should, the standard model of particles, the description of matter in terms of a perfect fluid is a very crude approximation. One should instead describe matter in terms of classical first and then of quantum fields. Apparently one might think that this is an overkill as quantum effects are important only on very small distances and they should be negligible at cosmological scales. Yet, since the contribution from fields enters the Einstein's equations via the stress-energy tensor which is quadratic in the fields themselves, even in absence of interaction, at a quantum level, the expectation value of this observable is ill-defined. This can be bypassed by what is known in quantum field theory as a regularization or as a renormalization procedure depending on the taste of the reader. In the eighties such procedure was always implemented subtracting the so-called zero-point energy using a procedure based on a mode expansion which can be made precise on Minkowski spacetime. When going to a curved background many assumed and expected that the procedure could be repeated almost slavishly leading to a famous statement that this subtraction procedure introduces a contribution to the stress-energy tensor proportional to the metric which can be interpreted as mimicking the cosmological constant. Yet the value estimated using the standard model of particles is 120 order of magnitude higher than all estimates on the value of Λ . This was known for many years as the cosmological constant problem and it still reported as such in many textbooks. Truth is that our current understanding of quantum field theory on curved backgrounds which has been developed starting from the end of the nineties has unveiled that the early estimates were imprecise, since they neglected that on curved backgrounds there exist several more renormalization freedoms than in Minkowski spacetime. These are of geometric nature and, when applied to the quantization of the stress-energy tensor, it turns out that its quantum counterpart is defined only up to multiples of the metric tensor. In other words, on every curved background, there is an intrinsic renormalization freedom proportional to $\Lambda g_{\mu\nu}$. Hence our current understanding of QFT does not allow to make any prediction on the value of Λ which has to be measured therefore experimentally.*

Anti-de Sitter Spacetime: Let us consider a spacetime (M, g) of dimension $\dim M > 2$ and a cosmological constant $\Lambda < 0$. Despite the observational evidences point towards a positive cosmological constant, the negative case is not of interest only for completeness or from a mathematical viewpoint. Maximally symmetric spacetimes with $\Lambda < 0$ are known as **anti-de Sitter** spacetimes and they are the building blocks of many models in theoretical physics going under the collective name of AdS/CFT correspondence with applications ranging from

condensed matter to quantum chromodynamics (seriously, ask Bacchetta!). Also in this case introductory material is hard to find and often hidden in string theory lecture notes. Therefore once more we recommend [HaEl73, §5.2].

Following the same rationale as before our starting point is Equation (200), which suggests to look for FLRW spacetimes whose scale factor is ruled by the first Friedmann equation (179). Yet, as we shall see, two out of the three possible spatial sections must be discarded.

Flat Spatial Sections, $\kappa = 0$: Let us consider Equation (181) setting $\rho(t) = \Lambda < 0$. It descends

$$H^2 = \frac{8\pi G\Lambda}{3} \implies \frac{d\ln(a(t))}{dt} = \pm \sqrt{\frac{8\pi G\Lambda}{3}}, \quad (217)$$

which must be discarded since the term under square root is negative. Hence this implies that the solutions of Equation (217) are complex exponential. This cannot be accepted since scale factors must be positive and smooth functions of the cosmological time t .

Closed Spatial Sections, $\kappa = 1$: The second option that we consider are FLRW spacetimes with spherical constant time hypersurfaces. Starting from Equation (179) setting $\rho(t) = \Lambda < 0$, we obtain

$$\dot{a} = \pm \sqrt{\frac{8\pi G\Lambda a^2}{3} - 1}, \quad (218)$$

which must be discarded since the term under square root is negative. Also in this case the solution of this ordinary differential equation is purely imaginary. This cannot be accepted since scale factors must be positive and smooth functions of the cosmological time t .

Open Spatial Sections, $\kappa = -1$: The last option that we consider are FLRW spacetimes with hyperbolic constant time hypersurfaces. Starting from Equation (179) setting $\rho(t) = \Lambda < 0$ and for simplicity an initial condition $a(t_0) = 0$ for $t_0 \in \mathbb{R}$, we obtain

$$\dot{a} = \pm \sqrt{\frac{8\pi G\Lambda a^2}{3} + 1} \implies a_{\pm}(t) = \pm \sqrt{\frac{3}{8\pi G|\Lambda|}} \sin \left(\sqrt{\frac{8\pi G|\Lambda|}{3}}(t - t_0) \right). \quad (219)$$

In this case we obtain two solutions which are actually describing the same spacetime if we observe that the hyperbolic sine is an odd function. More precisely, since $a(t)$ must be positive, setting for simplicity $t_0 = 0$, we should consider a_+ if $t > 0$ and a_- if $t < 0$. The ensuing line element reads

$$ds^2 = -dt^2 + \frac{3}{8\pi G\Lambda} \sin^2 \left(\sqrt{\frac{8\pi G\Lambda}{3}}(t - t_0) \right) (d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\varphi^2)). \quad (220)$$

This metric displays a periodic scale factor which vanishes for $t = t_0 + \sqrt{\frac{3}{8\pi G\Lambda}} n\pi$, $n \in \mathbb{Z}$. This is known as **four-dimensional anti-de Sitter spacetime (AdS₄)**. As in the case of de Sitter, we are not facing Big-Bang time singularities, but nonetheless AdS_4 displays a rather pathological behaviour since we can construct closed timelike curves, such as for example $\gamma(t) = (t, 0, 0, 0)$.

Anti-de Sitter as an homogeneous manifold: In the previous discussion, we have identified a possible candidate for replacing the rôle of Minkowski spacetime in four dimensions, although it is not clear whether it is maximally symmetric. As in the de Sitter case, an answer can be found using the theory of homogeneous manifolds. Here we will only review the main definition, leaving an interested reader to [Hel01].

Definition 145. Consider the $(n+1)$ -dimensional spacetime $(\mathbb{R}^{n+1}, g)$ whose metric in Cartesian coordinates $(X^0, \dots, X^n)$, $n \geq 2$, reads

$$ds^2 = -d(X^0)^2 - d(X^1)^2 + d(X^2)^2 + \dots + d(X^n)^2. \quad (221)$$

We call **(n-dimensional) anti-de Sitter spacetime** AdS_n , the hyperboloid

$$-(X^0)^2 - (X^1)^2 + \sum_{i=2}^n (X^i)^2 = -\ell^2, \quad (222)$$

where $\ell > 0$ is called de Sitter radius, endowed with the induced metric.

Remark 52. In the language of Lie groups we can recognize that AdS_n is a homogeneous manifold, namely it is diffeomorphic to the quotient $dS_n \simeq \frac{O(n-1,2)}{O(n-1,1)}$. Here $O(n-1,2)$ is the generalized orthogonal group, the Lie group of all linear transformations of an n -dimensional real vector space that leave invariant a nondegenerate, symmetric bilinear form of signature $(n-1, 2)$. In this respect we can immediately realize that $O(n-1, 2)$ is acting transitively on AdS_n . Consequently, being a subgroup of all isometries of $(\mathbb{R}^{n+1}, g)$ and being AdS_n endowed with the induced metric, we can conclude that $O(n-1, 2) \subseteq \text{Iso}(AdS_n, g|_{AdS_n})$. Yet, since $\dim O(n-1, 2) = \frac{n(n+1)}{2}$ and since $\dim \text{Iso}(M, g) \leq \frac{n(n+1)}{2}$ for every n -dimensional spacetime, we can conclude that $O(n-1, 2)$ is the full isometry group of AdS_n , which is, therefore, maximally symmetric.

Consider anti-de Sitter spacetime as per Definition 145 and the following change of coordinates on the hyperboloid in Equation (222):

$$\begin{cases} X^0 = \ell \cos\left(\frac{t}{\ell}\right) \cos \chi \\ X^1 = \ell \sin\left(\frac{t}{\ell}\right) \\ X^2 = \ell \sin\left(\frac{t}{\ell}\right) \sin \chi \cos \theta^1 \\ X^3 = \ell \sin\left(\frac{t}{\ell}\right) \sin \chi \sin \theta^1 \cos \theta^2 \\ \dots \\ X^n = \ell \sin\left(\frac{t}{\ell}\right) \sin \chi \sin \theta^1 \sin \theta^2 \dots \sin \theta^{n-2} \end{cases}, \quad (223)$$

where $\chi \in \mathbb{R}$ while $(\theta_1, \dots, \theta_{n-2})$ are coordinates on the unit $(n-2)$ -sphere $\mathbb{S}^{n-2}$. If we use these coordinates on the line element in Equation (221), the metric induced on AdS_n if $n = 4$ is equivalent to Equation (220) with the identification $\ell = \frac{3}{8\pi G|\Lambda|}$. One can observe that Equation (223) does not cover the whole AdS hyperboloid as in Equation (222) since the coordinate transformations are all defined in terms of trigonometric function. As an example $X^0 \in (-\ell, \ell)$

in this coordinate system, a constraint which is not necessary in Definition 145. In the following two exercises we display two different coordinate systems, also referred to as patches, which are more commonly used in the applications in theoretical physics

Exercise 146. Consider anti-de Sitter spacetime as per Definition 145 and the following change of coordinates on the hyperboloid in Equation (222):

$$\begin{cases} X^0 = \frac{\ell^2}{2r} \left(1 + \frac{r^2}{\ell^4} (\ell^2 + \|x\|_\delta^2 - t^2) \right) \\ X^1 = \frac{r}{\ell} t \\ X^i = \frac{r}{\ell} x^{i-1} \quad i \in \{2, \dots, n-1\} \\ X^n = \frac{\ell^2}{2r} \left(1 - \frac{r^2}{\ell^4} (\ell^2 - \|x\|_\delta^2 + t^2) \right) \end{cases}, \quad (224)$$

where $r \in [0, \infty)$, $x^{i-1} \in \mathbb{R}$ for all $i = 2, \dots, n-1$ and where $\|x\|_\delta^2 = \sum_{i=1}^{n-1} (x^{i-1})^2$. Show that the line element of the AdS_n metric reads

$$ds^2 = -\frac{r^2}{\ell^2} dt^2 + \frac{\ell^2}{r^2} dr^2 + \frac{r^2}{\ell^2} (d(x^1)^2 + \dots d(x^{n-2})^2), \quad (225)$$

or, introducing $z \doteq \frac{\ell^2}{r}$

$$ds^2 = -\frac{\ell^2}{z^2} (dt^2 + dz^2 + d(x^1)^2 + \dots d(x^{n-2})^2). \quad (226)$$

Observe that this metric is conformal to the Minkowski one, though one has to impose the constraint $z > 0$, hence covering only half of $\mathbb{R}^n$. This is known as **Poincaré patch** of AdS_n and it covers only half of the hyperboloid in Equation (222), since Equation (224) entails that $X^0 + X^n = \frac{\ell}{r}$ which needs to be positive since $r \in [0, \infty)$.

Exercise 147. Consider anti-de Sitter spacetime as per Definition 145 and the following change of coordinates on the hyperboloid in Equation (222):

$$\begin{cases} X^1 = \ell \cosh \rho \cos \tau \\ X^2 = \ell \cosh \rho \sin \tau \\ X^3 = \ell \sinh \rho \sin \chi \cos \theta^1 \\ X^4 = \ell \sinh \rho \sin \chi \sin \theta^1 \cos \theta^2 \\ \dots \\ X^n = \ell \sinh \rho \sin \chi \sin \theta^1 \sin \theta^2 \dots \sin \theta^{n-3} \end{cases} \quad (227)$$

where $\rho \in \mathbb{R}$, $\tau \in [0, 2\pi)$, while $(\theta^1, \dots, \theta^{n-3})$ are the standard angular coordinates on the unit $(n-3)$ -sphere $\mathbb{S}^{n-3}$. Show that the line element of the AdS_n metric reads

$$ds^2 = \ell^2 (-\cosh^2 \rho dt^2 + d\rho^2 + \rho^2 (d\mathbb{S}_{n-2}^2)), \quad (228)$$

where $d\mathbb{S}_{n-2}^2$ is the metric of the unit $(n-1)$ -sphere:

$$d\mathbb{S}_{n-2}^2 = d(\theta^1)^2 + \sin^2 \theta^1 (d(\theta^2)^2 + \sin^2(\theta^2) (\dots (d\theta^{n-2} + \sin^2(\theta^{n-2}) d(\theta^{n-2})^2) \dots)).$$

Observe that these coordinates cover the whole AdS_n and therefore this is known as the **global AdS patch**.

7 Black Holes

In this (last?) section of these notes, we construct explicit solutions of Einstein's equations, commenting on their physical significance. This allows to introduce one of the most famous consequences of general relativity: black holes. Yet, as we alluded in the previous sections, the task of constructing explicit solutions is particularly cumbersome in presence of an arbitrary stress-energy tensor and on a generic smooth manifold. For this reason we shall limit the attention to vacuum solutions, that is $T_{\mu\nu} = 0$ and we will consider highly symmetric background, since, as we have seen in Section 6, it is an assumption which greatly simplifies the complexity of Einstein's equations. Yet, as for isotropy and homogeneity in Definitions 136 and 135, we must establish first of all a geometric characterization of the symmetries that we wish to consider. For definiteness we shall only work with four spacetime dimensions, but all the following definitions and results admit a straightforward generalization to any dimension. We leave to our beloved readers the task of doing it.

Definition 148. *Let (M, g) be a Lorentzian spacetime such that $\dim M = 4$. We say that M is*

- **spherically symmetric:** *if there exists a subgroup of $\text{Diff}(M)$ which is isomorphic to $SO(3)$ and such that, for all $p \in M$, the orbits $\mathcal{O}_p = \{g(p) \mid g \in SO(3) \subset \text{Diff}(M)\}$ are diffeomorphic to 2-spheres $\mathbb{S}^2$.*
- **axially symmetric:** *if there exists a one-parameter group of isometries as per Definitions 73 and 67, whose orbits are closed spacelike curves as per Definition 117.*
- **stationary:** *if there exists a one-parameter group of isometries as per Definitions 73 and 67, whose orbits are timelike curves as per Definition 117.*
- **static:** *if it is stationary and if there exists a spacelike hypersurface Σ orthogonal to the orbits of the isometry*

Definition 148 might appear at first look as rather unsatisfactory. On the one hand, it establishes a global notion of symmetries while, for computational purposes, we have realized that it is often better to work with Killing fields as per Definition 80. On the other hand, while the notion of being spherically- or axially symmetric do coincide with our intuitive expectations, that of being stationary and static are somehow less intuitive. At a semantic level being stationary alludes to something which does not change in time, while static refers intuitively to stillness. If one has to think of a stationary, non static system, one can refer to a rigid body in

$\mathbb{R}^3$ which is rotating with constant velocity. On the contrary a static system is one where even rotations are not allowed. Yet this disambiguation is not obvious from Definition 148. Before entering more into the details of these aspects we need a rather important and powerful result from differential geometry, which might appear at first glance unrelated. The question can be phrased heuristically as follows: Suppose that we assign two vectors $v, w \in T_p M$. Under which conditions their integral lines allow to identify an embedded surface in M ?

Frobenius Theorem: Suppose now that M is a smooth manifold as per Definition 4 without a metric structure being necessarily assigned.

Definition 149. Given a smooth n -dimensional manifold M as per Definition 4 and given for any $p \in M$ $\mathcal{D}_p \subseteq T_p M$, a k -dimensional vector subspace with $1 \leq k \leq n$, we call **(tangent) distribution**

$$\mathcal{D} \doteq \bigcup_{p \in M} \mathcal{D}_p.$$

Given (N, ψ) , an immersed submanifold of M as per Definition 40, we call it **integral manifold of $\mathcal{D}$** if $d\psi(T_q N) = \mathcal{D}_{\psi(q)}$ for all $q \in N$, where $d\psi$ is the differential of ψ as per Definition 4.

Example 150. We consider two examples of distributions abiding by Definition 149, the first being sort of trivial:

1. Let $M \simeq \mathbb{R}^n$ and, endowing $\mathbb{R}^n$ with Cartesian coordinates $x^1, \dots, x^n$, for every $p \in \mathbb{R}^n$, we set

$$\mathcal{D}_p = \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^k} \right\},$$

where $1 \leq k < n$. This is distribution whose integral manifolds are diffeomorphic to $\mathbb{R}^k$ admitting $x^1, \dots, x^k$ as coordinates.

2. Let $M \simeq \mathbb{S}^2$ and for every $p \in \mathbb{S}^2$ we set

$$\mathcal{D}_p = \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial \varphi} \right\},$$

where we consider the standard angular coordinates on $\mathbb{S}^2$. This is a distribution whose integral manifolds are diffeomorphic to $\mathbb{S}^1$.

In the following we will be interested only in those distributions which are *smooth* that is, for each point $p \in M$, there exists a local chart U containing p and $X_1, \dots, X_k \in \Gamma(TM|_U)$ such that

$$\mathcal{D}_p = \text{span}_{\mathbb{R}} \{X_1(p), \dots, X_k(p)\}, \quad 1 \leq k \leq n = \dim M.$$

Having established a connection between tangent distributions and vector fields allows for a better control of their properties:

Definition 151. Given a smooth n -dimensional manifold M as well as a smooth distribution $\mathcal{D}$ as per Definition 149, we call it **involutive** if for every $p \in M$ and for every $X_1, \dots, X_k \in \Gamma(TM|_U)$ such that $\mathcal{D}_p = \text{span}_{\mathbb{R}}\{X_1(p), \dots, X_k(p)\}$, $1 \leq k \leq n = \dim M$,

$$[X_i, X_j](p) \in \mathcal{D}_p, \quad \forall i, j = 1, \dots, k$$

where $[\cdot, \cdot]$ denotes the Lie brackets as per Definition 36.

Definition 152. Given a smooth n -dimensional manifold M as well as a smooth distribution $\mathcal{D}$ as per Definition 149 we call it **integrable** if each $p \in M$ lies in an integral manifold of $\mathcal{D}$.

The next step consists of establishing a relation between the notions of involutive and integrable distributions.

Theorem 153. Let M be a smooth manifold and let $\mathcal{D}$ be a smooth and integrable distribution as per Definition 152. Then $\mathcal{D}$ is involutive as per Definition 151.

Proof. Let $p \in M$ while $X, Y \in \mathcal{D}$. Per hypothesis there exists an immersed submanifold (N, ψ) and $q \in N$ such that $\psi(q) = p$ while $d\psi(T_q N) = \mathcal{D}(p)$. Consequently $\exists \tilde{X}, \tilde{Y} \in \Gamma(TN)$ such that $d\psi \circ \tilde{X} = X \circ \psi$ and $d\psi \circ \tilde{Y} = Y \circ \psi$. Since N is a submanifold, it descends that $[\tilde{X}, \tilde{Y}] \in \Gamma(TN)$ and therefore Definition 152 entails that $d\psi([\tilde{X}, \tilde{Y}]_q) \in \mathcal{D}_p$. Hence, for all $f \in C^\infty(M)$,

$$\begin{aligned} d\psi([\tilde{X}, \tilde{Y}]_q)(f) &= [\tilde{X}, \tilde{Y}]_q(f \circ \psi) = \tilde{X}_q(\tilde{Y}(f \circ \psi)) - \tilde{Y}_q(\tilde{X}(f \circ \psi)) = \\ &= \tilde{X}_q((d\psi \circ \tilde{Y})(f)) - \tilde{Y}_q((d\psi \circ \tilde{X})(f)) = \tilde{X}_q(Y(f) \circ \psi) - \tilde{Y}_q(X(f) \circ \psi) = \\ &= d\psi(\tilde{X}_q(Y(f))) - d\psi(\tilde{Y}_q(X(f))) = X_{\psi(q)}(Y(f)) - Y_{\psi(q)}(X(f)) = [X, Y]_p(f), \end{aligned}$$

where $[\cdot, \cdot]$ denotes the Lie bracket as per Definition 36. It descends that $[X, Y]_{\psi(q)} = d\psi([\tilde{X}, \tilde{Y}]_q)$, namely $[X, Y] \in \mathcal{D}(p)$, which entails that $\mathcal{D}$ is involutive. $\square$

In the following we want to establish a relation between smooth tangential distributions and submanifolds of a smooth manifold M . To this end, we need the following auxiliary concepts:

- given a smooth manifold M as per Definition 4, we say that a local chart (U, φ) is a *cube* if there exist $a > 0$ and coordinates $\{x^1, \dots, x^n\}$, $n = \dim M$, such that $x^i(U) = (-a, a)$ for all $i = 1, \dots, n$,
- given a smooth n -dimensional manifold M and a k -dimensional distribution $\mathcal{D}$ thereon as per Definition 149, we say that a chart (U, φ) is *flat* for $\mathcal{D}$ if (U, φ) is a cube with respect to the coordinates $\{x^1, \dots, x^n\}$ and

$$\mathcal{D} \cap TM|_U = \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^k} \right\}.$$

This entails that for every $c_{k+1}, \dots, c_n \in (-a, a)$ the locus

$$V = \varphi^{-1}\{(x^1, \dots, x^n) \in (-a, a)^{\times n} \mid x^j = c^j, \text{ for all } j = k+1, \dots, n\},$$

is an integral manifold of $\mathcal{D}$. If a flat chart for $\mathcal{D}$ exists around every $p \in M$, we say that $\mathcal{D}$ is **completely integrable**.

Let us now study the inverse problem of Theorem 153, starting from two auxiliary results, the second being left as an exercise to our readers.

Proposition 154. *Let M, N be two smooth manifolds as per Definition 9 such that $\exists \psi \in C^\infty(M; N)$ and $p \in M$ for which $d\psi : T_p M \rightarrow T_{\psi(p)} N$ is an isomorphism. Then $\exists \tilde{U} \subseteq M$ such that $p \in \tilde{U}$ and ψ is a local diffeomorphism, namely $\psi : \tilde{U} \rightarrow \psi(\tilde{U})$ is smooth, injective and $\psi^{-1} \in C^\infty(\psi(\tilde{U}); \tilde{U})$.*

Proof. Observe that, since $d\psi$ is in isomorphism between $T_p M$ and $T_{\psi(p)} N$, then $\dim M = \dim N = n$. Let us consider a chart (U, φ) of M containing p and let (V, τ) be a chart of N such that $\psi(U) \subset V$. Under these assumptions the differential of the map $\tau \circ \psi \circ \varphi^{-1} : \varphi(U) \rightarrow \tau(\psi(U))$ is non singular at $\varphi(p) \in \mathbb{R}^n$. Hence, according to the implicit function theorem, $\exists \tilde{U} \subseteq \varphi(U)$ such that the restriction of $\tau \circ \psi \circ \varphi^{-1}$ to $\tilde{U}$ is a local diffeomorphism. Since also φ and τ are per definition local diffeomorphism, the sought result descends. $\blacksquare$  $\square$

Exercise 155. Let M be a smooth manifold and let $X \in \Gamma(TM)$ be such that $X(p) \neq 0$ for a given $p \in M$. Then, there exists a local chart (U, φ) of M containing p , as well as a coordinate system $\{x^1, \dots, x^n\}$, $n = \dim M$, centered at p and such that $X|_U = \frac{\partial}{\partial x^1}|_U$. [Hint: Use Proposition 154].

Theorem 156 (Frobenius). *Let M be a smooth manifold and let $\mathcal{D}$ be a smooth distribution as per Definition 149. If it is involutive as per Definition 151, then it is also integrable as per Definition 152.*

Proof. We proceed by induction with respect to the dimension of $\mathcal{D}_p$ for all $p \in M$, which denote by k . Starting from $k = 1$, Let us consider a local chart (V, ψ) such that $p \in V$ and such that we can find $X \in \Gamma(TM|_V)$ lying in $\mathcal{D}$. The existence is guaranteed by the smoothness of $\mathcal{D}$. Using Exercise 155, we can find a chart (U, φ) with $U \subseteq V$ as well as coordinates $\{x^1, \dots, x^n\}$ such that $X|_U = \frac{\partial}{\partial x^1}$. This entails the sought statement for $k = 1$. By the induction step let us assume the theorem to be true in $(k - 1)$ -dimensions and let us prove it for k . Given $p \in M$, being the distribution smooth, $\exists W \subseteq M$ containing p such that $\mathcal{D}|_W$ is generated by non vanishing $X_i \in \Gamma(TM|_W)$, $i = 1, \dots, k$. Using once more Exercise 155, we can find a local chart $(\tilde{U}, \psi)$ such that $p \in \tilde{U} \subseteq W$ and such that there exists a coordinate system $\{y^1, \dots, y^n\}$ for which $X_1|_{\tilde{U}} = \frac{\partial}{\partial y^1}$. In $\tilde{U}$ we can introduce the auxiliary and linearly independent vector fields

$$Y_1 = X_1, \quad Y_i = X_i - X_i(y^1)X_1, \quad i = 2, \dots, k.$$

Hence $\mathcal{D}$ is generated in $\tilde{U}$ also by $Y_1, \dots, Y_k$. Let $S \subset \tilde{U}$ be the locus of points such that $y^1 = 0$ and let $Z_i \doteq Y_i|_S$, $i = 2, \dots, k$. Observe that, per construction $Y_i(y^1) = 0$ and therefore the collection Z_i , $i = 2, \dots, k$ identifies $k - 1$ vector fields on S which are both smooth and linearly independent. Hence they generate a $(k - 1)$ -dimensional tangential distribution $\mathcal{D}'$ on S . Since $\mathcal{D}$ is per hypothesis involutive and since each Y_i , $i = 1, \dots, k$ is a linear combination of the vector fields X_i which generate $\mathcal{D}$, $\exists c_{ijl} \in C^\infty(\tilde{U})$, $i, j, l = 1, \dots, k$ such that, for all $i, j = 1, \dots, k$, $[Y_i, Y_j] = \sum_{l=1}^k c_{ijl} Y_l$.

Since $Y_i(y_1) = 0$ for all $i > 1$, it descends from Definition 36 that $[Y_i, Y_j](y_1) = 0$ for all $i, j \geq 2$. This implies in turn that $c_{ij1} = 0$ for all $i, j \geq 2$. In other words, if we consider the restriction to S , it holds that

$$[Z_i, Z_j] = \sum_{l=2}^k c_{ijl}|_S Z_l,$$

namely $\mathcal{D}'$ is involutive. Applying the inductive hypothesis, there must exist a local chart of M containing $p \in S$ as well as coordinates $\{z^2, \dots, z^n\}$, such that the level sets $z^j = \text{const.}$ are integrable manifolds of $\mathcal{D}'$.

We can now construct a cubic local chart starting from the natural projection map $\pi : \tilde{U} \rightarrow S$. Let us define the coordinate system

$$x^1 \doteq y^1 \quad x^j = z^j \circ \pi, \quad j = 2, \dots, d. \quad (229)$$

We need to show that the level sets $x^i = \text{const.}$, $i = k+1, \dots, d$ are integrable manifolds of $\mathcal{D}$. This is equivalent to showing that $Y_i(x^r) = 0$ for all $i = 1, \dots, k$ and all $r = k+1, \dots, d$ since this entails that the vectors $\frac{\partial}{\partial x^i}$, $i = 1, \dots, k$ identify a basis of $\mathcal{D}$ for all $p \in \tilde{U}$. Let us observe that Equation (229) entails that

$$\frac{\partial x^j}{\partial y^1} = \delta_{j1}, \quad \forall j = 1, \dots, n$$

namely, in $\tilde{U}$, $Y_1 = \frac{\partial}{\partial x^1}$, from which it descends $Y_1(x^r) = \frac{\partial x^1}{\partial x^r} = 0$ for all $r = k+1, \dots, d$. For any but fixed $j = 2, \dots, k$ and $r = k+1, \dots, n$ it holds that

$$\frac{\partial (Y_j(x^r))}{\partial x^1} = Y_1(Y_j(x^r)) = [Y_1, Y_j](x^r) = \sum_{l=2}^k c_{1jl} Y_l(x^r). \quad (230)$$

Observe that, in the second equality, we have implicitly used that $Y_1(x_r) = 0$ for all $r = k+1, \dots, n$, while the last one is a by product of $\mathcal{D}$ being involutive. Let $K \subseteq \tilde{U}$ be a level set such that $x^j = \text{const}$ for all $j = 2, \dots, n$. Consequently $Y_l(x^r)$ is a function of the lone coordinate x^1 e (230) becomes a system of $k-1$ ODEs in the variable x^1 . Once an initial condition is assigned, the solution exists and it is unique. We observe that the constant function 0 is indeed a possible solution of Equation (230) provided that the initial datum also vanishes. Recalling that S is the locus for which $y^1 = x^1 = 0$, it descends that $S \cap K$ consists of a single point for which $Y_l(x^r) = Z_l(z^r) = 0$ for all $l = 2, \dots, k$ and for all $r = k+1, \dots, n$. In this statement we have used the definition of both Z_l and of the coordinates x^r . In addition we have also exploited that the integral manifolds of $\mathcal{D}'$ in S , whose generators are the vector fields Z_l , are the level sets $z^r = \text{const}$, $r = k+1, \dots, n$. In other words the initial condition of Equation (230) is that all vector fields Y_j vanish. Hence the only possible solution on $\tilde{U}$ is $Y_l(x^r) = 0$, which concludes the inductive step.

At last let (N, ψ) be a connected, integrable manifold $\mathcal{D}$ such that $\psi(N)$ lies in the cubic open chart constructed above. Let $\tau : \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$ be the projection on the last $n-k$ coordinates. For every vector field $Y \in \mathcal{D}$, $d(\tau \circ \varphi)(Y) = 0$ and, therefore, for all $q \in N$, it holds that

$d(\tau \circ \varphi \circ \psi)(q) = 0$. Since N is connected $\tau \circ \varphi \circ \psi$ is the constant map and, hence, $\psi(N)$ lies in one of the level sets of $\mathcal{D}$.  $\square$

Definition 157. A maximal connected integral manifold (N, ψ) of a tangential distribution $\mathcal{D}$ of a smooth manifold M as per Definition 152 is a connected integral manifold of $\mathcal{D}$ whose image in M is not a proper submanifold of any other integrable manifold of $\mathcal{D}$.

We can now make precise a concept which is quite intuitive and it has been used already implicitly a few times in these notes.

Definition 158. Let M be an n -dimensional smooth manifold as per Definition 9. We call **foliation** of dimension $1 \leq k < n$ on M , a collection $\mathcal{F}$ of disjoint, connected, non-empty, immersed k -dimensional submanifolds of M , called **leaves**, whose union is the whole M and such that, for all $p \in M$, there exists a flat chart for $\mathcal{F}$ containing p .

One can realize that foliations are closely related to tangential distributions as the following exercise proves.

Exercise 159. Under the assumptions of Definition 158, prove that the union of all tangent spaces to the leaves of $\mathcal{F}$ form an involutive distribution as per Definition 151.

A far more important consequence of the notion of foliation is that a converse of Exercise 159 holds true. The following statement, which generalizes Theorem 156, has a very long and technical proof which can be found in [Lee13, Thm. 19.21 & Lem. 19.22].

Theorem 160 (Global Frobenius Theorem). Let $\mathcal{D}$ be an involutive distribution as per Definition 151. Then, the collection of all maximal, connected, integral manifolds of $\mathcal{D}$ identifies a foliation of the underlying manifold M as per Definition 158.

The metric of a spherically symmetric spacetime: Let us now revert our attention to Definition 148 and let us consider a four-dimensional, smooth and spherically symmetric spacetime (M, g) . Per definition, given $p \in M$, its orbit $\mathcal{O}_p$ is diffeomorphic to the 2-sphere. This entails that we can introduce $(\mathbb{S}^2, \iota_p)$ where $\iota_p : \mathbb{S}^2 \rightarrow \mathcal{O}_p$ identifies an immersed submanifold. Accordingly we can choose spherical coordinates (θ, φ) on $\mathbb{S}^2$ such that the line element associated to ι_p^*g reads $d\theta^2 + \sin^2 \theta d\varphi^2$. Following Example 85, we can individuate thereon three Killing vector field

$$\xi_1 = \frac{\partial}{\partial \varphi}, \quad \xi_2 = \sin \varphi \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi}, \quad \xi_3 = \cos \varphi \frac{\partial}{\partial \theta} - \cot \theta \sin \varphi \frac{\partial}{\partial \varphi},$$

whose Lie brackets are $[\xi_i, \xi_j] = \epsilon_{ijk} \xi_k$ where ϵ_{ijk} is the totally antisymmetric Levi-Civita symbol. In the language of Lie groups, it turns out that Killing fields individuate a Lie algebra isomorphic to that of $\mathfrak{so}(3)$, the set of skew-symmetric 3×3 matrices. Consequently, if we define

$$\mathcal{D}_p = d\iota_p(\text{span}_{\mathbb{R}}(\xi_1, \xi_2, \xi_3)) \subset T_p M,$$

where $d\iota_p$ is the differential as per Definition 4, then

$$\mathcal{D} = \bigcup_{p \in M} \mathcal{D}_p,$$

forms an involutive, smooth tangential distribution as per Definition 151. Using Theorem 160, we can infer that M can be foliated in terms of two-spheres $\mathbb{S}^2$. We want to exploit this property to constraint the form of the metric g on M . Yet, the presence of a foliation allows a priori only to establish that, for every $p \in M$, there exists a local chart (U, Φ) containing p and coordinates (t, r, θ, φ) such that on $U \cap \iota_p(\mathbb{S}^2)$ the line element of the induced metric reads

$$d\tilde{s}^2 = f^2(t_0, r_0) (d\theta^2 + \sin^2 \theta d\varphi^2),$$

where $(t_0, r_0) = (t(p), r(p))$ and where $f(t_0, r_0) > 0$. This result does not allow us to rule out that in the line element of g cross terms of the form $dt d\theta$, $dt d\varphi$ and so on and so forth do occur. This requires a separate analysis which can be summarized in the following theorem, see [Sch67].

Theorem 161. *Let (M, g) be a four-dimensional, spherically symmetric spacetime as per Definition 148. For every $p \in M$, there exists a local chart (U, Φ) containing p and a coordinate system (a, b, θ, φ) thereon such that the line element associated to g reads*

$$ds^2 = d\tau^2(a, b) + f^2(a, b) (d\theta^2 + \sin^2 \theta d\varphi^2), \quad (231)$$

where $f \in C^\infty(U; \mathbb{R}^+)$.

We can further simplify Equation (231) by means of the following steps:

1. Since f has to be a smooth and invertible function, else the metric would be singular, we can choose one of the two coordinates a, b in such a way to coincide with f . We consider therefore the change of coordinates $(a, b) \rightarrow (a, r)$ with $r = f^2(a, b)$ and thus Equation (231) becomes

$$ds^2 = d\tau^2(a, r) + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2),$$

2. If we focus on the component $d\tau^2(a, r)$ this can be decomposed as

$$d\tau^2(a, r) = -h_1^2(a, r) da^2 + h_2^2(a, r) dadr + h_3^2(a, r) dr^2,$$

where $h_1, h_2, h_3 \in C^\infty(U)$ are arbitrary barring the constraint that the metric must be non singular and Lorentzian. We have still the leeway of changing a and we do it in such a way that we can cancel the cross term proportional to $dadr$. In other words we introduce a new time variable $t = t(a, r)$ which entails that $dt = \partial_a t da + \partial_r t dr$. Since we want to end up with a line element of the form

$$d\tau^2(t, r) = -f_1^2 dt^2 + f_2^2 dr^2,$$

with $f_1, f_2 \in C^\infty(U)$, it must hold the identity

$$\begin{aligned} d\tau^2(t, r) &= -f_1^2 dt^2 + f_2^2 dr^2 = -f_1^2 (\partial_a t)^2 da^2 - 2f_1^2 \partial_a t \partial_r t da dr + \left(f_2^2 - f_1^2 (\partial_r t)^2 \right) dr^2 = \\ &= -h_1^2(a, r) da^2 + h_2^2(a, r) da dr + h_3^2(a, r) dr^2. \end{aligned}$$

This leads to the system of equations

$$\begin{cases} f_1 (\partial_a t) = h_1 \\ 2f_1^2 \partial_a t \partial_r t = h_2^2 \\ f_2^2 - f_1^2 (\partial_r t)^2 = h_3^2 \end{cases}$$

We should read this as three equations for three unknowns (f_1, f_2, t) in presence of three sources h_1, h_2, h_3 . Up to the assignment of an initial condition, this system always admits a solution.

In other words, observing that f_1 and f_2 must be strictly positive, we have shown that the metric of a four-dimensional spherically symmetric spacetime can always be put in the form

$$ds^2 = -e^{2\alpha(t,r)} dt^2 + e^{2\beta(t,r)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2), \quad (232)$$

where $f_1(t, r) = e^{\alpha(t,r)}$, $f_2(t, r) = e^{\beta(t,r)}$ with $\alpha, \beta \in C^\infty(U)$. This also known as **Birkhoff form** of the metric.

7.1 Schwarzschild Spacetime

Having established the most general form of the metric in a four dimensional, spherically symmetric manifold in Equation (232), our next step consists of imposing the vacuum Einstein's equations. As a starting point we use Equation (33) to compute all Christoffel symbols. The non-vanishing ones are

$$\begin{aligned} \Gamma_{tt}^t &= \partial_t \alpha, & \Gamma_{tr}^t &= \partial_r \alpha, & \Gamma_{tr}^t &= \Gamma_{rt}^t = \partial_t \beta e^{2(\beta-\alpha)}, \\ \Gamma_{tt}^r &= \partial_r \alpha e^{2(\alpha-\beta)}, & \Gamma_{tr}^r &= \Gamma_{rt}^r = \partial_t \beta, & \Gamma_{rr}^r &= \partial_r \beta, & \Gamma_{\theta\theta}^r &= -r e^{-2\beta}, & \Gamma_{\varphi\varphi}^r &= -r e^{-2\beta} \sin^2 \theta, \\ \Gamma_{r\theta}^\theta &= \Gamma_{\theta r}^\theta = r^{-1}, & \Gamma_{\varphi\varphi}^\theta &= -\frac{\sin(2\theta)}{2}, \\ \Gamma_{r\varphi}^\varphi &= \Gamma_{\varphi r}^\varphi = r^{-1}, & \Gamma_{\theta\varphi}^\varphi &= \Gamma_{\varphi\theta}^\varphi = \cot \theta. \end{aligned} \quad (233)$$

Using Equation (94) we can compute the components of the Ricci tensor. The non vanishing ones are

$$R_{tt} = \partial_t^2 \beta + (\partial_t \beta)^2 - \partial_t \alpha \partial_t \beta + e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right], \quad (234a)$$

$$R_{rr} = e^{2(\beta-\alpha)} \left[\partial_t^2 \beta + (\partial_t \beta)^2 - \partial_t \alpha \partial_t \beta \right] - \partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \beta, \quad (234b)$$

$$R_{tr} = R_{rt} = \frac{2}{r} \partial_t \beta, \quad (234c)$$

$$R_{\theta\theta} = e^{-2\beta} [r(\partial_r \beta - \partial_r \alpha) - 1] + 1, \quad (234d)$$

$$R_{\varphi\varphi} = e^{-2\beta} \sin^2 \theta [r(\partial_r \beta - \partial_r \alpha) - 1] + \sin^2 \theta. \quad (234e)$$

Since the vacuum Einstein's Equations can be rewritten as

$$R_{\mu\nu} = 0,$$

we can impose that all expressions above vanish. If we start from Equation (234c), it descends

$$R_{tr} = 0 \implies \partial_t \beta = 0 \implies \beta \equiv \beta(r).$$

Inserting this result in Equation (234a) we obtain

$$R_{tt} = e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \beta' \partial_r \alpha + \frac{2}{r} \partial_r \alpha \right] = 0, \quad (235)$$

where we have replaced $\partial_r \beta$ with β' to highlight that we have to consider a total derivative with respect to r . Similarly Equation (234b) becomes

$$\begin{aligned} R_{rr} &= -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \beta' \partial_r \alpha + \frac{2}{r} \beta' = 0 \implies \\ &\implies \frac{2}{r} \beta' = \partial_r^2 \alpha + (\partial_r \alpha)^2 - \beta' \partial_r \alpha. \end{aligned} \quad (236)$$

Inserting Equation (236) in Equation (235) we obtain

$$\begin{aligned} e^{2(\alpha-\beta)} \left[\frac{2}{r} \beta' + \frac{2}{r} \partial_r \alpha \right] &= 0 \implies \\ \implies \beta' + \partial_r \alpha &= 0 \implies \alpha(r, t) = -\beta(r) + f(t), \end{aligned} \quad (237)$$

where f is an arbitrary time dependent smooth function. Yet we can get rid of f observing that the line element in Equation (232) has now taken the form

$$ds^2 = e^{-2\beta(r)} e^{2f(t)} dt^2 + e^{2\beta(r)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2).$$

Hence if we define a new time coordinate $\tau \equiv \tau(t)$ such that $d\tau = e^{f(t)} dt$, or equivalently $\tau = \int_{t_0}^t ds e^{f(s)}$, it turns out that

$$ds^2 = e^{-2\beta(r)} d\tau^2 + e^{2\beta(r)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2).$$

Hence, with a slight abuse of notation, we shall still employ the symbol t for the time coordinate and we set $f = 0$ in Equation (237). Observing that Equation (234e) can be rewritten as

$$R_{\varphi\varphi} = \sin^2 \theta R_{\theta\theta} = 0,$$

we can infer that only Equation (234d) has not yet been used to extract some information on the coefficients. Using all constraints obtained so far, it can be rewritten as

$$R_{\theta\theta} = 2r \frac{d\beta}{dr} e^{-2\beta} - e^{-2\beta} + 1 = -r \frac{d(e^{-2\beta})}{dr} - e^{-2\beta} + 1 = 0,$$

which, renaming for simplicity $e^{-2\beta(r)}$ as $\lambda(r)$, yields

$$-r \frac{d\lambda}{dr} - \lambda + 1 = 0 \implies d \ln(1 - \lambda) = -d \ln(r).$$

In other words we can infer that

$$\lambda = e^{-2\beta(r)} = 1 - \frac{C}{r}, \quad (238)$$

where C is an *integration constant* which we shall write as $C = 2MG$ on the basis of the physical interpretation which we shall explain in the following. Since failing is not an option, we have succeeded in finding the **Schwarzschild metric** whose line element is

$$ds^2 = - \left(1 - \frac{2MG}{r} \right) dt^2 + \frac{dr^2}{1 - \frac{2MG}{r}} + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2). \quad (239)$$

Engrave this metric in your soul. The long procedure which has lead us here is also known as the following statement:

Theorem 162 (Birkhoff). *Any spherically symmetric, four dimensional, solution of the vacuum Einstein's equations with vanishing cosmological constant must be a submanifold of the Schwarzschild spacetime (M, g) where M is diffeomorphic to $\mathbb{R} \times I \times \mathbb{S}^2$, $I \doteq (2M, \infty)$ while g is as per Equation (239).*

It is important to highlight the following consequences of our construction:

1. The metric in Equation (239) allows us to establish the maximal domain of definition of the coordinates, namely $t \in \mathbb{R}$, $\theta \in (0, \pi)$, $\varphi \in (0, 2\pi)$, while $r \in (2M, \infty)$. As a matter of fact we can observe that, as $r \rightarrow 2M$, the metric becomes singular since $g_{tt} \rightarrow 0$ while g_{rr} diverges. As we will explain in Section 7.3 this pathological behaviour is not an intrinsic feature of the underlying spacetime, but of the chosen coordinate system. It is remarkable to notice that, if we restore the Newton constant G and the speed of light c , this limiting value for the coordinate r , known as **Schwarzschild radius** is

$$r = 3 \left(\frac{M}{M_\odot} \right) \text{Km},$$

where $M_\odot \simeq 2 \times 10^{33} g$ is a solar mass. In other words r_s is very small unless one considers masses orders of magnitude higher than the solar one,

2. The Schwarzschild metric possesses several notable geometric properties which we did not impose at the beginning. First and foremost, since the metric coefficients in Equation (239) do not depend explicitly on the time coordinate t , Example 83 guarantees us that $\xi \doteq \frac{\partial}{\partial t}$ is a Killing field. Furthermore, since $g(\partial_t, \partial_t) = -\left(1 - \frac{2M}{r}\right) < 0$ for all $r > 2M$, ξ is everywhere timelike and, therefore, according to Definition 148, (M, g) is a stationary spacetime. In addition, the constant time hypersurfaces are all orthogonal to ξ , since no cross term proportional to $dt dr$ or $dt d\theta$ and $dt d\varphi$ is present in Equation (239). Hence (M, g) is also automatically **static**.
3. At last we observe that, as $r \rightarrow \infty$ Equation (239) tends to the Minkowski metric. In other words (M, g) is *asymptotically Minkowskian*.

We have still not commented on the reason why the integration constant in Equation (239) has been written as $2MG$. This will come as a byproduct of studying geodesics in a Schwarzschild spacetime, the topic of the next section.

7.1.1 The motion of a test-particle

In order to interpret from a physical viewpoint the Schwarzschild metric in Equation (239), it is very instructive to look at the geodesic motion. As discussed in Section 3, this can be interpreted as looking at the dynamics of a point, test particle either massless or of positive mass $m > 0$.

Hence, in the following (M, g) shall denote the Schwarzschild spacetime and, on top of it we consider a curve $\gamma : I \rightarrow M$, $I \subseteq \mathbb{R}$. Contrary to Section 3 we shall use the symbol λ to denote the affine parameter along the curve, leaving t to indicate the time coordinate in Equation (239). Working in coordinates we can write

$$\gamma(\lambda) = (t(\lambda), r(\lambda), \theta(\lambda), \varphi(\lambda)),$$

and it must satisfy the geodesic equation (59b). Yet, this entails that we have to cope with a rather complicated system of coupled ordinary differential equations. In order to simplify the analysis we can use Corollary 93. Statement 1. herein entails that the tangent vector along a geodesic has a constant length. As pointed out in Remark 28, since the underlying manifold is Lorentzian we can normalize such length to $0, \pm 1$. From a physical viewpoint, the three cases correspond to considering point particles which are either massless (0), of positive (-1) or of negative mass (1). For physical reasons, related to implementing the principle of causality, we shall not analyze this last option, though mathematically allowed. Item 1. in Corollary 93 entails that

$$g(\dot{\gamma}(\lambda), \dot{\gamma}(\lambda)) = \kappa \implies -\left(1 - \frac{2MG}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \frac{1}{1 - \frac{2MG}{r}} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\left(\frac{d\theta}{d\lambda}\right)^2 + \sin^2 \theta \left(\frac{d\varphi}{d\lambda}\right)^2\right) = -\kappa, \quad (240)$$

where $\kappa = -1, 0$. At first glance finding solutions of this equation might appear hopeless, but we can exploit item 2. in Corollary 93 which guarantees us that, to each Killing field, we can

associate a conserved quantity along any geodesic. In the case under scrutiny we have two key symmetries, one with respect to time translations and one with respect to spatial rotations. Let us start from the latter. First of all, we observe the Equation (239) is symmetric under the reflection $\theta \mapsto \pi - \theta$. This entails that, if we consider a geodesic starting at $\theta = \frac{\pi}{2}$, $\theta(\lambda) = \frac{\pi}{2}$ for all $\lambda \in I$. In addition, if we start from a different value, we can always exploit the spherical symmetry to force the motion to happen on the equatorial plane. In other words, without loss of generality, we can set $\theta(\lambda)$ to be constant and this simplifies Equation (240) to

$$-\left(1 - \frac{2MG}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \frac{1}{1 - \frac{2MG}{r}} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\varphi}{d\lambda}\right)^2 = -\kappa.$$

Yet, following Example 83, since the metric coefficients in Equation (239) do not depend explicitly on t and on φ , we can conclude that $\xi_1 \doteq \frac{\partial}{\partial t}$ and $\xi_2 \doteq \frac{\partial}{\partial \varphi}$ are Killing fields. Hence item 2. of Corollary 93 and, in particular Equation (61), entail the two following conservation laws:

$$g_{\mu\nu}(\gamma(\lambda))\dot{\gamma}^\mu \xi_1^\nu = -\left(1 - \frac{2MG}{r}\right) \frac{dt}{d\lambda} = -E, \quad (241a)$$

$$g_{\mu\nu}(\gamma(\lambda))\dot{\gamma}^\mu \xi_2^\nu = r^2 \frac{d\varphi}{d\lambda} = L, \quad (241b)$$

where E, L are two constants. The symbols have been chosen because we want to interpret E either as the total energy ($\kappa = 0$) or as the total energy per unit mass ($\kappa = 1$), while L as the total angular momentum ($\kappa = 0$) or as the total angular momentum per unit mass ($\kappa = 1$). Henceforth, for simplicity we shall drop “per unit mass” from the text. Equation (240) boils down to a single ordinary differential equation

$$-\frac{E^2}{1 - \frac{2MG}{r}} + \frac{1}{1 - \frac{2MG}{r}} \left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2} = -\kappa,$$

or equivalently

$$\frac{1}{2} \left(\frac{dr}{d\lambda}\right)^2 + \frac{1}{2} \left(\frac{L^2}{r^2} + \kappa\right) \left(1 - \frac{2MG}{r}\right) = \mathcal{E}, \quad (242)$$

where $\mathcal{E} = \frac{E^2}{2}$. This equation can be read as the energy conservation for a particle of unit mass $m = 1$ with the first term on the left hand side playing the rôle of kinetic energy, the second one

$$V(r) = \frac{\kappa}{2} - \kappa \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}, \quad (243)$$

being the potential energy, while $\mathcal{E}$ is total energy. In this respect, one can realize that calling E energy density as in the literature is from many viewpoints an abuse of nomenclature. We can interpret all four terms in Equation (243) from the viewpoint of classical mechanics keeping in mind that the formula is exact, namely no approximation has been done to derive it:

1. $\frac{\kappa}{2}$ is a constant and, therefore, it is irrelevant at a dynamical level even when $\kappa \neq 0$ since we can reabsorb it in the definition of $\mathcal{E}$,
2. $\kappa \frac{GM}{r}$ contributes only when we consider massive particles, hence timelike geodesics, and it represents the Newtonian potential generated by a mass M . If we would have kept the integration constant as C in Equation (238), here, we would have been convinced to set it as $2GM$. In other words a timelike geodesic in a Schwarzschild spacetime is equivalent to the classical motion of a massive point particle in a gravitational well generated by a spherically symmetric mass M ,
3. $\frac{L^2}{r^2}$ is the standard contribution to the total energy from centrifugal forces,
4. $-\frac{GML^2}{r^3}$ is a *GR correction* which dominates at small r , namely whenever $r < GM$.

It is clear that, at this stage, it is quite hard to swallow the idea that M is a physical mass since we have constructed Schwarzschild spacetime as a solution of vacuum Einstein's equations. A model and an interpretation of M as a proper mass will be given in Section 7.2. Let us now study more in detail Equation (242) separating massless from massive particles.

$\kappa = 0$: In this case we are looking at the geodesic motion of massless particles on Schwarzschild spacetime such as photons. They are subject to the potential

$$V(r) = \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \quad (244)$$

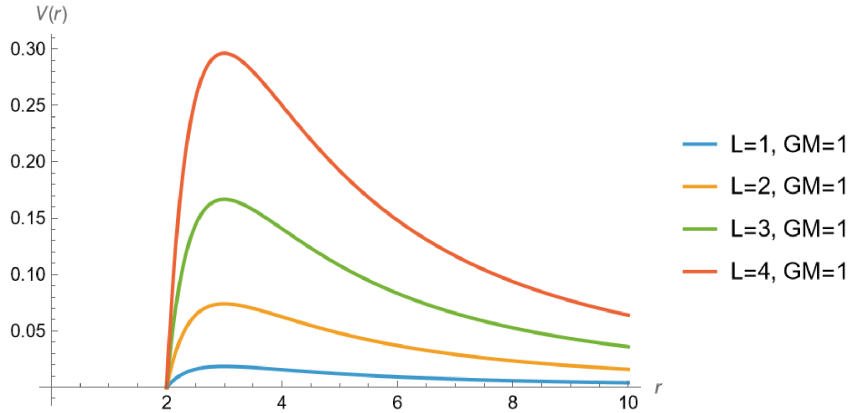


Figure 4: Plot of Equation (244) setting $GM = 1$ for different values of $L > 0$.

Unless $L = 0$, we can see that

$$V'(r) = -\frac{L^2}{r^3} + 3\frac{GML^2}{r^4} = 0 \implies r = r_c = 3GM,$$

while

$$V''(r) = 3\frac{L^2}{r^4} - 12\frac{GML^2}{r^5} \implies V''(r_c) = -\frac{L^2}{81(GM)^4} < 0.$$

In other words the potential tends to 0 as $r \rightarrow \infty$, it has a maximum at $r = 3GM$ and it is positive for all $r > 2GM$ while $V(2GM) = 0$. In other words we can separate the geodesic motion of a massless point particle coming from infinity towards $r \rightarrow 2M$ in three distinct regimes:

1. the total energy is $\mathcal{E} < V(r_c) = \frac{L^2}{54(GM)^2}$. In this case the particle does not have enough energy to overcome the potential barrier and it does not fall towards $r \rightarrow 2M$
2. the particle lies at $r = 3GM$ and it has vanishing initial velocity $\frac{dr}{d\lambda} = 0$. In this case the orbit is

$$\lambda \rightarrow (t_0 + \frac{L}{\sqrt{27}GM}\lambda, 3GM, \frac{\pi}{2}, \varphi_0 + \frac{L}{9(GM)^2}\lambda),$$

namely the particle never falls either towards infinity or towards $r \rightarrow 2M$.

3. the total energy is $\mathcal{E} > V(r_c) = \frac{L^2}{54(GM)^2}$. In this case the particle has enough energy to overcome the potential barrier and it falls towards $r \rightarrow 2M$.

$\kappa = 1$: In this case we are looking at the geodesic motion of massive particles on Schwarzschild spacetime such as electrons. They are subject to the potential

$$V(r) = \frac{1}{2} - \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \quad (245)$$

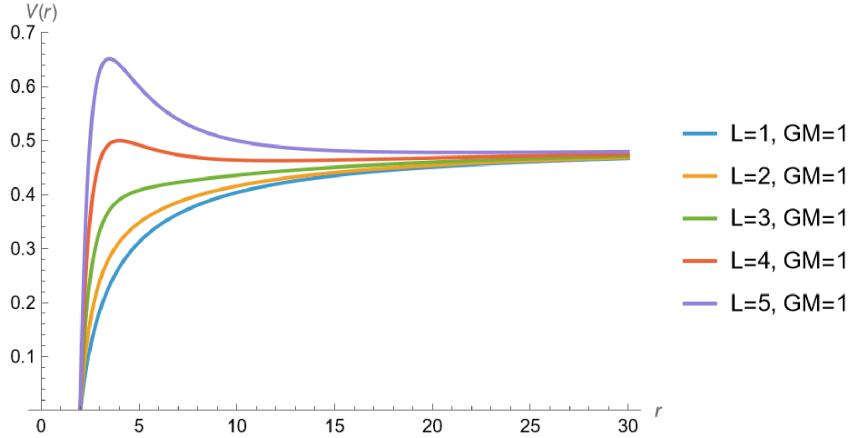


Figure 5: Plot of Equation (245) setting $GM = 1$ for different values of $L > 0$.

As we can infer both from Figure 7.1.1 and from Equation (245), at large values of r the Newtonian potential dominates barring the constant term $\frac{1}{2}$ which could be reabsorbed in the definition of $\mathcal{E}$. On the contrary for small or intermediate values of r all terms contribute and, therefore, we can look for circular orbits of the test particles, namely we evaluate

$$V'(r) = \frac{GM}{r^2} - \frac{L^2}{r^3} + 3\frac{GML^2}{r^4},$$

which vanishes if

$$r_{\pm} = \frac{L^2 \pm \sqrt{L^4 - 12(GLM)^2}}{2GM}. \quad (246)$$

Hence, if $L^2 < \sqrt{12}GM$ there are no extremal points coherently with Figure 7.1.1 with $L < 4$. Assuming to work in the regime $L^2 > \sqrt{12}GM$, in order to establish whether these are either maxima or minima of $V(r)$, we evaluate the second derivative:

$$V''(r) = -\frac{12GL^2M}{r^5} + \frac{3L^2}{r^4} - \frac{2GM}{r^3},$$

which entails that

$$V''(r_-) < 0 \quad \text{and} \quad V''(r_+) > 0.$$

In other words we can separate the following regimes describing the motion of a massive point particle on Schwarzschild spacetime:

1. if the angular momentum L is such that $L^2 < \sqrt{12}GM$, a particle coming from infinity will fall towards $r \rightarrow 2M$ regardless of its energy,
2. if $L^2 > \sqrt{12}GM$ and the energy $\mathcal{E} \geq \mathcal{E}_0$ where

$$\mathcal{E}_0 = \frac{1}{108} \left(36 + \frac{12\sqrt{L^4 - 12G^2L^2M^2}}{L^2} + \frac{L^2 - \sqrt{L^4 - 12G^2L^2M^2}}{(GM)^2} \right),$$

then a particle coming from $r \rightarrow \infty$ will also fall towards $r \rightarrow 2M$,

3. if $L^2 > \sqrt{12}GM$ and the energy $\mathcal{E} < \mathcal{E}_0$, then a particle coming from $r \rightarrow \infty$ shall not reach $r \rightarrow 2M$,
4. there are two circular geodesic orbits, namely assuming that $\frac{dr}{d\lambda} = 0$, the first one is unstable

$$\lambda \rightarrow \left(t_0 + \frac{2r_- V(r_-)}{r_- - 2MG} \lambda, r_-, \frac{\pi}{2}, \varphi_0 + \frac{L}{r_-^2} \lambda \right),$$

while the second one is stable:

$$\lambda \rightarrow \left(t_0 + \frac{2r_+ V(r_+)}{r_+ - 2MG} \lambda, r_+, \frac{\pi}{2}, \varphi_0 + \frac{L}{r_+^2} \lambda \right).$$

Observe that $r_+ > 6GM$, which therefore represents the lower bound for the existence of a stable circular orbit in a Schwarzschild spacetime.

Exercise 163. Discuss the limiting scenario of a massive test particle in Schwarzschild spacetime such that $L = \sqrt{12}GM$. Discuss whether the circular orbits for which $r_+ = r_- = 6GM$ are stable or unstable.

7.2 The Tolman-Volkhov-Oppenheimer metric

In the previous section, studying the motion of a point particle in a Schwarzschild spacetime, the large radius behaviour and the comparison with Newtonian gravity seems to justify to call the integration constant C in Equation (238) as $2GM$. Since Birkhoff theorem 162 is telling us that Schwarzschild spacetime is the maximal, connected spherically symmetric solution of Einstein's equation, one can therefore think of the following model: Consider a spherically symmetric mass distribution of total mass M , that is, for example, a non rotating star of radius $R > 2GM$. For all points (t, r, θ, φ) such that $r > R$, we can consider that the underlying metric is ruled by the vacuum Einstein's equations and therefore it is of Schwarzschild form as in Equation (239). In this scenario it is therefore natural to interpret M as the total mass of the body itself, justifying the interpretation of the previous section, but raising the following two questions:

1. what is the metric inside the star? If Schwarzschild is the *exterior* solution to the Einstein's equations, how does the inner one look like?
2. What does it happen if we consider a spherically symmetric mass distribution of total radius $R < 2GM$? In this case the Schwarzschild metric is not suitable to describe the whole spacetime outside the body itself since the coordinates are only meaningful for $r > 2GM$.

In this section we shall answer the first question leaving the second one to Section 7.3. In the following we shall need to make some preliminary, physically motivated assumptions, namely we assume that the underlying spacetime is spherically symmetric and static. In other words our starting point is Equation (232) though with the additional constraint that $\alpha \equiv \alpha(r)$ and $\beta \equiv \beta(r)$. Consequently the Einstein tensor reads

$$\begin{cases} G_{tt} = \frac{1}{r^2} e^{2(\alpha-\beta)} (2r\beta' - 1 + e^{2\beta}) \\ G_{rr} = \frac{1}{r^2} (2r\alpha' + 1 - e^{2\beta}) \\ G_{\theta\theta} = r^2 e^{-2\beta} [\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{1}{r}(\alpha' - \beta')] \\ G_{\varphi\varphi} = r^2 e^{-2\beta} \sin^2 \theta [\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{1}{r}(\alpha' - \beta')] \end{cases}, \quad (247)$$

where the prime symbol denotes once more a total derivative with respect to r . Since the Einstein's tensor is diagonal, compatibility with Equation (138) entails that the stress-energy tensor must enjoy this same property. Therefore a simple, albeit rather effective, option is to choose to model a spherically symmetric star as a perfect fluid, namely, recalling Equation (177)

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu},$$

where $u \equiv e^{-\alpha} \frac{\partial}{\partial t}$ since $P \equiv P(r)$ and $\rho \equiv \rho(r)$. Observe that the exponential factor in the expression of u is motivated by the requirement that $g(u, u) = -1$. Since $G_{\varphi\varphi} = \sin^2 \theta G_{\theta\theta}$, the

Einsteins' equations reduce to three independent components:

$$\frac{1}{r^2}e^{-2\beta} \left(2r\beta' - 1 + e^{2\beta} \right) = 8\pi G\rho, \quad (248a)$$

$$\frac{1}{r^2}e^{-2\beta} \left(2r\alpha' + 1 - e^{2\beta} \right) = 8\pi GP, \quad (248b)$$

$$r^2e^{-2\beta} \left[\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{1}{r}(\alpha' - \beta') \right] = 8\pi GP. \quad (248c)$$

Since the first equation does not depend on α and, mimicking Equation (239), it is convenient to define

$$e^{-2\beta(r)} \doteq 1 - \frac{2Gm(r)}{r} \implies m(r) \doteq \frac{r}{2G} \left(1 - e^{-2\beta(r)} \right), \quad (249)$$

which entails that Equation (248a) can be rewritten as

$$\frac{dm(r)}{dr} = 4\pi r^2 \rho \implies m(r) = 4\pi \int_0^r ds s^2 \rho(s). \quad (250)$$

If we consider as our model a spherically symmetric star of fixed total radius $R > 0$ and if we interpret, therefore, $\rho(r)$ in the stress-energy tensor as the mass density of the celestial body, it turns out that

$$M = 4\pi \int_0^R ds s^2 \rho(s), \quad (251)$$

can be interpreted as its *total Schwarzschild mass*. Observe that this constraint entails that the g_{rr} -component of Equation (232) matches that of Equation (239) at $r = R$. Yet our use of the nomenclature “Schwarzschild mass” is not accidental, since the total energy associated to $\rho(r)$ must be computed at an arbitrary but fixed time t integrating it with respect to h the induced metric on Σ_t , the constant time hypersurface. Its line element reads

$$ds^2(h) = h_{ij}dx^i dx^j = e^{2\beta(r)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2),$$

which entails that the *total energy* reads

$$E_\rho = 4\pi \int_0^R dr r^2 e^{2\beta(r)} \rho(r) = 4\pi \int_0^R dr \frac{r^2 \rho(r)}{\sqrt{1 - \frac{2Gm(r)}{r}}}.$$

It is interesting to observe that $E_\rho - M \geq 0$ and this difference should be ascribed to the binding gravitational forces between the constituents of the star, seen as a perfect fluid. Focusing instead on Equation (248b), writing it in terms of Equation (249) yields

$$\frac{d\alpha}{dr} = \frac{Gm(r) + 4\pi Gr^3 P}{r(r - 2Gm(r))}, \quad (252)$$

which should be combined with the conservation of the stress-energy tensor, namely $\nabla^\mu T_{\mu\nu} = 0$, which becomes in the case in hand

$$(\rho + P) \frac{d\alpha}{dr} = -\frac{dP}{dr}. \quad (253)$$

This identity, combined with Equation (252) yields the celebrated **Tolman-Oppenheimer-Volkov Equation**

$$\frac{dP}{dr} = -\frac{(\rho + P)(Gm(r) + 4\pi Gr^3 P)}{r(r - 2Gm(r))}. \quad (254)$$

Therefore, at the end of our analysis we are left with $\alpha(r)$, $m(r)$, $\rho(r)$ and $P(r)$ as four unknowns related by three equations, namely (250), (252) and (254). Hence we need one last datum, namely, similarly to Section 6, an equation of state relating P to ρ . An easier alternative consists of specifying globally the density function $\rho(r)$ and, in view of our experience with electrostatics, a simple, yet informative model consists of assuming a constant distribution of mass, namely, if we consider a star of radius $R > 0$, we set

$$\rho(r) = \begin{cases} \rho_0 & r \leq R \\ 0 & r > R \end{cases}. \quad (255)$$

Consequently we can read from Equation (250)

$$m(r) = \begin{cases} \frac{4\pi}{3}\rho_0 r^3 & r \leq R \\ M & r > R \end{cases}, \quad (256)$$

where M is the total Schwarzschild mass, see Equation (251). Focusing instead on Equation (254), we can obtain the pressure, namely

$$P(r) = \rho_0 \left[\frac{R\sqrt{R - 2GM} - \sqrt{R^3 - 2GM r^2}}{\sqrt{R^3 - 2GM r^2} - 3R\sqrt{R - 2GM}} \right]. \quad (257)$$

At last we can use Equation (252) to derive

$$\alpha(r) = \frac{1}{2} \ln \left(9 \frac{R^2(R - 2GM)}{R^3 - 2GM r^2} \right), \quad (258)$$

from which we can infer that the line element of the metric reads

$$ds^2 = - \left(\frac{3}{2} \sqrt{1 - \frac{2GM}{R}} - \frac{1}{2} \sqrt{1 - \frac{2GM r^2}{R^3}} \right) dt^2 + \frac{dr^2}{1 - \frac{8\pi G}{3} \rho_0 r^2} + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2). \quad (259)$$

Remark 53. Combining Equation (259) and Equation (239) we have constructed the full metric of a spherically symmetric star with a constant density. Similarly to what happens in electromagnetism, the metric coefficients are not smooth but only continuous at the boundary of the star reflecting the sharp jump in Equation (255).

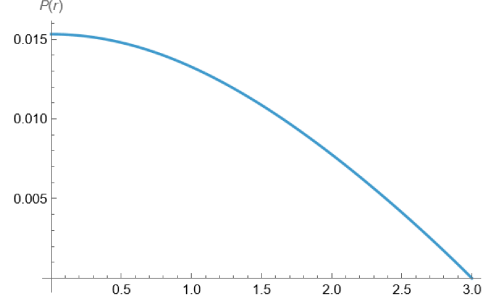


Figure 6: Plot of the pressure inside a star with a constant mass density, see Equation (257), here normalized so that $GM = 1$ and $R = 3$.

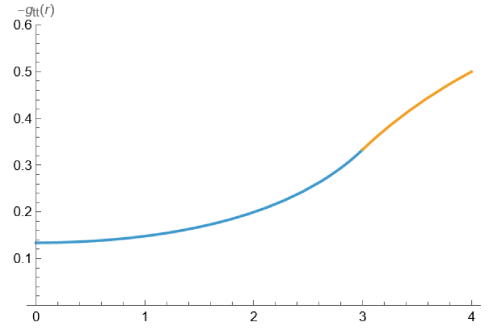


Figure 7: Plot of the g_{tt} -component of a spherically symmetric star with a constant density, here normalized so that $GM = 1$ and $R = 3$ in Equation (259). The orange line represents the contribution from Equation (239).

7.2.1 Experimental Evidences of the Schwarzschild metric

From a physical viewpoint, the scenario depicted in the previous section is that of a spherically symmetric star with a constant mass density. Yet, the metric at the exterior of the celestial object is independent from the density profile and one could think of testing it experimentally. It is evident that we are making implicitly a lot of assumptions, first and foremost since stars are rotating objects and therefore the assumption of spherical symmetry can hold true only for limited time intervals or whenever we can discard any departure from this idealized regime. Nonetheless, a lot of experimental tests of general relativity involve the Schwarzschild metric and it is therefore important to discuss them. We shall focus our attention on three of them.

The Gravitational Redshift: Since the Schwarzschild metric is time independent, we can consider thereon a static observer, that is its world line is a curve whose tangent vector U is such that its spatial components $U^i = 0$, $i = r, \theta, \varphi$. In addition we can choose the fourth component

so that $g(U, U) = -1$, namely $U^t = (1 - \frac{2GM}{r})^{-\frac{1}{2}}$. Suppose that this observer measures the frequency of a photon. This follows a null geodesic $\gamma : I \rightarrow M$, $I \subseteq \mathbb{R}$, whose components can be denoted by $x^\mu(\lambda)$ and the frequency reads

$$\omega = -g_{\mu\nu}U^\mu \frac{dx^\nu}{d\lambda} = \sqrt{1 - \frac{2GM}{r}} \frac{dt}{d\lambda} = \frac{E}{\sqrt{1 - \frac{2GM}{r}}}, \quad (260)$$

where E is defined as in Equation (241a). This entails that, since E is a constant, the frequency measure at different distances from the surface of the star or of the spherically symmetric body will change following the law

$$\frac{\omega_2}{\omega_1} = \sqrt{\frac{r_2(r_1 - 2GM)}{r_1(r_2 - 2GM)}}. \quad (261)$$

Observe that for $r_1, r_2 \gg 2GM$ this formula reduces to

$$\frac{\omega_2}{\omega_1} = 1 - \frac{GM}{r_1} + \frac{GM}{r_2},$$

which is the ratio expected in classical physics from the equivalence principle. This entails that photons falling towards a spherically symmetric body are blue-shifted. This phenomenon, known as *gravitational redshift* and based on Equation (261), has been verified experimentally only in 1960 by Pound and Rebka exploiting the Mössbauer effect. A reader interested in more details can refer to [Hen96].

The precession of the perihelion: One of the two most famous evidences of the validity of Einstein's theory has been the prediction of a phenomenon which in the early twentieth century was already observed and measured but lacked any theoretical support. More precisely the orbit of the planet Mercury was known not to be a perfect ellipsis around the Sun with the center of mass at one of the two focuses. This became manifest observing that the point closest to the Sun, known as *perihelion* (or *periapsis* if the most massive body is not the Sun), was not fixed but was precessing. This entails that the major axis of the ellipsis representing the orbit rotates about the center of mass.

Per se, the phenomenon does not come as a surprise since elliptic orbits are a by-product of the two body Newtonian problem, but, in the solar system, there are several massive objects which cause small deviations from the idealized regime. Yet these can be computed with a great degree of precision and, in the case of Mercury, this lead to observing a sizable deviation from what was expected from the Newtonian theory. At the times of the realization of this problem, in 1859 by Le Verrier, this was not seen as a major issue, since already the discovery of Neptune was a consequence of the observation of deviations of the orbit of Uranus from the expected behaviour. Therefore, it has been hypothesized the existence of new planet, named Vulcan (see Trekkies, where is it coming from), but it was not until around 1910 that astronomers excluded this option.

In order to address the problem under scrutiny within the framework of General Relativity, we have to assume that the Sun is spherically symmetric and therefore that the metric to its exterior is the Schwarzschild one in Equation (239). We look for an equation ruling the change of the distance of a test particle (Mercury) from the spherically symmetric body (the Sun) as a function of the angle φ . Recall that, on account of the underlying isometries, we can always set $\theta = \frac{\pi}{2}$. In Newtonian gravity we expect a dynamics, namely a function $r = r(\varphi)$, which is periodic, while, in presence of a precession, this feature should disappear.

Our starting point is Equation (241b), the conservation of the angular momentum, which can be combined with Equation (242), to obtain the phase portrait in the $r - \varphi$ plane. This is ruled by the differential equation

$$\left(\frac{dr}{d\varphi}\right)^2 + \frac{r^4}{L^2} - \frac{2GM}{L^2}r^3 + r^2 - 2GMr = \frac{2\mathcal{E}}{L^2}r^4. \quad (262)$$

If we introduce the new variable

$$x \doteq \frac{L^2}{GMr} \in \left(0, \frac{L^2}{2GM^2}\right), \quad (263)$$

Equation (262) becomes

$$\left(\frac{dx}{d\varphi}\right)^2 + \left(\frac{L}{GM}\right)^2 - 2x + x^2 - \frac{2(GM)^2}{L^2}x^3 = \frac{2\mathcal{E}L^2}{(GM)^2}. \quad (264)$$

If we differentiate with respect to φ , replace Equation (264) in the ensuing expression, we end up with a second order differential equation

$$\frac{d^2x}{d\varphi^2} - 1 + x = \frac{3(GM)^2}{L^2}x^2. \quad (265)$$

Under the assumption that $\frac{3(GM)^2}{L^2} \ll 1$, we can consider the non linear term as a perturbation and look for an expansion

$$x = x_0 + x_1 + \mathcal{O}(2),$$

where x_0 is the leading contribution

$$\frac{d^2x_0}{d\varphi^2} - 1 + x_0 = 0 \implies x_0 = 1 + e \cos \varphi, \quad (266)$$

which is the expected outcome in Newtonian gravity, interpreting e as the eccentricity of the elliptic orbit. At the same time the perturbation abides by

$$\frac{d^2x_1}{d\varphi^2} + x_1 = \frac{3(GM)^2}{L^2}x_0^2 = \frac{3(GM)^2}{L^2}(1 + e \cos \varphi)^2. \quad (267)$$

Using skills, luck or Mathematica 14.2 and setting the integration constants to $c_1 = -\frac{(1+3e^2)G^2M^2}{eL^2}$ and $c_2 = 0$, we can infer that a solution is given by

$$x_1 = \frac{3(GM)^2}{L^2} \left(-\frac{1}{6}e^2 \cos(2\varphi) + \frac{e^2}{2} + e\varphi \sin(\varphi) + 1 \right), \quad (268)$$

which, combined with Equation (266) yields

$$x = 1 + e \cos \varphi + \frac{3e(GM)^2}{L^2} \varphi \sin \varphi + \mathcal{O}(2).$$

A direct inspection of all terms in x_0 and x_1 unveils a periodicity in φ barring for $\frac{3e(GM)^2}{L^2} \varphi \sin \varphi$ which is therefore the candidate to be responsible for the precession. A cheap but effective trick consists of observing that, should the orbit be exactly elliptic, the variables x and φ would be related by the identity $x = 1 + e \cos \varphi$. Hence we can hypothesize that

$$x = 1 + e \cos \left(\left(1 - \frac{3(GM)^2}{L^2} \right) \varphi \right),$$

which coincides for small values of $\frac{3(GM)^2}{L^2}$ with $x_0 + x_1$ as one can infer from the Taylor expansion

$$1 + e \cos((1 - \alpha)\varphi) \sim 1 + e \cos \varphi + \alpha \varphi \sin \varphi + \mathcal{O}(\alpha^2).$$

This entails that, when the test particle has gone around a full orbit, the coordinate ϕ varies as (or, equivalently, the perihelion advances by an angle)

$$\Delta\varphi = \frac{6\pi(GM)^2}{L^2}.$$

In order to obtain a quantitative estimate observe that, for an elliptic orbit

$$r = \frac{(1 - e^2)a}{(1 + e \cos \varphi)},$$

where a is the semi-major axis. At the same time Equation (263) combined with Equation (266) entails that

$$x = 1 + e \cos \varphi \implies r = \frac{L^2}{GM(1 + e \cos \varphi)},$$

which, by comparison yields $L^2 \sim GM(1 - e^2)a$. In other words

$$\Delta\varphi = \frac{6\pi GM}{c^2(1 - e^2)a}, \quad (269)$$

where we have restored the explicit dependence on the speed of light by a dimensional argument. Recalling that for the system Sun-Mercury

$$\frac{GM_\odot}{c^2} \simeq 1.48 \times 10^5 \text{ cm}, \quad a \simeq 5.79 \times 10^{12} \text{ cm}, \quad e = 0.2056,$$

it descends that

$$\Delta\varphi_{\text{Mercury}} \simeq 0.103''/\text{orbit},$$

which amounts at around 43'' every century. This is fully compatible with the measured data.

The bending of light: The last experimental test of Einstein's theory that we shall discuss concerns the bending of light when passing close to a spherically symmetric body. Historically this has been one if not the first direct experimental evidence of general relativity and it is probably the most famous one together with the precession of the perihelion of Mercury. It has also been at the center of some movies and the author recommends the BBC single drama *Einstein and Eddington* from 2008. Also in this case the underlying background is described in terms of Equation (239) and, contrary to the previous scenario, we shall consider here a photon, hence a massless particle, moving close to the Sun. We wish to study how its trajectory deflects and, to this end once more, we start from Equations (241b) and (242), obtaining the phase portrait ruled by the ordinary differential equation

$$\frac{dr}{d\varphi} = \frac{L}{r^2} \left(\mathcal{E}^2 - \frac{L^2}{r^3} (r - 2GM) \right)^{\frac{1}{2}}. \quad (270)$$

Following the figure we consider the photon incoming from infinity along the x -axis and we look for an estimate of the deflection angle. From Equation (242) we can infer that the photon needs to have a minimal energy to overcome the gravitational barrier codified by $V(r)$ in Equation (244). Since V admits a maximum at $r = 3GM$, setting the kinetic term to vanish we obtain

$$\mathcal{E} = V(3GM) \implies \mathcal{E} = \frac{L^2}{54(GM)^2}.$$

In this case it is more convenient to reinstate $E = \sqrt{2}\mathcal{E}$ from Equation (241a) since the energy threshold can be expressed as

$$b^2 \doteq \left(\frac{L}{E} \right)^2 = 27(GM)^2,$$

where $b = \frac{L}{E}$ represents the (apparent) impact parameter for a massless particle far from the spherically symmetric body of mass M . The value $b_c = \sqrt{27}GM$ represents the critical value below which every photon is captured by the Sun. If we recall the analysis of Section 7.1.1, we can read the motion of a photon as that of a one-dimensional particle subject to the potential $V(r)$ and with an initial energy $\frac{E^2}{2}$. If the particle is not captured by the Sun, then there is a maximum value of r which acts as a turning point, namely the particle cannot get closer to the Sun than this value. To compute it we need to look for the highest root of the equation

$$\frac{E^2}{2} = V(R_0) = \frac{L^2}{2R_0^2} - \frac{GML^2}{R_0^3} \implies R_0^3 - \frac{b^2 R_0}{2} + 2GMb^2 = 0, \quad (271)$$

namely, using Cardano's formula for cubic equations,

$$R_0 = \frac{\sqrt[3]{\sqrt{3}\sqrt{27b^4GM^2 - b^6} - 9b^2GM}}{3^{2/3}} + \frac{b^2}{\sqrt[3]{3}\sqrt[3]{\sqrt{3}\sqrt{27b^4GM^2 - b^6} - 9b^2GM}}. \quad (272)$$

Luckily its exact value is not particularly important, but it only serves us to integrate Equation (270) as

$$\Delta\varphi = 2 \int_{R_0}^{\infty} \frac{dr}{\sqrt{\frac{r^4}{b^2} - r(r - 2GM)}}, \quad (273)$$

which cannot be computed analytically unless $M = 0$ since, in that case

$$\Delta\varphi|_{M=0} = 2 \arcsin\left(\frac{b}{R_0}\right) = \pi,$$

since $R_0 = b$ in the limit $M \rightarrow 0$. To simplify the evaluation of the integral we assume that the impact parameter b is computed at the closet possible distance, namely R_0 . Using Equation (271) we obtain that $b^2 = \frac{R_0^3}{R_0 - 2GM}$, which, replaced in Equation (273), yields

$$\Delta\varphi = 2 \int_{R_0}^{\infty} \frac{dr}{\sqrt{\frac{r^4}{R_0^3}(R_0 - 2GM) - r(r - 2GM)}}. \quad (274)$$

This is still an integral which we cannot solve analytically, but, since we are interested in studying the displacement from the behaviour of a light ray at $M = 0$, we can consider its Taylor expansion to first order in M . Hence we need to compute

$$\left. \frac{\partial \Delta\varphi}{\partial M} \right|_{M=0} = -2G \int_{R_0}^{\infty} dr \frac{\frac{2r^4}{R_0^3} - 2r}{2 \left(\frac{r^4}{R_0^2} - r^2 \right)^{3/2}} = \frac{4G}{R_0} = \frac{4G}{R_0}.$$

Since we can expand

$$\Delta\varphi = \Delta\varphi|_{M=0} + M \left. \frac{\partial \Delta\varphi}{\partial M} \right|_{M=0} = \pi + \frac{4GM}{R_0},$$

we end up with an estimate of the deflection, namely, restoring the speed of light

$$\delta\varphi = M \left. \frac{\partial \Delta\varphi}{\partial M} \right|_{M=0} = \frac{4GM}{c^2 R_0} \simeq \frac{4GM}{c^2 b}. \quad (275)$$

In the last equality we have replaced R_0 with the impact parameter using that, for small values of M , at leading order it holds $R_0 = b$. There are several experimental evidences for Equation (275), the first being that of sir Eddington. Despite several controversies on the measurements and on their precision, the dawn of the age of observations using radio frequencies has allowed to dispel any remaining doubt on this test of general relativity.

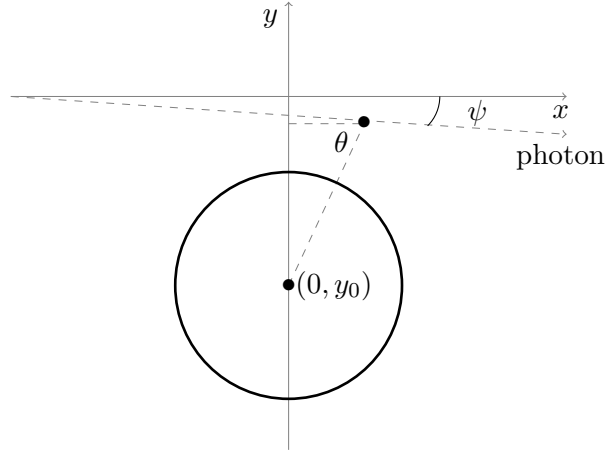
Remark 54. *It is quite interesting how in the popular folklore it is often stated that the bending of light itself is a proof of Einstein's theory since Newtonian gravity does not allow for it. Actually this is not true since already in the nineteenth century, deflection of light has been conjectured. In the pre-Plancks competition of 2023 I proposed an exercise in this direction which I here reproduce including its proof. (I feel magnanimous since Kolo Mouani has just scored his fifth goal with Juventus)*

Exercise 164. In 1801 Johann Georg von Soldner already predicted that light grazing the surface of the Sun would deflect. The following ideal model combines classical mechanics with special relativity to estimate the angle of deflection.

Hence, consider a two dimensional Cartesian plane $\mathbb{R}^2$ and a Cartesian references frame endowed with the standard Euclidean coordinates. Thereon a photon is traveling at the speed of light along the x -axis and he is grazing the surface of the Sun. We assume that the Sun can be modeled as a disk of radius $R_\odot$ and of mass $M_\odot$ centered at $(0, y_0)$ with $y_0 < 0$, $|y_0| \geq R_\odot$ while that the gravitational potential still reads $V = -\frac{GM_\odot}{r}$

- compute ψ , the deflection angle of the photon due to the action of the Newtonian gravitational force assuming that the gravitational mass of the photon coincides with the relativistic one,
- Knowing that $M_\odot = 1.98 \times 10^{30}\text{Kg}$ and $R_\odot \simeq 6.9 \times 10^8\text{m}$, evaluate ψ and compare the result with the prediction of the deflection angle from General Relativity, *i.e.* $\psi_{GR} = \frac{4GM_\odot}{R_\odot c^2}$ where G is the Newton constant while c is the speed of light.

Proof. We denote by $\underline{p} \doteq (p_x, p_y)$ the momentum of the photon and by $|p|$ the corresponding Euclidean length. As consequence the relativistic mass of the photon reads $m_{rel} = \frac{|p|}{c}$.



The dynamics is ruled by Newton's equations whose components are

$$\begin{cases} \frac{dp_x}{dt} = \frac{GM_\odot m_{rel}}{r^2} \cos \theta \\ \frac{dp_y}{dt} = \frac{GM_\odot m_{rel}}{r^2} \sin \theta \end{cases} ,$$

where $r^2 = (x^2 + (y_0 - y)^2)$ is the square of the distance between the position of the photon at any instant of time and the center of the Sun. Using the expression for m_{rel} and the trigonometric identities $\sin \theta = \frac{y - y_0}{r}$, $\cos \theta = \frac{x}{r}$, we can rewrite the equations of motion as

$$\begin{cases} \frac{dp_x}{dt} = \frac{GM_\odot |p|}{cr^3} x \\ \frac{dp_y}{dt} = \frac{GM_\odot |p|}{cr^3} (y - y_0) \end{cases} .$$

Assuming that $x \simeq ct$, we can replace $\frac{d}{dt}$ by $c\frac{d}{dx}$ and it descends

$$\begin{cases} \frac{dp_x}{dx} = \frac{GM_\odot|p|}{c^2 r^3} x \\ \frac{dp_y}{dx} = \frac{GM_\odot|p|}{c^2 r^3} (y - y_0) \end{cases}.$$

These equations can be integrated out explicitly computing the variation of the momentum along the whole trajectory:

$$\begin{cases} \Delta p_x = \frac{GM_\odot|p|}{c^2} \int_{\mathbb{R}} dx \frac{x}{r^3} = 0 \\ \Delta p_y = \frac{GM_\odot|p|}{c^2} \int_{\mathbb{R}} dx \frac{y-y_0}{r^3} = \frac{GM_\odot|p|}{c^2} \frac{x}{(y-y_0)\sqrt{(y-y_0)^2+x^2}} \Big|_{-\infty}^{\infty} = \frac{2GM_\odot|p|}{c^2(y-y_0)} \end{cases},$$

where the first integral is automatically vanishing since the integrand is odd with respect to the x -variable.

Hence we can now evaluate the deflection angle as

$$\sin \psi \simeq \psi = \frac{\Delta p_y}{|p|} = \frac{2GM_\odot}{c^2 R_\odot},$$

where we have replaced $(y - y_0)$ with $R_\odot$ in order to account for the grazing hypothesis.

Part 2) A direct comparison shows that $\psi_{GR} = 2\psi$ and, replacing the numerical values of $M_\odot$ and of $R_\odot$, we obtain $\psi \simeq 0,875''$. $\square$

7.3 The Kruskal Extension

In Sections 7.1 and 7.2 we have introduced and discussed thoroughly both the interior and the exterior metric generated by a spherically symmetric celestial body. Despite the assumptions on the analytic expression of the mass density inside the star, the exterior metric turns out to be universal. In a more physically realistic scenario, even keeping the assumption of spherical symmetry, hence discarding any rotation of a star, one cannot expect it to be in perfect equilibrium at all times. Both observations and more advanced models than the one considered in Section 7.2 suggest that, under suitable circumstances, gravitational and pressure forces might not lie in equilibrium. The first one might become dominant leading to a collapse of the star. This is a dynamical process certainly incompatible with the assumption of time independence taken in Section 7.2. Yet this is not the main issue here. The elephant in the room stems from the observation that, under these circumstances, nothing seems to prevent the star to shrink to a size $R < 2GM$. When this occurs, we face a serious difficulty since, at its exterior we can no longer use Equation (239) since the coordinate r is defined only in the interval $(2GM, \infty)$.

As a matter of fact the metric coefficients tend either to vanish or to diverge suggesting a scenario similar to that encountered in Section 6 when taking the limit $t \rightarrow 0^+$ in a FLRW Universe filled either with dust or radiation. Yet this is only an apparent similarity since, focusing on Equation (239), we can immediately conclude that $\lim_{r \rightarrow 2GM} R = \lim_{r \rightarrow 2GM} g^{\mu\nu} R_{\mu\nu} = 0$ since we are considering a vacuum solution to Einstein's equations.

Remark 55. *The reader should be warned that the statement “a spacetime is singular” is not so easy to make precise at a mathematical level. First and foremost, if a spacetime is everywhere smooth, then all diffeomorphism invariant scalars must not be divergent. It is not obvious how to classify them and, already in four dimensions, getting a complete list is rather difficult. If we consider the lone Riemann tensor, we can construct 16 of them, known as **Carminati-McLenaghan invariants**, [CaMcL91]. Yet, even if they do not diverge, one can construct invariants from the first covariant derivatives of the Riemann tensor and so on and so forth.*

*Nonetheless, if one finds at least one diffeomorphism invariant scalar which is divergent, it is certainly reasonable to refer to the underlying spacetime being singular where the divergence occurs. In the context of black hole physics, a rather efficient indicator of the presence of true singularities is the **Kretschmann scalar** $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$.*

*Even when it is possible to detect the non-smoothness of an underlying spacetime, we know from functional analysis that not all singularities are the same. As an example, one should compare the real valued functions $x^{-\frac{1}{2}}$ and $e^{\frac{1}{x^2}}$. They are both singular at $x = 0$, but the first one identifies a distribution lying in $\mathcal{D}'(\mathbb{R})$, while the second not. Hence the first one is to a certain extent sufficiently tame as a singularity, so that physically sensible models dealing with it can be formulated. The second one is instead rather hopeless. In differential geometry, for reasons which go beyond the scopes of these notes, it is convenient to consider as “tame” the **manifolds of bounded geometry**. Their definition in Lorentzian geometry is slightly involved, but in a Riemannian setting they are characterized as being those for which the scalar curvature and all its covariant derivatives to all orders are bounded functions.*

Hence the above digression points into the direction that working with the Schwarzschild metric as in Equation (239) is largely insufficient, since it forces us to cope with a singularity whose existence seems suspicious from a physical viewpoint. Another rather puzzling consequence of Equation (239) comes from the observation that the vector field $\frac{\partial}{\partial t}$ is everywhere timelike, but tends to become lightlike as $r \rightarrow 2M$:

$$g(\partial_t, \partial_t) = -\left(1 - \frac{2M}{r}\right).$$

Observe that, for simplicity, in this section we shall work with natural units, hence setting $G = c = 1$. This feature is best measured focusing on the $t - r$ plane on the structure of null curves, that, is we look at $\gamma : I \subseteq \mathbb{R} \rightarrow M$ such that

$$g(\dot{\gamma}, \dot{\gamma}) = -\left(1 - \frac{2M}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2M}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 = 0 \implies \frac{dt}{dr} = \pm \left(1 - \frac{2M}{r}\right)^{-1}. \quad (276)$$

Apparently this formula seems to reiterate our claim since the slope of the light cone in the $t - r$ plane diverges as $r \rightarrow 2M$. Yet it is also suggesting implicitly a road to bypass the problem. The following exercise teaches our readers why we are so sure a priori that we are dealing with a “fake” singularity:

Exercise 165. Consider a two dimensional manifold whose line element reads

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}},$$

where $t \in \mathbb{R}$ while $r \in (2M, \infty)$. Considering an arbitrary initial point (r_0, t_0) , prove that a massless particle falling towards $r \rightarrow 2M$ reaches the singularity in a finite proper time.

It seems therefore reasonable to consider new coordinates which are subordinated to the null geodesics lying in the $t - r$ plane. This procedure is not fail proof, but it is efficient in so many instances that it is convenient to start from it. Hence our starting point is Equation (276) which, upon integration, leads to introducing the **tortoise (or Regge-Wheeler) coordinates**

$$u = t - r_* \quad \text{and} \quad v = t + r_* \quad \text{with} \quad r_* \doteq r + 2M \ln \left(\frac{r}{2M} - 1 \right). \quad (277)$$

It would be convenient to use both u and v , but we start only by replacing (t, r, θ, φ) with the **Eddington-Finkelstein coordinates** (v, r, θ, φ) . Equation (239) takes the form

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dv^2 + 2dvdr + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2). \quad (278)$$

Although we have done nothing but a simple change of coordinates, we have gained some additional information. On the one hand, the determinant of the matrix associated to Equation (278) is non singular as $r \rightarrow 2M$ since

$$\det(g) = -r^4 \sin^2(\theta),$$

which seems to suggest that $g_{\mu\nu}$ is invertible. Yet, one should not be fooled by this feature since the same could have been said of the original form in Equation (239). The real advantage is that as $r \rightarrow 2M$ the metric is no longer degenerate even if g_{vv} tends to 0 contrary to the Schwarzschild form which lead to a divergence in g_{rr} . Observe that the inverse metric takes the matrix form

$$g^{\mu\nu} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 - \frac{2M}{r} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix},$$

which is non-degenerate as $r \rightarrow 2M$. An additional improvement can be observed looking at the equation for null geodesics in the $v - r$ plane. If we consider once more a curve $\gamma : I \subseteq \mathbb{R} \rightarrow M$, the defining equation reads

$$- \left(1 - \frac{2M}{r}\right) \left(\frac{dv}{d\lambda}\right)^2 + 2 \frac{dv}{d\lambda} \frac{dr}{d\lambda} = 0 \implies \frac{dv}{dr} = \begin{cases} 0 \\ \frac{2r}{r-2M} \end{cases}.$$

This entails that the curves $v = v_0$ are null geodesic for all values of r even if $r < 2M$. The second ODE can be integrated yielding $v = 2r + 4M \ln |r - 2M|$, which entails the existence of a

singularity at $r \rightarrow 2M$. Yet this does not mean that the light cones become degenerate, rather that null geodesics point at $r = 2M$ one along the v - and one along the r -axis. In this respect the submanifold $r = 2M$ becomes special since for $r < 2M$ all future pointing light rays (increasing v) all lead towards a decreasing value of r , while those past pointing behave the opposite way. In this respect, one can claim that, once inside the region bounded by the hypersurface $r = 2M$, light can no longer escape it, being trapped inside. For this reason we refer to $(v, 2M, \theta, \varphi)$ as the **event horizon** of Schwarzschild spacetime.

In other words we have shown that all issues with Equation (239) becoming ill-defined at $r = 2M$ are only apparent and therefore this represents the prototypical behaviour of what is known as a **coordinate singularity**. As a matter of fact, using the Eddington-Finkelstein coordinates removes any possible pathology. Yet we can observe that, with reference to Equation (277), we have chosen to work with the coordinate v rather than u . If we would have opted for the latter, the outcome would have been similar in the sense that no singularity would occur at $r = 2M$, but, for $r < 2M$, all past pointing light rays would not be able to escape this region.

It is therefore advisable to try to study these two options contemporary trying to unveil whether they can be combined together. Hence starting from Equations (239) and (277), we switch from the coordinates (t, r, θ, φ) to (u, v, θ, φ) . The new line element reads

$$ds^2 = - \left(1 - \frac{2M}{r(u, v)} \right) dudv + r^2(u, v) (d\theta^2 + \sin^2 \theta d\varphi^2), \quad (279)$$

where r is a function depending on u and v . In this coordinate system $u, v \in \mathbb{R}$, but we are not allowing to include points with $r < 2M$ since the identity

$$v - u = 2r + 4M \ln \left(\frac{r}{2M} - 1 \right),$$

entails that we have simply pushed $r \rightarrow 2M$ to infinity. This minor setback can be overcome introducing a new set of coordinates

$$U = -e^{\frac{u}{4M}} = -\sqrt{\frac{r-2M}{2M}} e^{\frac{r-t}{4M}} \quad \text{and} \quad V = e^{\frac{v}{4M}} = \sqrt{\frac{r-2M}{2M}} e^{\frac{r+t}{4M}}. \quad (280)$$

Consequently $-U, V \in (0, \infty)$ and Equation (279) becomes

$$ds^2 = -\frac{32M^3}{r} e^{-\frac{r}{2M}} dU dV + r^2(U, V) (d\theta^2 + \sin^2 \theta d\varphi^2). \quad (281)$$

On the one hand, we observe that nothing prevents us to extend the domain of U and V to the whole $\mathbb{R}$, the event horizon being located whenever either $U = 0$ or $V = 0$. On the other hand, they are lightlike coordinates since

$$g(\partial_U, \partial_U) = g(\partial_V, \partial_V) = 0.$$

Therefore it is more convenient to introduce a new set of coordinates (T, X, θ, φ) with $(T, X) \in \mathbb{R}^2$ such that

$$T = \frac{U + V}{2} = \sqrt{\frac{r-2M}{2M}} e^{\frac{r}{4M}} \sinh \left(\frac{t}{4M} \right) \quad \text{and} \quad X = \frac{V - U}{2} = \sqrt{\frac{r-2M}{2M}} e^{\frac{r}{4M}} \cosh \left(\frac{t}{4M} \right). \quad (282)$$

These are known as **Kruskal-Szekeres coordinates** and they lead to the maximal extension of Schwarzschild metric with a line element

$$ds^2 = \frac{32M^3}{r(T, X)} e^{-\frac{r}{2M}} (-dT^2 + dX^2) + r^2(T, X) (d\theta^2 + \sin^2 \theta d\varphi^2). \quad (283)$$

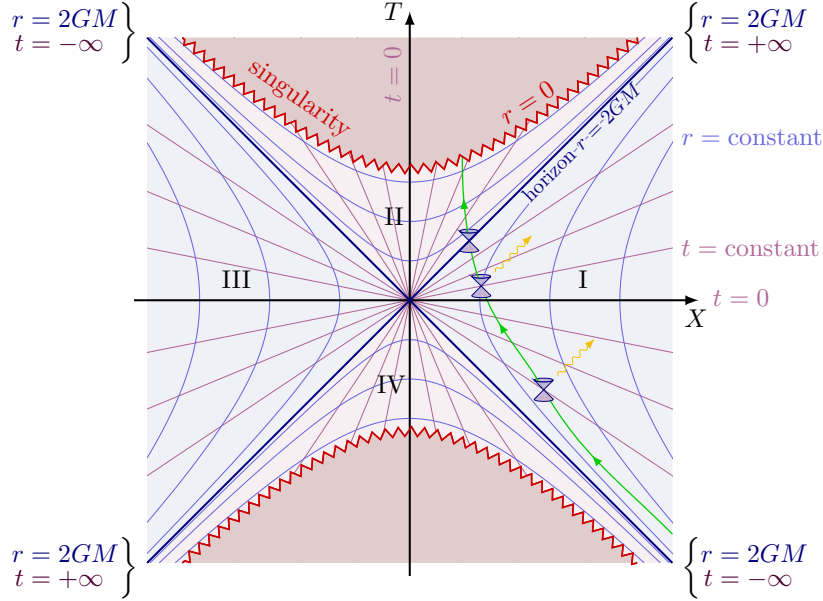


Figure 8: Kruskal Diagram of a maximally extended Schwarzschild spacetime.

We can draw several conclusions from Equation (283) and Figure 8 gives a pictorial description of the whole spacetime in what is called **Kruskal diagram**. It codifies all information of the underlying background in the $T - X$ plane, hence entailing that each point therein is actually a 2-sphere. All comments below refer to Figure 8.

- The conformal structure in the diagram is the same as that of the two-dimensional Minkowski spacetime, since, at constant (θ, φ) , Equation (283) is conformal to $-dT^2 + dX^2$. This entails that all lines of the form $T \pm X = \text{const.}$ are light rays.
- From Equation (282) we can infer that

$$T^2 - X^2 = \left(1 - \frac{r}{2M}\right) e^{\frac{r}{2M}},$$

which implies the constant r hypersurfaces are hyperbole in the $X - T$ plane, though with $T^2 - X^2 < 0$ if $r > 2M$

- From Equation (282) we can infer that

$$\frac{T}{X} = \tanh\left(\frac{t}{4M}\right), \quad (284)$$

which implies that constant time hypersurfaces in the original Schwarzschild spacetime are nothing but straight lines going through the origin in the Kruskal diagram. Yet the constraint $T^2 - X^2 < 0$ for $r > 2M$ entails that the absolute value of the slope must be smaller than 1.

- the region $r = 2M$ corresponds to the light rays $T = \pm X$ which individuate the **event horizon**. Their intersection, namely the 2-sphere $(0, 0, \theta, \varphi)$ is known as bifurcate horizon.

In these comments we have discussed the features of the Kruskal diagram focusing the attention on the those points which are contained also in the original Schwarzschild spacetime. Yet, as we can infer from Figure 8 these cover only a portion of the full Kruskal spacetime. Before commenting on them, we propose a very important exercise

Exercise 166. Let (M, g) be the Kruskal spacetime with line element as per Equation (283). Show that the Kretschmann scalar is

$$K \doteq R_{\mu\nu\rho\delta}R^{\mu\nu\rho\delta} = \frac{48M^2}{r^6(T, X)}.$$

Going back to Figure 8, we can therefore individuate 4 regions:

Region I: This covers the original Schwarzschild spacetime and it is also known as the static patch. Its boundary is the event horizon, both $T = X$ a $T = -X$ with $X \geq 0$.

Region II: This encompasses all points inside the event horizon and the causal structure unveils that no future directed causal curve stemming from a point herein can reach any other region of the Kruskal diagram. In other words we are giving sense to the folklore statement that *light cannot escape* and hence this is what is known as **black hole region**. Most notable Exercise 166 entails that at

$$r = 0 \implies T^2 - X^2 = 1 \text{ and } T > 0$$

the metric becomes **singular** with no hope of bypassing this issue. This is the famous **black hole singularity**.

Region III: This is the space reversed, $X \rightarrow -X$, counterpart of the physical region I . It is its exact mirror sharing all its properties, including being static. Yet it is not connected to region I except via the bifurcate horizon, although only spacelike curves can go through it. Therefore all sci-fi fan shall recognize the simplest example of a **wormhole**.

Region IV: This is the space reversed, $T \rightarrow -T$, counterpart of the physical region II and the causal structure unveils that no past directed causal curve stemming from a point herein can reach any other region of the Kruskal diagram. In other words we are giving sense to the

folklore statement that *all light escapes* from here in the future and hence this is what is known as **white hole region**. Hence the locus

$$T^2 - X^2 = 1 \text{ and } T < 0,$$

represents another geometric singularity which is not shadowed by the event horizon since all future directed light rays exit from region *I*. For this reason this is known as a **naked singularity**.

Remark 56. *There is a sharp difference between black and white holes. The first do exist, the second not. More precisely, there are several experimental evidences of the existence of black holes in our Universe, ranging from gravitational waves to the first image of a black hole, at the center of the Messier 87 galaxy, released in 2016 by the Event Horizon Telescope (EHT). It is fair to say that astronomical black holes are certainly quite different from the one realized in a spherically symmetric spacetime which is undoubtedly simplistic. Yet core features, such as the presence of a event horizon are universal and enter in the most general theory of black holes. From a theoretic viewpoint they are a solid prediction of Einstein's theory, since it has been proven using the initial value formulation of Einstein's equations that they form under rather general circumstances.*

On the contrary white holes do not have any experimental evidence and they appear to be a by-product of the mathematical formulation of the theory. From a physical viewpoint they are rather unnatural since they possess a singularity which is no way hidden by an event horizon. Therefore they should be directly "visible" to observers. This lead Roger Penrose to formulate the cosmic censorship hypothesis at the end of the sixties, which still resists in its weak formulation and it entails substantially the lack of any white hole. Hence, despite the rumbling of media and of some scientists appearing in TV, white holes are non physical. On the contrary, their existence within the mathematical framework of the theory seem to suggest that modifications might be necessary, though to which extent it is still unclear.

Killing Horizons In the construction of the Kruskal extension, see Equation (283), our first hint has been the observation that, as $r \rightarrow 2M$, some of the metric coefficients of Equation (239) either diverge or tend to vanish. Despite being both potentially problematic issues, the second one is more peculiar since it refers to $g_{tt} = g(\partial_t, \partial_t)$. Yet, contrary to ∂_r , which intervenes in the calculation of g_{rr} , the divergent coefficient as $r \rightarrow 2M$, ∂_t is a *timelike Killing vector field* whenever $r > 2M$.

Equation (239) seems to suggest that ∂_t is changing its character at the event horizon, becoming lightlike. Yet, to draw this conclusion, one needs to start from Equation (283) and in particular from Equation (284) which entails that $t = 4M \operatorname{arctanh}(\frac{T}{X})$ and that

$$\partial_t = \frac{1}{4M} (X\partial_T + T\partial_X). \quad (285)$$

Consequently

$$g(\partial_t, \partial_t) = \frac{2M}{r(T, X)} (X^2 - T^2),$$

which entails that

$$\partial_t \text{ is } \begin{cases} \text{timelike} & \text{if } |T| > |X| \implies r > 2M \\ \text{lightlike} & \text{if } T = \pm X \implies r = 2M \\ \text{spacelike} & \text{if } |T| < |X| \implies 0 < r < 2M \end{cases} .$$

In addition to this observation, the following exercise guarantees us that ∂_t keeps the same structural properties enjoyed in region I of the Kruskal diagram, also on the event horizon and on region II , see Figure 8.

Exercise 167. Let (M, g) be the Kruskal extension of Schwarzschild spacetime as per Equation (283). Prove that ∂_t as in Equation (285) identifies a Killing vector field, see Definition 80.

This exercise guarantees us that ∂_t keeps its property of identifying an isometry also in the full Kruskal extension, but the price to pay is that it changes its character from timelike to spacelike. The most notable consequence of this observation is that we can characterize the event horizon as the locus where ∂_t becomes lightlike. This prompts introducing a novel, geometric structure.

Definition 168. Let (M, g) be an n -dimensional spacetime, $n \geq 2$, as per Definition 52 and let $\chi \in \Gamma(TM)$ be a Killing vector field as per Definition 80. We say that an $(n-1)$ -dimensional, embedded submanifold (Σ, ι) as per Definition 40 is

1. a **null submanifold** if, for every $p \in \Sigma$, any vector $n_p \in T_{\iota(p)}M$ such that $g(n_p, v) = 0$ for all $v \in d\iota(T_p\Sigma)$, is lightlike, namely $g|_{\iota(p)}(n_p, n_p) = 0$.
2. a **Killing horizon** if χ is normal to Σ , namely $g(\chi(\iota(p)), v) = 0$ for all $v \in d\iota(T_p\Sigma)$ and in addition χ is lightlike on Σ , namely $g|_{\iota(p)}(\chi(\iota(p)), \chi(\iota(p))) = 0$.

One can realize by direct inspection that the event horizon in Kruskal spacetime is also a Killing horizon according to Definition 168. The reason for introducing this novel concept stems mainly by the possibility of associating to any Killing horizon a notable quantity which plays a distinguished rôle especially in understanding the thermodynamic properties of black holes. More precisely, given a Killing horizon as per Definition 168, we call **surface gravity** the value $\kappa \in \mathbb{R}$ such that

$$\chi^\mu|_\Sigma \nabla_\mu \chi^\nu|_\Sigma = \kappa \chi^\nu|_\Sigma. \quad (286)$$

Observe that, on account of Equation (49) $\nabla_{(\mu} \chi_{\nu)} = 0$, while the condition of χ being orthogonal to Σ as per Definition 168 translates to $\chi_{[\mu} \nabla_\nu \chi_{\sigma]} = 0$ for all $\mu, \nu, \sigma = 1, \dots, n$. These conditions can be combined to write the second one as

$$\chi_\mu \nabla_\nu \chi_\sigma + \chi_\nu \nabla_\sigma \chi_\mu - \chi_\sigma \nabla_\nu \chi_\mu = 0,$$

where we leave implicit the restriction to Σ to avoid a heavy notation. Contracting each side by $\nabla^\nu \chi^\sigma$ and using that $\nabla_\nu \chi_\sigma = -\nabla_\sigma \chi_\nu$ yields

$$\chi_\mu \nabla^\nu \chi^\sigma \nabla_\nu \chi_\sigma = -2(\nabla^\nu \chi^\sigma) \chi_\nu (\nabla_\sigma \chi_\mu).$$

Using twice Equation (286) it descends that

$$(\nabla^\nu \chi^\sigma) \chi_\nu (\nabla_\sigma \chi_\mu) = \chi_\nu (\nabla^\nu \chi^\sigma) (\nabla_\sigma \chi_\mu) = \kappa \chi^\sigma \nabla_\sigma \chi_\mu = \kappa^2 \chi_\mu.$$

Summarizing and relabeling dummy indices we have obtained that

$$\kappa^2 = -\frac{1}{2}(\nabla_\mu \chi_\nu)(\nabla^\mu \chi^\nu). \quad (287)$$

Remark 57. *It is interesting to observe that, given a Killing horizon as in Definition 168, since χ is normal to Σ at every point therein, it must obey to the geodesic equation. In this respect Equation (286) can be read as a measure of the failure of using the affine parameterization in the representation of χ in the underlying coordinate system.*

Example 169. Let us consider the Kruskal extension as per Equation (283) and the Killing field $\chi^\nu \partial_\nu = \frac{1}{4GM} (X \partial_T + T \partial_X)$ as in Equation (285) where we have restored the explicit dependence on G . In order to evaluate χ_ν , we need the inverse metric. Since the matrix encoding the components of Equation (283) is block diagonal, in order to compute its inverse we can limit our attention to $\frac{32(GM)^3}{r(T,X)} e^{-\frac{r}{2GM}} (-dT^2 + dX^2)$ from which it descends that

$$\begin{pmatrix} g^{TT} & g^{TX} \\ g^{XT} & g^{XX} \end{pmatrix} = \frac{r(T,X)}{32(GM)^3} e^{\frac{r}{2GM}} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

This entails that

$$\chi_\nu dx^\nu = -\frac{Xr(T,X)}{128(GM)^4} e^{\frac{r}{2GM}} dT + \frac{Tr(T,X)}{32(GM)^3} e^{\frac{r}{2GM}} dX.$$

Yet, it is more convenient to compute

$$\nabla_\mu \chi^\nu = \partial_\mu \chi^\nu + \Gamma_{\mu\rho}^\nu \chi^\rho,$$

where we can restrict the attention to $\nu, \rho = X, T$. Hence, using Equation (33), it holds that

$$\begin{aligned} \Gamma_{TT}^T &= \Gamma_{TX}^X = \Gamma_{XT}^X = \Gamma_{XX}^T = \frac{2GMT(r(T,X) + 2GM)}{r^2(T,X)} e^{-\frac{r(T,X)}{2GM}} \\ \Gamma_{TT}^X &= \Gamma_{TX}^T = \Gamma_{XT}^T = \Gamma_{XX}^X = -\frac{2GMX(r(T,X) + 2GM)}{r^2(T,X)} e^{-\frac{r(T,X)}{2GM}}. \end{aligned}$$

This leads to

$$\nabla_\mu \chi^\nu = \begin{cases} 0 & \text{if } \mu = X, \nu = X \\ \frac{1}{4GM} - \frac{(X^2 - T^2)(r(T,X) + 2GM)}{2r^2(T,X)} e^{-\frac{r(T,X)}{2GM}} & \text{if } \mu = T, \nu = X \\ 0 & \text{if } \mu = T, \nu = T \\ \frac{1}{4GM} - \frac{(X^2 - T^2)(r(T,X) + 2GM)}{2r^2(T,X)} e^{-\frac{r(T,X)}{2GM}} & \text{if } \mu = X, \nu = T \end{cases}.$$

On account of Equation (287), our next step consists of evaluating

$$(\nabla_\mu \chi_\nu)(\nabla^\mu \chi^\nu) = g^{\mu\rho} g_{\nu\lambda} (\nabla_\mu \chi^\lambda)(\nabla_\rho \chi^\nu) = g^{XX} g_{TT} (\nabla_X \chi^T)^2 + g^{TT} g_{XX} (\nabla_T \chi^X)^2 \quad (288)$$

Inserting this result in Equation (288) and recalling that the outcome has to be evaluated at the horizon $r(T, X) = 2M$ and $T = \pm X$, we obtain

$$(\nabla_\mu \chi_\nu)(\nabla^\mu \chi^\nu) = -\frac{1}{8(GM)^2}$$

Using at last Equation (287), this entails that the surface gravity on the event horizon of a Schwarzschild black hole is

$$\kappa_{Schwar.} = \frac{1}{4M}.$$

Remark 58. *One might be tempted to think that Killing horizons are a special instance of an event horizon even though we have given just one example of the latter. Yet, it turns out that actually this is quite the opposite. This can be seen with a rather simple, yet instructive example. Consider the four-dimensional Minkowski spacetime $(\mathbb{R}^4, \eta)$ and, working with the standard Cartesian coordinates (t, x, y, z) , we consider a boost $\chi \doteq t\partial_x + x\partial_t$. This individuates a Killing field and*

$$g(\chi, \chi) = t^2 - x^2,$$

from which we can infer that the hypersurfaces $(t, \pm t, y, z)$ are Killing horizons and, certainly, not event horizons. It is worth mentioning that these represent the boundary of two distinguished regions

$$\mathcal{W}_L = \{(t, x, y, z) \in \mathbb{R}^4 \mid x < 0 \text{ and } |t| < |x|\}, \quad \mathcal{W}_R = \{(t, x, y, z) \in \mathbb{R}^4 \mid x > 0 \text{ and } |t| < x\},$$

*which are known as the **left (L) and right (R) Rindler wedge**. They play a prominent rôle in the formalization of the Unruh effect.*

7.4 Rotating Black Holes

In the previous section we have introduced the first example of a black hole, see Equation (283). From a physical viewpoint, the underlying scenario is rather unrealistic since

1. the black hole is eternal, namely, being the metric static, either it exists at the beginning of time or it does not. There is no dynamical process leading to its formation, such as a collapsing star, which is expected to be a more realistic picture,
2. the black hole is spherically symmetric, hence static by Birkhoff theorem 162. This is also quite unrealistic since celestial bodies are expected to be rotating, a feature which is certainly incompatible with being static since, under time reversal, the direction of rotation would be the opposite.

Taking into account the first concern is rather prohibitive since we would have to develop a realistic model of a star collapse. There are several proposal in the literature, but they are quite long and they would require a significant detour from our main objective. On the contrary, allowing for a non-vanishing rotation seems more approachable and, indeed, this is the case. To give a time span to our readers, think that Schwarzschild wrote his metric as in Equation (239) in 1916, while the work of Kruskal and Szekeres is from 1960. The extension to a rotating scenario did not require that long, dating 1963 and it is due to *Roy Kerr*.

To enter in the technical details, in the following (M, g) shall denote a four-dimensional spacetime which is axially symmetric as per Definition 148. In this section we shall follow mainly [Tow97] which are old, but still very effective lecture notes. In contrast to Section 7.1 we shall not delve in all mathematical aspects of the derivation of the metric, limiting ourselves to mention the final outcome. We shall start directly with the explicit expression of the **Kerr metric**:

$$ds^2 = -\frac{\Delta - a^2 \sin^2 \theta}{\Sigma} dt^2 - 2a \sin^2 \theta \frac{r^2 + a^2 - \Delta}{\Sigma} dt d\varphi + \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \sin^2 \theta d\varphi^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2, \quad (289)$$

where

$$\begin{cases} \Sigma = r^2 + a^2 \cos^2 \theta \\ \Delta = r^2 - 2Mr + a^2 = (r - r_+)(r - r_-) \end{cases} \quad (290)$$

where $M \geq 0$ and $a \in \mathbb{R}$ are two parameters whose physical interpretation will be given below, while

$$r_{\pm} = M \pm \sqrt{M^2 - a^2}. \quad (291)$$

In Equation (289) we are employing the **Boyer-Lindquist** coordinates, according to which $t \in \mathbb{R}$, $r > 0$ while (θ, φ) are the standard coordinates over the 2-sphere. Let us observe some of the key features of the Kerr metric:

- the metric coefficients do not depend explicitly on t and on φ . Therefore, on account of Example 83, we can infer that both $\frac{\partial}{\partial t}$ and $\frac{\partial}{\partial \varphi}$ are Killing fields. The second one encodes the property of being axially symmetric, while, the first one entails that the spacetime is stationary, although this conclusion is slightly premature as we shall see below. Comparing with Definition 148, we can conclude that the Kerr metric is not static as constant time hypersurfaces are not orthogonal to $\frac{\partial}{\partial t}$ due to the presence of cross terms $dt d\varphi$.
- In the limit as $r \rightarrow \infty$ the metric tends to the Minkowski one, signaling that the background is *asymptotically flat* at spatial infinity. A precise mathematical definition of this concept can be given and it is at the heart of the global definition of a black hole region. An interested reader can refer to [Wal84, Chap. 8, 9 & 11].
- if we set $a = 0$ in Equation (239), we obtain the Schwarzschild metric as in Equation (239). Hence, taking into account the analysis of Section 7.1.1, this justifies the interpretation of M as the **black hole mass**.

- Although the spacetime is not static, there exists a manifest discrete isometry $(t, \varphi) \rightarrow (-t, -\varphi)$ which can be interpreted as stating that, if we invert both the direction of time and of the axis of rotation, the geometry is left unchanged. This is compatible with the interpretation of a rotating system.
- Contrary to Schwarzschild spacetime, the Kerr metric has only two Killing fields, but in addition to it, it possesses a *Killing tensor*, see Definition 82 whose expression in the Boyer-Lindquist coordinates is

$$K^{\mu\nu} = 2\Sigma l^{(\mu} n^{\nu)} + r^2 g^{\mu\nu}, \quad (292)$$

with

$$l^\mu = \left(\frac{r^2 + a^2}{\Delta}, 1, 0, \frac{a}{\Delta} \right) \text{ and } n^\mu = \left(\frac{r^2 + a^2}{\Sigma}, -\frac{\Delta}{2\Sigma}, 0, \frac{a}{2\Sigma} \right),$$

where Δ and Σ are defined as in Equation (290). Following Exercise 94, this entails that, given a geodesic $\gamma \in C^\infty(I; M)$ with $I \subseteq \mathbb{R}$, there exists an associate conserved quantity, known as *Carter constant*:

$$C = K^{\mu\nu} \dot{\gamma}_\mu \dot{\gamma}_\nu.$$

This property is not particularly useful to unveil the geometric properties of Kerr spacetime, but it becomes essential when one wants to solve the wave equation on this background, since it allows to use mode separation.

In addition to these considerations we can give an interpretation to the remaining parameter a . This is left as the following exercise, which is a special instance of what are known as *Komar integrals*.

Exercise 170. Let (M, g) be the Kerr spacetime as per Equation (289) and let $\chi \doteq \frac{\partial}{\partial \varphi}$. Show that, considering a 2-sphere $\mathbb{S}_R^2$ of radius R , it turns out that

$$\frac{1}{16\pi} \lim_{R \rightarrow \infty} \int_{\mathbb{S}^2} \omega_\chi = Ma, \quad (293)$$

where ω_χ is the 2-form of components $\epsilon_{\mu\nu\rho\sigma} \nabla^\rho \chi^\sigma$.

In order to convince ourselves that we are on the right track, if we consider Minkowski spacetime $\mathbb{R}^4$ and an observer therein rotating with constant velocity Ω , the metric in a frame comoving with him/her reads

$$ds^2 = -dt^2 + dr^2 + r^2(d\varphi - \Omega dt)^2 + dz^2,$$

where we have used cylindrical coordinates (t, r, θ, z) . In this case Equation (293) yields a result proportional to Ω hence justifying our interpretation of the parameter a in Equation (289) as being proportional to J , the **total angular momentum** of the black hole via

$$a = \frac{J}{M}. \quad (294)$$

Having established this interpretation of the parameters (a, M) we can look more closely at the metric coefficients in the Kerr metric which can become singular in two instances. The first one is when Δ as per Equation (290) vanishes, that is, if

$$r = r_{\pm} = M \pm \sqrt{M^2 - a^2}.$$

This entails the existence of three different regimes:

1. $M^2 - a^2 < 0$, namely Δ is never vanishing since it has only imaginary roots. Yet, as we can see, we can still get an additional singularity when $\Sigma = 0$, namely if $r = 0$ and $\theta = \frac{\pi}{2}$. Since the Kretschmann scalar diverges therein, this is not a coordinate singularity. The lack of any event horizon entails that this is naked and it is considered non physical. This scenario is usually discarded,
2. $M^2 - a^2 = 0$, namely $r_+ = r_-$. This is called an **extremal Kerr black hole** and it is considered a limiting case where only one horizon exists. In this respect this is very similar to Schwarzschild spacetime,
3. $M^2 - a^2 > 0$, namely there exists two horizons, an inner $r = r_-$ and an outer one $r = r_+$. These are coordinate singularities as it can be proven following the same strategy as in Section 7.3 and they shield a geometric singularity which corresponds to

$$\Sigma = 0 \implies r = 0 \text{ and } \theta = \frac{\pi}{2}. \quad (295)$$

The insurgence of two horizons is a peculiar phenomenon which can be fully ascribed to the presence of a non vanishing angular momentum J . This should be actually combined to an additional feature which was not present in Schwarzschild spacetime. Let us consider the Killing field $\frac{\partial}{\partial t}$ and let us evaluate

$$g(\partial_t, \partial_t) = \frac{a^2 \sin^2 \theta - \Delta}{\Sigma}. \quad (296)$$

If t represents a legitimate time coordinate, then ∂_t should be everywhere timelike. Since Σ is strictly positive, this amounts to requiring

$$\begin{aligned} \Delta - a^2 \sin^2 \theta = r^2 - 2Mr + a^2 \cos^2 \theta > 0 &\implies \\ \implies r < M - \sqrt{M^2 - a^2 \cos^2 \theta} \quad \text{or} \quad r > M + \sqrt{M^2 - a^2 \cos^2 \theta}. \end{aligned} \quad (297)$$

Since $0 \leq \cos^2 \theta \leq 1$, it turns out that ∂_t fails to be timelike in the region

$$(t, r, \theta, \varphi) \mid r_+ < r < M + \sqrt{M^2 - a^2 \cos^2 \theta}.$$

This is known as **ergosphere** and it is a unique feature of rotating spacetimes. The failure of ∂_t of being everywhere timelike is dramatic since it entails that Kerr is stationary with respect to it only in a portion of the region outside the outer event horizon. A natural question that one might ask is whether this can be ascribed to a bad choice of the Killing field. The following exercise shall dispel any doubt:

Exercise 171. Let (M, g) be the Kerr spacetime whose metric is as per Equation (289). Prove that there exists no linear combination of ∂_t and ∂_φ which is at the same time a Killing field and timelike for every $r > r_+$, where r_+ is as per Equation (291).

As a last feature, we observe that, if we consider a linear combination between ∂_t and ∂_φ , we are still working with a Killing field and, for $\kappa \in \mathbb{R}$,

$$g(\partial_t + \kappa \partial_\varphi, \partial_t + \kappa \partial_\varphi) = -\frac{\Delta - a^2 \sin^2 \theta}{\Sigma} - \frac{8Ma\kappa \sin^2 \theta r}{\Sigma} + \kappa^2 \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \sin^2 \theta.$$

Evaluating it at $r = r_+$, namely $\Delta = 0$, we obtain

$$g(\partial_t + \kappa \partial_\varphi, \partial_t + \kappa \partial_\varphi)|_{r=r_+} = \frac{\sin^2 \theta}{\Sigma|_{r=r_+}} (a^2 - 4Ma\kappa r_+ + \kappa^2 (r_+^2 + a^2)^2).$$

This is vanishing if

$$\kappa_\pm = \frac{2Mar_+ \pm \sqrt{4M^2 a^2 r_+^2 - a^2 (r_+^2 + a^2)^2}}{(r_+^2 + a^2)^2} = \frac{a}{r_+^2 + a^2}, \quad (298)$$

where we have used the identity $r_+^2 + a^2 = 2Mr_+$. In other words we have shown that $r = r_+$ is a **Killing horizon** with respect to the Killing field $\chi = \partial_t + \Omega_H \partial_\varphi$ with

$$\Omega_H = \frac{a}{r_+^2 + a^2},$$

r_+ being as in Equation (291). In order to give significance to Ω_H we observe that, if we consider an observer sitting at the outer horizon $r = r_+$ and if we set $\theta = \frac{\pi}{2}$, the trajectory of a lightlike particle along the horizon abides by

$$g_{tt} \left(\frac{dt}{d\lambda} \right)^2 + 2g_{t\varphi} \frac{dt}{d\lambda} \frac{d\varphi}{d\lambda} + g_{\varphi\varphi} \left(\frac{d\varphi}{d\lambda} \right)^2 = 0,$$

where g_{tt} , $g_{t\varphi}$ and $g_{\varphi\varphi}$ are the metric coefficients in Equation (289). This entails

$$\frac{d\varphi}{dt} = -\frac{g_{t\varphi}}{g_{\varphi\varphi}} \pm \sqrt{\left(\frac{g_{t\varphi}}{g_{\varphi\varphi}} \right)^2 - \frac{g_{tt}}{g_{\varphi\varphi}}},$$

which evaluated at $r = r_+$ yields

$$\frac{d\varphi}{dt} = \Omega_H = \frac{a}{r_+^2 + a^2}. \quad (299)$$

This justifies the interpretation of Ω_H as the *angular velocity* of/at the event horizon of a Kerr black hole.

Exercise 172. Let (M, g) be a Kerr black hole. Show that also the inner horizon, $r = r_-$ is a Killing horizon as per Definition 168 and show that the surface gravity is

$$\kappa_{\pm} = \frac{r_{\pm} - r_{\mp}}{2(r_{\pm}^2 + a^2)}.$$

Remark 59. In fully analogy with classical mechanics we can say that Ω_H is the angular velocity of an observer sitting at the outer event horizon and co-rotating with the Kerr black hole. This prompts the idea that one should rather consider $\chi_H \doteq \partial_t + \Omega_H \partial_\varphi$ as the natural timelike Killing field for $r > r_+$, in analogy to Schwarzschild spacetime where ∂_t is the generator of the Killing horizon. As Exercise 171 suggests this does not solve the problem of the ergosphere. More precisely χ_H has a complementary behaviour to that of ∂_t since close to the outer event horizon is timelike, but it become spacelike for large values of r . Hence χ_H becomes lightlike on an hypersurface at $r > r_+$ and this is called **speed of light surface**.

One last structural question concerning the Kerr solution concerns the existence of a generalization of Birkhoff theorem 162. We content ourselves with mentioning this important result omitting the proof which can be found in [Car71, Rob75].

Theorem 173 (Carter-Robinson). *Let (M, g) be a four-dimensional, asymptotically flat and axially symmetric and stationary solution of Einstein's vacuum equations. Then, if (M, g) is non singular on and outside the event horizon, then it lies in the two-parameter family of metrics as in Equation (289)*

Remark 60. One might wonder whether it is possible to construct additional black hole solutions beyond Schwarzschild and Kerr. The answer is of course affirmative and there exist several dozen examples in the literature. Among these, noteworthy is a class obtained considering a spacetime filled with an electromagnetic field. From a dynamical viewpoint, we are considering a coupled system consisting of Einstein and Maxwell equations. From a kinematic perspective, in addition to the total mass and to the angular velocity we have to include an additional parameter, the electric charge. These scenarios have been investigate thoroughly in the literature and the **Kerr-Newman metric** and, in the limit of $\Omega_H \rightarrow 0$, namely in absence of rotation, we obtain the **Reissner-Norström metric**. A reader interested in these topics can consult [Car19, Tow97]. From a structural viewpoint both these examples are very similar to a Kerr black hole and therefore we omit entering into their detailed analysis.

7.5 Black Hole Thermodynamics

The Kerr metric allows to gently introduce one of the most iconic features of black hole physics, which has been investigated in the seventies mainly by Jacob Bekenstein in [Bek72, Bek73], leading subsequently to the discovery of Hawking's radiation, the first and most celebrated exclusive phenomenon ascribed to quantum field theory on curved backgrounds [Haw75].

The starting point is the geodesic motion subordinated to the metric in Equation (289). If we consider a geodesic $\gamma : I \subseteq \mathbb{R} \rightarrow M$ and if we work with the Boyer-Lindquist coordinates,

the presence of two Killing fields, $\chi = \partial_t$ and $\xi = \partial_\varphi$ entails the existence of two conserved quantities along γ , namely, following Corollary 93, Equation (61) in particular,

$$\mathcal{E} = g_{\mu\nu}(\gamma(\lambda))\chi^\mu \dot{x}^\nu = -\frac{\Delta - a^2 \sin^2 \theta}{\Sigma} \frac{dt}{d\lambda} - 2a \sin^2 \theta \frac{r^2 + a^2 - \Delta}{\Sigma} \frac{d\varphi}{d\lambda}, \quad (300a)$$

$$\mathcal{L} = g_{\mu\nu}(\gamma(\lambda))\xi^\mu \dot{x}^\nu = -2a \sin^2 \theta \frac{r^2 + a^2 - \Delta}{\Sigma} \frac{dt}{d\lambda} + \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \sin^2 \theta \frac{d\varphi}{d\lambda}, \quad (300b)$$

where $\dot{x}^\mu$ represents the components of the tangent vector to γ . Focusing on Equation (300a), there is a reason why we have not employed the symbol E , but $\mathcal{E}$. If we consider a point particle of mass $m > 0$ moving along γ , that is in absence of external forces, we can interpret $p^\mu = m\dot{x}^\mu$ as the classical 4-momentum. This is a timelike vector field and so is χ far from the Kerr black hole, that is for $r \gg r_+$. Hence $m\mathcal{E} < 0$ and, at large distances from the event horizon, we can interpret $-\mathcal{E}$ as an energy density per unit mass. Yet, as shown in Equation (297), close to the outer event horizon, namely in the ergosphere region, χ becomes spacelike and therefore one can arrange p^μ so that $m\mathcal{E} > 0$. Hence this hinders any interpretation of $\mathcal{E}$ as being connected to energy therein.

Yet, this is not the issue. On the contrary, one can imagine an idealized scenario where a particle of mass m is launched toward the ergosphere with $\mathcal{E} = g_{\mu\nu}\chi^\mu \dot{x}^\nu < 0$. Observe that the trajectory is not geodesic since $\mathcal{E}$ needs to change sign while entering the ergoregion, which is incompatible with a conservation law. Yet we can still interpret $-\mathcal{E}$ as an energy density per unit mass. At a point inside the ergoregion the particle splits instantaneously in two different particles, either due to some internal forces, sheer injustice, unluckiness or the will of Cthulhu. The reason is not that relevant and it is more a matter of predisposition of the reader. Yet, when this occurs, the process realizes it in such a way that

$$p^\mu = p_{(1)}^\mu + p_{(2)}^\mu \implies \mathcal{E} = \mathcal{E}_{(1)} + \mathcal{E}_{(2)},$$

where the subscripts refer to the two particles, while the second equality implements Equation (300a). Fine tuning the process, one can arrange that the first particle falls towards the outer event horizon, that is $\mathcal{E}_{(1)} > 0$, while the second heads outside the ergoregion, where $-\mathcal{E}_{(2)} > -\mathcal{E}$. Apparently the particle has gained energy and this is known as **Penrose process**.

The most unsatisfactory feature of this reasoning is that it is unclear where the additional energy of the exiting particle is coming from. This can be understood recalling that we have not yet used Equation (300b) and the Killing vector field χ_H which is timelike close to the outer event horizon, see Remark 59. This entails that

$$p_{(1)}^\mu \chi_\mu = \mathcal{E}_{(1)} + \Omega_H \mathcal{L}_{(1)} < 0,$$

where Ω_H is the angular velocity of the black hole at $r = r_+$. In other words

$$\mathcal{L}_{(1)} < -\frac{\mathcal{E}_{(1)}}{\Omega_H} < 0, \quad (301)$$

where we have used that the particle falling inside the event horizon is such that $\mathcal{E}_{(1)} > 0$. When the particle enters the horizon, for an observer sitting outside it and infinitely far, this is contributing to the total mass and angular momentum of the system. In other words we expect still to see a Kerr metric, but with modified parameters, namely

$$M \rightarrow M + \delta M = M - \mathcal{E}_{(1)} \quad \text{and} \quad J \rightarrow J + \mathcal{L}_{(1)},$$

where $J = Ma$ is the total angular momentum of the black hole, see Equation (294). This entails that we can rewrite Equation (301) as

$$\delta J < \frac{\delta M}{\Omega_H}. \quad (302)$$

In other words, in the Penrose process we are *extracting energy in exchange for angular momentum* of the black hole. This prompts the idea that this phenomenon cannot go forever since, at some point, the black hole stops rotating, which, in the Kerr case, means that it tends to the Schwarzschild metric. In order to make this idea precise, observe that we can introduce the *irreducible mass*

$$M_{irr}^2[M, J] = \frac{1}{2} \left(M^2 + \sqrt{M^4 - J^2} \right) \quad (303)$$

Its variation reads

$$\begin{aligned} 2\delta M_{irr}^2 &= \frac{\partial 2M_{irr}^2}{\partial M} \delta M + \frac{\partial 2M_{irr}^2}{\partial J} \delta J = \left(2M + \frac{2M^3}{\sqrt{M^4 - J^2}} \right) \delta M - \frac{J}{\sqrt{M^4 - J^2}} \delta J = \\ &= \left(2M \frac{M + \sqrt{M^2 - a^2}}{\sqrt{M^2 - a^2}} \right) \delta M - \frac{a}{\sqrt{M^2 - a^2}} \delta J = \frac{a}{\sqrt{M^2 - a^2}} (\Omega_H^{-1} \delta M - \delta J). \end{aligned}$$

In other words Equation (302) holds true if and only if

$$\delta M_{irr} > 0.$$

Yet, from Equation (303) we infer that

$$M^2 = M_{irr}^2 + \frac{J^2}{4M_{irr}^2} \geq M_{irr}^2,$$

which entails that a Penrose process cannot ever make the mass of a Kerr black hole smaller than M_{irr} . In other words we can at most extract as much energy as the one ascribed to J , that is to rotation. A limit from above to this quantity can be obtained considering

$$M - M_{irr} = M \left(1 - \frac{1}{\sqrt{2}} \sqrt{1 + \sqrt{1 - \left(\frac{a}{M} \right)^2}} \right),$$

which, in the best case scenario, namely an extremal rotating black hole for which $a = M$, yields

$$\frac{M - M_{irr}}{M} = 1 - \frac{1}{\sqrt{2}} \sim 0.29.$$

A more enlightening consequence of Equation (302) comes from the following analysis. For an observer located far from the outer event horizon of a black hole, the structure which better characterizes it is its event horizon. From a geometric viewpoint the locus $r = r_+$ is a submanifold diffeomorphic to $\mathbb{R} \times \mathbb{S}^2$, where the $\mathbb{R}$ -direction can be seen as the collection of all integral lines of the Killing field ∂_t . These are null geodesics since we are dealing with a Killing horizon, see Equation (298). Hence an observer falling inside the Kerr black hole will intersect it only on 2-sphere and he/she will never be moving along it. In other words, for such an infalling observer, the area of such sphere represents the *size* of the black hole. To compute it, we observe that the induced metric thereon reads

$$ds^2|_{r=r_+} = \gamma_{ij}dx^i dx^j = (r_+^2 + a^2 \cos^2 \theta) d\theta^2 + \left(\frac{(r_+^2 + a^2)^2 \sin^2 \theta}{r_+^2 + a^2 \cos^2 \theta} \right) d\varphi^2, \quad (304)$$

where $x^1 = \theta$, while $x^2 = \varphi$. The area of such sphere, also referred to as the *area of the event horizon*, can be computed as

$$A \equiv A[M, J] = \int_0^\pi d\theta \int_0^{2\pi} d\varphi \sqrt{\gamma} = \int_0^\pi d\theta \int_0^{2\pi} d\varphi (r_+^2 + a^2) \sin \theta = 4\pi(r_+^2 + a^2). \quad (305)$$

If consider the variation of the area of the event horizon we obtain

$$\delta A = 8\pi \frac{a}{\Omega_H \sqrt{M^2 - a^2}} (\delta M - \Omega_H \delta J) = \frac{8\pi}{\kappa} (\delta M - \Omega_H \delta J), \quad (306)$$

where Ω_H is the angular velocity of the outer event horizon, see Equation (299), while $J = Ma$ and κ is the surface gravity at $r = r_+$ as per Exercise 172. This can be recast in a more convenient form, known as **First Law of Black Hole Thermodynamics**

$$\boxed{\delta M = \frac{\kappa}{8\pi} \delta A + \Omega_H \delta J,} \quad (307)$$

while, using Equation (302), we obtain the **Second Law of Black Hole Thermodynamics**

$$\boxed{\delta A \geq 0.} \quad (308)$$

The nomenclature *laws of thermodynamics* was first born from the observation that Equations (307) and (308) have a formal correspondence with the laws of thermodynamics in the sense that we can interpret $\Omega_H \delta J$ as the work done on the black hole by the particle absorbed via the Penrose process, which prompts the following correspondence:

$$\begin{aligned} M &\longrightarrow U = \text{internal energy density} \\ \frac{\kappa}{2\pi} &\longrightarrow T = \text{temperature} \\ \frac{A}{4} &\longrightarrow S = \text{entropy} \end{aligned}$$

It is fair to say that, at first glance and even after the first papers of Bekenstein, this correspondence was considered to be a mere formal coincidence since we have derived it in a very specific geometry, namely Kerr black holes, and using Penrose processes which are a very idealized mechanism. Nonetheless, in the seventies, Hawking proved the *black hole area theorem* which is nothing but Equation (308) under very mild assumptions, see [Wal84, Chap. 12.2]. This has elevated the laws of black hole thermodynamics from a mere curiosity holding true for Kerr and later for Kerr-Newman black holes to a structural property which is expected to hold true in full generality.

Another rather bizarre aspect of the above correspondence between geometric and mechanic parameters on the one hand and thermodynamics variables on the other hand is the association of $\frac{A}{4}$ to entropy. There is no apparent reason for the factor 1/4 and, a priori any proportionality constant might be as good as any other. Actually the reason lies in $\frac{\kappa}{2\pi}$ which is known in the literature as **Hawking temperature** and it is associated to the observation in [Haw75] that black holes emit a thermal radiation exactly at $T = \frac{\kappa}{2\pi}$.

Remark 61. *It is funny how history repeats itself even in small details such as ignoring people, ideas and part of whole theories. As a matter of fact, in thermodynamics, there exists a very disputed and not universally accepted zeroth law. Actually the same occurs in black hole thermodynamics, although this one can be elevate to a theorem, see [Tow97]. It states that, under suitable assumptions on the stress-energy tensor, the surface gravity is guaranteed to be constant on the event horizon of a black hole. What makes it less known in comparison to the first and to the second laws is that the assumptions on $T_{\mu\nu}$ are not satisfied by a large class of matter constituents, first and foremost classical scalar fields.*

Remark 62. *Since the eighties, one of the main consequences of the realization that the laws of black hole thermodynamics have a far deeper and universal meaning has been the quest for finding a microscopic derivation. More precisely, in classical mechanics, it is well-established that the laws of thermodynamics have been derived from empirical observations, but, they stem under suitable assumptions from statistical mechanics, hence from an understanding of the dynamics of the matter constituents. A counterpart of this step is absent in black hole physics, to the point that any theory which seeks to unify gravity with quantum field theory needs first and foremost to account for a derivation of the laws of black hole thermodynamics, in particular the second one. Candidates such as string theory, loop quantum gravity and so on and so forth have all claimed to be able to reproduce one way or the other the identity $S = \frac{A}{4}$. Yet this stays a necessary condition for a theory to be a viable candidate to quantum gravity and, by far, not a sufficient one as the past 50 years thought us.*

Nonetheless there is some merit to the obsession of deriving the entropy-to-area relation $S = \frac{A}{4}$. The reason stems from the observation that, in thermodynamics, the entropy is typically proportional to the volume of the physical system under investigation and, in statistical mechanics, S is a measure of the number of degrees of freedom accessible to the system. This seems rather intuitive: the larger the system, the bigger the underlying phase space. Yet, if the identity $S = \frac{A}{4}$ survives a microscopic description, this seems to suggest that, in gravity, the number of accessible degrees of freedom grows with the area. This has prompted the idea that, in gravity,

*all the information contained in suitable regions can be fully codified on its boundary. This was first remarked by G. 't Hooft in [tHoo93] and later rearranged by L. Susskind in [Sus95], which has to be credited with an eye-catching name **the holographic principle**. This has been the starting point for several models and theories among which the most famous is undoubtedly the AdS/CFT correspondence, but this is a tale for another set of lecture notes.*

7.6 Extra Stuff



1. Energy Conditions for the Stress-Energy Tensor,
2. Asymptotically flat spacetimes,
3. Hamiltonian formulation of general relativity,
4. The Bondi and ADM Mass; the positive energy theorem,
5. Penrose Singularity Theorems.

A Partition of Unity

In this short appendix, we introduce a technical tool which is simple and at the same time very powerful since it allows to reduce the analysis of global functions on a manifold and of operation on them to their counterpart on a local chart.

Definition 174. *Let M be an n -dimensional, $n \geq 1$, smooth manifold M as per Definition 9. Let $\{U_\alpha\}_{\alpha \in I}$, I being an index set, be a countable collection of open subsets $U_\alpha \subseteq M$ covering*

M , namely $\bigcup_{\alpha \in I} U_\alpha = M$. A **smooth partition of unity** subordinate to the cover is a family of functions $\psi_\alpha \in C^\infty(M; \mathbb{R})$, $\alpha \in I$ such that

1. $0 \leq \psi_\alpha(x) \leq 1$ for all $\alpha \in I$ and for all $x \in M$,
2. $\text{supp}(\psi_\alpha) \subset U_\alpha$ for all $\alpha \in I$,
3. The family of supports $\{\text{supp}(\psi_\alpha)\}_{\alpha \in I}$ is locally finite, meaning that every point has a neighborhood that intersects the elements of the collection for only finitely many values of α ,
4. $\sum_{\alpha \in I} \psi_\alpha(x) = 1$ for all $x \in M$.

Notice that the last condition in this definition is meaningful since item 3. entails that, at each $x \in M$, only a finite number of functions contributes to the sum. Hence convergence is not an issue. It is also worth stressing that it is not required that the functions ψ_α , $\alpha \in I$ are compactly supported, although, in the literature, this is often assumed. In the following we quote an existence result for such structure, leaving an interested reader to [Lee13, Thm 2.23] for the proof.

Proposition 175. *Let M be an n -dimensional, $n \geq 1$, smooth manifold M as per Definition 9. Let $\{U_\alpha\}_{\alpha \in I}$, I being an index set, be a countable collection of open subsets $U_\alpha \subseteq M$ covering M . There exists a partition of unity subordinate to this cover.*

B The Algebra of Differential Forms on $\mathbb{R}^n$

In this section we start discussing the theory of differential forms as devised by de Rham. As it should be clear now (now meaning, after having learned enough differential geometry), we start our construction from $\mathbb{R}^n$, $n \geq 1$ and we generalize it subsequently to an arbitrary smooth manifold. In order to be self contained, at the beginning there will be a significant overlap with concepts introduced in the first part of the course. Yet, we feel that, ultimately, the pros are more than the cons. In the following we follow alternatively [BT82] and [Mor01] (cacophony).

Definition 176. *Let $\{e_1, \dots, e_n\}$, $n > 0$, be a collection of arbitrary abstract elements. We call **alternating (or exterior) algebra** $\mathcal{A}_n$, the algebra over the reals generated by the abstract elements via the relation*

$$e_i \cdot e_j = -e_j \cdot e_i, \quad \forall i, j = 1, \dots, n$$

where $\cdot$ stands for the algebra product.

Henceforth we will omit writing explicitly the product symbol $\cdot$. It is immediate to observe the following consequences:

$$e_i^2 = 0, \quad \forall i = 1, \dots, n.$$

In addition we have required that $n > 1$ since, with only one generator, the algebra structure becomes trivial. In the following we will refer to $\mathcal{A}_1 = \text{span}_{\mathbb{R}}(e_1, \dots, e_n)$ which is not an algebra, but simply a vector space isomorphic to $\mathbb{R}^n$. A related concept is the following:

Definition 177. Let $\mathcal{A}$ be an algebra (over a commutative ring C) and let G be a group. We say that $\mathcal{A}$ is G -graded if, for every $g \in G$ there exists a C -module $\mathcal{A}_g \subseteq \mathcal{A}$ such that

1. $\mathcal{A}_g \mathcal{A}_h \subseteq \mathcal{A}_{gh}$, for all $g, h \in G$,
2. $\mathcal{A} = \bigoplus_{g \in G} \mathcal{A}_g$ as module.

In particular we call every element $a \in \mathcal{A}_g$, homogeneous of degree g . If $G = \mathbb{Z}$ and $\mathcal{A}_n = 0$ for all $n < 0$, we call it **positively graded**.

Proposition 178. For every $n > 1$, the alternating algebra $\mathcal{A}_n$ is positively graded

Proof. By using the algebra relations, one realizes that $\mathcal{A}_n$ is a vector space of dimension 2^n with a canonical basis $\{1, e_1, \dots, e_n, \bigcup_{i_1 < i_2} (e_{i_1} e_{i_2}), \bigcup_{i_1 < i_2 < i_3} (e_{i_1} e_{i_2} e_{i_3}), \dots, e_1 e_2 \dots e_n\}$. Hence, for any $k \in \mathbb{Z}$ we can set $(\mathcal{A}_n)_k = \{0\}$ if $k < 0$ or $k > n$, while $(\mathcal{A}_n)_0 = \mathbb{R}$ and $(\mathcal{A}_n)_k = \text{span}_{\mathbb{R}}(\bigcup_{i_1 < i_2 < \dots < i_k} (e_{i_1} e_{i_2} \dots e_{i_k}))$ if $0 < k \leq n$. Using Definition 176, both condition 1. and 2. in Definition 177 hold true either by direct inspection or using the defining relation $e_i e_j = -e_j e_i$ for all $i, j = 1, \dots, n$. $\square$

In order to simplify the notation we introduce

$$(\mathcal{A}_n)_k \equiv \mathcal{A}_n^k \implies \mathcal{A}_n = \bigoplus_{k=0}^n \mathcal{A}_n^k.$$

We are now ready to define the main structure of this section:

Definition 179. We call k -differential forms on $\mathbb{R}^n$ for $n \geq 1$ the elements of the vector space $\mathcal{A}^k(\mathbb{R}^n) \doteq C^\infty(\mathbb{R}^n; \mathcal{A}_n^k)$.

Notice that the notion of smoothness is meaningful since $\mathcal{A}_n^k$ is per construction a vector space of dimension $\binom{n}{k}$, hence isomorphic to $\mathbb{R}^{\binom{n}{k}}$ which can be endowed with the standard differential structure. The space of k -forms (differential will be henceforth implicit) admits several equivalent characterizations. The following exercise will be particularly instructive (if you solve it):

Exercise 180. There exists an isomorphism of topological vector spaces between $C^\infty(\mathbb{R}^n; \mathcal{A}_n^k)$ and $C^\infty(\mathbb{R}^n) \otimes \mathcal{A}_n^k$.

Warning: In the literature the following symbols often appear to indicate the same space:

$$\Omega^k(\mathbb{R}^n) \equiv \Lambda^k(\mathbb{R}^n) = \mathcal{A}^k(\mathbb{R}^n).$$

We might hop from one notation to the other depending on which one is more commonly used in the case in hand.

It is immediate to observe that the space of all possible differential forms inherits the alternating algebra structure.

Proposition 181. *Let $\mathcal{A}^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^n \mathcal{A}^k(\mathbb{R}^n)$. This is a positively graded alternating algebra if endowed with the **wedge product** structure $\wedge : \mathcal{A}^l(\mathbb{R}^n) \times \mathcal{A}^{l'}(\mathbb{R}^n) \rightarrow \mathcal{A}^{l+l'}(\mathbb{R}^n)$, $l, l' \in \mathbb{Z}$ completely and uniquely specified by the relation*

$$(f \otimes a) \wedge (f' \otimes b) = ff' \otimes (a \cdot b),$$

where $f, f' \in C^\infty(\mathbb{R}^n)$, while $a \in \mathcal{A}_n^l$, $b \in \mathcal{A}_n^{l'}$ and $\cdot$ is the algebra product in $\mathcal{A}_n$

The proof is fairly straightforward since it requires only a direct verification that the defining properties of a positively graded alternating algebra are met. In addition, we remark that, when working with k -forms it is often customary to introduce explicitly the *degree* map:

$$\deg : \mathcal{A}^l(\mathbb{R}^n) \rightarrow \mathbb{Z}, \quad \omega \mapsto \deg(\omega) = l, \quad (309)$$

from which it descends the famous relation between differential forms

$$\omega \wedge \omega' = (-1)^{\deg(\omega) \deg(\omega')} \omega' \wedge \omega. \quad \forall \omega \in \mathcal{A}^l(\mathbb{R}^n) \text{ and } \forall \omega' \in \mathcal{A}^{l'}(\mathbb{R}^n), \quad (310)$$

where both l and l' are arbitrary. Having introduced differential forms on $\mathbb{R}^n$, there are three natural questions to be asked:

1. what is the alternating algebra encoding as information?
2. does there exist a geometric characterization of k -forms?
3. what are k -forms good for?

In the next section we shall give an answer first of all to the second question starting from the simplest example of differentiable manifold, namely $\mathbb{R}^n$. Subsequently we shall generalize the whole construction to arbitrary differentiable manifolds, giving an answer to the third and most important question.

Before proceeding in our endeavor (which is also the name given to one of the space shuttles), we focus instead on the first question. As a matter of fact, Definition 176 in particular is nothing but the abstract algebraic realization of the canonical anticommutation relations. It goes without saying that, despite the temptation, the reason for introducing these structures does not stem from Fermions or from the Pauli exclusion principle. On the contrary, in many different instances, in mathematics one introduces operations which are anti-symmetric with respect to their arguments, the canonical example being the cross product between vectors. It is therefore natural to try to encode this property in the most abstract possible way within an algebraic structure. This is encoded by Definition 176 and 177.

It is worth emphasizing that, from the viewpoint of an algebraist, the alternating algebra is very informative and useful since it allows for an abstract characterization of one of the key operations in linear algebra: *the determinant*.

Definition 182. Let $\mathcal{A}_n^1 = \text{span}_{\mathbb{R}}\{e_1, e_2, \dots, e_n\}$ be the vector space, composed of homogeneous elements of $\mathcal{A}_n$ of degree 1. We call **determinant** the map

$$\det : \mathcal{L}(\mathcal{A}_n^1) \rightarrow \mathbb{R} \quad K \mapsto \det(K),$$

which is completely characterized by the defining relation

$$K(e_1) \cdot K(e_2) \cdot \dots \cdot K(e_n) = \det(K) e_1 \cdot e_2 \cdot \dots \cdot e_n, \quad (311)$$

where $\cdot$ is the algebra product as in Definition 176.

In order to convince ourselves that this is a meaningful definition, we must compare this with the standard notion of determinant of a matrix proving that they are equivalent.

Proposition 183. Let $\mathcal{K} \in M(n, \mathbb{R})$ and let $K \in \mathcal{L}(\mathcal{A}_n^1)$ be the linear map which is defined as the unique linear extension of the application

$$K(e_i) = \sum_{j=1}^n \mathcal{K}_{ij} e_j, \quad \forall i = 1, \dots, n$$

Then $\det(K) = \det_{M(n, \mathbb{R})}(\mathcal{K})$ where the right hand side stands for the determinant of a matrix.

Proof. We can proceed by direct inspection, since, using Definition 176, it holds

$$K(e_1) \cdot K(e_2) \cdot \dots \cdot K(e_n) = \sum_{j_1=1}^n \mathcal{K}_{1j_1} e_{j_1} \cdot \dots \cdot \sum_{j_n=1}^n \mathcal{K}_{nj_n} e_{j_n} = \sum_{\sigma \in S_n} \prod_{i=1}^n \text{sign}(\sigma) \mathcal{K}_{i\sigma(i)} e_1 \cdot \dots \cdot e_n,$$

where S_n is the permutation group of n -elements $(1, 2, \dots, n)$, while $\text{sign}(\sigma)$ is the sign of the permutation and $\sigma(i)$ is the i -th entry of the permutation $\sigma(1, 2, \dots, n)$. From standard linear algebra and from Equation (311), we obtain

$$\det(K) = \sum_{\sigma \in S_n} \prod_{i=1}^n \text{sign}(\sigma) \mathcal{K}_{i\sigma(i)} = \det_{M(n, \mathbb{R})}(\mathcal{K}),$$

which concludes the proof. $\square$

We leave to the reader as an exercise of linear algebra to prove a converse of Proposition 183, namely that every $n \times n$ real matrix induces an element of $\mathcal{L}(\mathcal{A}_n^1)$ and the determinants coincide. To conclude this first part we stress, that, while the notion of determinant might appear just a slightly more abstract generalization of the standard one, this will become especially useful in the theory of integration on manifolds.

B.1 Geometric realization of k -forms on $\mathbb{R}^n$

In the previous part, we have introduced k -differential forms $\mathcal{A}^k(\mathbb{R}^n)$ in Definition 179, endowing $\mathcal{A}^\bullet(\mathbb{R}^n)$ with the structure of an alternating, positively graded, algebra via the wedge product in Proposition 181. In this construction there is a purely algebraic component coming from $\mathcal{A}_n$ and a geometric one represented by $C^\infty(\mathbb{R}^n)$. In the following we show that one can give a concrete realization to the generating elements $\{e_1, e_2, \dots, e_n\}$ in terms of suitable geometric structures. This procedure will make more manifest the correlation between our approach to k -forms and the standard one. At the beginning the whole construct might appear baroque and ostentatious, but it will be clear at a later stage that it allows for a rather straightforward generalization to arbitrary differentiable manifolds (well... if you have come to my lectures on the theory of Lie groups, you should better know by now...).

The starting point of our analysis is the observation of the following non trivial interplay between vectors of $\mathbb{R}^n$ and functions lying in $C^\infty(\mathbb{R}^n)$, where smoothness is required only for simplicity, since it does not lead to a loss of information in the procedure. More precisely, observe that given the standard Euclidean coordinates $\{x_1, \dots, x_n\}$ of $\mathbb{R}^n$ and given a vector $v \in \mathbb{R}^n$ whose components with respect to the standard basis are $v_1, \dots, v_n$, then v identifies a *directional derivative* at the base point of the vector ($x = 0$ in this case) as follows:

$$\mathbb{R}^n \ni v \mapsto D_v : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R} \text{ where } f \mapsto D_v(f) = \sum_{i=1}^n v_i \partial_i f|_{x=0}, \quad (312)$$

This motivates the following definition, which aims at allowing to vary smoothly the base point of the vector and at keeping the interpretation of a vector as a directional derivative:

Definition 184. We call **smooth vector field** on $\mathbb{R}^n$ any linear map $v : C^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n)$ which satisfies the Leibnitz rule:

$$v(ff')(x) = f(x)v(f')(x) + f'(x)v(f)(x), \quad \forall x \in \mathbb{R}^n$$

We call the collection of all smooth vector fields $\mathfrak{X}(\mathbb{R}^n)$.

A standard result in differential geometry, see e.g. [Lee13, Chap.3], entails that, once chosen a coordinate system on $\mathbb{R}^n$, say $\{x_1, \dots, x_n\}$, not necessarily the Cartesian one, then, there must exist $v_i \in C^\infty(\mathbb{R}^n)$, $i = 1, \dots, n$ such that

$$v = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}. \quad (313)$$

This alternative viewpoint on the role of vectors has the main advantage of being easily generalized to arbitrary differentiable manifolds. Here we shall not focus on this aspect. On the contrary we emphasize that, while in Definition 184 the rationale is that vector fields act on an arbitrary smooth function to give another function, suitably interpreted as a derivative, there exists a dual, albeit equivalent, interpretation. This calls for defining the action of a smooth function on a vector field via the following structure.

Definition 185. Let $f \in C^\infty(\mathbb{R}^n)$. We call **differential** of f , the linear map

$$df : \mathfrak{N}(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n) \quad v \mapsto df(v) \doteq v(f).$$

If a reader is already familiar with the geometric notion of differential of a function (push-forward), we stress that it is equivalent to Definition 185. We will focus on this aspect when we will generalize this construction to arbitrary differentiable manifolds.

It is important to give a characterization of the differential in terms of an arbitrary but fixed coordinate system $\{x_1, \dots, x_n\}$. Per linearity it holds that, using Equation (313) and Definition 185, for any $f \in C^\infty(\mathbb{R}^n)$,

$$df(v) = \sum_{i=1}^n v_i df \left(\frac{\partial}{\partial x_i} \right) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i},$$

At the same time, if we take into account that each coordinate over $\mathbb{R}^n$ identifies a smooth map $x_i : \mathbb{R}^n \rightarrow \mathbb{R}$ which reads for each $x \in \mathbb{R}^n$ its i -th component with respect to the chosen coordinates, it holds, that, for all $v \in \mathfrak{N}(\mathbb{R}^n)$,

$$dx_j(v) = \sum_{i=1}^n v_i \frac{dx_j}{dx_i} = v_j.$$

Putting together these infos, we have shown that, for any coordinate system $\{x_1, \dots, x_n\}$ over $\mathbb{R}^n$,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i \tag{314}$$

We have finally all the ingredients necessary to give a geometric realization of k -forms. As a matter of fact, the differentials $\{dx_i\}_{i=1, \dots, n}$ are the natural candidate to play the role of the abstract basis $\{e_1, \dots, e_n\}$ in Definition 176.

Remark 63. In order to disentangle the abstract realization of k -differential forms from the concrete one, outlined in the last paragraph, we use a different notation. We call Ω_n the exterior algebra generated by $\{dx_1, \dots, dx_n\}$ and the vector space built out of elements of grade 1 is

$$\Omega^1 \equiv \Omega_n^1 = \text{span}_{\mathbb{R}}(dx_1, \dots, dx_n) \tag{315}$$

and consequently, in view of Definition 179, we have given a concrete and geometric realization of k -differential forms as

$$\Omega^k(\mathbb{R}^n) \doteq C^\infty(\mathbb{R}^n; \Omega_n^k) \equiv C^\infty(\mathbb{R}^n) \otimes \Omega_n^k,$$

where we used Exercise 180. With such notation, it is immediate to realize that Equation (314) identifies an element

$$df \in C^\infty(\mathbb{R}^n) \otimes \Omega^1 \equiv \Omega^1(\mathbb{R}^n).$$

Finally we recall that, in view of Proposition 181, if we introduce

$$\Omega^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^n \Omega^k(\mathbb{R}^n) \equiv C^\infty(\mathbb{R}^n) \otimes \Omega^\bullet, \quad \Omega^\bullet \doteq \bigoplus_{k=0}^n \Omega_n^k, \quad (316)$$

this is an exterior algebra with respect to the wedge product, which is still indicated as $\wedge$.

Exercise 186. Prove that the construction in Remark 63 does not depend on the choice of the coordinates $\{dx_1, \dots, dx_n\}$ in the sense, that different coordinates yield isomorphic algebras.

At this stage the question is the following: which is the advantage in having given a geometric realization of the alternating/exterior algebra?

B.2 The de Rham complex

In this section we give an answer to the question which concluded the previous one. The starting point is the observation that, in the process of constructing $\Omega^k(\mathbb{R}^n)$, $k = 0, \dots, n$, we have defined a new relevant operation, namely in Definition 185, we have implicitly introduced a map

$$d_0 : C^\infty(\mathbb{R}^n) \equiv \Omega^0(\mathbb{R}^n) \rightarrow \Omega^1(\mathbb{R}^n) \quad f \mapsto d_0(f) = df.$$

It is straightforward to observe that this operation can be generalized to arbitrary k -forms as follows:

Definition 187. We call **exterior differentiation** the linear map $d_k : \Omega^k(\mathbb{R}^n) \rightarrow \Omega^{k+1}(\mathbb{R}^n)$, $k \in \mathbb{N} \cup \{0\}$ such that, for any coordinate system $\{x_1, \dots, x_n\}$ on $\mathbb{R}^n$, it holds

1. if $k = 0$ $d_0(f) = df$ as in Equation (314),
2. if $k > 1$ then, for every $\omega \in \Omega^k(\mathbb{R}^n)$, decomposed as $\omega = \sum_{i_1 < i_2 < \dots < i_k} \omega_{i_1 \dots i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k}$,

$$d_k \omega = \sum_{j=1}^n \sum_{i_1 < i_2 < \dots < i_k} \frac{\partial \omega_{i_1 \dots i_k}(x)}{\partial x_j} dx_j \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

The $(k+1)$ -form $d_k \omega$, $\omega \in \Omega^k(\mathbb{R}^n)$ is called the **exterior derivative** of ω .

It is immediate to show that the exterior differentiation satisfies the following notable properties:

Corollary 188. Let d_k be the exterior differentiation. It holds that

1. for all $k \geq 0$, $d_{k+1} \circ d_k = 0$,
2. for all $k, k' \in \mathbb{N} \cup \{0\}$, it holds

$$d_{k+k'}(\omega \wedge \omega') = d_k \omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge d_{k'} \omega', \quad \omega \in \Omega^k(\mathbb{R}^n) \text{ and } \omega' \in \Omega^{k'}(\mathbb{R}^n).$$

Proof. Let us prove the two items separately.

1. Consider any $\omega \in \Omega^k(\mathbb{R}^n)$. From Definition 187, it descends that, chosen any basis of $\mathbb{R}^n$, $\{x_1, \dots, x_n\}$, we can write $\omega = \sum_{i_1 < i_2 < \dots < i_k} c_{i_1 \dots i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k}$. It descends

$$(d_{k+1} \circ d_k)(\omega) = \sum_{i,j=1}^n \sum_{i_1 < i_2 < \dots < i_k} \frac{\partial^2 \omega_{i_1 \dots i_k}(x)}{\partial x_i \partial x_j} dx_i \wedge dx_j \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k} = 0,$$

since $dx_i \wedge dx_j = -dx_j \wedge dx_i$, while the second derivative is symmetric under exchange of x_i and x_j by Schwartz theorem.

2. Since the exterior differentiation is linear, it suffices to prove the statement for $\omega = a(x) dx_{i_1} \wedge \dots \wedge dx_{i_k}$ and $\omega' = b(x) dx_{j_1} \wedge \dots \wedge dx_{j_{k'}}$, where $a, b \in C^\infty(\mathbb{R}^n)$ while $(i_1, \dots, i_k)$ (*resp.* $(j_1, \dots, j_{k'})$) are two arbitrary sequences of k (*resp.* k') increasing integers between 0 and n . Using Definition 187 it holds that

$$d_k \omega = \sum_{i=1}^n \frac{\partial a(x)}{\partial x_i} dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}, \quad d_{k'} \omega' = \sum_{i=1}^n \frac{\partial b(x)}{\partial x_i} dx_i \wedge dx_{j_1} \wedge \dots \wedge dx_{j_{k'}}.$$

Hence

$$\begin{aligned} d_k \omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge d_{k'} \omega' &= \sum_{i=1}^n \frac{\partial a(x)}{\partial x_i} b(x) dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_{k'}} + \\ &+ (-1)^{\deg \omega} \sum_{i=1}^n \frac{\partial b(x)}{\partial x_i} a(x) dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_i \wedge dx_{j_1} \wedge \dots \wedge dx_{j_{k'}} = \\ &= \sum_{i=1}^n \left(\frac{\partial a(x)}{\partial x_i} b(x) + \frac{\partial b(x)}{\partial x_i} a(x) \right) dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_{k'}}, \end{aligned}$$

where in the last equality we used that each wedge product is antisymmetric and that $\deg \omega = k$. A direct inspection unveils that the last expression is nothing but $d_{k+k'}(\omega \wedge \omega')$. $\square$

On account of the second property proven in this last corollary, it is said that the exterior differentiation is an *antiderivation*.

Remark 64. Observe that in many textbooks, the subscript k in the exterior differential is omitted. This is not a slight abuse of notation. On the contrary, since each $\Omega^k(\mathbb{R}^n) \hookrightarrow \Omega^\bullet(\mathbb{R}^n)$, we can introduce

$$d : \Omega^\bullet(\mathbb{R}^n) \rightarrow \Omega^\bullet(\mathbb{R}^n),$$

which is defined as follows: For all $\underline{\omega} \in \Omega^\bullet(\mathbb{R}^n)$, there exists $\omega_k \in \Omega^k(\mathbb{R}^n)$, $k = 0, \dots, n$ such that $\underline{\omega} = (\omega_0, \dots, \omega_n)$ and

$$d\underline{\omega} \doteq (0, d_0 \omega_0, \dots, d_{n-1} \omega_{n-1}).$$

A direct application of Corollary 188, entails that, for all $\omega \in \Omega^k(\mathbb{R}^n)$ and for all $\omega' \in \Omega^{k'}(\mathbb{R}^n)$ it holds

$$\underline{d}^2 = \underline{d} \circ \underline{d} = 0, \text{ and } \underline{d}(\omega \wedge \omega') = \underline{d}\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge \underline{d}\omega'. \quad (317)$$

The real (slight) abuse of notation originates from the habit of writing d in place of $\underline{d}$, using thus a symbol which appears in several different contexts with slightly different meanings.

In reading the above construction, one might have the feeling that the exterior differentiation is an operation which is not intrinsic on the space of differentiable forms of any order and it depends on the coordinate chosen on $\mathbb{R}^n$. In the following theorem we will prove that a uniqueness result which should dispel any doubt. Here we adapt to the case in hand [Lee13, Th. 14.24]:

Theorem 189. *Let $\Omega^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^n \Omega^k(\mathbb{R}^n)$. There exists a unique operator $\underline{d} = (d_0, \dots, d_n)$ acting component by component on any $\underline{\omega} \in \Omega^\bullet(\mathbb{R}^n)$ in such a way that the following four properties are satisfied:*

1. $\underline{d}$ is $\mathbb{R}$ -linear,
2. $\underline{d}^2 = 0$,
3. for all $\omega \in \Omega^k(M)$ and for all $\omega' \in \Omega^{k'}(\mathbb{R}^n)$

$$\underline{d}(\omega \wedge \omega') = \underline{d}\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge \underline{d}\omega'$$

4. for every $\underline{\omega} = (f, 0, \dots, 0)$ where $f \in C^\infty(\mathbb{R}^n) \equiv \Omega^0(\mathbb{R}^n)$, it holds

$$\underline{d}\omega = (0, d_0 f, 0, \dots, 0),$$

where $d_0 f = df$ coincides with the differential of f as per Definition 185.

Proof. Suppose that there exists a second operator $\underline{d}'$ which satisfies condition 1.-4. besides the one built in Remark 64. On account of item 4., $\underline{d}$ and $\underline{d}'$ coincide when acting on $\Omega^0(\mathbb{R}^n)$. Focusing on $\Omega^1(\mathbb{R}^n)$ and choosing any coordinate system $\{x_1, \dots, x_n\}$ on $\mathbb{R}^n$, linearity (item 2.) entails that it suffices to compare the two operators on $f(x)dx_i$, where $f(x) \in C^\infty(\mathbb{R}^n)$, while the index i is arbitrary but fixed. Yet item 3. guarantees that

$$\underline{d}(f(x)dx_i) = df \wedge dx_i = \underline{d}'(f(x)dx_i),$$

where we used implicitly that $\underline{d}^2 = (\underline{d}')^2 = 0$. We can proceed by induction. Suppose that the two operators coincide on k -forms. In order to check what happens on $(k+1)$ -forms, it suffices per linearity to check their action on $\omega_{k+1} = h(x)dx_{i_1} \wedge \dots \wedge dx_{i_{k+1}}$ where $h \in C^\infty(\mathbb{R}^n)$, while $(i_1, \dots, i_{k+1})$ is a collection of arbitrary $(k+1)$ integer indices in increasing order. If we define $\omega_k = h(x)dx_{i_1} \wedge \dots \wedge dx_{i_k}$, it holds $\omega_{k+1} = \omega_k \wedge dx_{i_{k+1}}$ from which it descends from item 3. that

$$\underline{d}\omega_{k+1} = \underline{d}\omega_k \wedge dx_{i_{k+1}} = \underline{d}'\omega_k \wedge dx_{i_{k+1}} = \underline{d}'\omega_{k+1},$$

where we used the induction hypothesis. As a consequence the operators $\underline{d}$ and $\underline{d}'$ coincide on all $\Omega^k(\mathbb{R}^n)$, $k \in \mathbb{N} \cup \{0\}$ and they are the same operator. $\square$

We can reinterpret the information extracted from the exterior differentiation in terms of suitable algebraic structures. We recall here

Definition 190. Let $C^\bullet$ be a graded vector space, i.e., there exists a countable family of vector spaces $\{C_k\}_{k \in \mathbb{N}}$, such that $C^\bullet \simeq \bigoplus_{k \in \mathbb{N}} C_k$. We call $C^\bullet$ a **differential (or cochain) complex** if there exists a map $\underline{d} : C^\bullet \rightarrow C^\bullet$, called the **differential** (of the complex), such that $\underline{d} = \{d_k\}_{k \in \mathbb{N}}$ is a collection of linear maps $d_{k-1} : C_{k-1} \rightarrow C_k$

$$\dots \longrightarrow C_{k-1} \xrightarrow{d_{k-1}} C_k \xrightarrow{d_k} C_{k+1} \longrightarrow \dots, \quad (318)$$

subject to the constraint $d_{k+1} \circ d_k = 0$ for all $k \in \mathbb{N}$.

Corollary 191. The pair $(\Omega^\bullet(\mathbb{R}^n), \underline{d})$ built out of differential forms and the exterior differentiation is a differential complex, called the **de Rham complex**

$$\dots \longrightarrow \Omega^{k-1}(\mathbb{R}^n) \xrightarrow{d_{k-1}} \Omega^k(\mathbb{R}^n) \xrightarrow{d_k} \Omega^{k+1}(\mathbb{R}^n) \longrightarrow \dots, \quad (319)$$

Proof. This is a direct consequence of Definition 179, Remark 63 together with Corollary 188. $\square$

We recall now the standard nomenclature used for differential forms, adapting it to a generic differential complex $(C^\bullet, \underline{d})$. We say that $c \in C_k$ is

- **closed (or a cocycle)** if $d_k c = 0$. We call $\mathbf{C}_d^k$ the vector subspace of C_k made of closed k -elements.
- **exact (or a coboundary)** if there exists $a \in C^{k-1}$ such that $c = d_{k-1} a$. The collection of all exact k -elements is a vector subspace of C_k indicated as $\mathbf{d}[\mathbf{C}_{k-1}]$ where the subscript $k-1$ can be dropped from the differential since no confusion can arise.

In view of these distinguished subspaces, we can introduce the following notable quotient of vector spaces:

Definition 192. Let $(C^\bullet, \underline{d})$ be any differential complex. For every $k \in \mathbb{N}$, we call **k-th cohomology** of the complex the vector space defined as the quotient

$$H_{\underline{d}}^k(C^\bullet) \doteq \frac{C_d^k}{\mathbf{d}[C_{k-1}]}.$$

If we consider in particular the de Rham complex, then, for every $k \in \mathbb{N} \cup \{0\}$, we call **k-th de Rham cohomology** of $\mathbb{R}^n$ the vector space defined as the quotient

$$H_{dR}^k(\mathbb{R}^n) \doteq \frac{\Omega_d^k(\mathbb{R}^n)}{\mathbf{d}[\Omega^{k-1}(\mathbb{R}^n)]}.$$

Exercise 193. Prove with a direct computation that, for all $n \in \mathbb{N}$, it holds

$$H_{dR}^k(\mathbb{R}^n) \simeq \begin{cases} \mathbb{R} & \text{if } k=0 \\ 0 & \text{otherwise} \end{cases},$$

where 0 stands for the trivial vector space.

Remark 65. Exercise 193 seems to indicate that de Rham cohomology is rather uninformative since it is trivial unless $k = 0$, namely the only functions (0-forms) with vanishing differential are the constant ones. Yet, observe that the notion of de Rham complex can be straightforwardly extended to any open subset $U \subseteq \mathbb{R}^n$ by restriction since, from Equation 316, it is natural to set

$$\Omega^\bullet(U) = \bigoplus_{k=0}^n \Omega^k(U) \quad \text{where} \quad \Omega^k(U) = \Omega^k(\mathbb{R}^n)|_U \equiv C^\infty(U) \otimes \Omega^k,$$

while exterior differentiation is left untouched. While the de Rham complex has a rather obvious behaviour under restriction, this is not the case for de Rham cohomology. As a matter of fact, let

- $\Omega_d^k(U)$ be the space of closed k -forms on U ,
- $d[\Omega^{k-1}(U)]$ be the space of exact $k-1$ forms on U .

We define

$$H_{dR}^k(U) \doteq \frac{\Omega_d^k(U)}{d[\Omega^{k-1}(U)]}.$$

The following exercise is very instructive

Exercise 194. Let $U \subset \mathbb{R}^n$, $n \geq 1$, be contractible, namely homotopous to a point. Prove that

$$H_{dR}^k(U) \simeq \begin{cases} \mathbb{R}^m & \text{if } k=0 \\ 0 & \text{otherwise} \end{cases},$$

where m stands for the number of different connected components of U .

In other words Exercise 194 seems to indicate that de Rham cohomology is a more sophisticated tool than one could infer at first glance. As a matter of fact the 0-th cohomology yields a topological information on the underlying space, counting the number of its disconnected components.

To conclude this subsection, we remark that we have learned that differential k -forms and the de Rham complex carry information on the topology of the underlying space U , here limited to being a subset of $\mathbb{R}^n$, though only through $k = 0$. On the one hand this is a consequence of the renown Poincaré lemma which we will prove later, while, on the other hand, it reminds

us why, on Minkowski spacetime, the formulation of Maxwell's equation via the Faraday tensor $F \in \Omega^2(\mathbb{R}^4)$ admits an equivalent description in terms of the vector potential. More precisely

$$d_2 F = 0 \iff [F] \in H_{dR}^2(\mathbb{R}^4) = \{0\} \implies F = d_1 A,$$

where $A \in \Omega^1(\mathbb{R}^n)$. Yet the de Rham complex from Corollary 191 tells us that A is not uniquely defined since we can make the identification $A \simeq A + d_0 \chi$, $\chi \in C^\infty(\mathbb{R}^4)$. This is nothing but the gauge freedom of electromagnetism stemming from the first of the two Maxwell's equations, *i.e.* $d_2 F = 0$. In addition, we have just got a glimpse of another important fact: This last equation is not dynamical, since it encodes only an information on the topology of the underlying space via an object $\underline{d}$, the exterior differentiation, built out of the differentiable structure of $\mathbb{R}^4$. This is the main reason why $dF = 0$ does not carry dynamical information on electromagnetism.

This leaves us with two additional natural questions:

1. does there exist a generalization of the de Rham complex for other gauge theories?
2. does there exist a generalization of the de Rham complex to an arbitrary differentiable manifold and which information is encoded in the associated de Rham cohomology?

We will start by giving an answer to the second question.

C Differential Forms on Manifolds

In the previous section we have introduced differential forms on $\mathbb{R}^n$. We have learned in Remark 63 and particularly in Equation (316) that we can realize differential k -forms as smooth maps from $\mathbb{R}^n$ to Ω_n^k . In the following we shall combine all structures that we have introduced to obtain a natural generalization of differential forms to manifolds.

First of all we need a slight generalization to arbitrary vector spaces of the structures outlined at the beginning of Section B.

Definition 195. *Let V be a real vector space of dimension $\dim V = n \geq 1$ and let φ be any isomorphism between V and $\mathbb{R}^n$. We call **alternating (or) exterior algebra of V** , $\mathcal{A}_{n,V}$ the unique algebra over the reals such that φ extends to an algebra isomorphism.*

Exercise 196. Prove that, under the hypotheses of Definition 195, $\mathcal{A}_{n,V}$ does not depend on the choice of φ , namely, the algebras obtained choosing two isomorphisms are isomorphic to each other.

As a consequence of Definition 195 $\mathcal{A}_{n,V}$ inherits all properties of the exterior algebra $\mathcal{A}_n$. In particular, in view of Proposition 178, it is positively graded and we can decompose it as

$$\mathcal{A}_{n,V} = \bigoplus_{k=0}^n \mathcal{A}_{n,V}^k.$$

We can now adapt this discussion to the specific scenario, we are interested in.

Definition 197. Let M be a smooth manifold of dimension $\dim M = n \geq 1$. for every $k \geq 1$, we call **bundle of alternating k -tensors on M** the vector bundle of rank $\binom{n}{k}$ $\Lambda^k(M) \doteq \bigsqcup_{p \in M} \Lambda_p^k$

where $\Lambda_p^k \equiv \mathcal{A}_{n, T_p^*M}^k$ is the vector space of positively graded elements of degree k of the alternating algebra associated to the vector space T_p^*M .

We can define the main object of investigation of this whole section:

Definition 198. Let M be a smooth manifold of dimension $\dim M = n \geq 1$. We call **k -differential forms**, $0 \leq k \leq n$, the elements of the vector space

$$\Omega^k(M) \doteq \Gamma^\infty(\Lambda^k(M)),$$

where we set $\Lambda^0(M)$ as the trivial line bundle $M \times \mathbb{R}$.

In view of this definition, it is desirable to have a concrete description of a k -form in a local trivialization of $\Lambda^k(M)$. Let us start from $k = 1$ so that $\Lambda^1(M) = T^*M$. In this case, mutatis mutandis, we can use Proposition 31 and its proof to infer that, taken any local chart (U, φ) , $U \subseteq M$ and an associated coordinate system $\{x_i\}_{i=1, \dots, n=\dim M}$, this induces both an associated trivialization of $T^*M \equiv T^*M[M, \pi, \mathbb{R}^n]$ and a coordinate system on $T^*M|_U$. In other words there exists a diffeomorphism $\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$ such that, for all $\omega \in \Omega^1(M)$,

$$\omega|_U = \sum_{i=1}^n \omega_i(x) dx_i,$$

where $\omega_i \in C^\infty(U)$ for all $i = 1, \dots, n$. Observe that, at each $p \in U$, $dx_i : T_p M \rightarrow T_{x_i(p)} \mathbb{R}$ is the differential of the local coordinates $x_i : M \rightarrow \mathbb{R}$, $i = 1, \dots, n$ and it is completely characterized by the relation

$$dx_i \left(\frac{\partial}{\partial x_j} \right) = \delta_{ij} \frac{d}{d\lambda},$$

where $\frac{d}{d\lambda}$ is the generator of $T_{x(p)} \mathbb{R}$ while $\left\{ \frac{\partial}{\partial x_j} \right\}_{j=1, \dots, n}$ is the natural basis of $T_p M$ induced by the coordinate system.

Remark 66. The geometric concept of differential (push-forward) here used is only apparently different from that of Section B and particularly of Definition 185, which is traded from calculus. As a matter of fact, if we restrict our attention to $M = \mathbb{R}^n$, given $f \in C^\infty(\mathbb{R}^n)$, the geometric differential, here indicated to avoid confusion with f_* , is a map

$$f_* : T_q \mathbb{R}^n \rightarrow T_{f(q)} \mathbb{R}.$$

Given a coordinate system $\{x_i\}_{i=1, \dots, n}$, we can write any $v \in T_q \mathbb{R}^n$ as $v = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}$, where each $v_i \in \mathbb{R}$, $i = 1, \dots, n$. Per construction

$$f_*(v) = \sum_{i=1}^n v_i \left. \frac{\partial f}{\partial x_i} \right|_q \frac{d}{d\lambda},$$

where $\frac{d}{d\lambda}$ is the generator of $T_{f(q)}\mathbb{R}$. One can now infer that, calling df the differential as per Proposition 185, it holds $f_*(v) = df(v)\frac{d}{d\lambda}$. In view of this simple connection, with a slight abuse of notation, henceforth we will equivalently speak of push-forward and of differential of a map $f : M \rightarrow N$ between two smooth manifolds, indicating it as df .

Having obtained a local description of 1-forms on a smooth manifold via Φ , we can exploit that at each $p \in U$, $\Phi : T_p^*U \rightarrow \mathbb{R}^n$ is a vector space isomorphism using it in the spirit of Definition 195 to induce on $\Lambda_p^\bullet \doteq \bigoplus_{k=0}^n \Lambda_p^k$ the structure of an algebra. This can be done inducing an isomorphism

$$\Phi^\bullet : \Lambda_p^\bullet \rightarrow \Omega_n^\bullet,$$

which is completely specified by the requirement that, for all $i, j = 1, \dots, n$,

$$(\Phi^\bullet)^{-1}(dx_i \wedge dx_j) = \Phi^{-1}(dx_i) \wedge \Phi^{-1}(dx_j),$$

where, with a slight abuse of notation we indicate with $\wedge$ the algebra product both of $\Omega^\bullet$ and of $\Lambda_p^\bullet$. Notice that the restriction of $\Phi^\bullet$ to the elements of fixed degree k becomes a vector space isomorphism $\Phi^\bullet|_k : \Lambda_p^k \rightarrow \Omega_p^k$ which coincides with Φ if $k = 1$. In view of these considerations, we obtain a local description of a k -form for $k > 1$, namely for all $\omega_k \in \Omega^k(M)$,

$$\omega_k|_U = \sum_{i_1 < i_2 < \dots < i_k} \omega_{i_1 \dots i_k} \Phi^{-1}(dx_{i_1}) \wedge \dots \wedge \Phi^{-1}(dx_{i_k}), \quad (320)$$

where each $\omega_{i_1 \dots i_k} \in C^\infty(U)$. As a consequence, we have proven the following

Proposition 199. *Let M be a smooth manifold. Then $\Omega^\bullet(M) \doteq \bigoplus_{k=0}^n \Omega^k(M)$ is an exterior algebra if endowed with the wedge product*

$$\wedge : \Omega^k(M) \times \Omega^{k'}(M) \rightarrow \Omega^{k+k'}(M), \quad (\omega, \omega') \mapsto \omega \wedge \omega',$$

such that for all local trivialization (U, Φ) of $\Lambda^k(M)$ and for all coordinate system $\{x_i\}_{i=1, \dots, n}$ subordinated to Φ , it holds that, if

$$\omega|_U = \sum_{i_1 < i_2 < \dots < i_k} \omega_{i_1 \dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k} \text{ and } \omega'|_U = \sum_{j_1 < j_2 < \dots < j_{k'}} \omega_{j_1 \dots j_{k'}} dx_{j_1} \wedge \dots \wedge dx_{j_{k'}},$$

$$(\omega \wedge \omega')|_U = \sum_{i_1 < i_2 < \dots < i_k} \sum_{j_1 < j_2 < \dots < j_{k'}} \omega_{i_1 \dots i_k} \omega_{j_1 \dots j_{k'}} dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_{k'}},$$

where, for simplicity of the notation, we omitted the symbol Φ^{-1} acting on the generic form dx_i .

The structure of k -differential forms is from a geometric viewpoint the natural generalization of the counterpart on $\mathbb{R}^n$, $n \geq 1$ discussed in Remark 63. At the same time the correspondence has its limit as one can infer noticing that $\Omega^k(M) \neq C^\infty(M) \otimes \Omega_n^k$ as one might naively expect at first glance. The reason lies in the global non trivial structure of the tangent and of the cotangent bundle. The following proposition makes this statement more precise.

Corollary 200. *Let M be a smooth manifold whose associated tangent bundle is trivial, namely it is diffeomorphic to $M \times \mathbb{R}^n$, $n = \dim M$. Then, for all $k \in \mathbb{N}$, it holds*

$$\Omega^k(M) \simeq C^\infty(M) \otimes \Omega_n^k,$$

where Ω_n^k is the algebra of homogeneous elements of degree k generated from Ω_n^1 as in (315).

Proof. If TM is diffeomorphic to $M \times \mathbb{R}^n$, $n = \dim M$, per duality also T^*M is diffeomorphic to $M \times \mathbb{R}^n$. Consequently $\Omega^1(M) = \Gamma^\infty(T^*M)$ is isomorphic to $C^\infty(M; \mathbb{R}^n)$ which is, in turn isomorphic to $C^\infty(M) \otimes \Omega_n^1$ up to the identification $\Omega_n^1 \equiv \mathbb{R}^n$. Consider now a global frame of T^*M , that is n -linearly independent sections $\omega_i \in \Gamma(T^*M)$. Consider in addition the collection $\{\omega_i \wedge \omega_j\}_{i < j}$. These form a global frame of $\Lambda^2(M)$. As a matter of fact, for any point $p \in M$, there exists a neighbourhood $U \subset M$, $p \in U$, and a coordinate system $\{x_j\}_{j=1, \dots, n}$ such that $\omega_i = f_i dx_i$ where $f_i \in C^\infty(U)$. As a consequence the 2-forms $\{f_i f_j dx_i \wedge dx_j\}_{i < j}$ are linearly independent since the coefficients never vanish and $\{dx_i \wedge dx_j\}$ is a basis of Ω_n^2 . The reasoning can be now repeated slavishly for all integers k . $\square$

To conclude this section we wish to analyze the behaviour of differential forms with respect to maps between smooth manifolds. This is nothing but the renown pull-back. Yet, as a preliminary step, we need to motivate why in Definition 197 we are talking of tensors, even though the structure involved is that of an alternating algebra.

Definition 201. *Let V be any finite dimensional real vector space. We call **alternating k -tensor** any $\alpha \in (V^*)^{\otimes k}$, $k \in \mathbb{N}$ such that, for all $v_1, \dots, v_k \in V$ and for every pair of indices i, j with $i \neq j$,*

$$\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = -\alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k).$$

The space of alternating k -tensors is indicated as $\Lambda^k(V^)$. In addition we call*

$$\text{Alt} : (V^*)^{\otimes k} \rightarrow \Lambda^k(V^*), \quad (321)$$

*the **alternation** projection (or antisymmetrization map). This is such that, for any collection $v_1, \dots, v_k \in V$ and for any $\alpha \in (V^*)^{\otimes k}$,*

$$(\text{Alt}(\alpha))(v_1, \dots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} (\text{sign}(\sigma)) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}),$$

where S_k is the permutation group of k -elements and $\text{sign}(\sigma)$ is the sign of the permutation $\sigma \in S_k$.

The following is a rather direct consequence of this last definition:

Corollary 202. *Let M be a smooth manifold. For every $p \in M$ there exists an isomorphism of vector spaces between Λ_p^k and $\Lambda^k(T_p^*M)$ which is completely characterized by the requirement*

$$dx_{i_1} \wedge \dots \wedge dx_{i_k} \mapsto \text{Alt}(dx_{i_1} \otimes \dots \otimes dx_{i_k}), \quad (322)$$

where Alt is the alternation projection as in Equation (321).

Proof. It suffices to observe that both Λ_p^k and $\Lambda^k(T_p^*M)$ are real vector spaces of dimension $\binom{n}{k}$ and the map defined via Equation (322) is manifestly injective. $\square$

We can now give an alternative characterization of differential forms. This is a nice exercise which allows a reader to recollect all given definitions.

Exercise 203. Prove that, for every $k \geq 1$, $\bigsqcup_{p \in M} \Lambda^k(T_p^*M)$ admits a natural structure of vector bundle of rank $\binom{n}{k}$ and that it is diffeomorphic to $\Lambda^k(M)$.

Remark 67. A notable consequence of this last exercise is that a differential form can now be seen as an antisymmetric covariant k -tensor field. This perspective is probably less natural from an algebraic viewpoint, but it is certainly the common interpretation of differential forms in many applications, especially in physics.

Henceforth we will use whenever necessary the interpretation of a differential k -form as a particular covariant k -tensor field without making explicit the diffeomorphism discussed in Exercise (203).

Proposition 204. Let M and N be two smooth manifolds and let $F \in C^\infty(M; N)$. For all $\omega \in \Omega^k(N)$, $k = 0, \dots, n = \dim N$, we call **pullback** of ω , the form $F^*\omega \in \Omega^k(M)$ such that, for each $p \in M$ and for every collection $v_1, \dots, v_k \in T_pM$

$$F^*\omega(v_1, \dots, v_k) \doteq \omega(dF(v_1), \dots, dF(v_k)),$$

where $dF : T_pM \rightarrow T_{F(p)}N$ is the differential map. It holds that

1. F^* is $\mathbb{R}$ -linear,
2. F^* is an algebra homomorphism, that is, for all $\omega \in \Omega^k(N)$ and $\omega' \in \Omega^{k'}(N)$,

$$F^*(\omega \wedge \omega') = F^*\omega \wedge F^*\omega'.$$

3. Given any coordinate system $\{y_1, \dots, y_n\}$ in $U \subseteq N$ and given any $\omega = \sum_{i_1 < i_2 < \dots < i_k} \omega_{i_1 \dots i_k} dy_{i_1} \wedge \dots \wedge dy_{i_k}$, with each $\omega_{i_1 \dots i_k} \in C^\infty(U)$ it holds that on $F^{-1}(U)$

$$F^*\omega = \sum_{i_1 < i_2 < \dots < i_k} (\omega_{i_1 \dots i_k} \circ F) d(y_{i_1} \circ F) \wedge \dots \wedge d(y_{i_k} \circ F).$$

Proof. The map F^* is linear per construction. The second property holds since, for every $p \in M$ and for every collection $v_1, \dots, v_{k+k'} \in T_pM$ it holds

$$\begin{aligned} F^*(\omega \wedge \omega')(v_1, \dots, v_{k+k'}) &= (\omega \wedge \omega')(dF(v_1), \dots, dF(v_{k+k'})) = \\ \frac{1}{(k+k')!} \sum_{\sigma \in S_{k+k'}} \text{sign}(\sigma) \omega(dF(v_{\sigma(1)}), \dots, dF(v_{\sigma(k)})) \omega'(dF(v_{\sigma(k+1)}), \dots, dF(v_{\sigma(k+k')})) &= \\ (F^*\omega \wedge F^*\omega')(v_1, \dots, v_{k+k'}). \end{aligned}$$

The third property is instead an immediate consequence of the second once we prove it for 1-forms. To this end it suffices to consider a basis element of the form $\omega_i = \alpha dy_i$, with $\alpha \in C^\infty(U)$. The definition of pullback entails that, for every $v \in T_q U$, where $p = F(q)$ and calling $\{x_i\}_{i=1, \dots, m = \dim M}$,

$$dF(v)|_{F(q)} = \sum_{i=1}^m \sum_{j=1}^n v_i \frac{\partial y_j \circ F}{\partial x_i} \frac{\partial}{\partial y_j}$$

and, consequently,

$$F^* \omega_i(v) = \omega_i(dF(v)) = \omega_i|_{F(p)}(dF(v)) = (\omega_i \circ F)d(y_i \circ F).$$

□

An application of this proposition, which is useful in several practical scenarios, is discussed in the following exercise

Exercise 205. Let M and N be two smooth manifolds of the same dimension n and let $F \in C^\infty(M; N)$. Let $\{x_i\}_{i=1, \dots, n}$ be a coordinate system in $U \subseteq M$ and let $\{y_j\}_{j=1, \dots, n}$ be a counterpart in $V \subseteq N$. Show that, if $U \cap F^{-1}(V) \neq \emptyset$, then, for every $u \in C^\infty(V)$ it holds

$$F^*(u dy_1 \wedge \dots \wedge dy_n) = (u \circ F) \det(DF) dx_1 \wedge \dots \wedge dx_n,$$

where $\det(DF)$ is the Jacobian determinant of F .

Another important structural property of the pull-back is the following:

Exercise 206. Let M, N be two smooth manifolds and let $F \in C^\infty(M; N)$ be a submersion, namely dF is everywhere surjective. Prove that, for all $k \in \mathbb{N} \cup \{0\}$, the pull-back $F^* : \Omega^k(N) \rightarrow \Omega^k(M)$ is injective.

C.1 The exterior differentiation on a manifold

As already outlined in Section B, one of the advantages of giving a geometric realization of differential forms on $\mathbb{R}^n$, $n \geq 1$ and of their algebra structure via the wedge product is the existence of a distinguished operator which maps k - into $(k+1)$ -forms, cf. Definition 187. A natural question is whether there exists a counterpart on the space of differential k -forms on an arbitrary manifold M . Intuitively the answer is affirmative and one could expect that, in every coordinate patch of M , which corresponds to a local trivialization of $\Lambda^k(M)$, it coincides with the one introduced in Definition 187.

In order to make precise this idea, the best course of action consists of generalizing to an arbitrary smooth manifold the universal characterization given in Theorem 189. We shall proceed in three separate steps: uniqueness, naturality and existence of the exterior differentiation.

Remark 68. *At this stage we have used for the first time the word naturality. This is a concept often invoked in algebra and geometry, which has a kind of heuristic and philosophic flavor, although it has a precise formulation in the theory of categories. To given an idea of the meaning of naturality, consider two sets A, B and a distinguished map $f : A \rightarrow B$. Let A and B be endowed with a map $d_A : A \rightarrow A$ and $d_B : B \rightarrow B$. These maps are called natural with respect to f if $f \circ d_A = d_B \circ f$.*

In many instances naturality plays an important role in understanding the formulation of models in physics.

We start from uniqueness:

Theorem 207. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. If existent, it is unique the operator $\underline{d} = (d_0, \dots, d_n) : \Omega^\bullet(M) \rightarrow \Omega^\bullet(M)$ acting component by component on any $\underline{\omega} \in \Omega^\bullet(M)$ in such a way that the following four properties are satisfied:*

1. $\underline{d}$ is $\mathbb{R}$ -linear,
2. $\underline{d}^2 = 0$,
3. $\underline{d}$ is an **antiderivation**, namely, for all $\omega \in \Omega^k(M)$ and for all $\omega' \in \Omega^{k'}(M)$

$$\underline{d}(\omega \wedge \omega') = \underline{d}\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge \underline{d}\omega',$$

where $\deg : \Omega^k(M) \rightarrow \mathbb{Z}$ is the natural generalization of Equation 309 to k -forms on M .

4. for every $\underline{\omega} = (f, 0, \dots, 0)$ where $f \in C^\infty(M) \equiv \Omega^0(M)$, it holds

$$\underline{d}\omega = (0, d_0 f, 0, \dots, 0),$$

where $d_0 f = df$ coincides with the differential of f , that is

$$df : \Gamma^\infty(TM) \rightarrow C^\infty(M), \quad X \mapsto df(X) \doteq X(f).$$

Proof. Suppose that the operator $\underline{d}$ exists. We can prove its uniqueness at the level of each component. We start by showing that, for any $k \in \mathbb{N} \cup \{0\}$, d_k acts locally. To this end, consider $\omega_1, \omega_2 \in \Omega^k(M)$ and suppose that there exists an open neighbourhood $U \subseteq M$ such that $\omega_1|_U - \omega_2|_U = 0$. Let us fix any $p \in U$ and, up to a shrinking of the open subset, we can assume that U admits a local chart. Let $\eta = \omega_1 - \omega_2$ and let $\psi \in C^\infty(M)$ be a bump function centered at p , cf. [BT82, Prop. 2.25]. Hence $\psi\eta = 0$ and, condition 1. and 3., entail

$$0 = d_k(\psi\eta) = d_0\psi \wedge \eta + \psi d_k\eta.$$

Since the bump function can be chosen to be constantly equal to 1 in a neighbourhood of p , the above identity entails that $d_k\eta|_p = 0$. In other words $d_k\omega_1|_p = d_k\omega_2|_p$. This entails that the operator d_k is completely specified by its action on a cover of M via local charts. Let (U, φ) be

any of such charts and let $\{x_i\}_{i=1,\dots,n}$ be a coordinate system in U . Using Equation (320), we know that

$$\omega|_U = \sum_{i_1 < i_2 < \dots < i_k} \omega_{i_1 \dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

where each $\omega_{i_1 \dots i_k} \in C^\infty(U)$. By using once more a bump function, we can use this expansion to construct $\tilde{\omega} \in \Omega^k(M)$ by replacing $\omega_{i_1 \dots i_k}$ and $\{x_i\}_{i=1,\dots,n}$ with $\tilde{\omega}_{i_1 \dots i_k} \in C^\infty(M)$ and $\tilde{x}_i \in C^\infty(M)$, for each $i = 1, \dots, n$. These are chosen to coincide respectively with $\omega_{i_1 \dots i_k}$ and $\{x_i\}_{i=1,\dots,n}$ on U . As a consequence the action of d_k on $\omega|_U$ coincides with that on $\tilde{\omega}$. Yet this is completely specified by using condition 1.-4. since

$$d_k \tilde{\omega} = \sum_{i_1 < i_2 < \dots < i_k} d_k(\tilde{\omega}_{i_1 \dots i_k} d\tilde{x}_{i_1} \wedge \dots \wedge d\tilde{x}_{i_k}) = \sum_{i_1 < i_2 < \dots < i_k} (d\tilde{\omega}_{i_1 \dots i_k}) \wedge d\tilde{x}_{i_1} \wedge \dots \wedge d\tilde{x}_{i_k}.$$

This entails that, on each coordinate neighbourhood of M , d_k is uniquely specified. $\square$

We can now establish the second part of the main result of this section, namely naturality on $\mathbb{R}^n$.

Theorem 208. *Let $F \in C^\infty(\mathbb{R}^m; \mathbb{R}^n)$ and let $\underline{d}$ and $\tilde{d}$ be the exterior differentiation respectively on $\Omega^\bullet(\mathbb{R}^m)$ and $\Omega^\bullet(\mathbb{R}^n)$. Then, for every $k \in \mathbb{N} \cup \{0\}$, the following diagram is commutative:*

$$\begin{array}{ccc} \Omega^k(\mathbb{R}^n) & \xrightarrow{F^*} & \Omega^k(\mathbb{R}^m) \\ \tilde{d}_k \downarrow & & \downarrow d_k \\ \Omega^{k+1}(\mathbb{R}^n) & \xrightarrow{F^*} & \Omega^{k+1}(\mathbb{R}^m) \end{array}$$

where F^* is the pullback as in Proposition 204.

Proof. In view of Remark 63, since $\Omega^k(\mathbb{R}^n)$ is a vector space it is sufficient to prove the statement of a k -form of $\alpha dx_1 \wedge \dots \wedge dx_k$ where $\alpha \in C^\infty(\mathbb{R}^n)$, while $\{x_i\}_{i=1,\dots,n}$ is an arbitrary coordinate system of $\mathbb{R}^n$. Using item 3. of Proposition 204, it holds

$$\begin{aligned} d_k(F^*(\alpha dx_1 \wedge \dots \wedge dx_k)) &= d_k((\alpha \circ F)d(x_1 \circ F) \wedge \dots \wedge d(x_k \circ F)) = \\ &= d(\alpha \circ F) \wedge d(x_1 \circ F) \wedge \dots \wedge d(x_k \circ F) \end{aligned}$$

At the same time, using that the pullback is an algebra homomorphism

$$\begin{aligned} F^*(\tilde{d}_k(\alpha dx_1 \wedge \dots \wedge dx_k)) &= F^*(\tilde{d}\alpha \wedge dx_1 \wedge \dots \wedge dx_k) = \\ &= F^*(\tilde{d}\alpha) \wedge d(x_1 \circ F) \wedge \dots \wedge d(x_k \circ F), \end{aligned}$$

where we used implicitly that differentiation over functions satisfies the chain rule on account of Definition 185. To conclude that the two expressions are the same, it suffices to observe that, on the one hand, per definition of pullback $F^*(d\alpha) = \tilde{d}\alpha \circ dF$, while, on the other hand that, the chain rule entails $d(\alpha \circ F) = \tilde{d}\alpha \circ dF$. $\square$

Theorem 209. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. There exists an operator $\underline{d} = (d_0, \dots, d_n) : \Omega^\bullet(M) \rightarrow \Omega^\bullet(M)$, called **exterior differentiation on M** acting component by component on any $\underline{\omega} \in \Omega^\bullet(M)$ and satisfying the same defining conditions as in Theorem 207.*

Proof. Suppose $\omega \in \Omega^k(M)$. In view of the proof of Theorem 207, we know that each operator d_k acts only locally. Hence, consider a smooth chart (U, φ) where $\varphi : U \rightarrow V \subset \mathbb{R}^n$. We define

$$d_k \omega|_U \doteq \varphi^* d_k|_{\mathbb{R}^n}((\varphi^{-1})^* \omega), \quad (323)$$

where $d_k|_{\mathbb{R}^n}$ is the exterior differentiation on $\mathbb{R}^n$, cf. Definition 187, here indicated with a subscript to avoid possible confusion. The maps φ^* and $(\varphi^{-1})^*$ are pullbacks as per Proposition 204. On account of Theorem 189 d_k inherits the condition 1.-4. from the counterpart on $\mathbb{R}^n$. Therefore we need to show that the given definition behaves well on overlapping coordinate patches. Hence, let (V, ψ) be another chart such that $U \cap V \neq \emptyset$. On account of Theorem 208 it holds that, being $\varphi \circ \psi^{-1}$ a diffeomorphism on $\psi(U \cap V) \subseteq \mathbb{R}^n$,

$$(\varphi \circ \psi^{-1})^* d_k|_{\mathbb{R}^n}((\varphi^{-1})^* \omega) = d_k|_{\mathbb{R}^n}((\varphi \circ \psi^{-1})^* \circ (\varphi^{-1})^* \omega) = d_k|_{\mathbb{R}^n}(\psi^* \omega),$$

where we used both that $(\varphi \circ \psi^{-1})^* = (\psi^{-1})^* \circ \varphi^*$ and that $\varphi \circ \varphi^{-1} = id|_U$ entails $(\varphi^*)^{-1} = (\varphi^{-1})^*$. Since, in addition,

$$(\varphi \circ \psi^{-1})^* d_k|_{\mathbb{R}^n}((\varphi^{-1})^* \omega) = (\psi^*)^{-1} \circ \varphi^* d_k|_{\mathbb{R}^n}((\varphi^{-1})^* \omega),$$

we have proven that

$$\varphi^* d_k|_{\mathbb{R}^n}((\varphi^{-1})^* \omega) = \psi^* d_k|_{\mathbb{R}^n}((\psi^{-1})^* \omega),$$

which implies that the given expression of $d_k \omega$ is well-behaved on overlapping patches. $\square$

To conclude the section, we extend the statement on the naturality of exterior differentiation to arbitrary smooth manifolds:

Theorem 210. *Let M and N be two smooth manifolds and let $F \in C^\infty(M; N)$ and let $\underline{d}^M$ and $\underline{d}^N$ be an exterior differentiation respectively on $\Omega^\bullet(M)$ and $\Omega^\bullet(N)$. Then, for every $k \in \mathbb{N} \cup \{0\}$, the following diagram is commutative:*

$$\begin{array}{ccc} \Omega^k(N) & \xrightarrow{F^*} & \Omega^k(M) \\ d_k^N \downarrow & & \downarrow d_k^M \\ \Omega^{k+1}(N) & \xrightarrow{F^*} & \Omega^{k+1}(M) \end{array}$$

where F^* is the pullback as in Proposition 204.

Proof. Since the exterior differentiation acts locally, it suffices to verify the commutativity of the diagram in an arbitrary chart (U, φ) , $U \subseteq M$ and (V, ψ) , $V \subseteq N$. Using Equation (323) on $U \cap F^{-1}(V)$ we obtain for an arbitrary $\omega \in \Omega^k(N)$,

$$\begin{aligned} F^*(d_k^N \omega)|_{U \cap F^{-1}(V)} &= F^* \psi^* d_k^N ((\psi^{-1})^* \omega) = \varphi^* \circ (\varphi^{-1})^* \circ F^* \psi^* d_k^N ((\psi^{-1})^* \omega) = \\ &= \varphi^* \circ (\psi \circ F \circ \varphi^{-1})^* d_k^N ((\psi^{-1})^* \omega) =_{N \rightarrow M} \varphi^* d_k^M ((\psi \circ F \circ \varphi^{-1})^* (\psi^{-1})^* \omega) = \\ &= \varphi^* d_k^M ((\varphi^{-1})^* F^* \omega) = d_k^M (F^* \omega)|_{U \cap F^{-1}(V)}, \end{aligned}$$

where for simplicity of notation we omitted in the intermediate steps the restriction $|_{U \cap V}$ and where $=_{N \rightarrow M}$ indicates that, in that equality, we used Theorem 208. $\square$

C.2 Orientation on a manifold

Finally we have met our goal of making precise the notion of differential k -forms on a smooth manifold. It is due time to use them to obtain some useful, cute, funny, good-looking, inspiring, awesome, tantalizing, results. The first one, that we discuss, might be quite surprising, but it has far reaching consequences. For example, whenever we integrate over $\mathbb{R}^n$, we always write the volume forms as $d^n x = dx_1 \dots dx_n$. What does it mean really? It might be surprising but the symbol is not only related to Lebesgue integration (obvious, come on!) but mainly to the notion of orientation. If you have no idea what does it mean, you can think of orientation as the mathematical procedure to make precise the idea behind the right hand rule that you learned in classical physics... bizarre isn't it?

We shall divide our analysis in two steps: first we discuss the notion of orientation on a generic vector space, having $\mathbb{R}^n$ in mind as guiding example and subsequently we discuss what happens on an arbitrary smooth manifold. Here we follow [BT82, Chap. 15].

Remark 69. *Let us consider the vector space $\mathbb{R}^n$. In this case we have a distinguished canonical basis*

$$e_i = (0, \dots, 0, 1, 0, \dots, 0)$$

$\downarrow$
 $i^{\text{th}}\text{-entry}$

and if we set $n = 3$ it is easy to observe that (e_1, e_2, e_3) is ordered according to the right hand rule. Hence, if we take any other orthonormal basis $\{e'_i\}_{i=1, \dots, n}$, we consider it oriented as the canonical one if the matrix $A \in O(n)$ such that $e'_i = Ae_i$ for all $i = 1, \dots, n$ has positive determinant.

It is important to stress that the peculiarity of this case rests in the existence of a distinguished basis which identifies a natural standard orientation. In many other concrete instances, this is not the case. Therefore one cannot expect to be able to identify a unique or a distinguished orientation.

In view of this remark, the following definition appears as natural

Definition 211. *Let V be a real vector space of dimension $\dim V = n$. We call **orientation** of V an equivalence class $[e_1, \dots, e_n]$ where $(e_1, \dots, e_n)$ is a basis of V and $(e_1, \dots, e_n) \sim (e'_1, \dots, e'_n)$ if the associated change of basis matrix has positive determinant*

In view of this definition we say that, given a real vector space V of dimension $\dim V = n$,

- a pair $(V, [e_1, \dots, e_n])$ is an **oriented vector space**.
- any representative $(e'_1, \dots, e'_n)$ of $[e_1, \dots, e_n]$ is called a *positively oriented basis*. Any other basis is referred to generically as being negatively oriented.

It is rather fascinating that there is close connection between the choice of an orientation on a vector space V and its associated exterior algebra and alternating k -tensors, cf. Definition 201.

Proposition 212. *Let V be a real vector space of dimension $\dim V = n \geq 1$. Then, for every alternating n -tensor $\alpha \in \Lambda^n(V^*)$, $\alpha \neq 0$, there exists an associated orientation, indicated as $\mathcal{O}_\alpha$ and defined as follows:*

- $\mathcal{O}_\alpha$ is the equivalence class of bases $[e_1, \dots, e_n]$ such that, for every representative

$$(e_1, \dots, e_n) \mapsto \alpha(e_1, \dots, e_n) > 0.$$

Proof. Let $\alpha \in \Lambda^n(V^*)$ and consider any basis $(e_1, \dots, e_n)$ such that $\alpha(e_1, \dots, e_n) \neq 0$. It must exist since $\alpha \neq 0$ and, without loss of generality, we can also assume that $\alpha(e_1, \dots, e_n) > 0$. If not, per linearity of α it suffices to change the sign to one of the elements of the basis. Consider the orientation $[e_1, \dots, e_n]$ and let $(e'_1, \dots, e'_n)$ be another representative. Let $B \in M(n, \mathbb{R})$ the matrix connecting the two basis. Per Definition 211 $\det(B) > 0$. Let $\{e^i\}_{i=1, \dots, n}$ be the dual basis of V^* and let $\mathcal{A}_n^n(V^*) = \text{span}_{\mathbb{R}}(e^1 \cdot \dots \cdot e^n)$ be the vector space built out the degree n elements of the exterior algebra $\mathcal{A}_n(V^*)$ generated by $\{e^1, \dots, e^n\}$ where we highlighted the explicit dependence on V^* . Adapting to the case in hand both Definition 201 and Corollary 202, we know that there must exist $\kappa > 0$ such that

$$\alpha = \kappa \text{Alt}(e^1 \otimes \dots \otimes e^n),$$

as well as an isomorphism $I : \Lambda^n(V^*) \rightarrow \mathcal{A}_n^n(V^*)$ such that

$$I(\alpha) = \kappa e^1 \cdot \dots \cdot e^n.$$

Consider now the second basis and its dual $\{(e')^i\}_{i=1, \dots, n}$. Proposition 183 entails the following chain of identities

$$I(\alpha') = \kappa (e')^1 \cdot \dots \cdot (e')^n = \det(B)^{-1} \kappa e^1 \cdot \dots \cdot e^n = \det(B^{-1}) I(\alpha),$$

where $\alpha' = \kappa \text{Alt}((e')^1 \otimes \dots \otimes (e')^n)$. Since I is an isomorphism it descends $\alpha = \det(B) \alpha'$ and consequently

$$\alpha(e'_1, \dots, e'_n) = \det(B) \alpha'(e'_1, \dots, e'_n) = \kappa \det B > 0.$$

□

Remark 70. *There are instances where it is convenient to consider also the degenerate case $n = 0$. In this scenario $\Lambda^0(V^*) = \mathbb{R}$ and the direct translation of Proposition 212 entails that $\mathcal{O}_\omega = [1] \equiv 1$ if $\Lambda^0(V^*) \ni \omega > 0$ while $\mathcal{O}_\omega = [-1] \equiv -1$ if $\Lambda^0(V^*) \ni \omega < 0$.*

Remark 71. *Observe that, since $\dim \Lambda^n(V^*) = 1$ for every finite dimensional real space V , the choice of an orientation for V is tantamount to choosing one of the two connected components of $\Lambda^n(V^*) \setminus \{0\}$. Hence, calling $\mathcal{O}(V)$ the collection of all possible orientations associated to V , there is a bijection $I : \mathcal{O}(V) \rightarrow \mathbb{Z}_2$.*

We can now ask ourselves how the notion of orientation can be generalized to an arbitrary smooth manifold. We can make a few preliminary observations:

- an orientation of a smooth manifold M , must correspond to the assignment of an orientation to $T_p M$ for all $p \in M$, indicated as $\mathcal{O}_p$,
- we expect this assignment $p \mapsto \mathcal{O}_p$ to be continuous in p . Yet, to make sense of this concept we need to induce a topological structure on the set of possible orientations. To this end observe, that, as outlined in Remark 71, for every finite dimensional vector space V , $\mathcal{O}(V)$ is in bijection with $\mathbb{Z}_2$ which is a topological Lie group with respect to the discrete topology. Yet, exploiting this information is far from being trivial.

In order to make precise these ideas, we start by generalizing the notion of orientation:

Definition 213. *Let M be an n -dimensional smooth manifold. We call it **orientable** if there exists a collection of local trivializations $\{\Phi_i = \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^n\}_{i=1,\dots,k}$ such that $\bigcup_{i=1}^k U_i = M$ and, for each $p \in U_i$, the linear isomorphism $\Phi|_{T_p M} : T_p M \rightarrow \{p\} \times \mathbb{R}^n$ is an orientation-preserving map with respect to the standard orientation of $\mathbb{R}^n$.*

From this definition we can infer that a manifold is orientable whether there exists at least one possible trivialization of the associated tangent bundle compatible with the standard orientation of $\mathbb{R}^n$. Yet nothing forces us to keep such orientation and we could make another choice. Hence it is convenient to speak of an **oriented manifold** $(M, \mathcal{O})$, to underline whenever a specific orientation has been selected.

It is rather obvious that Definition 213 does not offer a concrete way of checking whether a manifold is orientable or not. A more practical criterion is necessary and the quandary can be solved by generalizing Proposition 212 to an arbitrary smooth manifold.

Proposition 214. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. Any $\omega \in \Omega^n(M)$ such that $\omega(p) \neq 0$ for all $p \in M$ individuates a unique orientation of M . We call ω a **smooth orientation form**.*

Proof. Let (U, Φ) be any local trivialization of TM , chosen, without loss of generality so that (U, φ) is a chart of M . Let $\{e_i\}_{i=1,\dots,n}$ be the standard basis of $\mathbb{R}^n$ and let $\{(\varphi^{-1})_*(e_i)\}_{i=1,\dots,n}$ be the induced basis of $T_p U$ for each $p \in U$. Since $\dim \Lambda_p^n = 1$ for every $p \in U$, this entails that, calling $\{dx_1, \dots, dx_n\}$ the induced basis on $T_p^* U$, $\omega|_U = f dx_1 \wedge \dots \wedge dx_n$ where $f \in C^\infty(U)$.

Furthermore since ω is nowhere vanishing f must be of definite sign, chosen without loss of generality to be positive. Hence, for all $p \in U$,

$$((\varphi^{-1})^*\omega)(\varphi(p))(e_1, \dots, e_n) = \omega(p)((\varphi^{-1})_*(e_1), \dots, (\varphi^{-1})_*(e_n)) = f(p) > 0.$$

In addition, a direct computation shows that, for every other basis of $\mathbb{R}^n$, say $(e'_1, \dots, e'_n)$, falling in the same orientation class of the standard basis, it holds that $((\varphi^{-1})^*\omega)(\varphi(p))(e'_1, \dots, e'_n) > 0$ for every $p \in U$. By applying Proposition 212 we can conclude the proof. $\square$

Exercise 215. Let M be a smooth manifold of dimension $\dim M = n \geq 1$. For any orientation of M , there exists an associated smooth orientation form.

Proposition 214 is noteworthy for two reasons. The first is the connection between the existence of an orientation with that of a global top-form. The second is for its application to the case where the underlying smooth manifold is endowed with a Riemannian or with a Lorentzian structure as per Definition 50. The differences between the two scenarios are minimal and, hence, we shall focus only on the first case leaving to the reader the task of accounting for the other option. Where necessary we shall highlight the different formulae if we work with a Lorentzian signature

Hence, let $(M, \mathcal{O}, g)$ be an oriented Riemannian manifold of dimension $n = \dim M \geq 1$ and let $\{E_i\}_{i=1, \dots, n}$ be a (smooth) local frame over $U \subset M$. Without loss of generality we can require it to be a *local oriented orthonormal frame* that, for all $p \in U$,

$$[E_1|_p, \dots, E_n|_p] \in \mathcal{O} \quad \text{and} \quad g_p(E_i|_p, E_j|_p) = \delta_{ij} \quad \forall i, j = 1, \dots, n.$$

If g would have been Lorentzian, it suffices to replace δ_{ij} with the matrix elements of the Minkowski metric, namely η_{ij} . The following notable global result holds true:

Theorem 216. *Let $(M, \mathcal{O}, g)$ be an oriented Riemannian manifold of dimension $n = \dim M \geq 1$. Then, there exists a unique smooth orientation form $\omega_g \in \Omega^n(M)$ such that, for all local oriented orthonormal frame $\{E_i\}_{i=1, \dots, n}$, it holds*

$$\omega_g(E_1, \dots, E_n) = 1.$$

We call ω_g the **Riemannian volume form**.

Proof. Let us prove uniqueness. Consider ω_g abiding to the properties and conditions in the statement of the theorem. Let $\{E_i\}_{i=1, \dots, n}$ be any local oriented orthonormal frame in $U \subseteq M$ and let $\{E^i\}_{i=1, \dots, n}$ be 1-form on U , forming the dual basis of T_p^*M for every $p \in U$. There must exist $f \in C^\infty(U)$ such that

$$\omega_g|_U = f E^1 \wedge \dots \wedge E^n.$$

Yet, since $\omega_g|_U(E_1, \dots, E_n) = 1$, then $f = 1$. This determines ω_g uniquely in a cover of M and hence globally.

Let us prove existence. As above, let $\{E_i\}_{i=1,\dots,n}$ be any local oriented orthonormal frame in $U \subseteq M$ and let $\{E^i\}_{i=1,\dots,n}$ be 1-form on U , forming the dual basis of T_p^*M for every $p \in U$ and let

$$\omega_g|_U \doteq E^1 \wedge \cdots \wedge E^n.$$

We need to prove that this definition does not depend on the choice of the frame. Let $\{\tilde{E}_i\}_{i=1,\dots,n}$ be a second, local, oriented orthonormal frame in $U \subseteq M$ and let $\{\tilde{E}^i\}_{i=1,\dots,n}$ be 1-form on U , forming the dual basis of T_p^*M for every $p \in U$. Let $\tilde{\omega}_g|_U \doteq \tilde{E}^1 \wedge \cdots \wedge \tilde{E}^n$. In addition there must exist $A \in C^\infty(U; SO(n, \mathbb{R}))$ such that, for each $p \in U$, $\tilde{E}_i|_p = \sum_{j=1}^n A_{ij}(p) E_j|_p$. The function A is $SO(n)$ -valued since the two frames are orthonormal and consistently oriented. By Proposition 183 it holds that, at each $p \in U$

$$\tilde{E}^1|_p \wedge \cdots \wedge \tilde{E}^n|_p = \det(A(p)) E^1|_p \wedge \cdots \wedge E^n|_p = E^1|_p \wedge \cdots \wedge E^n|_p,$$

which entails that $\omega_g = \tilde{\omega}_g$. Hence the definition of the top form does not depend on the frame chosen. $\square$

A notable consequence of this theorem is its local version, which is probably well-known to the readers (if not, play dumb and nod... trust me... it is better... really... no kidding)

Corollary 217. *Let $(M, \mathcal{O}, g)$ be an oriented Riemannian or a Lorentzian manifold of dimension $n = \dim M \geq 1$. For any chart (U, φ) with an associated oriented coordinate system¹ $\{x_i\}_{i=1,\dots,n}$, it holds*

$$\omega_g|_U = \sqrt{|\det(g)|} dx_1 \wedge \cdots \wedge dx_n \equiv \sqrt{|g|} dx_1 \wedge \cdots \wedge dx_n,$$

where $\det(g)$ stands for the determinant of the matrix whose entries are $g_{ij} = g\left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right)$.

Proof. Let (U, φ) be any chart of M and let $\{x_i\}_{i=1,\dots,n}$ be any coordinate system. Since $\dim \Lambda_p^n = 1$ for any $p \in M$, there must exist $f \in C^\infty(U)$ such that

$$\omega_g|_U = f dx_1 \wedge \cdots \wedge dx_n.$$

To compute f , we can choose from the beginning U so that there exists an oriented orthonormal frame $\{E_i\}_{i=1,\dots,n}$. We call $\{E^i\}_{i=1,\dots,n}$ the dual basis so that, according to Theorem 216, $\omega_g = E^1 \wedge \cdots \wedge E^n$. In addition there exists $A \in C^\infty(U; GL(n, \mathbb{R}))$ such that $\frac{\partial}{\partial x_i}|_p = \sum_{j=1}^n A_{ij}(p) E_j|_p$.

This entails that, using Proposition 183, at each $p \in U$,

$$\omega_g\left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}\right) = f = \det(A(p)).$$

¹We call a coordinate system $\{x_i\}_{i=1,\dots,n}$ oriented if the associated basis of the tangent space $\left\{\frac{\partial}{\partial x_i}\right\}_{i=1,\dots,n}$ is oriented according to $\mathcal{O}$.

At the same time, observe that

$$g_{ij} = g \left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \right) = g \left(\sum_{k=1}^n A_{ik} E_k, \sum_{l=1}^n A_{jl} E_l \right) = \sum_{k=1}^n A_{ik} A_{jk}.$$

At a matrix level, this translates to $g = AA^T$ which entails $\det g = \det(A)^2$. Since, on a Riemannian manifold $\det g = |\det g|$ the proof is concluded. $\square$

Remark 72. *Theorem 216 and Corollary 217 have several practical consequences. Yet here, we wish to highlight that, in many papers, there is a tendency, when integrating functions on a Riemannian or on a Lorentzian manifold (see next section) to write the volume form either as $\sqrt{|g|} d^n x$ or as μ_g or even as $d\mu_g$. Besides the commonly accepted notation which avoids writing explicitly the wedge product, one should keep in mind that the first of the three writings is a real abuse of notation since it is valid only locally. Hence using μ_g or $d\mu_g$ is a better option, even though $\sqrt{|g|} d^n x$ is accepted almost universally even in the mathematics literature. As a matter of fact this is the notation that we adopt in the main body of these notes to make contact with the standard literature on general relativity, especially in Section 4.1.*

To conclude, observe also that the absolute value within the square root is usually omitted if one considers Riemannian manifolds.

Disclaimer: Henceforth we will adopt the same notation as in the literature, omitting to write explicitly the symbol $\mathcal{O}$ to indicate that a manifold is oriented. We will highlight it in words.

C.3 Integration on manifolds

All good things come to an end... but also the bad ones. This means that, regardless of what you think about the previous sections, everything is in place to solve one of the main open questions in differential geometry: how to integrate a function (or more generally a k -form) on a smooth manifold. We can give a complete answer, but a disclaimer is necessary. In the preceding discussions all manifold could be with empty or non empty boundary, since this played no role. It should not come as a surprise that, when discussing integration, this is instead a somehow important distinction. In the following we will only discuss the case where the underlying manifold M has empty boundary $\partial M = \emptyset$. The reason to lose in generality is to highlight the important structural aspects of integration on a manifold. The reader should keep in mind that, including a (smooth) boundary does not limit the range of applicability of the theory of integration... it only makes it more complicated since boundary terms do occur. A reader interested in understanding what happens in presence of boundaries, or even corners, can consult [Lee13, Chap. 16].

In the following we shall divide our analysis in three key steps: integration on $\mathbb{R}^n$ with forms, integration on smooth manifolds and integration on Riemannian manifolds.

Step 1: Integration of forms on $\mathbb{R}^n$

Consider $\mathbb{R}^n$, $n \geq 1$ and recall that a domain of integration is any bounded subset $D \subset \mathbb{R}^n$ such that $\mu(\partial D) = 0$ where μ is the Lebesgue measure. Consider $\Lambda^k(\mathbb{R}^n)$, cf. Definition 197 and, to be general, following Remark 63, we consider only continuous sections, that is

$$\Gamma(\Lambda^n(\mathbb{R}^n)) \equiv C^0(\mathbb{R}^n) \otimes \Omega_n,$$

Regardless of this extension, since $\Lambda^n(\mathbb{R}^n)$ is a rank 1 vector bundle, considering the standard Cartesian coordinates $\{x_i\}_{i=1,\dots,n}$, for any $\omega \in \Gamma(\Lambda^n(\mathbb{R}^n))$, there must exist $f_\omega \in C^0(\mathbb{R}^n)$ such that

$$\omega = f_\omega dx_1 \wedge \cdots \wedge dx_n.$$

Following the standard theory of Lebesgue integration, we can give a natural definition of what does it mean integrating ω .

Definition 218. Let $D \subset \mathbb{R}^n$ be a domain of integration and let $\omega \in C^0(\overline{D}) \otimes \Omega_n$. Considering the standard Cartesian coordinates and writing $\omega = f_\omega dx_1 \wedge \cdots \wedge dx_n$, with $f_\omega \in C^0(\overline{D})$, we call **integral of ω over D** ,

$$\int_D \omega = \int_D f_\omega dx_1 \wedge \cdots \wedge dx_n \doteq \int_D f_\omega d^n x.$$

Remark 73. It is immediate to notice that the hypothesis of D being a domain of integration and of considering forms lying in $C^0(\overline{D}) \otimes \Omega_n$ could be easily relaxed allowing for a generic open subset $U \subseteq \mathbb{R}^n$ while considering as integrand $\omega \in \Gamma_0(\Lambda^n(U))$. Here the subscript 0 indicates that we admit only n -forms of compact support. In this case we set

$$\int_U \omega = \int_D \omega$$

where D is any domain of integration such that $\text{supp}(\omega) \subset D \subset U$. It is immediate to observe that the definition does not depend on the choice of D . One might also wonder whether D exists since $\text{supp}(\omega)$ is in principle an arbitrary compact subset of $\mathbb{R}^n$. This is guaranteed by compactness which ensures that there exists a finite cover of $\text{supp}(\omega)$ in open balls of $\mathbb{R}^n$ with respect to the standard Euclidean metric. Each of these individuates a domain of integration and, by standard measure theory, a finite union of these sets shares the same property.

Disclaimer: Henceforth, we (in the sense of I since you have no right or reason or actually... way to choose) shall consider only the case of integrals of compactly supported n -forms on open subsets $U \subseteq \mathbb{R}^n$. In addition, for simplicity we shall only work with smooth forms, although continuity suffices. We shall use therefore

$$\Omega_0^k(U) \doteq \Gamma_0^\infty(\Lambda^k(U)). \quad \forall k \in \mathbb{N} \cup \{0\} \quad \text{and} \quad \forall U \subseteq \mathbb{R}^n$$

In the following proposition we show that the definition of integral of a top form is natural with respect to pull-back.

Proposition 219. Let $U, V \subseteq \mathbb{R}^n$ be two open subsets such that there exists an orientation preserving diffeomorphism $G : U \rightarrow V$. If $\omega \in \Gamma_0(\Lambda^n(V))$, then

$$\int_V \omega = \int_U G^* \omega,$$

where G^* stands for the pull-back, cf. Proposition 204.

Proof. Consider any domain of integration E such that $\text{supp}(\omega) \subset E \subset \overline{E} \subset V$. Since G is a diffeomorphism it holds that $D \doteq G^{-1}(E)$ is also a domain of integration such that $D \subset \overline{D} \subset U$. To avoid confusion, let us denote with $\{x_i\}_{i=1,\dots,n}$ the standard Euclidean coordinates on D and with $\{y_i \doteq x_i \circ G\}_{i=1,\dots,n}$ those on E . By using the formula for the change of coordinates within an integral, it holds

$$\begin{aligned} \int_E \omega &= \int_E f(y) d^n y = \int_D (f \circ G)(x) \det(DG) d^n x = \\ &= \int_D (f \circ G)(x) \det(DG) dx_1 \wedge \dots \wedge dx_n = \int_D G^* \omega, \end{aligned}$$

where $\det(DG)$ is the Jacobian determinant of G and where we used in the third equality that G is orientation preserving, while, in the last one, Exercise 205. $\square$

It is easy to check that, whenever G changes the orientation, then an overall minus sign has to be accounted for, namely

$$\int_V \omega = - \int_U G^* \omega.$$

Step 2: Integration of forms on manifolds

Using the results of the previous step we can start from Definition 218 constructing a well-defined integration of a top-form on a manifold. As one might expect the idea is to use a partition of unity argument, see Appendix A, so to reduce any integral of a top form on a generic smooth manifold to a sum of integrals locally defined on open subsets of $\mathbb{R}^n$. We proceed step by step showing that this idea can be made precise and rigorous.

Proposition 220. Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ and let $\omega \in \Omega_0^n(M) \doteq \Gamma_0^\infty(\Lambda^n(M))$ be such that there exists a single smooth chart (U, φ) obeying $\text{supp}(\omega) \subset U$. We call **integral of ω over M**

$$\int_M \omega \doteq \pm \int_{\varphi(U)} (\varphi^{-1})^* \omega,$$

where $(\varphi^{-1})^*$ is the pullback as per Proposition 204 while the sign $\pm$ depends whether $\varphi(U)$ is a positively oriented chart or not. Then the definition does not depend on the choice of the local chart provided that it contains the support of ω .

Proof. Suppose to consider two charts (U, φ) and (V, ψ) both containing $\text{supp}(\omega)$. Without loss of generality we can prove the statement only for positively oriented charts, the other cases following suit. Hence $\psi \circ \varphi^{-1}$ is a positively oriented diffeomorphism from $\varphi(U \cap V)$ to $\psi(U \cap V)$. We can apply Proposition 219 obtaining

$$\begin{aligned} \int_{\varphi(U)} (\varphi^{-1})^* \omega &= \int_{\varphi(U \cap V)} (\varphi^{-1})^* \omega = \int_{\psi(U \cap V)} (\varphi \circ \psi^{-1})^* (\varphi^{-1})^* \omega = \\ &= \int_{\psi(U \cap V)} (\psi^{-1})^* \circ \varphi^* \circ (\varphi^{-1})^* \omega = \int_{\psi(U \cap V)} (\psi^{-1})^* \omega = \int_{\psi(V)} (\psi^{-1})^* \omega, \end{aligned}$$

where we used that $(\varphi^{-1})^* = (\varphi^*)^{-1}$. $\square$

In order to extend this argument to the whole oriented manifold M , consider $\{(U_i, \varphi_i)\}_{i \in \mathbb{N}}$, a cover of M in open charts. Without loss of generality we can only analyze the case where all φ_i , $i \in \mathbb{N}$ are positively oriented and we observe that the support of every $\omega \in \Omega_0^n(M)$ intersects only a finite collection of elements of the cover. These preliminary observations lead to the following result:

Theorem 221. *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ and let $\{(U_i, \varphi_i)\}_{i \in \mathbb{N}}$, be a cover in open, orientation preserving, charts. Let $\{\psi_i\}_{i \in \mathbb{N}}$ be a partition of unity subordinated to the cover, see Proposition 175. Then, for any $\omega \in \Omega_0^n(M)$, we define the integral of ω on M as*

$$\int_M \omega \doteq \sum_{i=1}^{\infty} \int_M \psi_i \omega = \sum_{i=1}^{\infty} \int_{\varphi_i(U_i)} (\varphi_i^{-1})^* \psi_i \omega,$$

where (U_i, φ_i) is the local chart in which $\psi_i \omega$ has support. The definition of the integral does depend neither on the choice of open cover nor on that of partition of unity.

Proof. Consider $\{(V_i, \varphi'_i)\}_{i \in \mathbb{N}}$ a second cover of M in open, orientation preserving, charts and let $\{\phi_i\}_{i \in \mathbb{N}}$ be a partition of unity subordinated to it, see Proposition 175. Then, for any $\omega \in \Omega_0^n(M)$,

$$\int_M \omega = \sum_{i=1}^{\infty} \int_M \psi_i \omega = \sum_{i=1}^{\infty} \int_M \sum_{j=1}^{\infty} \phi_j \psi_i \omega = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \int_M \phi_j \psi_i \omega = \sum_{j=1}^{\infty} \int_M \sum_{i=1}^{\infty} \phi_j \psi_i \omega = \sum_{j=1}^{\infty} \int_M \phi_j \omega,$$

where each exchange between sums and integral is admissible since only a finite number of non-vanishing terms exists due to the compactness hypothesis. $\square$

The integral of forms over a manifold shares many properties of the standard integral of functions. The following exercise highlights them.

Exercise 222. Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ and let $\omega, \omega' \in \Omega_0^n(M)$. The following properties hold true:

1. Linearity: for all $a, b \in \mathbb{R}$

$$\int_M (a\omega + b\omega') = a \int_M \omega + b \int_M \omega',$$

2. Orientation reversal: Denoting with $-M$ the same manifold with opposite orientation,

$$\int_{-M} \omega = - \int_M \omega,$$

3. Positivity: if ω is positively oriented

$$\int_M \omega > 0$$

4. Diffeomorphism Invariance (or Naturality): If N is a smooth, oriented manifold and $F : N \rightarrow M$ is an orientation preserving diffeomorphism, then

$$\int_M \omega = \int_N F^* \omega.$$

Assuming that the dedicated reader has been eager to extend all results obtained up to now to smooth manifold with non empty boundary, we can prove the main result of the theory of integration on manifolds... the true, the one, the only inimitable, omnipresent Stokes theorem:

Theorem 223 (Stokes Theorem). *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$, with boundary ∂M endowed with the induced orientation. Then, for every $\omega \in \Omega_0^{n-1}(M)$,*

$$\int_M d_{n-1} \omega = \int_{\partial M} \omega,$$

where the right hand side is a short cut notation for indicating $\iota_{\partial M}^ \omega$, where $\iota_{\partial M} : \partial M \rightarrow M$ is the embedding map of the boundary ∂M into M . In addition, whenever $\partial M = \emptyset$ the right hand side is per definition vanishing.*

Proof. The proof is divided in three steps which we discuss separately.

Step 1: We consider the special case where the underlying manifold is the upper half space $M = \mathbb{H}^n \doteq \mathbb{R}^{n-1} \times [0, \infty)$. For any but fixed $\omega \in \Omega_0^{n-1}(\mathbb{H}^n)$ we can find $R > 0$ and an associated hyper-rectangle $K \doteq [-R, R]^{n-1} \times [0, R] \supset \text{supp}(\omega)$. By using the standard Euclidean coordinates $\{x_i\}_{i=1, \dots, n}$ with the convention that the domain of x_n is $[0, R]$, there must exist $\omega_i \in C_0^\infty(\mathbb{H}^n)$, $i = 1, \dots, n$ such that

$$\omega = \sum_{i=1}^n \omega_i dx_1 \wedge \dots \wedge \widehat{dx_i} \wedge \dots \wedge dx_n,$$

where the hat means that dx_i is omitted. By using Definition 187 it holds that

$$d_{n-1}\omega = \sum_{i,j=1}^n \frac{\partial \omega_i}{\partial x_j} dx_j \wedge dx_1 \wedge \dots \wedge \widehat{dx_i} \wedge \dots \wedge dx_n = \sum_{i=1}^n (-1)^{i-1} \frac{\partial \omega_i}{\partial x_i} dx_1 \wedge \dots \wedge dx_i \wedge \dots \wedge dx_n,$$

where we used implicitly Corollary 188. Standard Lebesgue theory of integration yields

$$\begin{aligned} \int_{\mathbb{H}^n} d_{n-1}\omega &= \sum_{i=1}^n (-1)^{i-1} \int_K \frac{\partial \omega_i}{\partial x_i} dx_1 \wedge \dots \wedge dx_n = \sum_{i=1}^n (-1)^{i-1} \int_0^R dx_n \int_{-R}^R dx_{n-1} \dots \int_{-R}^R dx_1 \frac{\partial \omega_i}{\partial x_i} = \\ &= \sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R dx_n \int_{-R}^R dx_{n-1} \dots \int_{-R}^R \widehat{dx_i} \dots \int_{-R}^R dx_1 \omega_i|_{-R}^R + (-1)^{n-1} \int_{-R}^R dx_{n-1} \dots \int_{-R}^R dx_1 \omega_n(x)|_0^R = \\ &= (-1)^n \int_{-R}^R dx_{n-1} \dots \int_{-R}^R dx_1 \omega_n(x_1, \dots, x_{n-1}, 0), \end{aligned}$$

where the hat symbol entails that the integral below is omitted and where ω_i vanishes at $x_i = \pm R$ for all $i = 1, \dots, n$ since ω is compactly supported. Observing that the boundary $\partial\mathbb{H}^n \doteq \{(x_1, \dots, x_n) \mid x_n = 0\}$, it turns out that the pull-back of ω to the boundary reads

$$\iota_{\partial M}^* \omega = \omega_n(x_1, \dots, x_{n-1}, 0) dx_1 \wedge \dots \wedge dx_{n-1}.$$

By integrating this form along the boundary we obtain the statement of the theorem, recalling that, if n is even the coordinates $(x_1, \dots, x_n)$ are positively oriented along $\partial\mathbb{H}^n$ while, if n is odd they are not and thus a minus sign has to be accounted for, cf. [Lee13, Example 15.26].

Step 2: We consider the special case where the underlying manifold is $M = \mathbb{R}^n$. In this case we can repeat the procedure of the preceding step, replacing the hyper-rectangle K with $K' = [-R, R]^n$, $R > 0$. Considering any $\omega \in \Omega_0^{n-1}(\mathbb{R}^n)$ it turns out that $\int_{\mathbb{R}^n} d_{n-1}\omega = 0$. This proves the statement of the theorem in this case.

Step 3: We consider a generic smooth manifold M but we take an arbitrary $\omega \in \Omega_0^{n-1}(M)$ such that $\text{supp}(\omega) \subset U$ where (U, φ) is a local chart, which we assume to be positively oriented. The other cases follow suit. Hence, by using Proposition 204 and Theorem 210, it descends

$$\int_M d_{n-1}\omega = \int_{\mathbb{H}^n} d_{n-1}(\varphi^{-1})^*\omega = \int_{\partial\mathbb{H}^n} (\varphi^{-1})^*\omega = \int_{\partial M} \omega,$$

where we used Step 1 in the last equality. Here we assumed that the chart intersects the boundary. If this were not the case, the same result could be obtained using Step 2 in place of Step 1.

Step 4: We consider both a generic smooth manifold M and an arbitrary $\omega \in \Omega_0^{n-1}(M)$. Consider now a cover of M via $\{(U_i, \varphi_i)\}_{i=1, \dots, n}$, local charts which are assumed to be positively oriented. The other case can be dealt with similarly. Consider $\{\psi_i\}_{i=1, \dots, n}$ a partition of unity subordinated to the cover, see Appendix A. It holds

$$\begin{aligned} \int_{\partial M} \omega &= \sum_{i=1}^n \int_{\partial M} \psi_i \omega = \sum_{i=1}^n \int_M d_{n-1}(\psi_i \omega) = \sum_{i=1}^n \int_M d_{n-1} \psi_i \wedge \omega + \psi_i d_{n-1} \omega = \\ &= \int_M d_{n-1} \left(\sum_{i=1}^n \psi_i \right) \wedge \omega + \left(\sum_{i=1}^n \psi_i \right) d_{n-1} \omega = \int_M d_{n-1} \omega, \end{aligned}$$

where we used that, being ω compactly supported, only a finite number of the elements in all sums are non vanishing. In addition we used that the differential is a linear operator and that $\sum_{i=1}^n \psi_i = 1$. $\square$

Corollary 224. *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ with empty boundary. Then, for every form $\omega \in \Omega_0^n(M)$ such that $\omega = d_{n-1}\eta$, $\eta \in \Omega_0^{n-1}(M)$, it holds*

$$\int_M \omega = 0.$$

Proof. This is a direct application of Stokes theorem. $\square$

Corollary 225. *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$. Then, for every form $\omega \in \Omega_0^{n-1}(M)$ such that $d_{n-1}\omega = 0$, it holds*

$$\int_{\partial M} \omega = 0.$$

Proof. This is a direct application of Stokes theorem. $\square$

Step 3: Integration of forms on a Riemannian/Lorentzian manifold

At last, we specialize our analysis to the case when the underlying background is either a Riemannian or a Lorentzian manifold (M, g) of dimension $\dim M = n \geq 1$, see Definition 50. The difference due to the underlying signature of the metric tensor are rather minimal. For this reason we shall focus in all proofs on Riemannian manifolds leaving to the readers the task of taking into account the small differences when the underlying background is Lorentzian. Our main building block is Theorem 216 which guarantees us the existence of a distinguished element in $\Omega^n(M)$, namely the Riemannian/Lorentzian volume form ω_g . This information can be combined with the definition of the integral of a top form in Proposition 220 as follows:

Definition 226. *Let (M, g) be an oriented Riemannian or Lorentzian manifold of dimension $\dim M = n \geq 1$ as per Definition 50. For any $f \in C_0^\infty(M)$, we call **integral of f over M***

$$\int_M f \omega_g,$$

where ω_g is the Riemannian/Lorentzian volume form as in Theorem 216, while the integral is interpreted in the sense of Proposition 220 reading $f \omega_g \in \Omega_0^n(M)$.

Remark 74. *We should stress that in Definition 226 we require f to be smooth only to keep the same attitude as in the previous steps and sections. It is quite straightforward to observe that one can certainly extend the class of integrands to $C_0^0(M)$.*

Disclaimer: In these notes we will adopt different equivalent notations for the integral over a Riemannian manifold, as it often occurs in the literature. As a matter of facts we shall write

$$\int_M f \omega_g \equiv \int_M f d\mu_g \equiv \int_M f \sqrt{|g|} d^n x,$$

where in the last equivalence we are referring implicitly to Corollary 217.

We prove the counterpart in this setting of a well-known property of Lebesgue integrals over $\mathbb{R}^n$.

Proposition 227. *Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. For any $f \in C_0^\infty(M)$ such that $f \geq 0$, it holds*

$$\int_M f d\mu_g \geq 0,$$

where equality holds if and only if $f = 0$. In addition if we set $f = 1$ and

$$\text{Vol}(M) \equiv \text{Vol}(M, g) \doteq \int_M d\mu_g < \infty,$$

we call $\text{Vol}(M)$ the **volume of M** .

Proof. Let $f \in C_0^\infty(M)$ be supported in a single chart (U, φ) . Then, choosing local coordinates $\{x_i\}_{i=1,\dots,n}$, we can use Corollary 217 to write

$$\int_M f d\mu_g = \int_{\varphi(U)} f(x) \sqrt{|g|} dx_1 \dots dx_n \geq 0,$$

whenever $f \geq 0$. At the same time the integral can vanish if and only if $f = 0$. To prove the same statement for a generic compactly supported function, it suffices to use the standard argument which calls for considering a cover of M in local charts $\{(U_i, \varphi_i)\}_{i=1,\dots,n}$ and a partition of unity $\{\psi_i\}_{i=1,\dots,n}$ subordinated to it, see Appendix A. Since each $\psi_i \in C_0^\infty(U_i)$ and $0 \leq \psi_i(x) \leq 1$, the proof is straightforward. $\square$

The following exercise shows once more how the integral of functions on a Riemannian manifold shares most of the properties of the Lebesgue counterpart on $\mathbb{R}^n$.

Exercise 228. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Prove that, for any $f \in C_0^\infty(M)$, it holds

$$\left| \int_M f d\mu_g \right| \leq \int_M |f| d\mu_g.$$

The careful reader should now dive in her/his past, with no fear of living again some of his or her own good dreams or nightmares... yes... I mean it... think about the course of electrodynamics and relativity. In that context, it emerged the possibility of associating on $\mathbb{R}^n$ to a k -form a distinguished $(n - k)$ -form. We discuss succinctly how this idea extends to a generic Riemannian manifold.

Definition 229. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$ and let us identify $\Omega^0(M) \equiv C^\infty(M)$. We call **Hodge operator** the map

$$\star : \Omega^0(M) \rightarrow \Omega^n(M), \quad f \mapsto \star f \doteq f \omega_g,$$

where ω_g is the Riemannian volume form.

The action of $\star$ can be extended as one should show via these exercises:

Exercise 230. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Prove that g induces a pairing between k -forms

$$\langle \cdot, \cdot \rangle_g : \Omega^k(M) \times \Omega^k(M) \rightarrow C^\infty(M),$$

which is pointwisely determined by $\langle \cdot, \cdot \rangle_{g,p} : \Lambda^k(T_p^*M) \times \Lambda^k(T_p^*M) \rightarrow \mathbb{R}$ such that for all $\{\omega_i\}_{i=1,\dots,k}$, $\{\eta_j\}_{j=1,\dots,k}$, with $\omega_i, \eta_j \in T_p^*M$,

$$\langle \omega_1 \wedge \dots \wedge \omega_k, \eta_1 \wedge \dots \wedge \eta_k \rangle = \det(A),$$

where A is the $k \times k$ real valued matrix whose entries are $A_{ij} = g^{-1}(\omega_i, \eta_j)$

Exercise 231. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. For any integer k such that $0 \leq k \leq n$, there exists a unique vector bundle isomorphism $\star : \Lambda^k(M) \rightarrow \Lambda^{n-k}(M)$ such that, for all $\eta, \eta' \in \Omega^k(M)$,

$$\eta \wedge \star \eta' = \langle \eta, \eta' \rangle_g \omega_g,$$

where $\langle \cdot, \cdot \rangle_g$ is the metric-induced pairing on $\Omega^k(M)$. Prove that $\star$ coincides with that of Definition 229 if $k = 0$.

Exercise 232. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Prove that the inverse of the Hodge operator, $\star^{-1} : \Omega^{n-k}(M) \rightarrow \Omega^k(M)$ obeys to the relation $\star^{-1} \doteq (-1)^{k(n-k)} \star$ where the right hand side should be interpreted as the Hodge operator acting on $(n-k)$ -forms.

Remark 75. Observe that, to be precise, one should always indicate with a subscript the space on which the Hodge operator acts, e.g. $\star_k : \Omega^k(M) \rightarrow \Omega^{n-k}(M)$. In addition, it is important to notice that, on Lorentzian manifolds, Exercise 232 is different due to the occurrence of an additional minus sign to account for the signature of the underlying metric.

With these ingredients we can discuss the counterpart of the divergence theorem on an arbitrary Riemannian manifold. At first we generalize the notion of **interior multiplication (by a vector field)** as follows: for every $\chi \in \Gamma(TM)$, we set

$$\iota_\chi : \Omega^k(M) \rightarrow \Omega^{k-1}(M), \quad \omega \mapsto \iota_\chi \omega \equiv \chi \lrcorner \omega, \quad (324)$$

which is the $(k-1)$ -form such that, for all $\{v_i\}_{i=1, \dots, k-1}$, $v_i \in \Gamma(TM)$,

$$(\iota_\chi \omega)(v_1, \dots, v_{k-1}) \doteq \omega(\chi, v_1, \dots, v_{k-1}).$$

We can prove a technical lemma.

Lemma 233. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$ and let

$$\beta : \Gamma(TM) \rightarrow \Omega^{n-1}(M), \quad \chi \mapsto \beta(\chi) = \chi \lrcorner \omega_g, \quad (325)$$

where $\lrcorner$ is defined in Equation (324). Let $S \subseteq M$ be an $(n-1)$ -dimensional, immersed hypersurface $(\iota_S : S \rightarrow M)$ with the orientation subordinated to the choice of a unit normal vector field N . Let h be the induced metric on S . For any vector field $\chi \in \Gamma(TM)$, calling $\chi|_S$ the restriction to S , it holds

$$\iota_S^*(\beta(\chi)) = \langle \chi|_S, N \rangle_g \omega_h,$$

where ι_S^* is the pullback as per Proposition 204 while $\langle \cdot, \cdot \rangle_h$ is the metric induced pairing on S as per Exercise 230.

Proof. Consider any $\chi \in \Gamma(TM)$ and indicate with $\chi|_S$ its restriction to S . Decompose this vector field as

$$\chi|_S = \chi^\perp + \chi^\top, \quad \chi^\perp \doteq \langle \chi|_S, N \rangle_g N, \quad \chi^\top \doteq \chi - \chi^\perp.$$

Using this decomposition and Equation (325), we obtain

$$\beta(\chi|_S) = \chi^\perp \lrcorner \omega_g + \chi^\top \lrcorner \omega_g.$$

Pulling back to S this expression, cf. Proposition 204, we get per linearity

$$\iota_S^* \beta(\chi|_S) = \langle \chi|_S, N \rangle_g \iota_S^*(N \lrcorner \omega_g) + \iota_S^*(\chi^\top \lrcorner \omega_g).$$

If we focus on the first term, there must exist $f \in C^\infty(S)$ such that $\iota_S^*(N \lrcorner \omega_g) = f\omega_h$. Yet $f = 1$ since, if we take an oriented orthonormal frame $(E_1, \dots, E_{n-1})$ of $T_p S$, $p \in S$, it descends that $(N, E_1, \dots, E_{n-1})$ is an oriented orthonormal frame of $T_p M$, where, with a slight abuse of notation, we omit to indicate the push-forward of the frame $\{E_i\}_{i=1, \dots, n-1}$ via ι_S . Hence

$$\iota_S^*(N \lrcorner \omega_g)(E_1, \dots, E_{n-1}) = \omega_g(N, E_1, \dots, E_{n-1}) = 1.$$

By Theorem 216, it descends that $\iota_S^*(N \lrcorner \omega_g)$ must coincide with the volume form on (S, h) , $h = \iota_S^* g$.

If we focus on the second term, consider once more an oriented orthonormal frame $(E_1, \dots, E_{n-1})$ of $T_p S$, $p \in S$. A direct evaluation shows that

$$\iota_S^*(\chi^\top \lrcorner \omega_g)(E_1, \dots, E_{n-1}) = \omega_g(\chi^\top, E_1, \dots, E_{n-1}) = 0,$$

since $\langle \chi^\top, N \rangle_g = 0$ and, consequently, at each $p \in S$, $\chi^\top$ must be a linear combination of $E_1, \dots, E_{n-1}$. $\square$

Definition 234. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. We call **divergence operator**

$$\operatorname{div} : \Gamma(TM) \rightarrow C^\infty(M), \quad \chi \mapsto \operatorname{div}(\chi) = \star^{-1} d_{n-1}(\beta(\chi)),$$

where $\star^{-1}$ is the inverse of the Hodge operator, while β is defined in Equation (325).

We can now prove the generalization of Gauss theorem to an arbitrary oriented smooth manifold.

Theorem 235 (Divergence Theorem). Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Then, for every $\chi \in \Gamma_0(TM)$,

$$\int_M \operatorname{div}(\chi) \omega_g = \int_{\partial M} \langle \chi, N \rangle_g \omega_h,$$

where ω_h is the metric-induced volume form on ∂M , N is the outward pointing unit vector field on ∂M , while $\langle \cdot, \cdot \rangle_g$ is the standard metric induced pairing on $\Gamma(TM)$.

Proof. Observe that, on account of Definition 234, $\operatorname{div}(\chi) = \star^{-1}d_{n-1}\beta(\chi) = \star d_{n-1}\beta(\chi)$. The last equality descends from Exercise 232. As a consequence of Definition 229 it holds that, since $\star(d_{n-1}\beta(\chi)) \in C^\infty(M)$,

$$\star(\star(d_{n-1}\beta(\chi))) = d_{n-1}\beta(\chi) = (\star d_{n-1}\beta(\chi))\omega_g.$$

Hence by Stokes Theorem 223 and by Lemma 233, it descends

$$\int_M (\operatorname{div}(\chi))\omega_g = \int_M (\star d_{n-1}\beta(\chi))\omega_g = \int_M d_{n-1}\beta(\chi) = \int_{\partial M} \iota_{\partial M}^* \beta(\chi) = \int_{\partial M} \langle \chi|_{\partial M}, N \rangle_g \omega_h.$$

□

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